MA 110
Lecture 01

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IIT Bombay

Spring 2025

Course Information

Course Code: MA 110

Course Name: Linear Algebra and Differential Equations

Instructors for the first part (Linear Algebra) are Professors Sudhir Ghorpade and Saurav Bhaumik.

Instructors for the second part (Differential Equations) are Professors Ronnie M. Sebastian and Mallesham Kummari.

Aim of the course: linear algebra

The first part of this core course aims at an introduction to the fundamental concepts of linear algebra, including systems of linear equations, matrices, linear transformations, vector spaces, eigenvalues, and eigenvectors.

Description/Syllabus of the Linear Algebra part

Vectors in $\mathbb{R}^n$, linear independence and dependence, linear span of a set of vectors, vector subspaces of $\mathbb{R}^n$, basis of a vector subspace. Systems of linear equations, matrices and Gauss elimination, row space, null space, and column space, rank of a matrix. Determinants and rank of a matrix in terms of determinants. Abstract vector spaces, linear transformations, matrix of a linear transformation, change of basis and similarity, rank-nullity theorem. Inner product spaces, Gram-Schmidt process, orthonormal bases, projections and least squares approximation. Eigenvalues and eigenvectors, characteristic polynomials, eigenvalues of special matrices (orthogonal, unitary, hermitian, symmetric, skew-symmetric, normal), algebraic and geometric multiplicity, diagonalization by similarity transformations, spectral theorem for real symmetric matrices, application to quadratic forms.

Basic Information (contd..)

Grading Policy

There will be two common quizzes (scheduled on 22 January and 12 February 2025) and one final exam.

The allotted 50 marks will be split as follows:

Common Quiz 1	:	10 marks
Common Quiz 2	:	10 marks
Final exam	:	30 marks

Attendance Policy

Attendance in lectures and tutorials is COMPULSORY. Students who do not meet 80% attendance will be awarded the DX grade.

Introduction to linear Algebra

Linear algebra deals with vectors and matrices and, more generally, with vector spaces and linear transformations.

Linear algebra is central to almost all areas of mathematics. For example, in geometry, basic objects such as lines, planes and rotations are represented in terms of linear algebra. Mathematical analysis applies linear algebra to function spaces.

Linear algebra is also used in most sciences and fields of engineering, because it allows modelling many natural phenomena, and computing efficiently with such models.

Introduction to linear algebra contd.

Linear algebra provides a vital arena where the interaction of Mathematics and machine computation is seen.

Many of the problems studied in Linear Algebra are amenable to systematic and even algorithmic solutions, and this makes them implementable on computers.

Numerous Applications within and outside Mathematics. For example, Google page rank algorithm is based on notions and results from Linear Algebra.

Notation

- $\mathbb{N} := \{1, 2, 3, \ldots\}$
- $\mathbb{Z} := \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
- $\mathbb{R} :=$ the set of all real numbers

For $n \in \mathbb{N}$, let us consider the **Euclidean space**

$$\mathbb{R}^n := \{(x_1, \ldots, x_n) : x_j \in \mathbb{R} \text{ for } j = 1, \ldots, n\}.$$

We let $\mathbf{0} := (0, \ldots, 0)$. Also, for $\boldsymbol{x} := (x_1, \ldots, x_n)$ and $\boldsymbol{y} := (y_1, \ldots, y_n)$ in $\mathbb{R}^n$, and for $\alpha \in \mathbb{R}$, we define

$$\begin{aligned} \text{(sum)} \quad \boldsymbol{x} + \boldsymbol{y} &:= (x_1 + y_1, \ldots, x_n + y_n) \in \mathbb{R}^n, \\ \text{(scalar multiple)} \quad \alpha\,\boldsymbol{x} &:= (\alpha\, x_1, \ldots, \alpha\, x_n) \in \mathbb{R}^n, \\ \text{(scalar product)} \quad \boldsymbol{x} \cdot \boldsymbol{y} &:= x_1 y_1 + \cdots + x_n y_n \in \mathbb{R}. \end{aligned}$$

Matrices

Let $m, n \in \mathbb{N}$. An $m \times n$ **matrix A** with real entries is a rectangular array of real numbers arranged in m rows and n columns, written as follows:

$$\mathbf{A} := \begin{bmatrix} a_{11} & \cdots & a_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{j1} & \cdots & \boxed{a_{jk}} & \cdots & a_{jn} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mk} & \cdots & a_{mn} \end{bmatrix} = [a_{jk}],$$

where $a_{jk} \in \mathbb{R}$ is called the (j,k)th **entry** of **A** for $j = 1, \ldots, m$ and $k = 1, \ldots, n$.

Let $\boxed{\mathbb{R}^{m \times n}}$ denote the set of all $m \times n$ matrices with real entries. If $\mathbf{A} := [a_{jk}]$ and $\mathbf{B} := [b_{jk}]$ are in $\mathbb{R}^{m \times n}$, then we say $\mathbf{A} = \mathbf{B} \iff a_{jk} = b_{jk}$ for all $j = 1, \ldots, m$ and $k = 1, \ldots, n$.

Let $0 \leq r < m$, $0 \leq s < n$. By deleting r rows and s columns from $\mathbf{A}$, we obtain an $(m-r)\times(n-s)$ **submatrix** of $\mathbf{A}$.

An $n \times n$ matrix, that is, an element of $\mathbb{R}^{n\times n}$, is called a **square matrix** of size n.

- A square matrix $\mathbf{A} = [a_{jk}]$ is called **symmetric** if $a_{jk} = a_{kj}$ for all j, k.
- A square matrix $\mathbf{A} = [a_{jk}]$ is called **skew-symmetric** if $a_{jk} = -a_{kj}$ for all j, k.
- A square matrix $\mathbf{A} = [a_{jk}]$ is called a **diagonal matrix** if $a_{jk} = 0$ for all $j \neq k$.
- A diagonal matrix $\mathbf{A} = [a_{jk}]$ is called a **scalar matrix** if all diagonal entries of $\mathbf{A}$ are equal.

Two important scalar matrices are the **identity matrix I** in which all diagonal elements are equal to 1, and the **zero matrix O** in which all diagonal elements are equal to 0.

A square matrix $\mathbf{A} = [a_{jk}]$ is called **upper triangular** if $a_{jk} = 0$ for all $j > k$, and **lower triangular** if $a_{jk} = 0$ for all $j < k$.

Note: A matrix $\mathbf{A}$ is upper triangular as well as lower triangular if and only if $\mathbf{A}$ is a diagonal matrix.

Examples

The matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ is symmetric, while the matrix

$\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$ is skew-symmetric.

Note: Every diagonal entry of a skew-symmetric matrix is 0 since $a_{jj} = -a_{jj} \implies a_{jj} = 0$ for $j = 1, \ldots, n$.

Examples

The matrix $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ is diagonal, while $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is a scalar matrix.

The matrix $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{bmatrix}$ is upper triangular,

while the matrix $\begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & 4 \end{bmatrix}$ is lower triangular.

A **row vector a** of length n is a matrix with only one row consisting of n real numbers; it is written as follows:

$$\mathbf{a} = \begin{bmatrix} a_1 & \cdots & a_k & \cdots & a_n \end{bmatrix},$$

where $a_k \in \mathbb{R}$ for $k = 1, \ldots, n$. Here $\mathbf{a} \in \mathbb{R}^{1 \times n}$.

A **column vector b** of length n is a matrix with only one column consisting of n real numbers; it is written as follows:

$$\mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_k \\ \vdots \\ b_n \end{bmatrix},$$ where $b_k \in \mathbb{R}$ for $k = 1, \ldots, n$. Here $\mathbf{b} \in \mathbb{R}^{n \times 1}$.

When $n = 1$, we may identify $[\alpha] \in \mathbb{R}^{1 \times 1}$ with $\alpha \in \mathbb{R}$.

Operations on Matrices

Let $m, n \in \mathbb{N}$, and let $\mathbf{A} := [a_{jk}]$ and $\mathbf{B} := [b_{jk}]$ be $m \times n$ matrices. Then the $m \times n$ matrix $\mathbf{A} + \mathbf{B} := [a_{jk} + b_{jk}]$ is called the **sum** of $\mathbf{A}$ and $\mathbf{B}$. Also, if $\alpha \in \mathbb{R}$, then the $m \times n$ matrix $\alpha\mathbf{A} := [\alpha a_{jk}]$ is called the **scalar multiple** of $\mathbf{A}$ by α.

These operations follow the usual rules: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$, $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$, which we write as $\mathbf{A} + \mathbf{B} + \mathbf{C}$, $\alpha(\mathbf{A} + \mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}$, $(\alpha + \beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{A}$ and $\alpha(\beta\mathbf{A}) = (\alpha\beta)\mathbf{A}$, which we write as $\alpha\beta\mathbf{A}$. Also, we write $(-1)\mathbf{A}$ as $-\mathbf{A}$, and $\mathbf{A} + (-\mathbf{B})$ as $\mathbf{A} - \mathbf{B}$.

The **transpose** of an $m \times n$ matrix $\mathbf{A} := [a_{jk}]$ is the $n \times m$ matrix $\mathbf{A}^{\mathsf{T}} := [a_{kj}]$ (in which the rows and the columns of $\mathbf{A}$ are interchanged).

Clearly, $(\mathbf{A}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{A}$, $(\mathbf{A} + \mathbf{B})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}} + \mathbf{B}^{\mathsf{T}}$ and $(\alpha\mathbf{A})^{\mathsf{T}} = \alpha\mathbf{A}^{\mathsf{T}}$.

Note: A square matrix $\mathbf{A}$ is symmetric $\iff \mathbf{A}^{\mathsf{T}} = \mathbf{A}$.

In particular, the preceding operations can be performed on row vectors, and also on column vectors since they are particular types of matrices.

Notice that the sum $\mathbf{a}_1 + \mathbf{a}_2$ of two row vectors $\mathbf{a_1}$ and $\mathbf{a_2}$ of the same length follows the parallelogram law, and so does the sum $\mathbf{b}_1 + \mathbf{b}_2$ of two column vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ of the same length. (Note: All vectors 'originate' from the zero vector.)

Also, note that the transpose of a row vector is a column vector, and vice versa.

We shall often write a column vector $\mathbf{b} := \begin{bmatrix} b_1 \\ \vdots \\ b_k \\ \vdots \\ b_n \end{bmatrix} \in \mathbb{R}^{n\times 1}$, as

$\begin{bmatrix} b_1 & \cdots & b_k & \cdots & b_n \end{bmatrix}^\mathsf{T}$ in order to save space.

Let $m, n \in \mathbb{N}$. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$.

If $\mathbf{a}_1, \ldots, \mathbf{a}_m \in \mathbb{R}^{1\times n}$, then

$$\alpha_1 \mathbf{a}_1 + \cdots + \alpha_m \mathbf{a}_m \in \mathbb{R}^{1\times n}$$

is called a **(finite) linear combination** of $\mathbf{a}_1, \ldots, \mathbf{a}_m$.

Similarly, if $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^{n\times 1}$, then

$$\alpha_1 \mathbf{b}_1 + \cdots + \alpha_m \mathbf{b}_m \in \mathbb{R}^{n\times 1}$$

is a **(finite) linear combination** of $\mathbf{b}_1, \ldots, \mathbf{b}_m$.

In particular, for $k = 1, \ldots, n$, consider the column vector $\mathbf{e}_k := \begin{bmatrix} 0 & \cdots & 1 & \cdots & 0 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n \times 1}$, where the kth entry is 1 and all other entries are 0.

If $\mathbf{b} = \begin{bmatrix} b_1 & \cdots & b_k & \cdots & b_n \end{bmatrix}^{\mathsf{T}}$ is any column vector of length n, then it follows that $\mathbf{b} = b_1\mathbf{e}_1 + \cdots + b_k\mathbf{e}_k + \cdots + b_n\mathbf{e}_n$, which is a linear combination of $\mathbf{e}_1, \ldots, \mathbf{e}_n$. The vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are known as the **basic column vectors** in $\mathbb{R}^{n \times 1}$.

Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m \times n}$.

Then $\mathbf{a}_j := \begin{bmatrix} a_{j1} & \cdots & a_{jn} \end{bmatrix} \in \mathbb{R}^{1 \times n}$ is called the jth **row vector of A** for $j = 1, \ldots, m$, and we write $\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$.

Also, $\mathbf{c}_k := \begin{bmatrix} a_{1k} & \cdots & a_{mk} \end{bmatrix}^{\mathsf{T}}$ is called the kth **column vector of A** for $k = 1, \ldots, n$, and we write $\mathbf{A} = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_n \end{bmatrix}$.

Examples

Let $\mathbf{A} := \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}$ and $\mathbf{B} := \begin{bmatrix} 1 & 0 & 2 \\ -1 & 4 & 1 \end{bmatrix}$.

Then $\mathbf{A} + \mathbf{B} := \begin{bmatrix} 3 & 1 & 1 \\ -1 & 7 & 2 \end{bmatrix}$ and $5\mathbf{A} = \begin{bmatrix} 10 & 5 & -5 \\ 0 & 15 & 5 \end{bmatrix}$.

The row vectors of $\mathbf{A}$ are $\begin{bmatrix} 2 & 1 & -1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 3 & 1 \end{bmatrix}$.

The column vectors of $\mathbf{A}$ are $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Also, $\mathbf{A}^{\mathsf{T}} = \begin{bmatrix} 2 & 0 \\ 1 & 3 \\ -1 & 1 \end{bmatrix}$.

Matrix multiplication

So if $m, n, p \in \mathbb{N}$, $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m \times n}$ and $\mathbf{B} := [b_{jk}] \in \mathbb{R}^{n \times p}$, then $\mathbf{AB} \in \mathbb{R}^{m \times p}$, and for $j = 1, \ldots, m$; $k = 1 \ldots, p$,

$$\mathbf{AB} = [c_{jk}], \quad \text{where } c_{jk} := \mathbf{a}_j \mathbf{b}_k = \sum_{\ell=1}^{n} a_{j\ell} b_{\ell k}.$$

Note that the (j, k)th entry of $\mathbf{AB}$ is a product of the jth row vector of $\mathbf{A}$ with the kth column vector of $\mathbf{B}$ as shown below:

$$\begin{bmatrix} & & & & \\ a_{j1} & \cdots & a_{j\ell} & \cdots & a_{jn} \\ & & & & \end{bmatrix} \begin{bmatrix} & b_{1k} & \\ & \vdots & \\ & b_{\ell k} & \\ & \vdots & \\ & b_{nk} & \end{bmatrix}$$

Clearly, the product $\mathbf{AB}$ is defined only when the number of columns of $\mathbf{A}$ is equal to the number of rows of $\mathbf{B}$.

Note that $\mathbf{AI} = \mathbf{A}, \mathbf{IA} = \mathbf{A}, \mathbf{AO} = \mathbf{O}$ and $\mathbf{OA} = \mathbf{O}$ whenever these products are defined.

Examples

(i) Let $\mathbf{A} := \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}_{2\times 3}$ and $\mathbf{B} := \begin{bmatrix} 1 & 6 & 0 & 2 \\ 2 & -1 & 1 & -2 \\ 2 & 0 & -1 & 1 \end{bmatrix}_{3\times 4}$.

Then $\mathbf{AB} = \begin{bmatrix} 2 & 11 & 2 & 1 \\ 8 & -3 & 2 & -5 \end{bmatrix}_{2\times 4}$.

(ii) In general, $\mathbf{AB} \neq \mathbf{BA}$. For example, if $\mathbf{A} := \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then $\mathbf{AB} := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, while $\mathbf{BA} := \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Note that $\mathbf{BA} = \mathbf{O}$, while $\mathbf{AB} = \mathbf{B} \neq \mathbf{O}$. Since $\mathbf{A} \neq \mathbf{I}$, we see that the so-called cancellation law does not hold.

Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m \times n}$, and let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the basic column vectors in $\mathbb{R}^{n \times 1}$. Then for $k = 1, \ldots, n$,

$$\mathbf{A}\,\mathbf{e}_k = \begin{bmatrix} a_{1k} \\ \vdots \\ a_{jk} \\ \vdots \\ a_{mk} \end{bmatrix}, \text{ which is the } k\text{th column of } \mathbf{A}.$$

Properties of Matrix Multiplication

Consider matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\alpha \in \mathbb{R}$. Then it is easy to see that $(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}$, $\mathbf{C}(\mathbf{A}+\mathbf{B}) = \mathbf{CA}+\mathbf{CB}$ and $(\alpha\mathbf{A})\mathbf{B} = \alpha\mathbf{AB} = \mathbf{A}(\alpha\mathbf{B})$, if sums & products are well-defined.

Matrix multiplication also satisfies the **associative law**:

Proposition

Let $m, n, p, q \in \mathbb{N}$. If $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times p}$ and $\mathbf{C} \in \mathbb{R}^{p\times q}$, then $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$ (which we shall write as $\mathbf{ABC}$).

Proof. Let $\mathbf{A} := [a_{jk}]$, $\mathbf{B} := [b_{jk}]$ and $\mathbf{C} := [c_{jk}]$. Also, let $(\mathbf{AB})\mathbf{C} := [\alpha_{jk}]$ and $\mathbf{A}(\mathbf{BC}) := [\beta_{jk}]$. Then

$$\alpha_{jk} = \sum_{i=1}^{p}\left(\sum_{\ell=1}^{n} a_{j\ell}b_{\ell i}\right)c_{ik} = \sum_{\ell=1}^{n} a_{j\ell}\left(\sum_{i=1}^{p} b_{\ell i}c_{ik}\right) = \beta_{jk}$$

for $j = 1, \ldots, m$ and $k = 1, \ldots, q$. Hence the result. $\square$

Also, the transpose of a product is the product of the transposes in the reverse order:

Proposition

Let $m, n, p \in \mathbb{N}$. If $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$, then $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$.

Proof. Let $\mathbf{A} := [a_{jk}]$, $\mathbf{B} := [b_{jk}]$ and $\mathbf{AB} := [c_{jk}]$.

Also, let $\mathbf{A}^{\mathsf{T}} := [a'_{jk}]$, $\mathbf{B}^{\mathsf{T}} := [b'_{jk}]$ and $(\mathbf{AB})^{\mathsf{T}} := [c'_{jk}]$. Then

$$c_{jk} = \sum_{\ell=1}^{n} a_{j\ell} b_{\ell k} \quad \text{and so} \quad c'_{jk} = c_{kj} = \sum_{\ell=1}^{n} a_{k\ell} b_{\ell j}$$

for $j = 1, \ldots, m; k = 1, \ldots, p$. Suppose $\mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}} := [d_{jk}]$. Then

$$d_{jk} = \sum_{\ell=1}^{n} b'_{j\ell} a'_{\ell k} = \sum_{\ell=1}^{n} b_{\ell j} a_{k\ell} = c'_{jk}$$

for $j = 1, \ldots, m; k = 1, \ldots, p$. Hence the result. $\square$

Matrix Multiplication Revisited

Let $m, n, p \in \mathbb{N}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$. For $j = 1, \ldots, m$, let $\mathbf{a}_j := \begin{bmatrix} a_{j1} & \cdots & a_{jn} \end{bmatrix}$ be the jth row of $\mathbf{A}$, and for $k = 1, \ldots, p$, let $\mathbf{b}_k := \begin{bmatrix} b_{1k} \\ \vdots \\ b_{nk} \end{bmatrix}$ be the kth column of $\mathbf{B}$. Then $\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$, in terms of its rows. Also, $\mathbf{B} = \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{bmatrix}$, in terms of its columns. Note: $\mathbf{AB}$ has m rows and p columns.

Rows of AB

Fix $j \in \{1, \ldots, m\}$, and consider the jth row of $\mathbf{AB}$, namely, $\mathbf{a}_j\mathbf{B} = [\mathbf{a}_j\mathbf{b}_1 \cdots \mathbf{a}_j\mathbf{b}_p]$. For $k = 1, \ldots, p$, the kth entry of the jth row of $\mathbf{AB}$ is $\mathbf{a}_j\mathbf{b}_k = a_{j1}b_{1k} + \cdots + a_{jn}b_{nk}$, where $b_{1k}, \ldots, b_{nk}$ are the kth entries of the n row vectors of $\mathbf{B}$.

Thus we see that for $j = 1, \ldots, m$, the jth row of $\mathbf{AB}$ is a linear combination of the n row vectors of $\mathbf{B}$ with coefficients $a_{j1}, \ldots, a_{jn}$ provided by the jth row of $\mathbf{A}$.

Columns of **AB**

Fix $k \in \{1, \ldots, p\}$, and consider the kth column of $\mathbf{AB}$, namely, $\mathbf{Ab}_k = \begin{bmatrix} \mathbf{a}_1\mathbf{b}_k \\ \vdots \\ \mathbf{a}_m\mathbf{b}_k \end{bmatrix}$. For $j = 1, \ldots, m$, the jth entry of the kth column of $\mathbf{AB}$ is $\mathbf{a}_j\mathbf{b}_k = a_{j1}b_{1k} + \cdots + a_{jn}b_{nk}$, that is, $b_{1k}a_{j1} + \cdots + b_{nk}a_{jn}$, where $a_{j1}, \ldots, a_{jn}$ are the jth entries of the n columns of $\mathbf{A}$.

Thus we see that for $k = 1, \ldots, n$, the kth column of $\mathbf{AB}$ is a linear combination of the n column vectors of $\mathbf{A}$ with coefficients $b_{1k}, \ldots, b_{nk}$ provided by the kth column of $\mathbf{B}$.

These descriptions of the rows and columns of $\mathbf{AB}$ are useful.

Example

As we have seen,

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 6 & 0 & 2 \\ 2 & -1 & 1 & -2 \\ 2 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 11 & 2 & 1 \\ 8 & -3 & 2 & -5 \end{bmatrix}, \quad \text{where}$$

$$\begin{aligned}
\begin{bmatrix} 2 & 11 & 2 & 1 \end{bmatrix} &= 2\begin{bmatrix} 1 & 6 & 0 & 2 \end{bmatrix} + 1\begin{bmatrix} 2 & -1 & 1 & -2 \end{bmatrix} \\
&\quad -1\begin{bmatrix} 2 & 0 & -1 & 1 \end{bmatrix}, \\
\begin{bmatrix} 8 & -3 & 2 & -5 \end{bmatrix} &= 0\begin{bmatrix} 1 & 6 & 0 & 2 \end{bmatrix} + 3\begin{bmatrix} 2 & -1 & 1 & -2 \end{bmatrix} \\
&\quad +1\begin{bmatrix} 2 & 0 & -1 & 1 \end{bmatrix}, \\
\begin{bmatrix} 2 \\ 8 \end{bmatrix} &= 1\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \\
\begin{bmatrix} 11 \\ -3 \end{bmatrix} &= 6\begin{bmatrix} 2 \\ 0 \end{bmatrix} - 1\begin{bmatrix} 1 \\ 3 \end{bmatrix} + 0\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \text{etc.}
\end{aligned}$$

.

Linear System

Let $m, n \in \mathbb{N}$. A **linear system** of m equations in the n unknowns $x_1, ..., x_n$ is given by

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 && (1)\\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 && (2)\\
\vdots \qquad \vdots \qquad \vdots \qquad\quad &\vdots \quad \vdots\\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, && (m)
\end{aligned}
$$

where $a_{jk} \in \mathbb{R}$ for $j = 1, \ldots, m; k = 1, \ldots, n$ and also $b_j \in \mathbb{R}$ for $j = 1, \ldots, m$ are given.

Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m \times n}$, $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n \times 1}$ and $\mathbf{b} := \begin{bmatrix} b_1 & \cdots & b_m \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{m \times 1}$. Using matrix multiplication, we write the linear system as

$$\mathbf{A}\mathbf{x} = \mathbf{b}.$$

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Linear System

Let $m, n \in \mathbb{N}$. A **linear system** of m equations in the n unknowns $x_1, ..., x_n$ is given by

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a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \qquad (1)\\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \qquad (2)\\
\vdots \qquad \vdots \qquad \vdots \qquad\quad &\vdots \quad \vdots\\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \qquad (m)
\end{aligned}
$$

where $a_{jk} \in \mathbb{R}$ for $j = 1, \ldots, m$; $k = 1, \ldots, n$ and also $b_j \in \mathbb{R}$ for $j = 1, \ldots, m$ are given.

Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m \times n}$, $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n \times 1}$ and $\mathbf{b} := \begin{bmatrix} b_1 & \cdots & b_m \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{m \times 1}$. Using matrix multiplication, we write the linear system as

$$\mathbf{A}\mathbf{x} = \mathbf{b}.$$

The $m \times n$ matrix $\mathbf{A}$ is known as the **coefficient matrix** of the linear system.

A column vector $\mathbf{x}_0 \in \mathbb{R}^{n \times 1}$ is called a **solution** of the above linear system if it satisfies $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$.

Case (i) **Homogeneous Linear System**: $\mathbf{b} := \mathbf{0}$, that is, $b_1 = \cdots = b_m = 0$.
A homogenous linear system always has a solution, namely the zero solution $\mathbf{0} := \begin{bmatrix} 0 & \cdots & 0 \end{bmatrix}^{\mathsf{T}}$ since $\mathbf{A}\mathbf{0} = \mathbf{0}$.

Also, if $r \in \mathbb{N}$ and $\mathbf{x}_1, \ldots, \mathbf{x}_r$ are solutions of such a system, then so is their linear combination $\alpha_1 \mathbf{x}_1 + \cdots + \alpha_r \mathbf{x}_r$, since $\mathbf{A}(\alpha_1 \mathbf{x}_1 + \cdots + \alpha_r \mathbf{x}_r) = \alpha_1 \mathbf{A}\mathbf{x}_1 + \cdots + \alpha_r \mathbf{A}\mathbf{x}_r = \mathbf{0} + \cdots + \mathbf{0} = \mathbf{0}$.

Case (ii) **General Linear System**: $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is arbitrary.

A nonhomogenous linear system, that is, where $\mathbf{b} \neq \mathbf{0}$, may not have a solution, may have only one solution or may have (infinitely) many solutions.

Examples

(i) The linear system $x_1 + x_2 = 1$, $2x_1 + 2x_2 = 1$ does not have a solution.

(ii) The linear system $x_1 + x_2 = 1$, $x_1 - x_2 = 0$ has a unique solution, namely $x_1 = 1/2 = x_2$.

(iii) The linear system $x_1 + x_2 = 1$, $2x_1 + 2x_2 = 2$ has (infinitely) many solutions, namely $x_1 = \alpha$, $x_2 = 1 - \alpha$, $\alpha \in \mathbb{R}$.

Important Note:

Let S denote the set of all solutions of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$. If $\mathbf{x}_0$ is a particular solution of the general system $\mathbf{Ax} = \mathbf{b}$, then the set of all solutions of the general system $\mathbf{Ax} = \mathbf{b}$ is given by $\{\mathbf{x}_0 + \mathbf{s} : \mathbf{s} \in S\}$ since

$\mathbf{s} \in S \implies \mathbf{A}(\mathbf{x}_0 + \mathbf{s}) = \mathbf{Ax}_0 + \mathbf{As} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, and also

$\mathbf{Ax} = \mathbf{b} \implies \mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{Ax} - \mathbf{Ax}_0 = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so that $(\mathbf{x} - \mathbf{x}_0) \in S$, that is, $\mathbf{x} = \mathbf{x_0} + \mathbf{s}$ for some $\mathbf{s} \in S$.

We shall, therefore, address the problem of finding all solutions of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$, and one particular solution of the corresponding general system $\mathbf{Ax} = \mathbf{b}$.

A Special Case

Suppose the coefficient matrix $\mathbf{A}$ is upper triangular and its diagonal elements are nonzero. Then the linear system

$$\begin{array}{rcll}
a_{11}x_1 + a_{12}x_2 + \cdots + \cdots + \cdots + \cdots + a_{1n}x_n & = & b_1 & (1) \\
a_{22}x_2 + a_{23}x_3 + \cdots + \cdots + \cdots + a_{2n}x_n & = & b_2 & (2) \\
\ddots \qquad\qquad\qquad\qquad \vdots & \vdots & \vdots & \\
a_{(n-1)(n-1)}x_{n-1} + a_{(n-1)n}x_n & = & b_{n-1} & (n-1) \\
a_{nn}x_n & = & b_n & (n)
\end{array}$$

can be solved by back substitution as follows.

$$
\begin{aligned}
x_n &= b_n/a_{nn} \\
x_{n-1} &= (b_{n-1} - a_{(n-1)n}x_n)/a_{(n-1)(n-1)}, \text{ where } x_n = b_n/a_{nn} \\
\vdots\ &\ \vdots\ \ \vdots \\
x_2 &= (b_2 - a_{2n}x_n - \cdots - a_{23}x_3)/a_{22}, \text{where } x_n = \cdots, x_3 = \cdots, \\
x_1 &= (b_1 - a_{1n}x_n - \cdots - \cdots - a_{12}x_2)/a_{11}, \text{where } x_n = \cdots, x_2 = \cdots
\end{aligned}
$$

Here the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution and the general system $\mathbf{Ax} = \mathbf{b}$ has a unique solution.

Taking a cue from this special case of an upper triangular matrix, we shall attempt to transform any $m \times n$ matrix to an upper triangular form. In this process, we successively attempt to eliminate the unknown x_1 from the equations $(m), \ldots, (2)$, the unknown x_2 from the equations $(m), \ldots, (3)$, and so on.

Example

Consider the linear system

$$\begin{aligned} x_1 - x_2 + x_3 &= 0 \\ -x_1 + x_2 - x_3 &= 0 \\ 10x_2 + 25x_3 &= 90 \\ 20x_1 + 10x_2 &= 80. \end{aligned}$$

Eliminating x_1 from the 4th, 3rd and 2nd equations,

$$\begin{aligned} x_1 - x_2 + x_3 &= 0 \\ 0 &= 0 \\ 10x_2 + 25x_3 &= 90 \\ 30x_2 - 20x_3 &= 80. \end{aligned}$$

Interchanging the 2nd and the 3rd equations,

$$\begin{aligned} x_1 - x_2 + x_3 &= 0 \\ 10x_2 + 25x_3 &= 90 \\ 0 &= 0 \\ 30x_2 - 20x_3 &= 80. \end{aligned}$$

Eliminating x_2 from the 4th equation, and then interchanging the 3rd and the 4th equations,

$$\begin{aligned} x_1 - x_2 + x_3 &= 0 \\ 10x_2 + 25x_3 &= 90 \\ -95x_3 &= -190 \\ 0 &= 0. \end{aligned}$$

Now back substitution gives $x_3 = 2$, $x_2 = (90 - 25x_3)/10 = 4$ and $x_1 = -x_3 + x_2 = 2$, that is, $\mathbf{x} = \begin{bmatrix} 2 & 4 & 2 \end{bmatrix}^T$.

The above process can be carried out without writing down the entire linear system by considering the **augmented matrix**

$$[\mathbf{A}|\mathbf{b}] := \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{12} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_n \end{array}\right].$$

This $m \times (n+1)$ matrix completely describes the linear system. In the above example,

$$[\mathbf{A}|\mathbf{b}] = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ 0 & 10 & 25 & 90 \\ 20 & 10 & 0 & 80 \end{array}\right].$$

Subtracting 20 times the first row from the 4th row, and adding the first row to the second row, we obtain

$$\xrightarrow{R_4-20R_1,\, R_2+R_1,} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 10 & 25 & 90 \\ 0 & 30 & -20 & 80 \end{array}\right].$$

Interchanging the 2nd and the 3rd rows, we obtain

$$\xrightarrow{R_2 \longleftrightarrow R_3,} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 10 & 25 & 90 \\ 0 & 0 & 0 & 0 \\ 0 & 30 & -20 & 80 \end{array}\right].$$

Finally, subtracting 3 times the 2nd row from the 4th row and interchanging the 3rd and the 4th rows, we arrive at

$$\xrightarrow{R_4-3R_2,\, R_3 \longleftrightarrow R_4} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 10 & 25 & 90 \\ 0 & 0 & -95 & -190 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The upper triangular nature of the 3×3 matrix on the top left enables back substitution.

Row Echelon Form

We shall now consider a general form of a matrix for which the method of back substitution works.

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, that is, $\mathbf{A}$ is an $m \times n$ matrix with real entries.

A row of $\mathbf{A}$ is said to be **zero** if all its entries are zero.

If a row is not zero, then its first nonzero entry (from the left) is called the **pivot**.Thus all entries to the left of a pivot equal 0.

Suppose $\mathbf{A}$ has r nonzero rows and $m - r$ zero rows.
Then $0 \leq r \leq m$. Clearly, $r = 0 \iff \mathbf{A} = \mathbf{O}$.

Example

If $\mathbf{A} = \begin{bmatrix} 0 & \boxed{1} & 4 \\ 0 & 0 & 0 \\ \boxed{5} & 6 & 7 \end{bmatrix}$, then $m = n = 3$ and $r = 2$.

The pivot in the 1st row is 1 and the pivot in the 3rd row is 5.

A matrix $\mathbf{A}$ is said to be in a row echelon form (REF)[1] if the following conditions are satisfied.

(i) The nonzero rows of $\mathbf{A}$ precede the zero rows of $\mathbf{A}$.

(ii) If $\mathbf{A}$ has r nonzero rows, where $r \in \mathbb{N}$, and the pivot in row 1 appears in the column k_1, the pivot in row 2 appears in the column k_2, and so on the pivot in row r appears in the column k_r, then $\boxed{k_1 < k_2 < \cdots < k_r}$.

Examples

The matrices $\begin{bmatrix} 0 & \boxed{1} & 4 \\ 0 & 0 & 0 \\ \boxed{5} & 6 & 7 \end{bmatrix}$, $\begin{bmatrix} 0 & \boxed{1} & 4 \\ \boxed{5} & 6 & 7 \\ 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & \boxed{1} & 4 \\ 0 & \boxed{5} & 7 \\ 0 & 0 & 0 \end{bmatrix}$

are not in REF. The matrix $\begin{bmatrix} \boxed{5} & 6 & 7 \\ 0 & \boxed{1} & 4 \\ 0 & 0 & 0 \end{bmatrix}$ is in a REF.

[1] In French, echelon means level.

Pivotal Columns

(i) Suppose a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ is in a REF. If $\mathbf{A}$ has exactly r nonzero rows, then there are exactly r pivots. A column of $\mathbf{A}$ containing a pivot, is called a **pivotal column**. Thus there are exactly r pivotal columns, and so $0 \le r \le n$.
(We have already noted that $0 \le r \le m$.)

(ii) In a pivotal column, all entries below the pivot equal 0.

Here is a typical example of how back substitution works when a matrix $\mathbf{A}$ is in a REF. Let

$$\mathbf{A} := \begin{bmatrix} 0 & \boxed{a_{12}} & a_{13} & a_{14} & a_{15} & a_{16} \\ 0 & 0 & 0 & \boxed{a_{24}} & a_{25} & a_{26} \\ 0 & 0 & 0 & 0 & \boxed{a_{35}} & a_{36} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{b} := \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}.$$

where a_{12}, a_{24}, a_{35} are nonzero. They are the pivots.

Here $m = 4$, $n = 6$, $r = 3$, pivotal columns: $\boxed{2}$, $\boxed{4}$ and $\boxed{5}$.

Suppose there is $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_6 \end{bmatrix}^\mathsf{T} \in \mathbb{R}^{6\times 1}$ such that $\mathbf{Ax} = \mathbf{b}$. Then $0x_1 + \cdots + 0x_6 = b_4$, that is, b_4 must be equal to 0.

Next, $a_{35}x_5 + a_{36}x_6 = b_3$, that is, $x_5 = (b_3 - a_{36}x_6)/a_{35}$, where we can assign an arbitrary value to the unknown $\boxed{x_6}$.

Next, $a_{24}x_4 + a_{25}x_5 + a_{26}x_6 = b_2$, that is, $x_4 = (b_2 - a_{25}x_5 - a_{26}x_6)/a_{24}$, where we back substitute the values of x_5 and x_6.

Finally, $a_{12}x_2 + a_{13}x_3 + a_{14}x_4 + a_{15}x_5 + a_{16}x_6 = b_1$, that is, $x_2 = (b_1 - a_{13}x_3 - a_{14}x_4 - a_{15}x_5 - a_{16}x_6)/a_{12}$, where we can assign an arbitrary value to the variable $\boxed{x_3}$, and back substitute the values of x_4, x_5 and x_6.
Also, we can assign an arbitrary value to the variable $\boxed{x_1}$.

The variables x_1, x_3 and x_6 to which we can assign arbitrary values correspond to the nonpivotal columns $\boxed{1}$, $\boxed{3}$ and $\boxed{6}$.

Suppose an $m \times n$ matrix $\mathbf{A}$ is in a REF, and there are r nonzero rows. Let the r pivots be in the columns $k_1, \ldots, k_r$ with $k_1 < \cdots < k_r$, and let the columns $\ell_1, \ldots, \ell_{n-r}$ be nonpivotal. Then $\boxed{x_{k_1}, \ldots, x_{k_r}}$ are called the **pivotal variables** and $\boxed{x_{\ell_1}, \ldots, x_{\ell_{n-r}}}$ are called the **free variables**.

Let $\mathbf{b} := [b_1 \cdots b_r \; b_{r+1} \cdots b_m]^\mathsf{T}$, and consider the linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}.$$

Important Observations

1. The linear system has a solution $\iff b_{r+1} = \cdots = b_m = 0$. This is known as the **consistency condition**.
2. Let the consistency condition $b_{r+1} = \cdots = b_m = 0$ be satisfied. Then we obtain a **particular solution** $\mathbf{x}_0 := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T}$ of the linear system by letting $x_k := 0$ if $k \in \{\ell_1, \ldots, \ell_{n-r}\}$, and then by determining the pivotal variables $x_{k_1}, \ldots, x_{k_r}$ by back substitution.

3. We obtain $n-r$ **basic solutions** of the homogeneous linear system $\mathbf{Ax}=\mathbf{0}$ as follows. Fix $\ell \in \{\ell_1,\ldots,\ell_{n-r}\}$. Define $\mathbf{s}_\ell := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$ by $x_k := 1$ if $k=\ell$, while $x_k := 0$ if $k \in \{\ell_1,\ldots,\ell_{n-r}\}$ but $k \neq \ell$. Then determine the pivotal variables $x_{k_1},\ldots,x_{k_r}$ by back substitution.

4. Let $\mathbf{s} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n\times 1}$ be any solution of the homogeneous system, that is, $\mathbf{As}=\mathbf{0}$. Then $\mathbf{s}$ is a linear combination of the $n-r$ basic solutions $\mathbf{s}_{\ell_1},\ldots,\mathbf{s}_{\ell_{n-r}}$. To see this, let $\mathbf{y} := \mathbf{s} - x_{\ell_1}\mathbf{s}_{\ell_1} - \cdots - x_{\ell_{n-r}}\mathbf{s}_{\ell_{n-r}}$. Then $\mathbf{Ay} = \mathbf{As} - x_{\ell_1}\mathbf{As}_{\ell_1} - \cdots - x_{\ell_{n-r}}\mathbf{As}_{\ell_{n-r}} = \mathbf{0}$, and moreover, the kth entry of $\mathbf{y}$ is 0 for each $k \in \{\ell_1,\ldots,\ell_{n-r}\}$. It then follows that $\mathbf{y}=\mathbf{0}$, that is, $\mathbf{s} = x_{\ell_1}\mathbf{s}_{\ell_1} + \cdots + x_{\ell_{n-r}}\mathbf{s}_{\ell_{n-r}}$. Thus we find that the general solution of the homogeneous system is given by

$$\boxed{\mathbf{s} = \alpha_1\mathbf{s}_{\ell_1} + \cdots + \alpha_{n-r}\mathbf{s}_{\ell_{n-r}}, \text{ where } \alpha_1,\ldots,\alpha_{n-r} \in \mathbb{R}}.$$

MA 110:
Lecture 03

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Spring 2025

Suppose an $m \times n$ matrix $\mathbf{A}$ is in a REF, and there are r nonzero rows. Let the r pivots be in the columns $k_1, \ldots, k_r$ with $k_1 < \cdots < k_r$, and let the columns $\ell_1, \ldots, \ell_{n-r}$ be nonpivotal. Then $\boxed{x_{k_1}, \ldots, x_{k_r}}$ are called the **pivotal variables** and $\boxed{x_{\ell_1}, \ldots, x_{\ell_{n-r}}}$ are called the **free variables**.

Let $\mathbf{b} := [b_1 \cdots b_r \; b_{r+1} \cdots b_m]^{\mathsf{T}}$, and consider the linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}.$$

Important Observations

1. The linear system has a solution $\iff b_{r+1} = \cdots = b_m = 0$. This is known as the **consistency condition**.

2. Let the consistency condition $b_{r+1} = \cdots = b_m = 0$ be satisfied. Then we obtain a **particular solution** $\mathbf{x}_0 := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$ of the linear system by letting $x_k := 0$ if $k \in \{\ell_1, \ldots, \ell_{n-r}\}$, and then by determining the pivotal variables $x_{k_1}, \ldots, x_{k_r}$ by back substitution.

3. We obtain $n - r$ **basic solutions** of the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ as follows. Fix $\ell \in \{\ell_1, \ldots, \ell_{n-r}\}$. Define $\mathbf{s}_\ell := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T}$ by $x_k := 1$ if $k = \ell$, while $x_k := 0$ if $k \in \{\ell_1, \ldots, \ell_{n-r}\}$ but $k \neq \ell$. Then determine the pivotal variables $x_{k_1}, \ldots, x_{k_r}$ by back substitution.
4. Let $\mathbf{s} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T} \in \mathbb{R}^{n \times 1}$ be any solution of the homogeneous system, that is, $\mathbf{As} = \mathbf{0}$. Then $\mathbf{s}$ is a linear combination of the $n - r$ basic solutions $\mathbf{s}_{\ell_1}, \ldots, \mathbf{s}_{\ell_{n-r}}$. To see this, let $\mathbf{y} := \mathbf{s} - x_{\ell_1}\mathbf{s}_{\ell_1} - \cdots - x_{\ell_{n-r}}\mathbf{s}_{\ell_{n-r}}$. Then $\mathbf{Ay} = \mathbf{As} - x_{\ell_1}\mathbf{As}_{\ell_1} - \cdots - x_{\ell_{n-r}}\mathbf{As}_{\ell_{n-r}} = \mathbf{0}$, and moreover, the kth entry of $\mathbf{y}$ is 0 for each $k \in \{\ell_1, \ldots, \ell_{n-r}\}$. It then follows that $\mathbf{y} = \mathbf{0}$, that is, $\mathbf{s} = x_{\ell_1}\mathbf{s}_{\ell_1} + \cdots + x_{\ell_{n-r}}\mathbf{s}_{\ell_{n-r}}$. Thus we find that the general solution of the homogeneous system is given by

   $\boxed{\mathbf{s} = \alpha_1\mathbf{s}_{\ell_1} + \cdots + \alpha_{n-r}\mathbf{s}_{\ell_{n-r}}, \text{ where } \alpha_1, \ldots, \alpha_{n-r} \in \mathbb{R}}$.

5. The general solution of $\mathbf{Ax} = \mathbf{b}$ is given by

$$\boxed{\mathbf{x} = \mathbf{x}_0 + \alpha_1 \mathbf{s}_{\ell_1} + \cdots + \alpha_{n-r} \mathbf{s}_{\ell_{n-r}}, \text{ where } \alpha_1, \ldots, \alpha_{n-r} \in \mathbb{R},}$$

provided the consistency condition is satisfied.

Example

$$\text{Let} \quad \mathbf{A} := \begin{bmatrix} 0 & \boxed{2} & 1 & 0 & 2 & 5 \\ 0 & 0 & 0 & \boxed{3} & 5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{b} := \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix}.$$

As we have seen earlier, here $m = 4$, $n = 6$, $r = 3$, pivotal columns: $\boxed{2}$, $\boxed{4}$ and $\boxed{5}$, and nonpivotal columns: $\boxed{1}$, $\boxed{3}$, $\boxed{6}$.

Since $b_4 = 0$, the linear system $\mathbf{Ax} = \mathbf{b}$ is consistent.

For a particular solution of $\mathbf{Ax} = \mathbf{b}$, let $x_1 = x_3 = x_6 = 0$.
Then $x_5 + 2x_6 = 2 \implies x_5 = 2$,
$3x_4 + 5x_5 + 0x_6 = 1 \implies x_4 = -3$,
$2x_2 + x_3 + 0x_4 + 2x_5 + 5x_6 = 0 \implies x_2 = -2$.
Thus $\mathbf{x}_0 := \begin{bmatrix} 0 & -2 & 0 & -3 & 2 & 0 \end{bmatrix}^\mathsf{T}$ is a particular solution.

Basic solutions of $\mathbf{Ax} = \mathbf{0}$:
$x_1 = 1, x_3 = x_6 = 0$ gives $\mathbf{s}_1 := \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^\mathsf{T}$,
$x_3 = 1, x_1 = x_6 = 0$ gives $\mathbf{s}_3 := \begin{bmatrix} 0 & -1/2 & 1 & 0 & 0 & 0 \end{bmatrix}^\mathsf{T}$,
$x_6 = 1, x_1 = x_3 = 0$ gives $\mathbf{s}_6 := \begin{bmatrix} 0 & -1/2 & 0 & 10/3 & -2 & 1 \end{bmatrix}^\mathsf{T}$.

The general solution of $\mathbf{Ax} = \mathbf{b}$ is given by
$\mathbf{x} = \mathbf{x}_0 + \alpha_1 \mathbf{s}_1 + \alpha_3 \mathbf{s}_3 + \alpha_6 \mathbf{s}_6$, that is,
$x_1 = \alpha_1,\ x_2 = -2 - (\alpha_3 + \alpha_6)/2,\ x_3 = \alpha_3,\ x_4 = -3 + 10\alpha_6/3,$
$x_5 = 2(1 - \alpha_6),\ x_6 = \alpha_6$, where $\alpha_1, \alpha_3, \alpha_6 \in \mathbb{R}$.

Conclusion

Suppose an $m \times n$ matrix $\mathbf{A}$ is in a REF, and let r be the number of nonzero rows of $\mathbf{A}$. If $\mathbf{b} \in \mathbb{R}^{m\times 1}$, then the linear system $\mathbf{Ax} = \mathbf{b}$ has

(i) no solution if one of $b_{r+1}, \ldots, b_m$ is nonzero.

(ii) a unique solution if $b_{r+1} = \cdots = b_m = 0$ and $r = n$.

(iii) infinitely many solutions if $b_{r+1} = \cdots = b_m = 0$ and $r < n$. (In this case, there are $\boxed{n - r \text{ free variables}}$ which give $\boxed{n - r \text{ degrees of freedom}}$.)

Considering the case $\mathbf{b} = \mathbf{0} \in \mathbb{R}^{m\times 1}$ and recalling that $r \leq m$, we obtain the following important results.

Proposition

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ be in REF with r nonzero rows. Then the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution if and only if $r = n$. In particular, if $m < n$, then $\mathbf{Ax} = \mathbf{0}$ has a nonzero solution.

Gauss Elimination Method (GEM)

We have seen how to solve the linear system $\mathbf{Ax} = \mathbf{b}$ when the matrix $\mathbf{A}$ is in a row echelon form (REF).

We now explain the **Gauss Elimination Method** (GEM) by which we can transform any $\mathbf{A} \in \mathbb{R}^{m \times n}$ to a REF.
This involves the following two **elementary row operations** (EROs):

Type I: Interchange of two rows

Type II: Addition of a scalar multiple of a row to another row

We shall later consider the following elementary row operation:

Type III: Multiplication of a row by a nonzero scalar

Consider the matrix $T_{i,j}$, which is obtained from the identity matrix by interchanging the row i and the row j. Then

$$T_{i,j} = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & & 1 & & \\ & & & \ddots & & & \\ & & 1 & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$$

Then for any matrix A, the result of interchanging the row i and row j of A is given by the matrix $T_{i,j}A$.

For any constant α, consider the matrix $L_{i,j}(\alpha)$, which is obtained from the identity matrix by adding α in the (i,j)-th position. So, $L_{ij}(\alpha) = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & \alpha & & \\ & & & \ddots & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$

If A is a matrix, then the result of adding α times the row j to the row i is given by the matrix $L_{i,j}(\alpha)A$.

First we remark that if the augmented matrix $[\mathbf{A}|\mathbf{b}]$ is transformed to a matrix $[\mathbf{A}'|\mathbf{b}']$ by any of the EROs, then $\mathbf{A}\mathbf{x} = \mathbf{b} \iff \mathbf{A}'\mathbf{x} = \mathbf{b}'$ for $\mathbf{x} \in \mathbb{R}^{n\times 1}$, that is, the linear systems $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{A}'\mathbf{x} = \mathbf{b}'$ have the same solutions.

This follows by noting that an interchange of two equations does not change the solutions, neither does an addition of an equation to another, nor does a multiplication of an equation by a nonzero scalar, since these operations can be undone by similar operations, namely, interchange of the equations in the reverse order, subtraction of an equation from another, and division of an equation by a nonzero scalar.

Consequently, we are justified in performing EROs on the augmented matrix $[\mathbf{A}|\mathbf{b}]$ in order to obtain all solutions of the given linear system.

Transformation to REF

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, that is, let $\mathbf{A}$ be an $m \times n$ matrix with entries in $\mathbb{R}$. If $\mathbf{A} = \mathbf{O}$, the zero matrix, then it is already in REF.

Suppose $\mathbf{A} \neq \mathbf{O}$.

(i) Let column k_1 be the first nonzero column of $\mathbf{A}$, and let some nonzero entry p_1 in this column occur in the jth row of $\mathbf{A}$. Interchange row j and row 1. Then $\mathbf{A}$ is transformed to

$$\mathbf{A}' := \begin{bmatrix} 0 & \cdots & 0 & \boxed{p_1} & * & \cdots & * \\ 0 & \cdots & 0 & * & * & \cdots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & * & * & \cdots & * \end{bmatrix},$$

where $*$ denotes a real number. Note: p_1 becomes the chosen pivot in row 1. (This choice may not be unique.)

(ii) Since $p_1 \neq 0$, add suitable scalar multiples of row 1 of $\mathbf{A}'$ to rows 2 to m of $\mathbf{A}'$, so that all entries in column k_1 below the pivot p_1 are equal to 0. Then $\mathbf{A}'$ is transformed to

$$\mathbf{A}'' := \begin{bmatrix} 0 & \cdots & 0 & \boxed{p_1} & * & \cdots & * \\ 0 & \cdots & 0 & \boxed{0} & * & \cdots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & \boxed{0} & * & \cdots & * \end{bmatrix}.$$

(iii) Keep row 1 of $\mathbf{A}''$ intact, and repeat the above process for the remaining $(m-1) \times n$ submatrix of $\mathbf{A}''$ to obtain

$$\mathbf{A}''' := \begin{bmatrix} 0 & \cdots & 0 & \boxed{p_1} & * & * & * & * & \cdots & * \\ 0 & \cdots & 0 & \boxed{0} & \cdots & 0 & \boxed{p_2} & * & \cdots & * \\ 0 & \cdots & 0 & \boxed{0} & \cdots & 0 & \boxed{0} & * & \cdots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & \boxed{0} & \cdots & 0 & \boxed{0} & * & \cdots & * \end{bmatrix},$$

where $p_2 \neq 0$ and occurs in column k_2 of $\mathbf{A}'''$, where $k_1 < k_2$.

Note: p_2 becomes the chosen pivot in row 2. (Again, this choice may not be unique.)

(iv) Keep rows 1 and 2 of $\mathbf{A}'''$ intact, and repeat the above process till the remaining submatrix has no nonzero row. The resulting $m \times n$ matrix is in REF with pivots $p_1, \ldots, p_r$ in columns $k_1, \ldots, k_r$, and the last $m - r$ rows are zero rows, where $1 \leq r \leq m$.

Notation

$R_i \longleftrightarrow R_j$ will denote the interchange of the ith row R_i and the jth row R_j for $1 \leq i, j \leq m$ with $i \neq j$.

$R_i + \alpha R_j$ will denote the addition of α times the jth row R_j to the ith row R_i for $1 \leq i, j \leq m$ with $i \neq j$.

αR_j will denote the multiplication of the jth row R_j by the nonzero scalar α for $1 \leq j \leq m$.

Remark

A matrix $\mathbf{A}$ may be transformed to different REFs by EROs.

For example, we can transform $\mathbf{A} := \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}$ by EROs to

$\begin{bmatrix} 1 & 3 \\ 0 & -6 \end{bmatrix}$ as well as to $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, both of which are REFs.

Examples

(i) Consider the linear system

$$\begin{aligned} 3x_1 + 2x_2 + \ \ x_3 &= 3 \\ 2x_1 + \ \ x_2 + \ \ x_3 &= 0 \\ 6x_1 + 2x_2 + 4x_3 &= 6. \end{aligned}$$

We can check that

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 0 \\ 6 & 2 & 4 & 6 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} \boxed{3} & 2 & 1 & 3 \\ 0 & \boxed{-1/3} & 1/3 & -2 \\ 0 & -2 & 2 & 0 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{ccc|c} \boxed{3} & 2 & 1 & 3 \\ 0 & \boxed{-1/3} & 1/3 & -2 \\ 0 & 0 & 0 & 12 \end{array}\right].$$

Here $m = 3 = n$, $r = 2$ and $b'_{r+1} = b'_3 = 12 \neq 0$. Hence the given linear system has no solution.

(ii) Consider the linear system

$$\begin{aligned} x_1 - x_2 + x_3 &= 0 \\ -x_1 + x_2 - x_3 &= 0 \\ 10x_2 + 25x_3 &= 90 \\ 20x_1 + 10x_2 &= 80. \end{aligned}$$

As we have already seen in Lecture 2,

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ 0 & 10 & 25 & 90 \\ 20 & 10 & 0 & 80 \end{array}\right] \xrightarrow{\text{EROs}} \left[\begin{array}{ccc|c} \boxed{1} & -1 & 1 & 0 \\ 0 & \boxed{10} & 25 & 90 \\ 0 & 0 & \boxed{-95} & -190 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Here $m = 4$, $n = 3$, $r = 3$, pivotal columns: $\boxed{1}$, $\boxed{2}$, $\boxed{3}$.

Since $b'_{r+1} = b'_4 = 0$ and $r = n$, the linear system has a unique solution, namely $\mathbf{x}_0 := \begin{bmatrix} 2 & 4 & 2 \end{bmatrix}^{\mathsf{T}}$, which we had obtained by back substitution in Lecture 2.

(iii) Consider the linear system

$$\begin{aligned} 3.0x_1 + 2.0x_2 + 2.0x_3 - 5.0x_4 &= 8.0 \\ 0.6x_1 + 1.5x_2 + 1.5x_3 - 5.4x_4 &= 2.7 \\ 1.2x_1 - 0.3x_2 - 0.3x_3 + 2.4x_4 &= 2.1. \end{aligned}$$

We can check that

$$[\mathbf{A}|\mathbf{b}] = \left[\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 0.6 & 1.5 & 1.5 & -5.4 & 2.7 \\ 1.2 & -0.3 & -0.3 & 2.4 & 2.1 \end{array}\right]$$

$$\xrightarrow{\text{EROs}} \left[\begin{array}{cccc|c} \boxed{3} & 2 & 2 & -5 & 8 \\ 0 & \boxed{1.1} & 1.1 & -4.4 & 1.1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] =: [\mathbf{A}'|\mathbf{b}'].$$

Here $m = 3, n = 4, r = 2$, pivotal columns: $\boxed{1}$, $\boxed{2}$,
nonpivotal columns: $\boxed{3}$, $\boxed{4}$.

Since $b'_{r+1} = b'_3 = 0$, the linear system $\mathbf{A}'\mathbf{x} = \mathbf{b}'$ has a solution.

For a particular solution of $\mathbf{A}'\mathbf{x} = \mathbf{b}'$, let $x_3 = x_4 = 0$. Then
$1.1\,x_2 = 1.1 \implies x_2 = 1$,
$3x_1 + 2x_2 = 8 \implies x_1 = 2$,
Thus $\mathbf{x}_0 := \begin{bmatrix} 2 & 1 & 0 & 0 \end{bmatrix}^{\mathsf{T}}$ is a particular solution.

Since $r = 2 < 4 = n$, the linear system has many solutions.
For basic solutions of $\mathbf{A}'\mathbf{x} = \mathbf{0}'$, where $\mathbf{0}' = \mathbf{0}$,
let $x_3 = 1, x_4 = 0$, so that $\mathbf{s}_3 := \begin{bmatrix} 0 & -1 & 1 & 0 \end{bmatrix}^{\mathsf{T}}$,
and $x_4 = 1, x_3 = 0$, so that $\mathbf{s}_4 := \begin{bmatrix} -1 & 4 & 0 & 1 \end{bmatrix}^{\mathsf{T}}$,

The general solution of $\mathbf{A}'\mathbf{x} = \mathbf{b}'$ is given by
$\mathbf{x} = \mathbf{x}_0 + \alpha_3\mathbf{s}_3 + \alpha_4\mathbf{s}_4$, that is,
$x_1 = 2 - \alpha_4,\ x_2 = 1 - \alpha_3 + 4\alpha_4,\ x_3 = \alpha_3,\ x_4 = \alpha_4$, where
α_3, α_4 are arbitrary real numbers. These are precisely the solutions of the given linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$.

Proposition

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution if and only if any REF of $\mathbf{A}$ has n nonzero rows. In particular, if $m < n$, then $\mathbf{Ax} = \mathbf{0}$ has a nonzero solution.

Proof. We saw that these results hold if $\mathbf{A}$ itself is in REF. Since every $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be transformed to a REF $\mathbf{A}'$ by EROs, and since the solutions of the linear system $\mathbf{Ax} = \mathbf{0}$ and the transformed system $\mathbf{A}'\mathbf{x} = \mathbf{0}'$, where $\mathbf{0}' = \mathbf{0}$, are the same, the desired results hold. □

Note: Suppose an $m \times n$ matrix $\mathbf{A}$ is transformed by EROs to different REFs $\mathbf{A}'$ and $\mathbf{A}''$. Suppose $\mathbf{A}'$ has r' nonzero rows and $\mathbf{A}''$ has r'' nonzero rows. Then $0 \leq r', r'' \leq \min\{m, n\}$. The above result implies that $r' = n \iff r'' = n$. We shall later see that $r' = r''$ always.

A Challenge Problem

Let $\mathbf{A} \in \mathbb{R}^{9\times 4}$ and $\mathbf{B} \in \mathbb{R}^{7\times 3}$. Is there $\mathbf{X} \in \mathbb{R}^{4\times 7}$ such that $\mathbf{X} \neq \mathbf{O}$ but $\mathbf{AXB} = \mathbf{O}$?

Inverse of a Square Matrix

We now introduce a special kind of square matrices.

Let $\mathbf{A}$ be a square matrix of size $n \in \mathbb{N}$, that is, let $\mathbf{A} \in \mathbb{R}^{n \times n}$. We say that $\mathbf{A}$ is **invertible** if there is $\mathbf{B} \in \mathbb{R}^{n \times n}$ such that $\mathbf{AB} = \mathbf{I} = \mathbf{BA}$, and in this case, $\mathbf{B}$ is called an **inverse** of $\mathbf{A}$.

Examples

The matrix $\mathbf{A} := \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ is invertible. To see this, let $\mathbf{B} := \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$, and check $\mathbf{AB} = \mathbf{I} = \mathbf{BA}$. On the other hand, the nonzero matrix $\mathbf{A} := \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is not invertible. To see this, let $\mathbf{B} := \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ and note that $\mathbf{AB} = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \neq \mathbf{I}$.

If $\mathbf{A}$ is invertible, then it has a unique inverse. In fact, if $\mathbf{AC} = \mathbf{I} = \mathbf{BA}$, then $\mathbf{C} = \mathbf{IC} = (\mathbf{BA})\mathbf{C} = \mathbf{B}(\mathbf{AC}) = \mathbf{BI} = \mathbf{B}$ by the associativity of the matrix multiplication.

If $\mathbf{A}$ is invertible, its inverse will be denoted by $\mathbf{A}^{-1}$, and so $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} = \mathbf{A}\mathbf{A}^{-1}$. Clearly, $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$. If $\mathbf{A}$ is invertible and if one can guess its inverse, then it is easy to verify that it is in fact the inverse of $\mathbf{A}$. Here is a case in point.

Proposition

Let $\mathbf{A}$ be a square matrix. Then $\mathbf{A}$ is invertible if and only if $\mathbf{A}^\mathsf{T}$ is invertible. In this case, $(\mathbf{A}^\mathsf{T})^{-\mathbf{1}} = (\mathbf{A}^{-\mathbf{1}})^\mathsf{T}$.

Proof. Suppose $\mathbf{A}$ is invertible and $\mathbf{B}$ is its inverse. Then $\mathbf{AB} = \mathbf{I} = \mathbf{BA}$, and so $\mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T} = \mathbf{I}^\mathsf{T} = \mathbf{A}^\mathsf{T}\mathbf{B}^\mathsf{T}$. Since $\mathbf{I}^\mathsf{T} = \mathbf{I}$, we see that $\mathbf{A}^\mathsf{T}$ is invertible and $(\mathbf{A}^\mathsf{T})^{-\mathbf{1}} = \mathbf{B}^\mathsf{T} = (\mathbf{A}^{-\mathbf{1}})^\mathsf{T}$.

Next, if $\mathbf{A}^\mathsf{T}$ is invertible, then $\mathbf{A} = (\mathbf{A}^\mathsf{T})^\mathsf{T}$ is invertible. $\square$

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Spring 2025

Recall: A square matrix $\mathbf{A} \in \mathbb{R}^{\mathbf{n}\times\mathbf{n}}$ is said to be invertible if there is $\mathbf{B} \in \mathbb{R}^{n\times n}$ such that

$$\mathbf{AB} = \mathbf{I} = \mathbf{BA},$$

and in this case, $\mathbf{B}$ is called an inverse of $\mathbf{A}$.

We have seen examples of square matrices that are invertible and also those that are not invertible. Further we noted that:

- If a square matrix $\mathbf{A} \in \mathbb{R}^{\mathbf{n}\times\mathbf{n}}$ is invertible, then it has a unique inverse, and it is denoted by $\mathbf{A}^{-1}$
- If a square matrix $\mathbf{A} \in \mathbb{R}^{\mathbf{n}\times\mathbf{n}}$ is invertible, then so is its transpose $\mathbf{A}^\mathsf{T}$ and in this case,

$$(\mathbf{A}^\mathsf{T})^{-\mathbf{1}} = (\mathbf{A}^{-\mathbf{1}})^\mathsf{T}.$$

We now relate the invertibility of a square matrix $\mathbf{A}$ to the solutions of the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Proposition

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then $\mathbf{A}$ is invertible if and only if the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution.

Proof. Suppose $\mathbf{A}$ is invertible. Then by definition, there is $\mathbf{B} \in \mathbb{R}^{n \times n}$ such that $\mathbf{BA} = \mathbf{I}$. If $\mathbf{x} \in \mathbb{R}^{n \times 1}$ satisfies $\mathbf{Ax} = \mathbf{0}$, then $\mathbf{x} = \mathbf{BAx} = \mathbf{B}(\mathbf{Ax}) = \mathbf{B0} = \mathbf{0}$. Thus the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution.

Conversely, suppose the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution. Let $\mathbf{y} = \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n \times 1}$. We transform the augmented matrix $[\mathbf{A}|\mathbf{y}]$ to a matrix $[\mathbf{A}'|\mathbf{y}']$, where $\mathbf{A}'$ is in REF. By our previous result, $\mathbf{A}'$ has n nonzero rows, and so back substitution gives a unique $\mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n \times 1}$ such that $\mathbf{A}'\mathbf{x} = \mathbf{y}'$. Hence $\mathbf{Ax} = \mathbf{y}$.

Further, the process of the back substitution shows that the entries $x_1, \ldots, x_n$ of $\mathbf{x}$ are given as follows:

$$\begin{array}{rcl}
x_n & = & c'_{nn} y'_n \\
x_{n-1} & = & c'_{(n-1)(n-1)} y'_{n-1} + c'_{(n-1)n} y'_n \\
\vdots & \vdots & \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\
x_2 & = & c'_{22} y'_2 + \cdots + \cdots + \cdots + c'_{2n} y'_n \\
x_1 & = & c'_{11} y'_1 + c'_{12} y'_2 + \cdots + \cdots + \cdots + c'_{1n} y'_n,
\end{array}$$

where $\mathbf{y}' = \begin{bmatrix} y'_1 & \cdots & y'_n \end{bmatrix}^{\mathsf{T}}$ and $c'_{jk} \in \mathbb{R}$ for $j, k = 1, \ldots, n$.

Also, since $\mathbf{y}'$ is obtained from $\mathbf{y}$ by performing EROs (which are of the type $R_i \longleftrightarrow R_j$, $R_i + \alpha R_j$ and αR_j) on $[\mathbf{A}|\mathbf{y}]$, we see that each $y'_1, \ldots, y'_n$ is a linear combination of the entries $y_1, \ldots, y_n$ of $\mathbf{y}$. As a result, each $x_1, \ldots, x_n$ is a linear combination of $y_1, \ldots, y_n$.

Thus there is $c_{jk} \in \mathbb{R}$ for $j, k = 1, \ldots, n$ (not depending on $y_1, \ldots, y_n$) such that

$$
\begin{array}{rcl}
x_1 & = & c_{11}y_1 + c_{12}y_2 + \cdots + c_{1n}y_n \\
x_2 & = & c_{21}y_1 + c_{22}y_2 + \cdots + c_{2n}y_n \\
\vdots & \vdots & \vdots \qquad\qquad \vdots \qquad\qquad\quad \vdots \\
x_n & = & c_{n1}y_1 + c_{n2}y_2 + \cdots + c_{nn}y_n.
\end{array}
$$

Define $\mathbf{C} := \left[c_{jk}\right] \in \mathbb{R}^{n\times n}$. Then $\mathbf{x} = \mathbf{Cy}$, and so $\mathbf{ACy} = \mathbf{A}(\mathbf{Cy}) = \mathbf{Ax} = \mathbf{y}$. Letting $\mathbf{y} := \mathbf{e}_k \in \mathbb{R}^{n\times 1}$, we see that $(\mathbf{AC})\mathbf{e_k} = \mathbf{e_k}$ for $k = 1, \ldots, n$. Hence $\mathbf{AC} = \mathbf{I}$. We still need to show that $\mathbf{CA} = \mathbf{I}$. For this, consider the linear system $\mathbf{Cx} = \mathbf{0}$. Note that $\mathbf{Cx} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{ACx} = \mathbf{A}(\mathbf{Cx}) = \mathbf{A0} = \mathbf{0}$. Thus the linear system $\mathbf{Cx} = \mathbf{0}$ has only the zero solution.

Hence by what we have proved above, there is $\mathbf{D} \in \mathbb{R}^{n\times n}$ such that $\mathbf{CD} = \mathbf{I}$. Now, $\mathbf{D} = \mathbf{ID} = (\mathbf{AC})\mathbf{D} = \mathbf{A}(\mathbf{CD}) = \mathbf{AI} = \mathbf{A}$. Thus $\mathbf{AC} = \mathbf{I} = \mathbf{CA}$, and so $\mathbf{A}$ is invertible. $\square$

Corollary

Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. If there is $\mathbf{B} \in \mathbb{R}^{n\times n}$ such that either $\mathbf{BA} = \mathbf{I}$ or $\mathbf{AB} = \mathbf{I}$, then $\mathbf{A}$ is invertible, and $\mathbf{A}^{-1} = \mathbf{B}$.

Proof. Let $\mathbf{B} \in \mathbb{R}^{n\times n}$ be such that $\mathbf{BA} = \mathbf{I}$. If $\mathbf{x} \in \mathbb{R}^{n\times 1}$ satisfies $\mathbf{Ax} = \mathbf{0}$, then $\mathbf{x} = \mathbf{BAx} = \mathbf{B}(\mathbf{Ax}) = \mathbf{B0} = \mathbf{0}$. Thus the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution. By the previous proposition, $\mathbf{A}$ is invertible. Then there is $\mathbf{C} \in \mathbb{R}^{\mathbf{n}\times\mathbf{n}}$ such that $\mathbf{AC} = \mathbf{I}$, and $\mathbf{B} = \mathbf{C}$. Hence $\mathbf{A}^{-1} = \mathbf{B}$.

Next, let $\mathbf{B} \in \mathbb{R}^{n\times n}$ be such that $\mathbf{AB} = \mathbf{I}$. Then $\mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}} = \mathbf{I}^{\mathsf{T}} = \mathbf{I}$. By what we have just proved, $\mathbf{A}^{\mathsf{T}}$ is invertible, and $(\mathbf{A}^{\mathsf{T}})^{-1} = \mathbf{B}^{\mathsf{T}}$. Hence $\mathbf{A} = (\mathbf{A}^{\mathsf{T}})^{\mathsf{T}}$ is invertible, and $\mathbf{A}^{-1} = (\mathbf{B}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{B}$. $\square$

Note: The above result is a definite improvement over requiring the existence of a matrix $\mathbf{B}$ satisfying both $\mathbf{BA} = \mathbf{I}$ and $\mathbf{AB} = \mathbf{I}$ for the invertibility of a square matrix $\mathbf{A}$.

Proposition

Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices. Then $\mathbf{AB}$ is invertible if and only if $\mathbf{A}$ and $\mathbf{B}$ are invertible, and then $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Proof. Let $\mathbf{A}$ and $\mathbf{B}$ be invertible. Using the associativity of matrix multiplication, we easily see that

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{B} = \mathbf{I}.$$

Hence $\mathbf{AB}$ is invertible and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ by the previous corollary.

Conversely, let $\mathbf{AB}$ be invertible. Then there is $\mathbf{C}$ such that $(\mathbf{AB})\mathbf{C} = \mathbf{I} = \mathbf{C}(\mathbf{AB})$. Since $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C} = \mathbf{I}$, we see that $\mathbf{A}$ is invertible, and $\mathbf{A}^{-1} = \mathbf{BC}$. Also, since $(\mathbf{CA})\mathbf{B} = \mathbf{C}(\mathbf{AB}) = \mathbf{I}$, we see that $\mathbf{B}$ is invertible and $\mathbf{B}^{-1} = \mathbf{CA}$, again by the previous corollary. $\square$

Proposition

Let A be an $n \times m$ matrix.
(i) If there is an $m \times n$ matrix B such that $BA = I_{m \times m}$ then $n \leq m$.
(ii) If there is an $n \times m$ matrix C such that $AC = I_{n \times n}$ then $m \leq n$.
(iii) If there are matrices B and C such that $BA = I$, $AC = I$ then $m = n$ and $B = C$.

Proof. Indeed, if $BA = I$ and if $x \in \mathbb{R}^{n \times 1}$ is a vector such that $Ax = 0$ then $x = Ix = BAx = 0$. On the other hand if $n > m$ then there is at least one nonzero solution to $Ax = 0$. This proves (i). For (ii), note that $AC = I$ implies $I = C^T A^T$, so by (i), as A^T is of order $n \times m$, $m \leq n$. For (iii), if $BA = I = AC$ then by (i) and (ii) we know $n = m$. Again, $B = BI = B(AC) = (BA)C = IC = C$. $\square$

Row Canonical Form (RCF)

As we have seen, a matrix **A** may not have a unique REF. However, a special REF of **A** turns out to be unique.

An $m \times n$ matrix **A** is said to be in a **row canonical form** (RCF) or a **reduced row echelon form** (RREF) if
(i) it is in a row echelon form (REF),
(ii) all pivots are equal to 1 and
(iii) in each pivotal column, all entries above the pivot are (also) equal to 0.

For example, the matrix

$$\mathbf{A} := \begin{bmatrix} 0 & \boxed{1} & * & 0 & 0 & * \\ 0 & 0 & 0 & \boxed{1} & 0 & * \\ 0 & 0 & 0 & 0 & \boxed{1} & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is in a RCF, where $*$ denotes any real number.

Note: If $\mathbf{A}$ is in REF, then in each pivotal column, all entries below the pivot are 0. If $\mathbf{A}$ is in fact in RCF and has r nonzero rows, then the $r \times r$ submatrix formed by the first r rows and the r pivotal columns is the $r \times r$ identity matrix $\mathbf{I}$.

Suppose an $m \times n$ matrix $\mathbf{A}$ is in RCF and has r nonzero rows. If $r = n$, then it has n pivotal columns, that is, all its columns are pivotal, and so $\mathbf{A} = \mathbf{I}$ if $m = n$, and $\mathbf{A} = \begin{bmatrix} \mathbf{I} \\ \\ \mathbf{O} \end{bmatrix}$ if $m > n$, where $\mathbf{I}$ is the $n \times n$ identity matrix and $\mathbf{O}$ is the $(m - n) \times n$ zero matrix.

To transform an $m \times n$ matrix to a RCF, we first transform it to a REF by elementary row operations of type I and II. Then we multiply a row containing a pivot p by $1/p$ (which is an elementary row operation of type III), and then we add a suitable nonzero multiple of this row to each preceding row.

Every matrix has a unique RCF. (Proof by induction on n)

Example

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 16 \end{bmatrix} \xrightarrow{\text{EROs}} \begin{bmatrix} \boxed{1} & 3 & -2 & 0 & 2 & 0 \\ 0 & 0 & \boxed{-1} & -2 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 & \boxed{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

which is in REF,

$$\xrightarrow{\text{EROs}} \begin{bmatrix} \boxed{1} & 3 & 0 & 4 & 2 & 6 \\ 0 & 0 & \boxed{1} & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & \boxed{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{EROs}} \begin{bmatrix} \boxed{1} & 3 & 0 & 4 & 2 & 0 \\ 0 & 0 & \boxed{1} & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

which is in RCF.

Recall: A square matrix $\mathbf{A}$ is invertible if and only if the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution.

Proposition

An $n \times n$ matrix is invertible if and only if it can be transformed to the $n \times n$ identity matrix by EROs.

Proof. Suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible. Using EROs, transform $\mathbf{A}$ to a matrix $\mathbf{A}' \in \mathbb{R}^{n \times n}$ such that $\mathbf{A}'$ is in a RCF. Since $\mathbf{A}$ is invertible, the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution. Hence $\mathbf{A}'$ has n nonzero rows, and so each of the n columns of $\mathbf{A}'$ is pivotal. Also, the number of rows of $\mathbf{A}$ is equal to the number of its columns, that is, $m = n$. Therefore $\mathbf{A}' = \mathbf{I}$.

Conversely, suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ can be transformed to the $n \times n$ identity matrix $\mathbf{I}$ by EROs. Since $\mathbf{Ix} = \mathbf{0} \implies \mathbf{x} = \mathbf{0}$ for $\mathbf{x} \in \mathbb{R}^{n \times 1}$, we see that the linear system $\mathbf{Ax} = \mathbf{0}$ has only the zero solution. Hence $\mathbf{A}$ is invertible. $\square$

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Spring 2025

Recall:

Proposition

An $n \times n$ matrix is invertible if and only if it can be transformed to the $n \times n$ identity matrix by EROs.

Proof. Suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible. Using EROs, transform $\mathbf{A}$ to a matrix $\mathbf{A}' \in \mathbb{R}^{n \times n}$ such that $\mathbf{A}'$ is in a RCF. Since $\mathbf{A}$ is invertible, the linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has only the zero solution. Hence $\mathbf{A}'$ has n nonzero rows, and so each of the n columns of $\mathbf{A}'$ is pivotal. Also, the number of rows of $\mathbf{A}$ is equal to the number of its columns, that is, $m = n$. Therefore $\mathbf{A}' = \mathbf{I}$.

Conversely, suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ can be transformed to the $n \times n$ identity matrix $\mathbf{I}$ by EROs. Since $\mathbf{I}\mathbf{x} = \mathbf{0} \implies \mathbf{x} = \mathbf{0}$ for $\mathbf{x} \in \mathbb{R}^{n \times 1}$, we see that the linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has only the zero solution. Hence $\mathbf{A}$ is invertible. $\square$

Remark.

Suppose an $n \times n$ square matrix $\mathbf{A}$ is invertible. In order to solve the linear system $\mathbf{Ax} = \mathbf{b}$ for a given $\mathbf{b} \in \mathbb{R}^{n\times 1}$, we may transform the augmented matrix $[\mathbf{A}|\mathbf{b}]$ to $[\mathbf{I}\,|\,\mathbf{c}]$ by EROs. Now $\mathbf{Ax} = \mathbf{b} \iff \mathbf{Ix} = \mathbf{c}$ for $\mathbf{x} \in \mathbb{R}^{n\times 1}$. Hence $\mathbf{Ac} = \mathbf{b}$. Thus $\mathbf{c}$ is the unique solution of $\mathbf{Ax} = \mathbf{b}$. This observation is the basis of an important method to find the inverse of a square matrix.

Gauss-Jordan Method for Finding the Inverse of a Matrix

Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be an invertible matrix. Consider the basic column vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n \in \mathbb{R}^{n\times 1}$. Then $\begin{bmatrix}\mathbf{e}_1 & \cdots & \mathbf{e}_n\end{bmatrix} = \mathbf{I}$. Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be the unique elements of $\mathbb{R}^{n\times 1}$ be such that $\mathbf{Ax}_1 = \mathbf{e}_1, \ldots, \mathbf{Ax}_n = \mathbf{e}_n$, and define $\mathbf{X} := \begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix}$. Then

$$\mathbf{AX} = \mathbf{A}\begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\mathbf{Ax}_1 & \cdots & \mathbf{Ax}_n\end{bmatrix} = \begin{bmatrix}\mathbf{e}_1 & \cdots & \mathbf{e}_n\end{bmatrix} = \mathbf{I}.$$

By an earlier result, it follows that $\mathbf{X} = \mathbf{A}^{-1}$.

Hence to find $\mathbf{A}^{-1}$, we may solve the n linear systems $\mathbf{A}\mathbf{x}_1 = \mathbf{e}_1, \ldots, \mathbf{A}\mathbf{x}_n = \mathbf{e}_n$ simultaneously by considering the $n \times 2n$ augmented matrix

$$[\mathbf{A}|\mathbf{e}_1 \cdots \mathbf{e}_n] = [\mathbf{A} \,|\, \mathbf{I}]$$

and transform $\mathbf{A}$ to its RCF, namely to $\mathbf{I}$, by EROs. Thus if $[\mathbf{A} \,|\, \mathbf{I}]$ is transformed to $[\mathbf{I} \,|\, \mathbf{X}]$, then $\mathbf{X}$ is the inverse of $\mathbf{A}$.

Remark To carry out the above process, we need not know beforehand that the matrix $\mathbf{A}$ is invertible. This follows by noting that $\mathbf{A}$ can be transformed to the identity matrix by EROs if and only if $\mathbf{A}$ is invertible. Hence the process itself reveals whether $\mathbf{A}$ is invertible or not.

Example

Let

$$\mathbf{A} := \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}.$$

We use EROs to transform $[\mathbf{A} \mid \mathbf{I}]$ to $[\mathbf{I} \mid \mathbf{X}]$, where $\mathbf{X} \in \mathbb{R}^{3\times 3}$.

$$\left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \end{array}\right] \to$$

$$\left[\begin{array}{ccc|ccc} \boxed{-1} & 1 & 2 & 1 & 0 & 0 \\ 0 & \boxed{2} & 7 & 3 & 1 & 0 \\ 0 & 0 & \boxed{-5} & -4 & -1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 0 & 0 \\ 0 & 1 & 3.5 & 1.5 & 0.5 & 0 \\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0.6 & 0.4 & -0.4 \\ 0 & 1 & 0 & -1.3 & -0.2 & 0.7 \\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -0.7 & 0.2 & 0.3 \\ 0 & 1 & 0 & -1.3 & -0.2 & 0.7 \\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2 \end{array}\right] = [\mathbf{I} \mid \mathbf{X}].$$

Thus $\mathbf{A}$ is invertible and

$$\mathbf{A}^{-1} = \mathbf{X} = \frac{1}{10}\begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}.$$

Linear Dependence

Let $n \in \mathbb{N}$. We shall work entirely with row vectors in $\mathbb{R}^{1 \times n}$ (of length n), or entirely with column vectors in $\mathbb{R}^{n \times 1}$(of length n), both of which will be referred to as 'vectors'.

We have already considered a linear combination

$$\alpha_1 \mathbf{a}_1 + \cdots + \alpha_m \mathbf{a}_m$$

of vectors $\mathbf{a}_1, \ldots, \mathbf{a}_m$, where $\alpha_1, \ldots, \alpha_m$ are scalars.

A set S of vectors is called **linearly dependent** if there is $m \in \mathbb{N}$, there are (distinct) vectors $\mathbf{a}_1, \ldots, \mathbf{a}_m$ in S and there are scalars $\alpha_1, \ldots, \alpha_m$, not all zero, such that

$$\alpha_1 \mathbf{a}_1 + \cdots + \alpha_m \mathbf{a}_m = \mathbf{0}.$$

It can be seen that S is linearly dependent $\iff$ either $\mathbf{0} \in S$ or a vector in S is a linear combination of other vectors in S.

Examples

(i) Let $S := \left\{ \begin{bmatrix} 1 & 2 \end{bmatrix}^{\mathsf{T}}, \begin{bmatrix} 2 & 1 \end{bmatrix}^{\mathsf{T}}, \begin{bmatrix} 1 & -1 \end{bmatrix}^{\mathsf{T}} \right\} \subset \mathbb{R}^{2\times 1}$. Then the set S is linearly dependent since

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \text{ Clearly, } \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

(ii) Let
$S := \left\{ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 2 & 3 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & -3 & 3 \end{bmatrix} \right\} \subset \mathbb{R}^{1\times 3}$.
Then the set S is linearly dependent since
$\begin{bmatrix} 0 & -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} - 2\begin{bmatrix} 2 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 1 & 2 \end{bmatrix}$. Clearly,

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} - 2\begin{bmatrix} 2 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 1 & 2 \end{bmatrix} - \begin{bmatrix} 0 & -3 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}.$$

In (i) above, S is a set of 3 vectors in $\mathbb{R}^{2\times 1}$, and in (ii) above, S is a set of 4 vectors in $\mathbb{R}^{1\times 3}$. These examples illustrate an important phenomenon to which we now turn. First we prove the following crucial result.

Proposition

Let S be a set of s vectors, each of which is a linear combination of elements of a (fixed) set of r vectors. If $s > r$, then the set S is linearly dependent.

Proof. Let $S := \{\mathbf{x}_1, \ldots, \mathbf{x}_s\}$, and suppose each vector in S is a linear combination of elements of the set $\{\mathbf{y}_1, \ldots, \mathbf{y}_r\}$ of r vectors and $s > r$. Then

$$\mathbf{x}_j = \sum_{k=1}^{r} a_{jk}\mathbf{y}_k \quad \text{for } j = 1, \ldots, s, \text{ where } a_{jk} \in \mathbb{R}.$$

Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{s \times r}$. Then $\mathbf{A}^\mathsf{T} \in \mathbb{R}^{r \times s}$. Since $r < s$, the linear system $\mathbf{A}^\mathsf{T}\mathbf{x} = \mathbf{0}$ has a nonzero solution, that is, there are $\alpha_1, \ldots, \alpha_s$, not all zero, such that

$$\mathbf{A}^\mathsf{T} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_s \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{s1} \\ \vdots & \vdots & \vdots \\ a_{1r} & \cdots & a_{sr} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_s \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{r \times 1},$$

that is, $\sum_{j=1}^{s} a_{jk}\alpha_j = 0$ for $k = 1, \ldots, r$. It follows that

$$\sum_{j=1}^{s} \alpha_j \mathbf{x}_j = \sum_{j=1}^{s} \alpha_j \Big(\sum_{k=1}^{r} a_{jk} \mathbf{y}_k \Big) = \sum_{k=1}^{r} \Big(\sum_{j=1}^{s} a_{jk}\alpha_j \Big) \mathbf{y}_k = \mathbf{0}.$$

Since not all $\alpha_1, \ldots, \alpha_n$ are zero, S is linearly dependent. □

Corollary

Let $n \in \mathbb{N}$ and S be a set of vectors of length n. If S has more than n elements, then S is linearly dependent.

Proof. If S is a set of column vectors of length n, then each element of S is a linear combination of the n column vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$. Similarly, if S is a set of row vectors of length n, then each element of S is a linear combination of the n row vectors $\mathbf{e}_1^{\mathsf{T}}, \ldots, \mathbf{e}_n^{\mathsf{T}}$. Hence the desired result follows from the crucial result we just proved. □

Linear Independence

A set S of vectors is called **linearly independent** if it is not linearly dependent, that is,

$$\alpha_1 \mathbf{a}_1 + \cdots + \alpha_m \mathbf{a}_m = \mathbf{0} \implies \alpha_1 = \cdots = \alpha_m = 0,$$

whenever $\mathbf{a}_1, \ldots, \mathbf{a}_m$ are (distinct) vectors in S and $\alpha_1, \ldots, \alpha_m$ are scalars. We may also say that the vectors in S are linearly independent.

Linearly independent sets are important because each one of them gives us data that we cannot obtain from any linear combination of the others. In this sense, each element of a linearly independent set is indispensable!

Examples

(i) The empty set is linearly independent vacuously.

(ii) Let S be the subset of $\mathbb{R}^{n\times 1}$ consisting of the basic column vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$. Then S is linearly independent. To see this, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$ and $\alpha_1\mathbf{e}_1 + \cdots + \alpha_n\mathbf{e}_n = \mathbf{0}$. Then the jth entry α_j of $\alpha_1\mathbf{e}_1 + \cdots + \alpha_n\mathbf{e}_n$ is equal to 0 for $j = 1, \ldots, n$.

(iii) Let S denote the subset of $\mathbb{R}^{1\times 4}$ consisting of the vectors $\begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}, \begin{bmatrix}1 & 1 & 0 & 0\end{bmatrix}, \begin{bmatrix}1 & 1 & 1 & 0\end{bmatrix}$ and $\begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}$.

Then S is linearly independent. To see this, let $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$ be such that

$\alpha_1 \begin{bmatrix}1 & 0 & \cdots & 0\end{bmatrix} + \alpha_2 \begin{bmatrix}1 & 1 & 0 & 0\end{bmatrix} + \alpha_3 \begin{bmatrix}1 & 1 & 1 & 0\end{bmatrix} +$
$\alpha_4 \begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix} = \begin{bmatrix}0 & 0 & 0 & 0\end{bmatrix}$.

Then $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 0$, $\alpha_2 + \alpha_3 + \alpha_4 = 0$, $\alpha_3 + \alpha_4 = 0$ and $\alpha_4 = 0$, that is, $\alpha_4 = \alpha_3 = \alpha_2 = \alpha_1 = 0$.

How to Decide Linear Independence of Column Vectors?

Suppose $\mathbf{c}_1, \ldots, \mathbf{c}_n$ are column vectors each of length m. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be the matrix having $\mathbf{c}_1, \ldots, \mathbf{c}_n$ as its n columns. Then for $x_1, \ldots, x_n \in \mathbb{R}$,

$$x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{A}\mathbf{x}.$$

Hence the subset $S := \{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ of $\mathbb{R}^{m \times 1}$ is linearly independent if and only if the linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has only the zero solution. This is the case if and only if in any REF $\mathbf{A}'$ of $\mathbf{A}$, there are n nonzero rows, as we have seen in Lecture 3.

Hence if $\mathbf{A} \in \mathbb{R}^{m \times n}$ is transformed to a REF $\mathbf{A}'$, and $\mathbf{A}'$ has r nonzero rows, then the columns of $\mathbf{A}$ form a linearly independent subset of $\mathbb{R}^{m \times 1}$ if $r = n$, and they form a linearly dependent subset $\mathbb{R}^{m \times 1}$ if $r < n$.

Row Rank of a Matrix

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The **row rank** of $\mathbf{A}$ is the maximum number of linearly independent row vectors of $\mathbf{A}$. Thus the row rank of $\mathbf{A}$ is equal to r if and only if there is a linearly independent set of r rows of $\mathbf{A}$ and any set of $r+1$ rows of $\mathbf{A}$ is linearly dependent.

Let r be the row rank of $\mathbf{A}$. Then $r = 0$ if and only if $\mathbf{A} = \mathbf{O}$. Since the total number of rows of $\mathbf{A}$ is m, we see that $r \leq m$. Also, since the row vectors of $\mathbf{A}$ form a subset of $\mathbb{R}^{1 \times n}$, no more than n of them can be linearly independent. Thus $r \leq n$.

Let $\mathbf{a}_1, \ldots, \mathbf{a}_m$ be any m vectors in $\mathbb{R}^{1 \times n}$. Clearly, they are linearly independent if and only if the matrix $\mathbf{A}$ formed with these vectors as row vectors has row rank m, and they are linearly dependent if the row rank of $\mathbf{A}$ is less than m.

Examples

(i) Let $\mathbf{A} := \begin{bmatrix} 3 & 0 & 2 & 2 \\ -3 & 0 & 12 & 27 \\ -21 & 21 & 0 & 15 \end{bmatrix}$.

The row vectors of $\mathbf{A}$ are $\mathbf{a}_1 := \begin{bmatrix} 3 & 0 & 2 & 2 \end{bmatrix}$, $\mathbf{a}_2 := \begin{bmatrix} -3 & 0 & 12 & 27 \end{bmatrix}$ and $\mathbf{a}_3 := \begin{bmatrix} -21 & 21 & 0 & 15 \end{bmatrix}$.

Let $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ be such that $\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \alpha_3 \mathbf{a}_3 = \mathbf{0}$.

This means

$$\begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{bmatrix} \begin{bmatrix} 3 & 0 & 2 & 2 \\ -3 & 0 & 12 & 27 \\ -21 & 21 & 0 & 15 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix},$$

that is, $3\alpha_1 - 3\alpha_2 - 21\alpha_3 = 0$, $21\alpha_3 = 0, 2\alpha_1 + 12\alpha_2 = 0$ and $2\alpha_1 + 27\alpha_2 + 15\alpha_3 = 0$. Clearly, $\alpha_3 = 0$, and the two equations $3\alpha_1 - 3\alpha_2 = 0, 2\alpha_1 + 12\alpha_2 = 0$ show that $\alpha_1 = \alpha_2 = 0$ as well. Thus the 3 rows of $\mathbf{A}$ are linearly independent. Hence the row rank of $\mathbf{A}$ is 3.

(ii)

Let $\mathbf{A} := \begin{bmatrix} 3 & 0 & 2 & 2 \\ -3 & 21 & 12 & 27 \\ -21 & 21 & 0 & 15 \end{bmatrix}$.

The row vectors of $\mathbf{A}$ are $\mathbf{a}_1 := \begin{bmatrix} 3 & 0 & 2 & 2 \end{bmatrix}$, $\mathbf{a}_2 := \begin{bmatrix} -3 & 21 & 12 & 27 \end{bmatrix}$ and $\mathbf{a}_3 := \begin{bmatrix} -21 & 21 & 0 & 15 \end{bmatrix}$. Observe that $6\mathbf{a}_1 - \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$. Hence the three row vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are not linearly independent. But the set $\{\mathbf{a}_1, \mathbf{a}_2\}$ is linearly independent since $\mathbf{a}_1 \neq \mathbf{0}, \mathbf{a}_2 \neq \mathbf{0}$, and they are not scalar multiples of each other. (The same holds for the sets $\{\mathbf{a}_2, \mathbf{a}_3\}$ and $\{\mathbf{a}_3, \mathbf{a}_1\}$.) Hence the row rank of $\mathbf{A}$ is 2.

We used the relation $6\mathbf{a}_1 - \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$ above to determine the row rank of $\mathbf{A}$. It is difficult to think of such a relation out of nowhere. We shall develop a systematic approach to find the row rank of a matrix.

First we prove the following preliminary results.

Proposition

If a matrix $\mathbf{A}$ is transformed to a matrix $\mathbf{A}'$ by elementary row operations, then the row ranks of $\mathbf{A}$ and $\mathbf{A}'$ are equal, that is, EROs do not alter the row rank of a matrix.

Proof. ERO of type I: $R_i \longleftrightarrow R_j$ with $i \neq j$: $\mathbf{A}$ and $\mathbf{A}'$ have the same set of row vectors. So there is nothing to prove.

ERO of type II: $R_i + \alpha R_j$ with $i \neq j$: Suppose the set $\{\mathbf{a}_1, \ldots, \mathbf{a}_i, \ldots, \mathbf{a}_j, \ldots, \mathbf{a}_m\}$ of all row vectors of $\mathbf{A}$ contains a linearly independent subset $S := \{\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_s}\}$ having s elements. We claim that the set $\{\mathbf{a}_1, \ldots, \mathbf{a}_i + \alpha\, \mathbf{a}_j, \ldots, \mathbf{a}_j, \ldots \mathbf{a}_m\}$ of all row vectors of $\mathbf{A}'$ also contains a linearly independent set S' containing s elements. If $\mathbf{a}_i \notin S$, then we let $S' := S$. Next, suppose $\mathbf{a}_i \in S$. Then we may replace $\mathbf{a}_i$ suitably either by $\mathbf{a}_i + \alpha \mathbf{a}_j$ or by $\mathbf{a}_j$ in the set S to form a linearly independent set S'. The last statement follows by considering the cases $\mathbf{a}_i + \alpha \mathbf{a}_j = \mathbf{0}$, $\mathbf{a}_j = \mathbf{0}$, and by observing that if $\mathbf{a}_i + \alpha\, \mathbf{a}_j$ as well as $\mathbf{a}_j$ were linear combinations of vectors in $S \setminus \{\mathbf{a}_i\}$, then so would be $\mathbf{a}_i = (\mathbf{a}_i + \alpha\, \mathbf{a}_j) - \alpha\, \mathbf{a}_j$, and this would contradict the linear independence of S. Note that the converse claim also holds.

ERO of type III: $\alpha\, R_j$ with $\alpha \neq 0$: $\{\mathbf{a}_j, \mathbf{a}_{j_1}, \ldots \mathbf{a}_{j_s}\}$ is linearly independent $\iff$ $\{\alpha\, \mathbf{a}_j, \mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_s}\}$ is linearly independent.

Thus the maximum number of linearly independent rows of $\mathbf{A}$ is the same as the maximum number of linearly independent rows of $\mathbf{A}'$, that is, the row ranks of $\mathbf{A}$ and $\mathbf{A}'$ are equal. $\square$

Proposition

Let a matrix $\mathbf{A}'$ be in REF. Then the nonzero rows of $\mathbf{A}'$ are linearly independent, and so the row rank of $\mathbf{A}'$ is equal to the number of nonzero rows of $\mathbf{A}'$.

Proof. Let the number of nonzero rows of $\mathbf{A}'$ be r. Let the pivots $p_1, \ldots, p_r$ in these rows be in columns $k_1, \ldots, k_r$, where $1 \leq k_1 < \cdots < k_r \leq n$. Suppose $\alpha_1 \mathbf{a}_1 + \cdots + \alpha_r \mathbf{a}_r = \mathbf{0}$, where $\alpha_1, \ldots, \alpha_r \in \mathbb{R}$.

Assume for a moment that not all $\alpha_1, \ldots, \alpha_r$ are zero, and let α_j be the first nonzero number among them. Now the k_jth component of $\alpha_1 \mathbf{a}_1 + \cdots + \alpha_r \mathbf{a}_r = \alpha_j \mathbf{a}_j + \cdots + \alpha_r \mathbf{a}_r$ is equal to $\alpha_j p_j$ since all entries in the k_jth column below the pivot p_j are equal to 0. Hence $\alpha_j p_j = 0$. But $p_j \neq 0$, and so $\alpha_j = 0$, contrary to our assumption. Thus $\alpha_1 = \cdots = \alpha_r = 0$. This shows that the first r rows of $\mathbf{A}'$ are linearly independent.

Also, since the last $m - r$ rows of $\mathbf{A}'$ are zero rows, any $r + 1$ row vectors of $\mathbf{A}'$ will contain the vector $\mathbf{0}$, and so they will not be linearly independent. Hence the row rank of $\mathbf{A}'$ is r. □

We have now obtained an important result which tells us how to find the row rank of a matrix.

Proposition

The row rank of a matrix is equal to the number of nonzero rows in any row echelon form of the matrix.

Proof. Let $\mathbf{A}$ be a $m \times n$ matrix. By using EROs of type I and type II, we transform $\mathbf{A}$ to a row echelon form $\mathbf{A}'$. Then the row rank of $\mathbf{A}$ is equal to the row rank of $\mathbf{A}'$, and it is equal to the number of nonzero rows of $\mathbf{A}'$. $\square$

The above proposition implies that if $\mathbf{A}'$ and $\mathbf{A}''$ are two row echelon forms of a matrix $\mathbf{A}$, then they have the same number of nonzero rows; this number is equal to the row rank of $\mathbf{A}$.

Since each nonzero row of a matrix in a REF has exactly one pivot, we see that the row rank of a matrix is equal to the number of pivots in any row echelon form of the matrix.

MA 110:
Lecture 06

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Spring 2025

Let S be a set of vectors (in $\mathbb{R}^{1\times n}$ or in $\mathbb{R}^{n\times 1}$). Recall:

- S is linearly dependent if a nontrivial linear combination of finitely many vectors in S is zero, i.e., there are (distinct) vectors $\mathbf{a}_1, \ldots, \mathbf{a}_m$ in S and scalars $\alpha_1, \ldots, \alpha_m$, not all zero, such that $\alpha_1\mathbf{a}_1 + \cdots + \alpha_m\mathbf{a}_m = \mathbf{0}$.
- S is linearly independent if it is not linearly dependent.
- Crucial Result: If S has s vectors, and each of them is a linear combination of a (fixed) set of r vectors, with $s > r$, then S is linearly dependent.
- In particular, if S has more than n vectors, then S is linearly dependent.
- Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. The row rank of $\mathbf{A}$ is the maximum number of linearly independent row vectors of $\mathbf{A}$.
- Elementary row operations do not alter the row rank.
- row-rank($\mathbf{A}$) $=$ number of nonzero rows in a REF of $\mathbf{A}$ $=$ the number of pivots in a REF of $\mathbf{A}$.

Example

In a previous lecture, we had seen that the matrix

$$\mathbf{A} := \begin{bmatrix} 3 & 2 & 2 & -5 \\ 0.6 & 1.5 & 1.5 & -5.4 \\ 1.2 & -0.3 & -0.3 & 2.4 \end{bmatrix}$$

can be transformed to a row echelon form

$$\mathbf{A}' := \begin{bmatrix} \boxed{3} & 2 & 2 & -5 \\ 0 & \boxed{1.1} & 1.1 & -4.4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

by elementary row transformations of type I and type II. Since the number of nonzero rows of $\mathbf{A}'$ is 2, we see that the row rank of $\mathbf{A}$ is 2. This shows that the 3 row vectors of $\mathbf{A}$ are linearly dependent.

Column Rank of a Matrix

Definition

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The column rank of $\mathbf{A}$ is the maximum number of linearly independent column vectors of $\mathbf{A}$.

Clearly, column-rank$(\mathbf{A})$ = row-rank$(\mathbf{A}^\mathsf{T})$.

Proposition

The column rank of a matrix is equal to its row rank.

Proof. Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m \times n}$. Let r = row-rank$(\mathbf{A})$ and s :=column-rank$(\mathbf{A})$. We will show that $s \le r$ and $r \le s$. Let $\mathbf{a}_1, \ldots, \mathbf{a}_m$ denote the row vectors of $\mathbf{A}$, so that

$$\mathbf{a}_j := \begin{bmatrix} a_{j1} & \cdots & a_{jk} & \cdots & a_{jn} \end{bmatrix} \quad \text{for } j = 1, \ldots, m.$$

Among these m row vectors of $\mathbf{A}$, there are r row vectors $\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_r}$ which are linearly independent, and each row vector of $\mathbf{A}$ is a linear combination of these r row vectors.

For simplicity, let $\mathbf{b}_1 := \mathbf{a}_{j_1}, \ldots, \mathbf{b}_r := \mathbf{a}_{j_r}$. Then for $j = 1, \ldots, m$ and $\ell = 1, \ldots, r$, there is $\alpha_{j\ell} \in \mathbb{R}$ such that

$$\mathbf{a}_j = \alpha_{j1}\mathbf{b}_1 + \cdots + \alpha_{j\ell}\mathbf{b}_\ell + \cdots + \alpha_{jr}\mathbf{b}_r \quad \text{for } j = 1, \ldots, m.$$

If $\mathbf{b}_\ell := \begin{bmatrix} b_{\ell 1} & \cdots & b_{\ell k} & \cdots & b_{\ell n} \end{bmatrix}$ for $\ell = 1, \ldots, r$, then the above equations can be written componentwise as follows:

$$\begin{array}{ccccccccccc} a_{1k} & = & \alpha_{11}b_{1k} & + & \cdots & + & \alpha_{1\ell}b_{\ell k} & + & \cdots & + & \alpha_{1r}b_{rk} \\ \vdots & & \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{jk} & = & \alpha_{j1}b_{1k} & + & \cdots & + & \alpha_{j\ell}b_{\ell k} & + & \cdots & + & \alpha_{jr}b_{rk} \\ \vdots & & \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{mk} & = & \alpha_{m1}b_{1k} & + & \cdots & + & \alpha_{m\ell}b_{\ell k} & + & \cdots & + & \alpha_{mr}b_{rk} \end{array}$$

for $k = 1, \ldots, n$. These m equations show that the kth column of $\mathbf{A}$ can be written as follows:

$$\begin{bmatrix} a_{1k} \\ \vdots \\ a_{jk} \\ \vdots \\ a_{mk} \end{bmatrix} = b_{1k} \begin{bmatrix} \alpha_{11} \\ \vdots \\ \alpha_{j1} \\ \vdots \\ \alpha_{m1} \end{bmatrix} + b_{2k} \begin{bmatrix} \alpha_{12} \\ \vdots \\ \alpha_{j2} \\ \vdots \\ \alpha_{m2} \end{bmatrix} + \cdots + b_{rk} \begin{bmatrix} \alpha_{1r} \\ \vdots \\ \alpha_{jr} \\ \vdots \\ \alpha_{mr} \end{bmatrix}.$$

for $k = 1, \ldots, n$. Thus each of the n columns of $\mathbf{A}$ is a linear combination of the r column vectors $\begin{bmatrix} \alpha_{11} & \cdots & \alpha_{j1} & \cdots & \alpha_{m1} \end{bmatrix}^{\mathsf{T}}$, $\begin{bmatrix} \alpha_{12} & \cdots & \alpha_{j2} & \cdots & \alpha_{m2} \end{bmatrix}^{\mathsf{T}}, \ldots, \begin{bmatrix} \alpha_{1r} & \cdots & \alpha_{jr} & \cdots & \alpha_{mr} \end{bmatrix}^{\mathsf{T}}$.

By a crucial result we have proved earlier, no more than r columns of $\mathbf{A}$ can be linearly independent. Hence $s \leq r$, i.e., column-rank($\mathbf{A}$) $\leq$ row-rank($\mathbf{A}$).

Applying the above result to $\mathbf{A}^{\mathsf{T}}$ in place of $\mathbf{A}$, we obtain $r \leq s$. Thus $s = r$, as desired. □

In view of the above result, we define the **rank** of a matrix $\mathbf{A}$ to be the common value of the row rank of $\mathbf{A}$ and the column rank of $\mathbf{A}$, and we denote it by $\boxed{\operatorname{rank}\mathbf{A}}$. Consequently, $\boxed{\operatorname{rank}\mathbf{A}^{\mathsf{T}} = \operatorname{rank}\mathbf{A}}$.

Example

$$\text{Let } \mathbf{A}' := \begin{bmatrix} \boxed{1} & 2 & 3 & 4 & 5 & 6 \\ 0 & 0 & 0 & \boxed{7} & 8 & 9 \\ 0 & 0 & 0 & 0 & 0 & \boxed{10} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since $\mathbf{A}'$ is in REF, its row rank is equal to the nonzero rows, namely 3. Also, we can easily check that the columns $\mathbf{c}_1, \mathbf{c}_4$ and $\mathbf{c}_6$ are linearly independent, and so the column rank of $\mathbf{A}'$ is at least 3. Further, each of the columns $\mathbf{c}_2 = 2\mathbf{c}_1$, $\mathbf{c}_3 = 3\mathbf{c}_1$ and $\mathbf{c}_5 = (3/7)\mathbf{c}_1 + (8/7)\mathbf{c}_4$ is linear combination of the columns $\mathbf{c}_1, \mathbf{c}_4$ and $\mathbf{c}_6$, and so by a crucial result, the column rank of $\mathbf{A}'$ is at most 3. Hence the column rank of $\mathbf{A}'$ is also 3.

Let us restate some results proved earlier in terms of the newly introduced notion of the rank of a matrix.

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$.

1. If $\mathbf{A}$ is transformed to $\mathbf{A}'$ by EROs, then $\operatorname{rank} \mathbf{A}' = \operatorname{rank} \mathbf{A}$.
2. If $\mathbf{A}$ is transformed to $\mathbf{A}'$ by EROs and $\mathbf{A}'$ is in REF, then

$$\begin{aligned} \operatorname{rank} \mathbf{A} &= \text{number of nonzero rows of } \mathbf{A}' \\ &= \text{number of pivotal columns of } \mathbf{A}'. \end{aligned}$$

(Note: Each nonzero row has exactly one pivot, and a zero row has no pivot.)

3. Let $m = n$. Then

$$\mathbf{A} \text{ is invertible} \iff \operatorname{rank} \mathbf{A} = n.$$

(Recall: A square matrix is invertible if and only if it can be transformed to the identity matrix by EROs.)

Subspace, Basis, Dimension

From now on we shall consider only the $\boxed{\text{column vectors}}$.

Let $n \in \mathbb{N}$. We have denoted the set of all column vectors whose entries are real numbers and whose length is n by $\mathbb{R}^{n\times 1}$.

Definition

A nonempty subset V of $\mathbb{R}^{n\times 1}$ is called a vector subspace, or simply a subspace of $\mathbb{R}^{n\times 1}$ if

(i) $\mathbf{a}, \mathbf{b} \in V \implies \mathbf{a} + \mathbf{b} \in V$, and

(ii) $\alpha \in \mathbb{R}$, $\mathbf{a} \in V \implies \alpha\,\mathbf{a} \in V$.

Note: The zero column vector $\mathbf{0}$ is in every subspace V since $V \neq \emptyset$ and $\mathbf{0} = \mathbf{a} + (-1)\mathbf{a}$ for each $\mathbf{a} \in V$.

Each linear combination of elements of a subspace V is in V.

$\{\mathbf{0}\}$ is the smallest and $\mathbb{R}^{n\times 1}$ is the largest subspace of $\mathbb{R}^{n\times 1}$.

Two Important Examples of Subspaces Related to a Matrix

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$.

Definition

The **null space** of $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) := \{\mathbf{x} \in \mathbb{R}^{n \times 1} : \mathbf{A}\mathbf{x} = \mathbf{0}\}.$$

Definition

The **column space** of $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) := \text{ the set of all linear combinations of columns of } \mathbf{A}.$$

We shall later find an interesting relationship between the null space $\mathcal{N}(\mathbf{A})$ and the column space $\mathcal{C}(\mathbf{A})$.

Basis and dimension of a subspace

Let V be a subspace of $\mathbb{R}^{n\times 1}$. Since every element of $\mathbb{R}^{n\times 1}$ is a linear combination of the vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$, we see that no linearly independent subset of V contains more than n vectors.

Definition

A subset S of V is called a basis of V if S is linearly independent and S has maximum possible number of elements among linearly independent subsets of V.

Clearly, a basis of V has at most n elements.

Definition

Let $S \subset \mathbb{R}^{n\times 1}$. The set of all linear combinations of elements of S is denoted by $\operatorname{span} S$ and called the span of S.

Proposition

Let V be a subspace of $\mathbb{R}^{n\times 1}$, and let $S \subset V$. Then S is a basis for $V \iff S$ is linearly independent and $\operatorname{span} S = V$.

Proof. Suppose S is a basis for V. By the definition of a basis, S is linearly independent. Let S have r elements. If $\mathbf{x} \in V$, but $\mathbf{x}$ is not a linear combination of elements of S, then $S \cup \{\mathbf{x}\}$ would be a linearly independent subset of V containing $r+1$ elements, which would be a contradiction.

Conversely, suppose S is linearly independent and $\operatorname{span} S = V$. Let S have r elements. Since every element of V is a linear combination of elements of S, no more than r elements of V can be linearly independent. Hence S is a basis for V. $\square$

Corollary

Let V be a subspace of $\mathbb{R}^{n\times 1}$ and let S, S' be two bases of V. Then S and S' have the same cardinality.

Proof. It is enough to prove that $|S| \leq |S'|$, because we can reverse the roles of S and S' and prove the other inequality. By the last proposition, every element of S is a linear combination of vectors from S'. If $|S| > |S'|$, then S would be linearly dependent. As S is a basis, it is linearly independent, which implies $|S| \leq |S'|$. $\square$

Definition

The dimension of V is defined as the number of elements in a basis of V. It is denoted by $\dim V$.

Example: Clearly, $\dim \mathbb{R}^{n\times 1} = n$ and $\dim\{\mathbf{0}\} = 0$.

Corollary

Let V be a subspace of $\mathbb{R}^{n\times 1}$. Every linearly independent subset of V can be enlarged to a basis for V.

Proof. Let S be a linearly independent subset of V. If $\operatorname{span} S = V$, then by the previous result, S is a basis for V. Suppose $\operatorname{span} S \neq V$. Then there is $\mathbf{x}_1 \in V$ such that $\mathbf{x}_1$ is not a linear combination of elements of S. Now $S_1 := S \cup \{\mathbf{x}_1\}$ is a linearly independent subset of V. If $\operatorname{span} S_1 = V$, then as before, S_1 is a basis for V. If $\operatorname{span} S_1 \neq V$, then there is $\mathbf{x}_2 \in V$ such that $\mathbf{x}_2$ is not a linear combination of elements of S_1. Now $S_2 := S_1 \cup \{\mathbf{x}_2\}$ is a linearly independent subset of V.

This process must end after a finite number of steps since the number of elements of any linearly independent subset of V is less than or equal to $\dim V$, and $\dim V \leq n$. $\square$

Remark: All things we have defined above for column vectors can also be defined for row vectors.

Another immediate consequence of the earlier result is the unique representation of every element of a subspace in terms of the elements belonging to a basis of that subspace.

Proposition

Let $S := \{\mathbf{c}_1, \ldots, \mathbf{c}_r\}$ be a basis for a subspace V of $\mathbb{R}^{n\times 1}$, and let $\mathbf{x} \in V$. Then there are unique $\alpha_1, \ldots, \alpha_r \in \mathbb{R}$ such that $\mathbf{x} = \alpha_1\mathbf{c}_1 + \cdots + \alpha_r\mathbf{c}_r$.

Proof. Since V is a basis for V, we obtain $V = \operatorname{span} S$, and so the vector $\mathbf{x}$ is a linear combination of vectors in S, that is, there are scalars $\alpha_1, \ldots, \alpha_r$ such that $\mathbf{x} = \alpha_1\mathbf{c}_1 + \cdots + \alpha_r\mathbf{c}_r$. Now suppose $\mathbf{x} = \beta_1\mathbf{c}_1 + \cdots + \beta_r\mathbf{c}_r$ for some $\beta_1, \ldots, \beta_r \in \mathbb{R}$. Then

$$(\alpha_1 - \beta_1)\mathbf{c}_1 + \cdots + (\alpha_r - \beta_r)\mathbf{c}_r = \mathbf{0}.$$

Since the set S is linearly independent, $\alpha_1 - \beta_1 = \cdots = \alpha_r - \beta_r = 0$, that is, $\beta_1 = \alpha_1, \ldots, \beta_r = \alpha_r$. This proves the uniqueness. $\square$

Let us recall some definitions and results from the last lecture.

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. The row rank of $\mathbf{A}$ is the maximum number of linearly independent row vectors of $\mathbf{A}$. The column rank of $\mathbf{A}$ is the maximum number of linearly independent column vectors of $\mathbf{A}$. The two are equal, and we denote both by $\boxed{\operatorname{rank}\mathbf{A}}$. Further, if $\mathbf{A}'$ is any REF of $\mathbf{A}$, then the number of nonzero rows of $\mathbf{A}'$ is equal to $\operatorname{rank}\mathbf{A}'$ and it is equal to $\operatorname{rank}\mathbf{A}$.

The column space of $\mathbf{A}$ is denoted by $\mathcal{C}(\mathbf{A})$ and the null space of $\mathbf{A}$ by $\mathcal{N}(\mathbf{A})$.

Proposition

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$, and let $\operatorname{rank}\mathbf{A} = r$. Then $\dim \mathcal{C}(\mathbf{A}) = r$ and $\dim \mathcal{N}(\mathbf{A}) = n - r$.

Proof. Since the column rank of $\mathbf{A}$ is equal to r, there are r linearly independent columns $\mathbf{c}_{k_1}, \ldots, \mathbf{c}_{k_r}$ of $\mathbf{A}$, and any other column of $\mathbf{A}$ is a linear combination of these r columns.

Let $\mathbf{x} \in \mathcal{C}(\mathbf{A})$. Then $\mathbf{x}$ is a linear combination of columns of $\mathbf{A}$, each of which in turn is a linear combination of $\mathbf{c}_{k_1}, \ldots, \mathbf{c}_{k_r}$. Thus $\mathbf{x}$ is a linear combination of $\mathbf{c}_{k_1}, \ldots, \mathbf{c}_{k_r}$. This shows that $\operatorname{span}\{\mathbf{c}_{k_1}, \ldots, \mathbf{c}_{k_r}\} = \mathcal{C}(A)$. Hence $\{\mathbf{c}_{k_1}, \ldots, \mathbf{c}_{k_r}\}$ is a basis for $\mathcal{C}(\mathbf{A})$ and $\dim \mathcal{C}(\mathbf{A}) = r$.

To find the dimension of $\mathcal{N}(\mathbf{A})$, let us transform $\mathbf{A}$ to a REF $\mathbf{A}'$ by EROs of type I and type II. Since the row rank of $\mathbf{A}$ is equal to r, the matrix $\mathbf{A}'$ has exactly r nonzero rows and exactly r pivotal columns.

Let the $n - r$ nonpivotal columns be denoted by $\mathbf{c}_{\ell_1}, \ldots, \mathbf{c}_{\ell_{n-r}}$. Then $x_{\ell_1}, \ldots, x_{\ell_{n-r}}$ are the free variables. For each $\ell \in \{\ell_1, \ldots, \ell_{n-r}\}$, there is a basic solution $\mathbf{s}_\ell$ of the homogeneous equation $\mathbf{Ax} = \mathbf{0}$, and every solution of this homogeneous equation is a linear combination of these $n - r$ basic solutions. Let S denote the set of these $n - r$ basic solutions. Then $\operatorname{span} S = \mathcal{N}(\mathbf{A})$.

We claim that the set S of the $n-r$ basic solutions is linearly independent. To see this, we note that each basic solution is equal to 1 in one of the free variables and it is equal to 0 in the other free variables. Let $\alpha_1, \ldots, \alpha_{n-r} \in \mathbb{R}$ be such that

$$\mathbf{x} := \alpha_1 \mathbf{s}_{\ell_1} + \cdots + \alpha_{n-r} \mathbf{s}_{\ell_{n-r}} = \mathbf{0}.$$

For $j = 1, \ldots, n$, let x_j denote the jth entry of $\mathbf{x}$. Then for each $\ell \in \{\ell_1, \ldots, \ell_{n-r}\}$, we see that $\alpha_\ell \cdot 1 = x_\ell = 0$. Hence S is linearly independent. Thus S is a basis for $\mathcal{N}(\mathbf{A})$ and $\dim \mathcal{N}(\mathbf{A}) = n - r$, the number of elements in S. □

The dimension of the null space $\mathcal{N}(\mathbf{A})$ of $\mathbf{A}$ is called the **nullity** of $\mathbf{A}$. Since $r + (n-r) = n$, we have proved the following **Rank-Nullity Theorem**.

Theorem

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then $\operatorname{rank} \mathbf{A} + \operatorname{nullity} \mathbf{A} = n$.

MA 110:
Lecture 07

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Spring 2025

Review of last lecture

We have discussed the following important notions.

- Rank of a matrix
- Vector subspaces (of $\mathbb{R}^{n\times 1}$).
- Basis of a subspace
- Dimension of a subspace
- Span of a subset of a vector subspace.
- Null space and the column space of a matrix.

And we proved several important results such as:

- Characterizations of a basis of a vector subspace
- Rank-Nullity Theorem
- Fundamental Theorem for Linear Systems:

Remark: The notion of a vector subspace of $\mathbb{R}^{1\times m}$ is defined similarly. The corresponding notions of basis, dimension, span, etc. are defined in an identical manner.

Let us restate two earlier results which are in conformity with the Rank-Nullity Theorem. Let $\mathbf{A}$ be an $n \times n$ matrix. Then

$$\boxed{\mathbf{A} \text{ is invertible} \iff \text{nullity}\, \mathbf{A} = 0 \iff \text{rank}\, \mathbf{A} = n}.$$

Further, $\text{rank}\, \mathbf{A} = n \iff \mathcal{C}(\mathbf{A}) = \mathbb{R}^{n\times 1}$.

We are now in a position to state and prove a comprehensive result regarding solutions of a system of m linear equations in n unknowns that we started with, namely

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \qquad (1)\\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \qquad (2)\\
\vdots \qquad \vdots \qquad \vdots \qquad\quad &\vdots \quad \vdots\\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \qquad (m)
\end{aligned}$$

As usual, we write this as $\mathbf{Ax} = \mathbf{b}$, where
$\mathbf{A} := [a_{jk}] \in \mathbb{R}^{m\times n}$, $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$ and $\mathbf{b} := \begin{bmatrix} b_1 & \cdots & b_m \end{bmatrix}^{\mathsf{T}}$.

Theorem (Fundamental Theorem for Linear Systems: FTLS)

Let $m, n \in \mathbb{N}$ and $\mathbf{A}$ be an $m \times n$ matrix with real entries. Suppose $\boxed{\text{rank}\,\mathbf{A} = r}$.

(i) $\boxed{\text{Homogeneous Linear System}}$: $\mathbf{A}\mathbf{x} = \mathbf{0}$ $\boxed{\text{(H)}}$

The solution space $\{\mathbf{x} \in \mathbb{R}^{n\times 1} : \mathbf{A}\mathbf{x} = \mathbf{0}\}$ of (H) is a subspace of $\mathbb{R}^{n\times 1}$ of dimension $\boxed{n - r}$.

In particular, $r = n$ if and only if $\mathbf{0}$ is the only solution of (H).

If $r < n$, then there are linearly independent solutions $\mathbf{x}_1, \ldots, \mathbf{x}_{n-r}$ of (H) and every solution of (H) is a unique linear combination of these $\mathbf{x}_1, \ldots, \mathbf{x}_{n-r}$.

(ii) $\boxed{\text{General Linear System}}$: $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \in \mathbb{R}^{m\times 1}$ $\boxed{\text{(G)}}$

(G) has a solution if and only if $\text{rank}[\mathbf{A}|\mathbf{b}] = r$. In this case, let $\mathbf{x}_0$ be a particular solution of (G). If $\mathbf{x}$ is a solution of (G), then $\mathbf{x} = \mathbf{x}_0 + \mathbf{x}_h$, where $\mathbf{x}_h$ is a solution of (H) above.

Proof. (i) The solution space of the homogeneous linear system (H) is just the nullspace $\mathcal{N}(\mathbf{A})$ of $\mathbf{A}$, and we have seen that its dimension, that is, the nullity of $\mathbf{A}$, is equal to $n - r$.

We note that $r = n \iff n - r = 0$, that is, the dimension of $\mathcal{N}(\mathbf{A})$ is zero; in other words, $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$. This means that $\mathbf{0}$ is the only solution of (H).

Let now $r < n$. Then $\mathcal{N}(\mathbf{A})$ has a basis consisting of $n - r$ elements, say $\mathbf{x}_1, \ldots, \mathbf{x}_{n-r}$. Hence every element of the solution space is a unique linear combination of the elements in this basis.

(ii) Let $\mathbf{b} \in \mathbb{R}^{n \times 1}$. Let $\mathbf{c}_1, \ldots, \mathbf{c}_n$ be the n columns of $\mathbf{A}$. Then

$$\mathbf{A}\mathbf{x} = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n \quad \text{for } \mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n \times 1}.$$

Hence $\mathbf{A}\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{R}^{n \times 1}$ if and only if $\mathbf{b}$ is a linear combination of the columns of $\mathbf{A}$, that is, $\mathbf{b} \in \mathcal{C}(\mathbf{A})$.

Since every column of $\mathbf{A}$ is also a column of the augmented matrix $[\mathbf{A}|\mathbf{b}]$, the column space $\mathcal{C}(\mathbf{A})$ of $\mathbf{A}$ is contained in the column space $\mathcal{C}([\mathbf{A}|\mathbf{b}])$ of $[\mathbf{A}|\mathbf{b}]$. It follows that $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ if and only if $\mathcal{C}([\mathbf{A}|\mathbf{b}]) = \mathcal{C}(\mathbf{A})$, that is, the column rank of $[\mathbf{A}|\mathbf{b}]$ is equal to the column rank of $\mathbf{A}$. So $\operatorname{rank}[\mathbf{A}|\mathbf{b}] = \operatorname{rank} \mathbf{A} = r$.

Let $\mathbf{x}_0$ be a particular solution of (G), that is, let $\mathbf{x}_0 \in \mathbb{R}^{n\times 1}$ satisfy $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$. Then for any $\mathbf{x} \in \mathbb{R}^{n\times 1}$, we see that $\mathbf{A}\mathbf{x} = \mathbf{b}$ if and only if $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{A}\mathbf{x} - \mathbf{A}\mathbf{x}_0 = \mathbf{b} - \mathbf{b} = \mathbf{0}$, that is, $\mathbf{x}$ is a solution of (G) if and only if $\mathbf{x}_h := \mathbf{x} - \mathbf{x}_0$ is a solution of (H). The proof is complete. $\square$

Remark

The above theorem is of immense theoretical importance. It tells us precisely when solutions exist, and also describes the nature of solutions of a linear system of equations.
For example, it says that when there is a nonzero solution of a homogeneous linear system, there are infinitely many solutions.

Further, when a homogeneous system has infinitely many solutions, it says that they can be described by a one parameter family, or a two parameter family etc.

It may seem that to implement the results of the above theorem, we must first find the rank of the coefficient matrix $\mathbf{A}$ of the linear system. This is not necessary.

We have already seen that we may directly proceed to find the solutions of the linear system by considering the augmented matrix $[\mathbf{A}|\mathbf{b}]$ and transform the coefficient matrix $\mathbf{A}$ to a row echelon form by the Gauss Elimination Method and then use Back Substitution . This process itself reveals all possibilities.

In particular, when the rank r of $\mathbf{A}$ is less than the number n of variables, we have shown how to construct a set S of basic solutions of an homogeneous linear system. This set S is in fact a basis of the solution space of the system. That is the reason for using the terminology 'basic solutions'.

Definition

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The row space of A, denoted $\mathcal{R}(\mathbf{A})$, is defined as the subspace of $\mathbb{R}^{1 \times n}$ spanned by the row vectors of $\mathbf{A}$.

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Here are some important observations.

- The row-rank of $\mathbf{A}$ is precisely the dimension of $\mathcal{R}(\mathbf{A})$.
- If $\mathbf{A}' \in \mathbb{R}^{m \times n}$ is obtained from $\mathbf{A}$ by an elementary row operation, then $\mathcal{R}(\mathbf{A}) = \mathcal{R}(\mathbf{A}')$.
- If $\mathbf{A}' \in \mathbb{R}^{m \times n}$ is in REF, then the nonzero rows of $\mathbf{A}'$ form a basis of $\mathcal{R}(\mathbf{A}')$.
- A basis of $\mathcal{R}(\mathbf{A})$ is given by the nonzero rows of its REF.
- If $\mathbf{A}'$ is obtained from $\mathbf{A}$ by an elementary row operation, then $\mathcal{C}(\mathbf{A})$ need not be equal to $\mathcal{C}(\mathbf{A}')$.

Lemma

The columns of $\mathbf{A}$ *corresponding to the pivotal columns of its REF form a basis of* $\mathcal{C}(\mathbf{A})$.

Proof. Let $\mathbf{A}'$ be an REF of $\mathbf{A}$. Suppose the columns $\mathbf{c}'_{\mathbf{i_1}}, \cdots, \mathbf{c}'_{\mathbf{i_r}}$ of $\mathbf{A}'$ are the pivotal columns. Let $\mathbf{c}_{\mathbf{i_1}}, \cdots, \mathbf{c}_{\mathbf{i_r}}$ be the corresponding columns of $\mathbf{A}$. It is enough to prove that $\mathbf{c}_{\mathbf{i_1}}, \cdots, \mathbf{c}_{\mathbf{i_r}}$ are linearly independent. If they were not so, then there would be constants $\alpha_{i_1}, \cdots, \alpha_{i_r}$, not all zero such that

$$\alpha_{i_1}\mathbf{c}_{\mathbf{i_1}} + \cdots + \alpha_{i_r}\mathbf{c}_{\mathbf{i_r}} = \mathbf{0}.$$

Consider the vector $\mathbf{x} = (x_1, \ldots, x_n)^T$ such that $x_j = \alpha_j$ if j is one of $i_1, \ldots, i_r$, and $x_j = 0$ other wise. Then $\mathbf{Ax} = \mathbf{0}$, thus $\mathbf{A}'\mathbf{x} = \mathbf{0}$ as well. But this means,

$$\alpha_{i_1}\mathbf{c}'_{\mathbf{i_1}} + \cdots + \alpha_{i_r}\mathbf{c}'_{\mathbf{i_r}} = \mathbf{0},$$

which is a contradiction because $\mathbf{c}'_{\mathbf{i_1}}, \cdots, \mathbf{c}'_{\mathbf{i_r}}$ being the pivotal columns of a matrix in REF, are linearly independent. $\square$

Example: Consider the 5×6 matrix $\mathbf{A}$ and its REF $\mathbf{A}'$ given by

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix}, \quad \mathbf{A}' = \begin{bmatrix} \boxed{1} & 3 & -2 & 0 & 2 & 0 \\ 0 & 0 & \boxed{-1} & -2 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 & \boxed{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Then $\operatorname{rank} \mathbf{A} = 3$. A basis of the row space $\mathcal{R}(\mathbf{A})$ is given by

$$\left\{ \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & -1 & -2 & 0 & -3 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 6 \end{bmatrix} \right\}$$

whereas a basis for the column space $\mathcal{C}(\mathbf{A})$ is given by

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ -5 \\ 0 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 0 \\ 15 \\ 18 \end{bmatrix} \right\}.$$

Determinants

You already know formulas for determinants of 1×1, 2×2 and 3×3 matrices. Let us recall them.

$$\det \begin{bmatrix} a_1 \end{bmatrix} = a_1, \quad \det \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} = a_1 b_2 - a_2 b_1 \quad \text{and}$$

$$\det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2),$$

which is also equal to
$a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$.

We shall presently give formulas for the determinant of an $n \times n$ matrix, that is, of a matrix of size n, where $n \in \mathbb{N}$, and we shall explore their use in matrix theory.

Let $n \in \mathbb{N}$ and $\mathbf{A} := \begin{bmatrix} a_{11} & \cdots & a_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{j1} & \cdots & \boxed{a_{jk}} & \cdots & a_{jn} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nk} & \cdots & a_{nn} \end{bmatrix} \in \mathbb{R}^{n \times n}$.

The determinant of $\mathbf{A}$ is a real number defined inductively as follows. For $n := 1$, define $\det \mathbf{A} := a_{11}$. Let $n \geq 2$, and suppose we have defined the determinant of any $(n-1)\times(n-1)$ matrix. For $j, k = 1, \ldots, n$, let $\mathbf{A}_{jk}$ denote the submatrix of $\mathbf{A}$ obtained by deleting the jth row and the kth column of $\mathbf{A}$, and let $M_{jk} := \det \mathbf{A}_{jk}$, called the (j, k)th **minor** of $\mathbf{A}$. Define

$$\det \mathbf{A} := a_{11}M_{11} - a_{12}M_{12} + \cdots + (-1)^{1+k} a_{1k}M_{1k} + \cdots + (-1)^{1+n} a_{1n}M_{1n}.$$

This is also known as the **expansion for the determinant of A in terms of the first row of A**.

An immediate consequence of our definition is the following.

Proposition

If an $n \times n$ matrix $\mathbf{A}$ is lower triangular, then the determinant of $\mathbf{A}$ is the product of its diagonal entries.

Proof. We prove it by induction on n. For $n = 1$, it is the definition. Suppose we know the proposition for all $(n-1) \times (n-1)$ matrices. Now $\det \mathbf{A} = a_{11} M_{11}$ since $a_{12} = \cdots = a_{1n} = 0$, etc. Now $\mathbf{A_{1,1}}$ is lower triangular and thus by the induction hypothesis, $M_{1,1}$ is the product of the diagonal entries of $\mathbf{A}$ other than a_{11}. $\square$

Next, it can be proved by induction on the size n of a matrix that $\det \mathbf{A}$ is equal to the following expansions in terms of the jth row of $\mathbf{A}$, and also in terms of the kth column of $\mathbf{A}$:

$$
\begin{aligned}
\det \mathbf{A} &= \sum_{k=1}^{n} (-1)^{j+k} a_{jk} M_{jk} \quad \text{for each } j \in \{1, \ldots, n\} \\
\det \mathbf{A} &= \sum_{j=1}^{n} (-1)^{j+k} a_{jk} M_{jk} \quad \text{for each } k \in \{1, \ldots, n\}
\end{aligned}
$$

(For a proof, see Kreyszig, Appendix 4, page A81.)

Proposition

Let $\mathbf{A}$ be a square matrix. Then $\det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A}$.

Proof. This is obvious if $n = 1$. Let now $n \geq 2$, and assume this property for all $(n-1)\times(n-1)$ matrices. Note that $(\mathbf{A}^{\mathsf{T}})_{jk} = (\mathbf{A}_{kj})^{\mathsf{T}}$ for all $j, k = 1, \ldots, n$, that is, the submatrix obtained by deleting the jth row and the kth column of $\mathbf{A}^{\mathsf{T}}$ is the same as the transpose of the submatrix obtained by deleting the kth row and the jth column of $\mathbf{A}$. (For example,

$$\mathbf{A} := \begin{bmatrix} a_{11} & \boxed{a_{12}} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \implies \mathbf{A}^{\mathsf{T}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ \boxed{a_{12}} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix},$$

so that $(\mathbf{A}^{\mathsf{T}})_{21} = \begin{bmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}^{\mathsf{T}} = (\mathbf{A}_{12})^{\mathsf{T}}$.)

Let $\mathbf{A} := [a_{jk}]$ and $\mathbf{A}^\mathsf{T} := [a'_{jk}]$. Then $a'_{jk} = a_{kj}$ and $M'_{jk} := \det(\mathbf{A}^\mathsf{T})_{jk} = \det(\mathbf{A}_{kj})^\mathsf{T} = \det \mathbf{A}_{kj} = M_{kj}$ by the inductive hypothesis for $j, k = 1, \ldots, n$. Expanding $\det \mathbf{A}^\mathsf{T}$ in terms of its first row, and $\det \mathbf{A}$ in terms of its first column,

$$\begin{aligned} \det \mathbf{A}^\mathsf{T} &= a'_{11}M'_{11} - a'_{12}M'_{12} + \cdots + (-1)^{1+n}a'_{1n}M'_{1n} \\ &= a_{11}M_{11} - a_{21}M_{21} + \cdots + (-1)^{n+1}a_{n1}M_{n1} \\ &= \det \mathbf{A}. \quad \square \end{aligned}$$

Corollary

If $\mathbf{A}$ is upper triangular, then the determinant of $\mathbf{A}$ is the product of its diagonal entries.

Proof. Let $\mathbf{A}$ be upper triangular. Then $\mathbf{A}^\mathsf{T}$ is lower triangular and has the same diagonal entries as those of $\mathbf{A}$. $\square$

Let us write $\boxed{\mathbf{A} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_n \end{bmatrix}}$ in terms of its n columns.

Crucial Properties of the **determinant function** $\mathbf{A} \longmapsto \det \mathbf{A}$ from $\mathbb{R}^{n\times n}$ to $\mathbb{R}$:

1. Let $k \in \{1, \ldots, n\}$ and $\boxed{\mathbf{c}_k = \alpha\, \mathbf{c}'_k + \beta\, \mathbf{c}''_k}$, where $\alpha, \beta \in \mathbb{R}$.

Then $\det \mathbf{A} = \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \alpha\, \mathbf{c}'_k + \beta\, \mathbf{c}''_k & \cdots & \mathbf{c}_n \end{bmatrix}$ is equal to $\alpha \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}'_k & \cdots & \mathbf{c}_n \end{bmatrix} + \beta \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}''_k & \cdots & \mathbf{c}_n \end{bmatrix}$.

This is proved by expanding $\det \mathbf{A}$ in terms of its kth column. In particular,
$\det(\alpha \mathbf{A}) = \det \begin{bmatrix} \alpha\mathbf{c}_1 & \cdots & \alpha\mathbf{c}_k & \cdots & \alpha\mathbf{c}_n \end{bmatrix} = \alpha^n \det \mathbf{A}$.

2. Let $n \geq 2$. If $\boxed{\mathbf{c}_k = \mathbf{c}_\ell}$ for some $k \neq \ell$, then $\det \mathbf{A} = 0$, that is, if 2 columns of a matrix are identical, then its determinant is equal to 0. This is clear for $n = 2$, and for $n \geq 3$, this is proved by expanding $\det \mathbf{A}$ in terms of a column $\mathbf{c}_p$ of $\mathbf{A}$, where $p \neq k$ and $p \neq \ell$, and then using induction on n.

3. $\boxed{\det \mathbf{I} = 1}$, that is, the determinant of the identity matrix is equal to 1. This is obvious.

Proposition

Let $\mathbf{A}$ be a square matrix.

(i) If two columns of $\mathbf{A}$ are interchanged, then $\det \mathbf{A}$ gets multiplied by -1.

(ii) Addition of a multiple of a column to another column of $\mathbf{A}$ does not alter $\det \mathbf{A}$.

(iii) Multiplication of a column of $\mathbf{A}$ by a scalar α results in the multiplication of $\det \mathbf{A}$ by α.

Proof: Let $\mathbf{A} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix}$, where $k \neq \ell$.

(i) Define $\alpha := \det \begin{bmatrix} \mathbf{c}_1 & \cdots & (\mathbf{c}_k + \mathbf{c}_\ell) & \cdots & (\mathbf{c}_k + \mathbf{c}_\ell) & \cdots \mathbf{c}_n \end{bmatrix}$.
Then $\alpha = 0$ since the matrix has two identical columns.

On the other hand, $\alpha = \beta + \gamma$, where
$\beta := \det \begin{bmatrix} \mathbf{c}_1 & \cdots & (\mathbf{c}_k + \mathbf{c}_\ell) & \cdots & \mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix}$ and
$\gamma := \det \begin{bmatrix} \mathbf{c}_1 & \cdots & (\mathbf{c}_k + \mathbf{c}_\ell) & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix}$.

In turn, $\beta := \beta_1 + \beta_2$ and $\gamma = \gamma_1 + \gamma_2$, where

$$\begin{aligned}
\beta_1 &= \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix}, \\
\beta_2 &= \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_\ell & \cdots & \mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix}, \\
\gamma_1 &= \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix}, \\
\gamma_2 &= \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_\ell & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix}.
\end{aligned}$$

But $\beta_1 = 0 = \gamma_2$ since two columns are identical. Since $0 = \alpha = \beta + \gamma = \beta_2 + \gamma_1$, we see that $\gamma_1 = -\beta_2$, that is, $\det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix}$ is equal to $-\det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_\ell & \cdots & \mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix}$, as desired.

(ii) Suppose α times the ℓth column of $\mathbf{A}$ is added to the kth column of $\mathbf{A}$. Then $\det \begin{bmatrix} \mathbf{c}_1 & \cdots & (\mathbf{c}_k + \alpha\,\mathbf{c}_\ell) & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix}$ is equal to $\det \mathbf{A} + \alpha \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_\ell & \cdots & \mathbf{c}_\ell & \cdots \mathbf{c}_n \end{bmatrix} = \det \mathbf{A}$.

(iii) Suppose the kth column of $\mathbf{A}$ is multiplied by α. Then $\det \begin{bmatrix} \mathbf{c}_1 & \cdots & \alpha\,\mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix} = \alpha \det \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix} = \alpha \det \mathbf{A}$.

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Spring 2025

Corollary

Let $\mathbf{A}$ be a square matrix.

(i) If two rows of $\mathbf{A}$ are interchanged, then $\det \mathbf{A}$ gets multiplied by -1.

(ii) Addition of a multiple of a row to another row of $\mathbf{A}$ does not alter $\det \mathbf{A}$.

(iii) Multiplication of a row of $\mathbf{A}$ by a scalar α results in the multiplication of $\det \mathbf{A}$ by α.

Proof. Since the columns of $\mathbf{A}^{\mathsf{T}}$ are the rows of $\mathbf{A}$, and since $\det \mathbf{A} = \det \mathbf{A}^{\mathsf{T}}$, these results follow from the previous proposition. $\square$

The above corollary can be used to find $\det \mathbf{A}$ as follows. Transform $\mathbf{A}$ to $\mathbf{A}'$ by EROs of type I and type II, where $\mathbf{A}'$ is in REF, keeping track of the number of row interchanges.

Now $\mathbf{A}'$ is an upper triangular matrix. Let p be the number of row interchanges, and let $a'_{11}, \ldots, a'_{nn}$ be the diagonal entries of $\mathbf{A}'$. Then $\det \mathbf{A} = (-1)^p \det \mathbf{A}' = (-1)^p a'_{11} \cdots a'_{nn}$.

Corollary

A square matrix $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.

Proof. Note that $\mathbf{A}$ is invertible if and only if $\mathbf{A}'$ is. Now as $\mathbf{A}'$ is a square matrix in REF, it is invertible if and only if there are no zero rows. If there are no zero rows then each column is pivotal with pivots given by precisely the diagonal entries. Thus the formula above shows that $\det \mathbf{A} = \det \mathbf{A}' \neq 0$. Conversely, if the k-th row is zero in $\mathbf{A}'$, then the diagonal term $a_{kk} = 0$. Then the same formula above shows that $\det \mathbf{A} = \det \mathbf{A}' = 0$. $\square$

Example

$$\text{Let } \mathbf{A} := \begin{bmatrix} 0 & 2 & 0 & -1 \\ 1 & 2 & 1 & -1 \\ 0 & 0 & 3 & 2 \\ 1 & -2 & 1 & -2 \end{bmatrix} \xrightarrow{R_1 \longleftrightarrow R_2} \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 3 & 2 \\ 1 & -2 & 1 & -2 \end{bmatrix}$$

$$\xrightarrow{R_4 - R_1} \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 3 & 2 \\ 0 & -4 & 0 & -1 \end{bmatrix} \xrightarrow{R_4 + 2R_2} \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & -3 \end{bmatrix} = \mathbf{A}'.$$

Since there is only one row interchange while transforming $\mathbf{A}$ to $\mathbf{A}'$, since $\mathbf{A}'$ is in REF, and since $\det \mathbf{A}' = 1 \cdot 2 \cdot 3 \cdot (-3) = -18$, we see that $\det \mathbf{A} = (-1)(-18) = 18$.

Remark We have given several criteria for the invertibility of an $n \times n$ matrix $\mathbf{A}$. We list them in the order we proved them.

(i) $\text{nullity}(\mathbf{A}) = 0$.
(ii) There is a matrix $\mathbf{B}$ such that either $\mathbf{BA} = \mathbf{I}$ or $\mathbf{AB} = \mathbf{I}$.
(iii) The RCF of $\mathbf{A}$ is $\mathbf{I}$.
(iv) $\operatorname{rank} \mathbf{A} = n$.
(v) $\det \mathbf{A} \neq 0$.

Determinant and Rank

We now relate the rank of a matrix with determinants of its submatrices.

Lemma

Let $\mathbf{A}$ *be an* $m \times n$ *matrix, and* $r \in \mathbb{N}$. *Then*

$$\operatorname{rank}\mathbf{A} \geq r \iff \exists \text{ an } r \times r \text{ submatrix } \mathbf{B} \text{ of } \mathbf{A} \text{ with } \det\mathbf{B} \neq 0$$

Proof. Suppose $\operatorname{rank}\mathbf{A} \geq r$. Since $\operatorname{rank}\mathbf{A}$ equals the column rank of $\mathbf{A}$, there are r linearly independent columns of $\mathbf{A}$. Let $\mathbf{C}$ denote the $m \times r$ submatrix of $\mathbf{A}$ consisting of these r columns. Then the column rank of $\mathbf{C}$ is r, and so the row rank of $\mathbf{C}$ is also r. Hence there are r linearly independent rows of $\mathbf{C}$. Let $\mathbf{B}$ denote the $r \times r$ submatrix of $\mathbf{C}$ consisting of these r rows. These r rows of $\mathbf{B}$ are linearly independent, and so $\operatorname{rank}\mathbf{B} = r$. Hence $\mathbf{B}$ is invertible, and so $\det\mathbf{B} \neq 0$.

Conversely, suppose $\mathbf{B}$ is an $r \times r$ submatrix of $\mathbf{A}$ such that $\det \mathbf{B} \neq 0$. Then $\mathbf{B}$ is invertible, and so $\operatorname{rank} \mathbf{B} = r$. Hence the r rows of $\mathbf{B}$, and consequently, the corresponding r rows of $\mathbf{A}$ are linearly independent. Hence $\operatorname{rank} \mathbf{A} \geq r$. $\square$

Corollary (Determinantal Characterization of Rank)

Let $\mathbf{A}$ be an $m \times n$ matrix, and $r \in \mathbb{N}$. Then $r = \operatorname{rank} \mathbf{A}$ if and only if the following two conditions are satisfied.

(i) there is an $r \times r$ submatrix $\mathbf{B}$ of $\mathbf{A}$ such that $\det \mathbf{B} \neq 0$,

(ii) $\det \mathbf{C} = 0$ for every $(r+1)\times(r+1)$ submatrix $\mathbf{C}$ of $\mathbf{A}$.

Proof. This is an immediate consequence of the above lemma.

We remark that although the above result is of theoretical interest, it does not give a practically useful method for finding the rank of a matrix $\mathbf{A}$. On the other hand, transformation of $\mathbf{A}$ to a Row Echelon Form quickly reveals its rank.

Example

Let $\mathbf{A} := \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$. Since $\det \begin{bmatrix} 3 & 0 \\ -6 & 42 \end{bmatrix} \neq 0$ and since the determinants of all 3×3 submatrices of $\mathbf{A}$ are equal to 0, we see that $\operatorname{rank} \mathbf{A} = 2$.

This also follows by noting that $\mathbf{A}$ can be transformed by EROs to

$$\mathbf{A}' = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

which is in REF, and by noting that $\operatorname{rank} \mathbf{A}$ is equal to the row rank of $\mathbf{A}'$, which is 2.

We now consider another classical method of finding solutions of $\mathbf{A}\mathbf{x} = \mathbf{b}$, where $\mathbf{A}$ is invertible.

Proposition (Cramer's Rule)

Let $\mathbf{A} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_n \end{bmatrix} \in \mathbb{R}^{n \times n}$ be invertible, and let $\mathbf{b} \in \mathbb{R}^{n \times 1}$. For $k = 1, \ldots, n$, let $\mathbf{B}_k := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{b} & \cdots & \mathbf{c}_n \end{bmatrix}$ be the matrix obtained by replacing the kth column $\mathbf{c}_k$ of $\mathbf{A}$ by the right side $\mathbf{b}$ of $\mathbf{A}\mathbf{x} = \mathbf{b}$. Then the unique solution $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$ of the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given by

$$x_k := \frac{\det \mathbf{B}_k}{\det \mathbf{A}} \quad \text{for } k = 1, \ldots, n.$$

Proof. If $\mathbf{A}\mathbf{x} = \mathbf{b}$, then $\mathbf{b} = x_1\mathbf{c}_1 + \cdots + x_k\mathbf{c}_k + \cdots + x_n\mathbf{c}_n$. By the first two crucial properties of the determinant function, $\det \mathbf{B}_k = \det \begin{bmatrix} \mathbf{c}_1 & \cdots & x_1\mathbf{c}_1 + \cdots + x_k\mathbf{c}_k + \cdots + x_n\mathbf{c}_n & \cdots & \mathbf{c}_n \end{bmatrix}$ $= x_k \det \mathbf{A}$ for $k = 1, \ldots, n$. Since $\mathbf{A}$ is invertible, $\det \mathbf{A} \neq 0$. Hence the result. □

Let $\mathbf{A} := \begin{bmatrix} 3 & -2 & 1 \\ -2 & 1 & 4 \\ 1 & 4 & -5 \end{bmatrix}$, $\mathbf{b} := \begin{bmatrix} 13 \\ 11 \\ -31 \end{bmatrix}$. Then $\det \mathbf{A} = -60$.

Also,

$$\det \mathbf{B}_1 = \det \begin{bmatrix} 13 & -2 & 1 \\ 11 & 1 & 4 \\ -31 & 4 & -5 \end{bmatrix} = -60,$$

$$\det \mathbf{B}_2 = \det \begin{bmatrix} 3 & 13 & 1 \\ -2 & 11 & 4 \\ 1 & -31 & -5 \end{bmatrix} = 180,$$

$$\det \mathbf{B}_3 = \det \begin{bmatrix} 3 & -2 & 13 \\ -2 & 1 & 11 \\ 1 & 4 & -31 \end{bmatrix} = -240.$$

Hence the unique solution of the linear system $\mathbf{Ax} = \mathbf{b}$ is given by $x_1 := 1,\ x_2 = -3,\ x_3 = 4$, that is, $\mathbf{x} = \begin{bmatrix} 1 & -3 & 4 \end{bmatrix}^\mathsf{T}$.

Note: Cramer's Rule is rarely used for solving linear systems; the preferred method is the GEM. But Cramer's Rule is of theoretical interest, especially in solutions of differential eqns.

Formula for the Inverse of a Matrix

Let $\mathbf{A} := [a_{jk}] \in \mathbb{R}^{n\times n}$ with $n \geq 2$. Recall that $\mathbf{A}_{jk}$ denotes the submatrix of $\mathbf{A}$ obtained by deleting the jth row and the kth column of $\mathbf{A}$, and $M_{jk} := \det \mathbf{A}_{jk}$, the (j,k)th minor of $\mathbf{A}$, for $j,k = 1,\ldots,n$. We define $C_{jk} := (-1)^{j+k} M_{jk}$, $j,k = 1,\ldots,n$. It is called the **cofactor** of the entry a_{jk}. Then the expansion of $\det \mathbf{A}$ in terms of the kth column is given by

$$\det \mathbf{A} = \sum_{\ell=1}^{n} a_{\ell k} C_{\ell k}, \quad \text{where } k \in \{1,\ldots,n\}.$$

Define $\mathbf{C} := [C_{jk}] \in \mathbb{R}^{n\times n}$. It is called the **cofactor matrix** of $\mathbf{A}$.

Theorem

Let $\mathbf{A}$ be a square matrix. Then $\mathbf{C}^{\mathsf{T}}\mathbf{A} = (\det \mathbf{A})\mathbf{I} = \mathbf{A}\mathbf{C}^{\mathsf{T}}$. In particular, if $\det \mathbf{A} \neq 0$, then $\mathbf{A}$ is invertible and

$$\mathbf{A}^{-1} = \mathbf{C}^{\mathsf{T}} / \det \mathbf{A}.$$

Proof. Let $\mathbf{D} := \mathbf{C}^{\mathsf{T}}\mathbf{A} = [d_{jk}]$ say. By the definition of matrix multiplication, the (j,k)th entry of $\mathbf{D}$ is $d_{jk} = \sum_{\ell=1}^{n} C_{\ell j} a_{\ell k}$.

If $j = k$, then $d_{kk} = \sum_{\ell=1}^{n} C_{\ell k} a_{\ell k} = \det \mathbf{A}$, being the expansion in terms of its kth column of $\mathbf{A}$.

Let now $j \neq k$. Write $\mathbf{A} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_j & \cdots \mathbf{c}_n \end{bmatrix}$ in terms of its columns, and let $\mathbf{B}$ denote the matrix obtained by replacing the jth column $\mathbf{c}_j$ by the kth column $\mathbf{c}_k$ of $\mathbf{A}$, that is, $\mathbf{B} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k & \cdots & \mathbf{c}_k & \cdots \mathbf{c}_n \end{bmatrix} = [b_{jk}]$, say. Then $\det \mathbf{B} = 0$ since two columns are identical. Expanding $\det \mathbf{B}$ in terms of its jth column, $\det \mathbf{B} = \sum_{\ell=1}^{n} b_{\ell j} C_{\ell j} = \sum_{\ell=1}^{n} a_{\ell k} C_{\ell j}$. Thus $d_{jk} = \sum_{\ell=1}^{n} C_{\ell j} a_{\ell k} = \det \mathbf{B} = 0$ if $j \neq k$.

This shows that $\mathbf{C}^{\mathsf{T}}\mathbf{A} = (\det \mathbf{A})\mathbf{I}$. Similarly, we can prove $\mathbf{A}\mathbf{C}^{\mathsf{T}} = (\det \mathbf{A})\mathbf{I}$, and so $\mathbf{C}^{\mathsf{T}}\mathbf{A} = (\det \mathbf{A})\mathbf{I} = \mathbf{A}\mathbf{C}^{\mathsf{T}}$.

In case $\det \mathbf{A} \neq 0$, we see that $\mathbf{A}$ is invertible, and

$$\mathbf{A}^{-1} = \mathbf{C}^{\mathsf{T}} / \det \mathbf{A}.$$

$\square$

Multiplicativity of Determinant Function

Proposition

Let $\mathbf{A}$, $\mathbf{B}$ be $n \times n$ matrices. Then $\det(\mathbf{AB}) = (\det \mathbf{A})(\det \mathbf{B})$.

Proof. Suppose first $\mathbf{A}$ is not invertible. Then $(\det \mathbf{A}) = 0$. Also, $\mathbf{AB}$ is not invertible; otherwise there would be $\mathbf{C}$ such that $(\mathbf{AB})\mathbf{C} = \mathbf{I}$, that is, $\mathbf{A}(\mathbf{BC}) = \mathbf{I}$, which is impossible since $\mathbf{A}$ is not invertible. Hence $\det(\mathbf{AB}) = 0 = (\det \mathbf{A})(\det \mathbf{B})$.

Next, suppose $\mathbf{A}$ is invertible. Then we can transform $\mathbf{A}$ to a diagonal matrix $\mathbf{A}'$ (having nonzero diagonal elements) by elementary row transformations of type I and type II. Then $\det \mathbf{A}' = \det \mathbf{A}$ if an even number of row interchanges are involved, and $\det \mathbf{A}' = -\det \mathbf{A}$ otherwise.

We observe that the same elementary row operations transform $\mathbf{AB}$ to $\mathbf{A}'\mathbf{B}$.

To see this, we can use Q. 2.3 in Tut Sheet 2: Making an elementary row operation is equivalent to multiplying on the left by the corresponding elementary matrix! And of course $\mathbf{E}(AB) = (EA)B$ where E is any elementary matrix.

Hence $\det \mathbf{A}'\mathbf{B} = \det \mathbf{AB}$ if an even number of row interchanges are involved, and $\det \mathbf{A}'\mathbf{B} = -\det \mathbf{AB}$ otherwise.

Thus it is enough to show that

$$\det(\mathbf{AB}) = (\det \mathbf{A})(\det \mathbf{B}) \quad \text{when } \mathbf{A} \text{ is a diagonal matrix.}$$

But this is easily seen because

$$\mathbf{A} := \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \alpha_n \end{bmatrix} \text{ and } \mathbf{B} := \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_n \end{bmatrix} \implies \mathbf{AB} := \begin{bmatrix} \alpha_1\mathbf{b}_1 \\ \alpha_2\mathbf{b}_2 \\ \vdots \\ \alpha_n\mathbf{b}_n \end{bmatrix},$$

where $\mathbf{b}_1, \ldots, \mathbf{b}_n$ denote the rows of $\mathbf{B}$. Hence

$$\det(\mathbf{AB}) = \alpha_1\alpha_2\cdots\alpha_n \det \mathbf{B} = (\det \mathbf{A})(\det \mathbf{B}). \qquad \square$$

Corollary

If $\mathbf{A}$ is invertible, then $\det \mathbf{A}^{-1} = \dfrac{1}{\det \mathbf{A}}$.

Proof. $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \implies (\det \mathbf{A})(\det \mathbf{A}^{-1}) = \det \mathbf{I} = 1.$ $\square$

Example

Let $\mathbf{A} := \begin{bmatrix} 13 & 0 & 0 \\ 11 & 1 & 0 \\ -31 & 22 & -5 \end{bmatrix}$. Then $\det \mathbf{A}^{-1} = \dfrac{1}{\det \mathbf{A}} = -\dfrac{1}{65}$.

Linear Transformations

Just as we can define a continuous function from a subset of $\mathbb{R}^n$ to $\mathbb{R}^m$, we now define a 'linear' function from a subspace of $\mathbb{R}^{n\times 1}$ to $\mathbb{R}^{m\times 1}$.

Let V be a subspace of $\mathbb{R}^{n\times 1}$, and let W be a subspace of $\mathbb{R}^{m\times 1}$. A **linear transformation** or a **linear map** from V to W is a function $T : V \to W$ which 'preserves' the operations of addition and scalar multiplication, that is, for all $\mathbf{x}, \mathbf{y} \in V$ and $\alpha \in \mathbb{R}$,

$$T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \quad \text{and} \quad T(\alpha\,\mathbf{x}) = \alpha\, T(\mathbf{x}).$$

It follows that if $T : V \to W$ is linear, then $T(\mathbf{0}) = \mathbf{0}$, and T 'preserves' linear combinations of vectors in V, that is,

$$T(\alpha_1\mathbf{x}_1 + \cdots + \alpha_k\mathbf{x}_k) = \alpha_1 T(\mathbf{x}_1) + \cdots + \alpha_k T(\mathbf{x}_k)$$

for all $\mathbf{x}_1, \ldots, \mathbf{x}_k \in V$ and $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$.

Model Example

Let $V := \mathbb{R}^{n\times 1}$, $W := \mathbb{R}^{m\times 1}$ and $\mathbf{A}$ be an $m\times n$ matrix, that is, $\mathbf{A} \in \mathbb{R}^{m\times n}$. Define $T_{\mathbf{A}} : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ by

$$T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\,\mathbf{x} \quad \text{for } \mathbf{x} \in V.$$

The properties of matrix multiplication show that $T_{\mathbf{A}}$ is linear.

Conversely, suppose $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ is linear. We show that $T = T_{\mathbf{A}}$ for some $\mathbf{A} \in \mathbb{R}^{m\times n}$. Let $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n\times 1}$. Then $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$, where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the basic column vectors in $\mathbb{R}^{n\times 1}$. Since T is linear, we obtain

$$T(\mathbf{x}) = x_1\,T(\mathbf{e}_1) + \cdots + x_n\,T(\mathbf{e}_n).$$

Define $\mathbf{c}_k := T(\mathbf{e}_k)$ for $k = 1, \ldots, n$, and $\mathbf{A} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_n \end{bmatrix}$. Then $T(\mathbf{x}) = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \mathbf{A}\,\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^{n\times 1}$. Thus $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $T = T_{\mathbf{A}}$. (Note: kth column of $\mathbf{A}$ is $T(\mathbf{e}_k)$.)

Thus every linear transformation $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ is given by

$$T\left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}\right) := \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix} \quad \text{for} \quad \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n\times 1},$$

where $a_{11}, \ldots, a_{1n}, \ldots, a_{m1}, \ldots, a_{mn} \in \mathbb{R}$.
Similarly, one can define a linear map $T : \mathbb{R}^{1\times n} \to \mathbb{R}^{1\times m}$, and find that for $\begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{1\times n}$,

$$T\left(\begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}\right) := \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n & \cdots & a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Remark: Let D be an open subset of $\mathbb{R}^{1\times 2}$, $[x_0, y_0] \in D$, and let a function $f : D \to \mathbb{R}$ be differentiable at $[x_0, y_0]$. Then the total derivative of f at $[x_0, y_0]$ is a linear map (which depends on f) given by $T([x,\ y]) = \alpha x + \beta y$ for $[x,\ y] \in \mathbb{R}^{1\times 2}$, where $\alpha := f_x(x_0, y_0)$ and $\beta := f_y(x_0, y_0)$.

Let $\mathbf{A}$, $\mathbf{B} \in \mathbb{R}^{m\times n}$ and $\alpha, \beta \in \mathbb{R}$. Then $\alpha\mathbf{A} + \beta\mathbf{B} \in \mathbb{R}^{m\times n}$ and

$$T_{\alpha\mathbf{A}+\beta\mathbf{B}}(\mathbf{x}) = (\alpha\mathbf{A} + \beta\mathbf{B})\mathbf{x} = \alpha T_{\mathbf{A}}(\mathbf{x}) + \beta T_{\mathbf{B}}(\mathbf{x})$$

for $\mathbf{x} \in \mathbb{R}^{n\times 1}$. We write this as follows:

$$T_{\alpha\mathbf{A}+\beta\mathbf{B}} = \alpha T_{\mathbf{A}} + \beta T_{\mathbf{B}}.$$

Next, let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$. Then $\mathbf{AB} \in \mathbb{R}^{m\times p}$, and

$$T_{\mathbf{AB}}(\mathbf{x}) = (\mathbf{AB})\mathbf{x} = \mathbf{A}(\mathbf{Bx}) = T_{\mathbf{A}}(\mathbf{Bx}) = T_{\mathbf{A}}(T_{\mathbf{B}}(\mathbf{x})) = T_{\mathbf{A}} \circ T_{\mathbf{B}}(\mathbf{x})$$

for $\mathbf{x} \in \mathbb{R}^{p\times 1}$ by the associativity of matrix multiplication. Thus

$$T_{\mathbf{AB}} = T_{\mathbf{A}} \circ T_{\mathbf{B}}.$$

This says that the linear map associated with the product $\mathbf{AB}$ of matrices $\mathbf{A}$ and $\mathbf{B}$ is the composition of the linear maps associated with $\mathbf{A}$ and associated with $\mathbf{B}$ in the same order. This partially justifies the definition of matrix multiplication.

Examples

Let $\mathbf{A} \in \mathbb{R}^{2\times 2}$. Then $T_{\mathbf{A}} : \mathbb{R}^{2\times 1} \to \mathbb{R}^{2\times 1}$.

(i) Let $\mathbf{A} := \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. Then $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ stretches each vector by a factor of 2.

(ii) Let $\mathbf{A} := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} x_2 & x_1 \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ is the reflection in the line $x_1 = x_2$.

(iii) Let $\mathbf{A} := \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \mapsto \begin{bmatrix} -x_1 & -x_2 \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ is the reflection in the origin.

(iv) Let $\mathbf{A} := \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, where $\theta \in (-\pi, \pi]$. Then

$T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} x_1\cos\theta - x_2\sin\theta & x_1\sin\theta + x_2\cos\theta \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ is the rotation through an angle θ.

These are geometric interpretations of matrices.

MA110: Lecture 09

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Spring 2025

Recall: Linear Transformations

Just as we can define a continuous function from a subset of $\mathbb{R}^n$ to $\mathbb{R}^m$, we now define a 'linear' function from a subspace of $\mathbb{R}^{n\times 1}$ to $\mathbb{R}^{m\times 1}$.

Let V be a subspace of $\mathbb{R}^{n\times 1}$, and let W be a subspace of $\mathbb{R}^{m\times 1}$. A **linear transformation** or a **linear map** from V to W is a function $T : V \to W$ which 'preserves' the operations of addition and scalar multiplication, that is, for all $\mathbf{x}, \mathbf{y} \in V$ and $\alpha \in \mathbb{R}$,

$$T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \quad \text{and} \quad T(\alpha\,\mathbf{x}) = \alpha\, T(\mathbf{x}).$$

It follows that if $T : V \to W$ is linear, then $T(\mathbf{0}) = \mathbf{0}$, and T 'preserves' linear combinations of vectors in V, that is,

$$T(\alpha_1\mathbf{x}_1 + \cdots + \alpha_k\mathbf{x}_k) = \alpha_1 T(\mathbf{x}_1) + \cdots + \alpha_k T(\mathbf{x}_k)$$

for all $\mathbf{x}_1, \ldots, \mathbf{x}_k \in V$ and $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$.

Model Example

Let $V := \mathbb{R}^{n\times 1}$, $W := \mathbb{R}^{m\times 1}$ and $\mathbf{A}$ be an $m\times n$ matrix, that is, $\mathbf{A} \in \mathbb{R}^{m\times n}$. Define $T_{\mathbf{A}} : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ by

$$T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\,\mathbf{x} \quad \text{for } \mathbf{x} \in V.$$

The properties of matrix multiplication show that $T_{\mathbf{A}}$ is linear.

Conversely, suppose $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ is linear. We show that $T = T_{\mathbf{A}}$ for some $\mathbf{A} \in \mathbb{R}^{m\times n}$. Let $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n\times 1}$. Then $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$, where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the basic column vectors in $\mathbb{R}^{n\times 1}$. Since T is linear, we obtain

$$T(\mathbf{x}) = x_1 T(\mathbf{e}_1) + \cdots + x_n T(\mathbf{e}_n).$$

Define $\mathbf{c}_k := T(\mathbf{e}_k)$ for $k = 1, \ldots, n$, and $\mathbf{A} := \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_n \end{bmatrix}$. Then $T(\mathbf{x}) = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \mathbf{A}\,\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^{n\times 1}$. Thus $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $T = T_{\mathbf{A}}$. (Note: kth column of $\mathbf{A}$ is $T(\mathbf{e}_k)$.)

Thus every linear transformation $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ is given by

$$T\left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}\right) := \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix} \quad \text{for} \quad \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n\times 1},$$

where $a_{11}, \ldots, a_{1n}, \ldots, a_{m1}, \ldots, a_{mn} \in \mathbb{R}$.

Similarly, one can define a linear map $T : \mathbb{R}^{1\times n} \to \mathbb{R}^{1\times m}$, and find that for $\begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{1\times n}$,

$$T\left(\begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}\right) := \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n & \cdots & a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Remark: Let D be an open subset of $\mathbb{R}^{1\times 2}$, $[x_0, y_0] \in D$, and let a function $f : D \to \mathbb{R}$ be differentiable at $[x_0, y_0]$. Then the total derivative of f at $[x_0, y_0]$ is a linear map (which depends on f) given by $T([x,\ y]) = \alpha x + \beta y$ for $[x,\ y] \in \mathbb{R}^{1\times 2}$, where $\alpha := f_x(x_0, y_0)$ and $\beta := f_y(x_0, y_0)$.

Let $\mathbf{A}$, $\mathbf{B} \in \mathbb{R}^{m \times n}$ and $\alpha, \beta \in \mathbb{R}$. Then $\alpha\mathbf{A} + \beta\mathbf{B} \in \mathbb{R}^{m \times n}$ and

$$T_{\alpha\mathbf{A}+\beta\mathbf{B}}(\mathbf{x}) = (\alpha\mathbf{A} + \beta\mathbf{B})\mathbf{x} = \alpha T_{\mathbf{A}}(\mathbf{x}) + \beta T_{\mathbf{B}}(\mathbf{x})$$

for $\mathbf{x} \in \mathbb{R}^{n \times 1}$. We write this as follows:

$$T_{\alpha\mathbf{A}+\beta\mathbf{B}} = \alpha T_{\mathbf{A}} + \beta T_{\mathbf{B}}.$$

Next, let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$. Then $\mathbf{AB} \in \mathbb{R}^{m \times p}$, and

$$T_{\mathbf{AB}}(\mathbf{x}) = (\mathbf{AB})\mathbf{x} = \mathbf{A}(\mathbf{Bx}) = T_{\mathbf{A}}(\mathbf{Bx}) = T_{\mathbf{A}}(T_{\mathbf{B}}(\mathbf{x})) = T_{\mathbf{A}} \circ T_{\mathbf{B}}(\mathbf{x})$$

for $\mathbf{x} \in \mathbb{R}^{p \times 1}$ by the associativity of matrix multiplication. Thus

$$T_{\mathbf{AB}} = T_{\mathbf{A}} \circ T_{\mathbf{B}}.$$

This says that the linear map associated with the product $\mathbf{AB}$ of matrices $\mathbf{A}$ and $\mathbf{B}$ is the composition of the linear maps associated with $\mathbf{A}$ and associated with $\mathbf{B}$ in the same order. This partially justifies the definition of matrix multiplication.

Examples

Let $\mathbf{A} \in \mathbb{R}^{2\times 2}$. Then $T_{\mathbf{A}} : \mathbb{R}^{2\times 1} \to \mathbb{R}^{2\times 1}$.

(i) Let $\mathbf{A} := \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. Then $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ stretches each vector by a factor of 2.

(ii) Let $\mathbf{A} := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} x_2 & x_1 \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ is the reflection in the line $x_1 = x_2$.

(iii) Let $\mathbf{A} := \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \mapsto \begin{bmatrix} -x_1 & -x_2 \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ is the reflection in the origin.

(iv) Let $\mathbf{A} := \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, where $\theta \in (-\pi, \pi]$. Then

$T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} x_1\cos\theta - x_2\sin\theta & x_1\sin\theta + x_2\cos\theta \end{bmatrix}^{\mathsf{T}}$.

$T_{\mathbf{A}}$ is the rotation through an angle θ.

These are geometric interpretations of matrices.

While constructing an $m\times n$ matrix which represents a transformation from $\mathbb{R}^{n\times 1}$ to $\mathbb{R}^{m\times 1}$, we made an explicit use of the standard basis of $\mathbb{R}^{n\times 1}$ consisting of the basic column vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ (in this order). Also, we implicitly used the standard basic column vectors $\mathbf{e}_1, \ldots, \mathbf{e}_m$ in $\mathbb{R}^{m\times 1}$ (in this order) when we wrote $\mathbf{A} := \begin{bmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{bmatrix}$.

More generally, let an ordered basis $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ of $\mathbb{R}^{n\times 1}$ and an ordered basis $F := (\mathbf{y}_1, \ldots, \mathbf{y}_m)$ of $\mathbb{R}^{m\times 1}$ be given. Then there are unique $a_{11}, \ldots, a_{1n}, \ldots, a_{m1}, \ldots, a_{mn} \in \mathbb{R}$ such that

$$\begin{array}{rcl}
T(\mathbf{x}_1) & = & a_{11}\mathbf{y}_1 + \cdots + a_{j1}\mathbf{y}_j + \cdots + a_{m1}\mathbf{y}_m, \\
\vdots & & \qquad\qquad \vdots \qquad\qquad\quad \vdots \\
T(\mathbf{x}_k) & = & a_{1k}\mathbf{y}_1 + \cdots + a_{jk}\mathbf{y}_j + \cdots + a_{mk}\mathbf{y}_m, \\
\vdots & & \qquad\qquad \vdots \qquad\qquad\quad \vdots \\
T(\mathbf{x}_n) & = & a_{1n}\mathbf{y}_1 + \cdots + a_{jn}\mathbf{y}_j + \cdots + a_{mn}\mathbf{y}_m.
\end{array}$$

The $m \times n$ matrix $[a_{jk}]$ is called the **matrix of the linear transformation** $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ with respect to the ordered basis $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ of $\mathbb{R}^{n\times 1}$ and the ordered basis $F := (\mathbf{y}_1, \ldots, \mathbf{y}_m)$ of $\mathbb{R}^{m\times 1}$. This matrix is denoted by $\boxed{\mathbf{M}_F^E(T)}$.

Note: The kth column of $\mathbf{M}_F^E(T)$ is $\begin{bmatrix} a_{1k} & \cdots & a_{mk} \end{bmatrix}^{\mathsf{T}}$, where $T(\mathbf{x}_k) = a_{1k}\mathbf{y}_1 + \cdots + a_{jk}\mathbf{y}_j + \cdots + a_{mk}\mathbf{y}_m$ for $k = 1, \ldots, n$.

The $m \times n$ matrix $\mathbf{M}_F^E(T)$ represents the linear map T in the following sense. For $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$,

$$\begin{aligned} T(\alpha_1\mathbf{x}_1 + \cdots + \alpha_n\mathbf{x}_n) &= \sum_{k=1}^{n} \alpha_k T(\mathbf{x}_k) = \sum_{k=1}^{n} \alpha_k \left(\sum_{j=1}^{m} a_{jk}\mathbf{y}_j \right) \\ &= \sum_{j=1}^{m} \left(\sum_{k=1}^{n} a_{jk}\alpha_k \right) \mathbf{y}_j = \beta_1\mathbf{y}_1 + \cdots + \beta_m\mathbf{y}_m, \end{aligned}$$

where $\beta_j := \sum_{k=1}^{n} a_{jk}\alpha_k$ for $j = 1, \ldots, m$. Thus

$$T\left(\begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix}\begin{bmatrix}\alpha_1\\ \vdots\\ \alpha_n\end{bmatrix}\right)=\begin{bmatrix}\mathbf{y}_1 & \cdots & \mathbf{y}_m\end{bmatrix}\begin{bmatrix}\beta_1\\ \vdots\\ \beta_m\end{bmatrix},$$

$$\text{while}\quad \mathbf{M}_F^E(T)\begin{bmatrix}\alpha_1\\ \vdots\\ \alpha_n\end{bmatrix}=\begin{bmatrix}a_{11} & \cdots & a_{1n}\\ \vdots & \vdots & \vdots\\ a_{m1} & \cdots & a_{mn}\end{bmatrix}\begin{bmatrix}\alpha_1\\ \vdots\\ \alpha_n\end{bmatrix}=\begin{bmatrix}\beta_1\\ \vdots\\ \beta_m\end{bmatrix}.$$

Conversely, suppose we are given an $m \times n$ matrix $\mathbf{A}$. Define $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ as follows. For $\mathbf{x} := \alpha_1\mathbf{x}_1 + \cdots + \alpha_n\mathbf{x}_n$, let

$$T(\mathbf{x}) := \beta_1\mathbf{y}_1 + \cdots + \beta_m\mathbf{y}_m, \quad \text{where} \begin{bmatrix}\beta_1\\ \vdots\\ \beta_m\end{bmatrix} := \mathbf{A}\begin{bmatrix}\alpha_1\\ \vdots\\ \alpha_n\end{bmatrix}.$$

Then T is a linear map, and $\mathbf{M}_F^E(T) = \mathbf{A}$.
In particular, this holds if E and F are the standard ordered bases of $\mathbb{R}^{n\times 1}$ and $\mathbb{R}^{m\times 1}$ respectively.

Examples

(i) Consider the map $T : \mathbb{R}^{2\times 1} \to \mathbb{R}^{3\times 1}$ defined by $T(\mathbf{x}) := \begin{bmatrix} x_1 - x_2 & -x_1 + 2x_2 & x_2 \end{bmatrix}^{\mathsf{T}}$ for $\mathbf{x} := \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}}$.

Then T is a linear map. If $E := (\mathbf{e}_1, \mathbf{e}_2)$ is the standard ordered basis for $\mathbb{R}^{2\times 1}$ and $F := (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ is the standard ordered basis for $\mathbb{R}^{3\times 1}$, then $\mathbf{M}_F^E(T) = \begin{bmatrix} 1 & -1 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$.

On the other hand, let $E' := (\mathbf{e}_1', \mathbf{e}_2')$, where $\mathbf{e}_1' := \mathbf{e}_1$ and $\mathbf{e}_2' := \mathbf{e}_1 + \mathbf{e}_2$, and let $F' := (\mathbf{e}_1', \mathbf{e}_2', \mathbf{e}_3')$, where $\mathbf{e}_1' := \mathbf{e}_1$, $\mathbf{e}_2' := \mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{e}_3' := \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$. Then

$$T(\mathbf{e}_1') = \begin{bmatrix} 1 & -1 & 0 \end{bmatrix}^{\mathsf{T}} = 2\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^{\mathsf{T}} - \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^{\mathsf{T}} = 2\mathbf{e}_1' - \mathbf{e}_2',$$

$$T(\mathbf{e}_2') = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^{\mathsf{T}} = -\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^{\mathsf{T}} + \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^{\mathsf{T}} = -\mathbf{e}_1' + \mathbf{e}_3'.$$

Hence $\mathbf{M}_{F'}^{E'}(T) = \begin{bmatrix} 2 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$.

(ii) Consider the map $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{3\times 1}$ defined by $T(\mathbf{x}) := \begin{bmatrix} 2.9x_1 + 0.6x_2 - 0.1x_3 & 2.9x_1 + 1.6x_2 - 1.1x_3 & 2.5x_1 + x_2 + 1.5x_3 \end{bmatrix}^{\mathsf{T}}$ for $\mathbf{x} := \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^{\mathsf{T}}$. Then T is a linear map.

If $E := (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ is the standard ordered basis for $\mathbb{R}^{3\times 1}$, then

$$\mathbf{M}_E^E(T) = \frac{1}{10}\begin{bmatrix} 29 & 6 & -1 \\ 29 & 16 & -11 \\ 25 & 10 & 15 \end{bmatrix}.$$

On the other hand, let $E' := (\mathbf{e}_1', \mathbf{e}_2', \mathbf{e}_3')$, where $\mathbf{e}_1' := \begin{bmatrix} -1 & 3 & -1 \end{bmatrix}^{\mathsf{T}}, \mathbf{e}_2' := \begin{bmatrix} 1 & -1 & 3 \end{bmatrix}^{\mathsf{T}}, \mathbf{e}_3' := \begin{bmatrix} 2 & 1 & 4 \end{bmatrix}^{\mathsf{T}}$.
Then it can be checked that

$$T(\mathbf{e}_1') = \mathbf{e}_1', \quad T(\mathbf{e}_2') = 2\,\mathbf{e}_2', \quad T(\mathbf{e}_3') = 3\,\mathbf{e}_3'.$$

Hence $\mathbf{M}_{E'}^{E'}(T) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, which is a diagonal matrix!

Remark: We have shown that if $T : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$ is linear map, and if E and F are ordered bases of $\mathbb{R}^{n\times 1}$ and $\mathbb{R}^{m\times 1}$ respectively, then T is represented by an $m\times n$ matrix $\mathbf{M}_F^E(T)$ with respect to E and F. Now let V and W be subspaces of dimension n and m of some possibly higher dimensional spaces of vectors, and let T be a linear map from V to W. Even in this case, if E and F are ordered bases of V and W respectively, then the linear map T from V to W is represented, with respect to E and F, by an $m\times n$ matrix. This matrix is denoted by $\boxed{\mathbf{M}_F^E(T)}$.

Thus if $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ and $F := (\mathbf{y}_1, \ldots, \mathbf{y}_m)$, and we let $\mathbf{A} := \mathbf{M}_F^E(T)$, then for $\mathbf{x} = \alpha_1\mathbf{x}_1 + \cdots + \alpha_n\mathbf{x}_n \in V$ and $\mathbf{y} = \beta_1\mathbf{y}_1 + \cdots + \beta_m\mathbf{y}_m \in W$, we see that

$$T(\mathbf{x}) = \mathbf{y} \iff \mathbf{A}\begin{bmatrix}\alpha_1\\ \vdots\\ \alpha_n\end{bmatrix} = \begin{bmatrix}\beta_1\\ \vdots\\ \beta_m\end{bmatrix}.$$

Remark: Let V be a subspace of $\mathbb{R}^{n\times 1}$, W be a subspace of $\mathbb{R}^{m\times 1}$, and let $T : V \to W$ be a linear map. Two important subspaces associated with T are as follows.

(i) $\mathcal{N}(T) := \{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\}$, called the **null space** of T,

(ii) $\mathcal{I}(T) := \{T(\mathbf{x}) : \mathbf{x} \in V\}$, called the **image space** of T.

We note that
a linear map T is one-one $\iff \mathcal{N}(T) = \{\mathbf{0}\}$, and
a linear map T is onto $\iff \mathcal{I}(T) = W$.

Further, if $V := \mathbb{R}^{n\times 1}$, $W := \mathbb{R}^{m\times 1}$, and $\mathbf{A} \in \mathbb{R}^{m\times n}$, then

$$\begin{aligned}
\mathcal{N}(T_{\mathbf{A}}) &= \{\mathbf{x} \in \mathbb{R}^{n\times 1} : \mathbf{A}\mathbf{x} = \mathbf{0}\} = \mathcal{N}(\mathbf{A}), \\
\mathcal{I}(T_{\mathbf{A}}) &= \{\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{R}^{n\times 1}\} = \mathcal{C}(\mathbf{A}).
\end{aligned}$$

The last equality follows by noting that if $\mathbf{A} = \begin{bmatrix}\mathbf{c}_1 & \cdots & \mathbf{c}_n\end{bmatrix}$, then $\mathbf{A}\mathbf{x} = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n$ for $\mathbf{x} := \begin{bmatrix}x_1 & \cdots & x_n\end{bmatrix} \in \mathbb{R}^{n\times 1}$.

Example

Let $\mathbf{A} := \begin{bmatrix} 1 & -1 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2}$. Then $T_{\mathbf{A}} : \mathbb{R}^{2\times 1} \to \mathbb{R}^{3\times 1}$.

In fact, $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} x_1 - x_2 & -x_1 + 2x_2 & x_2 \end{bmatrix}^{\mathsf{T}}$ for all $\begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{2\times 1}$. Clearly, $\mathcal{N}(T_{\mathbf{A}}) = \{\mathbf{0}\}$. Also,

$$\mathcal{I}(T_{\mathbf{A}}) = \left\{ \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{3\times 1} : y_1 + y_2 - y_3 = 0 \right\}.$$

To see this, note that $(x_1 - x_2) + (-x_1 + 2x_2) - x_2 = 0$ for all $\begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{2\times 1}$, and if $\begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{3\times 1}$ satisfies $y_1 + y_2 - y_3 = 0$, then we may let $x_1 := y_1 + y_3$, $x_2 := y_3$, so that $x_1 - x_2 = y_1$, $-x_1 + 2x_2 = y_2$ and $x_2 = y_3$, that is, $T_{\mathbf{A}}(\begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}}) = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathsf{T}}$.

Note: $\mathcal{I}(T_{\mathbf{A}})$ is a plane through the origin $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^{\mathsf{T}}$ in $\mathbb{R}^{3\times 1}$.

Complex Numbers

In our development of matrix theory, we have so far used real numbers as scalars, and we have considered matrices whose entries are real numbers. Now we introduce an extension of $\mathbb{R}$ which has all the properties that $\mathbb{R}$ has (and one more).

A **complex number** is a 2×2 matrix $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$, where $a, b \in \mathbb{R}$. The set of all complex numbers is denoted by $\mathbb{C}$. Addition and multiplication in $\mathbb{C}$ are defined as in $\mathbb{R}^{2\times 2}$. Thus

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} c & d \\ -d & c \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ -(b+d) & a+c \end{bmatrix}$$

and

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} c & d \\ -d & c \end{bmatrix} = \begin{bmatrix} ac-bd & ad+bc \\ -(ad+bc) & ac-bd \end{bmatrix}.$$

These algebraic operations possess all the usual properties such as associativity, distributivity and commutativity.

Moreover, the map $a \longmapsto \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ from $\mathbb{R}$ to $\mathbb{C}$ is one-one.

Hence we identify $a \in \mathbb{R}$ with $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \in \mathbb{C}$. We define

$$i := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \text{so that } i^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We write $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} a + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} b \in \mathbb{C}$ as $a + ib$.

It follows that $(a + ib) + (c + id) = (a + c) + i(b + d)$ and $(a + ib)(c + id) = (ac - bd) + i(ad + bc)$.

Let $z \in \mathbb{C}$. Then $z = x + iy$ for unique $x, y \in \mathbb{R}$. Then x is called the **real part** of z, and it is denoted by $\mathfrak{R}(z)$, while y is called the **imaginary part** of z, and it is denoted by $\mathfrak{I}(z)$.

The complex number $x - iy$ is called the **conjugate** of $z = x + iy$, and it is denoted by $\overline{z}$.

We shall use complex numbers as scalars and consider matrices whose entries are complex numbers.

An $m \times n$ matrix with complex entries is an element of $\mathbb{C}^{m\times n}$. In particular, a row vector of length n belongs to $\mathbb{C}^{1\times n}$ and a column vector of length m belongs to $\mathbb{C}^{m\times 1}$.

For $\mathbf{A} := [a_{jk}] \in \mathbb{C}^{m\times n}$, define $\mathbf{A}^* := [\overline{a_{kj}}]$. Then $\mathbf{A}^* \in \mathbb{C}^{n\times m}$. It is called the **adjoint** of $\mathbf{A}$. We note that $(\alpha\mathbf{A} + \beta\mathbf{B})^* = \overline{\alpha}\mathbf{A}^* + \overline{\beta}\mathbf{B}^*$ for $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{m\times n}$ and $\alpha, \beta \in \mathbb{C}$.

- A square matrix $\mathbf{A} = [a_{jk}]$ is called **self-adjoint** if $\mathbf{A}^* = \mathbf{A}$, that is, if $a_{jk} = \overline{a_{kj}}$ for all j, k.
- A square matrix $\mathbf{A} = [a_{jk}]$ is called **skew self-adjoint** if $a_{jk} = -\overline{a}_{kj}$ for all j, k.

Note: Every diagonal entry of a self-adjoint matrix is real since $a_{jj} = \overline{a_{jj}} \implies a_{jj} \in \mathbb{R}$ for $j = 1, \ldots, n$. On the other hand, the real part of every diagonal entry of a skew self-adjoint matrix is equal to 0 since $a_{jj} = -\overline{a_{jj}} \implies \Re(a_{jj}) = 0$ for $j = 1, \ldots, n$.

Note: If $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{C}^{n\times 1}$ is a column vector, then $\mathbf{x}^* = \begin{bmatrix} \overline{x_1} & \cdots & \overline{x_n} \end{bmatrix} \in \mathbb{C}^{1\times n}$ is a row vector, and $\mathbf{x}^*\mathbf{x} = |x_1|^2 + \cdots + |x_n|^2$. It follows that $\mathbf{x}^*\mathbf{x} = 0 \iff \mathbf{x} = \mathbf{0}$.

A matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ defines a linear transformation from $\mathbb{C}^{n\times 1}$ to $\mathbb{C}^{m\times 1}$, and every linear transformation from $\mathbb{C}^{n\times 1}$ to $\mathbb{C}^{m\times 1}$ can be represented by a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ (with respect to an ordered basis for $\mathbb{C}^{n\times 1}$ and an ordered basis for $\mathbb{C}^{m\times 1}$).

Similarly, we can consider vector subspaces of $\mathbb{C}^{n\times 1}$, and the concepts of linear dependence of vectors and of the span of a subset carry over to $\mathbb{C}^{n\times 1}$. The Fundamental Theorem for Linear Systems remains valid for matrices with complex entries.

Having thus completed our discussion of solution of a linear system, we shall turn to solution of an 'eigenvalue problem' associated with a matrix. In this development, the role of complex numbers will turn out to be important.

For future use, we define the **absolute value** of a complex number $z = x + iy \in \mathbb{C}$ by

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

Then the **triangle inequality** $|z_1 + z_2| \leq |z_1| + |z_2|$ holds for all $z_1, z_2 \in \mathbb{C}$.
(Recall: $\|(x_1, y_1) + (x_2, y_2)\| \leq \|(x_1, y_1)\| + \|(x_2, y_2)\|$ for all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$.)

Also, note that

$$\max\{|\Re(z)|, |\Im(z)|\} \ \leq \ |z| \ \leq \ |\Re(z)| + |\Im(z)| \quad \text{for all } z \in \mathbb{C}.$$

We shall use complex numbers as scalars and consider matrices whose entries are complex numbers.

The set of all $m \times n$ matrices with entries in $\mathbb{C}$ is denoted by $\mathbb{C}^{m\times n}$. In particular, $\mathbb{C}^{1\times n}$ is the set of all row vectors of length n, while $\mathbb{C}^{m\times 1}$ is the set of all column vectors of length m.

For $\mathbf{A} := [a_{jk}] \in \mathbb{C}^{m\times n}$, define $\mathbf{A}^* := [\overline{a_{kj}}]$. Then $\mathbf{A}^* \in \mathbb{C}^{n\times m}$. It is called the **conjugate transpose** or the **adjoint** of $\mathbf{A}$. Note: $(\alpha\mathbf{A} + \beta\mathbf{B})^* = \overline{\alpha}\mathbf{A}^* + \overline{\beta}\mathbf{B}^*$ for $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{m\times n}$ and $\alpha, \beta \in \mathbb{C}$. In case $m = n$, then $(\mathbf{AB})^* = \mathbf{B}^*\mathbf{A}^*$.

- A square matrix $\mathbf{A} = [a_{jk}]$ is called **Hermitian** or **self-adjoint** if $\mathbf{A}^* = \mathbf{A}$, that is, if $a_{jk} = \overline{a_{kj}}$ for all j, k.
- A square matrix $\mathbf{A} = [a_{jk}]$ is called **skew-Hermitian** or **skew self-adjoint** if $a_{jk} = -\overline{a}_{kj}$ for all j, k.

Note: Every diagonal entry of a self-adjoint matrix is real since $a_{jj} = \overline{a_{jj}} \implies a_{jj} \in \mathbb{R}$ for $j = 1, \ldots, n$. On the other hand, the real part of every diagonal entry of a skew self-adjoint matrix

Note: If $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{C}^{n \times 1}$ is a column vector, then $\mathbf{x}^* = \begin{bmatrix} \overline{x_1} & \cdots & \overline{x_n} \end{bmatrix} \in \mathbb{C}^{1 \times n}$ is a row vector, and $\mathbf{x}^*\mathbf{x} = |x_1|^2 + \cdots + |x_n|^2$. It follows that $\mathbf{x}^*\mathbf{x} = 0 \iff \mathbf{x} = \mathbf{0}$.

A matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$ defines a linear transformation from $\mathbb{C}^{n \times 1}$ to $\mathbb{C}^{m \times 1}$, and every linear transformation from $\mathbb{C}^{n \times 1}$ to $\mathbb{C}^{m \times 1}$ can be represented by a matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$ (with respect to an ordered basis for $\mathbb{C}^{n \times 1}$ and an ordered basis for $\mathbb{C}^{m \times 1}$).

Similarly, we can consider vector subspaces of $\mathbb{C}^{n \times 1}$, and the concepts of linear dependence of vectors and of the span of a subset carry over to $\mathbb{C}^{n \times 1}$. The Fundamental Theorem for Linear Systems remains valid for matrices with complex entries.

Having thus completed our discussion of solution of a linear system, we shall turn to solution of an 'eigenvalue problem' associated with a matrix. In this development, the role of complex numbers will turn out to be important.

MA110:
Lecture 10

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Spring 2025

Matrix of a Linear Transformation

General Case: Let V and W be vector subspaces of dimension n and m, and let $T : V \to W$ be a linear map. Suppose $E := (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $F := (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ are ordered bases of V and W respectively. Then the matrix of T with respect to E and F is the $m \times n$ matrix $A = (a_{jk})$ determined by

$$T(\mathbf{v}_k) = \sum_{j=1}^{m} a_{jk}\mathbf{w}_j \quad \text{for } k = 1, \ldots, n.$$

This matrix is denoted by $\boxed{\mathbf{M}_F^E(T)}$. Note that the kth column of this matrix corresponds to the coefficients of $T(v_k)$ when expressed as a linear combination of $\mathbf{w}_1, \ldots, \mathbf{w}_m$. Also, for $\mathbf{v} = \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n \in V$, $\mathbf{w} = \beta_1\mathbf{w}_1 + \cdots + \beta_m\mathbf{w}_m \in W$,

$$T(\mathbf{v}) = \mathbf{w} \iff \mathbf{A}\begin{bmatrix}\alpha_1\\ \vdots \\ \alpha_n\end{bmatrix} = \begin{bmatrix}\beta_1\\ \vdots \\ \beta_m\end{bmatrix}.$$

Composites of Linear Transformation

Suppose U is another vector space of dimension p and $D = (\mathbf{u}_1, \ldots, \mathbf{u}_p)$ is an ordered basis of U. If $\mathbf{S} : U \to V$ is a linear map, then we can form the composite of S and T, i.e.,

$$T \circ S : U \to W$$

Now if $B = (b_{k\ell})$ is the $n \times p$ matrix of S w.r.t. D and E, i.e., $B = \mathbf{M}_E^D(S)$, then for $\ell = 1, \ldots, p$,

$$S(u_\ell) = \sum_{k=1}^{n} b_{k\ell}\mathbf{v}_k \quad \text{and hence} \quad (T \circ S)(u_\ell) = \sum_{k=1}^{n} b_{k\ell} T(\mathbf{v}_k);$$

substituting for $T(\mathbf{v}_k)$ from the earlier equation, we see that

$$(T \circ S)(u_\ell) = \sum_{k=1}^{n} b_{k\ell} \sum_{j=1}^{m} a_{jk}\mathbf{w}_j = \sum_{j=1}^{m} \left(\sum_{k=1}^{n} a_{jk} b_{k\ell} \right) \mathbf{w}_j$$

Thus we see that $\boxed{\mathbf{M}_F^D(T \circ S) = AB = \mathbf{M}_F^E(T)\mathbf{M}_E^D(S).}$

Remark: Let V be a subspace of $\mathbb{R}^{n\times 1}$, W be a subspace of $\mathbb{R}^{m\times 1}$, and let $T : V \to W$ be a linear map. Two important subspaces associated with T are as follows.

(i) $\mathcal{N}(T) := \{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\}$, called the **null space** of T,

(ii) $\mathcal{I}(T) := \{T(\mathbf{x}) : \mathbf{x} \in V\}$, called the **image space** of T.

We note that
a linear map T is one-one $\iff \mathcal{N}(T) = \{\mathbf{0}\}$, and
a linear map T is onto $\iff \mathcal{I}(T) = W$.

Further, if $V := \mathbb{R}^{n\times 1}$, $W := \mathbb{R}^{m\times 1}$, and $\mathbf{A} \in \mathbb{R}^{m\times n}$, then

$$\begin{aligned} \mathcal{N}(T_{\mathbf{A}}) &= \{\mathbf{x} \in \mathbb{R}^{n\times 1} : \mathbf{A}\mathbf{x} = \mathbf{0}\} = \mathcal{N}(\mathbf{A}), \\ \mathcal{I}(T_{\mathbf{A}}) &= \{\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{R}^{n\times 1}\} = \mathcal{C}(\mathbf{A}). \end{aligned}$$

The last equality follows by noting that if $\mathbf{A} = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_n \end{bmatrix}$, then $\mathbf{A}\mathbf{x} = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n$ for $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{n\times 1}$.

Example

Let $\mathbf{A} := \begin{bmatrix} 1 & -1 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2}$. Then $T_{\mathbf{A}} : \mathbb{R}^{2\times 1} \to \mathbb{R}^{3\times 1}$.

In fact, $T_{\mathbf{A}} : \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \longmapsto \begin{bmatrix} x_1 - x_2 & -x_1 + 2x_2 & x_2 \end{bmatrix}^{\mathsf{T}}$ for all $\begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{2\times 1}$. Clearly, $\mathcal{N}(T_{\mathbf{A}}) = \{\mathbf{0}\}$. Also,

$$\mathcal{I}(T_{\mathbf{A}}) = \left\{ \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{3\times 1} : y_1 + y_2 - y_3 = 0 \right\}.$$

To see this, note that $(x_1 - x_2) + (-x_1 + 2x_2) - x_2 = 0$ for all $\begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{2\times 1}$, and if $\begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{3\times 1}$ satisfies $y_1 + y_2 - y_3 = 0$, then we may let $x_1 := y_1 + y_3$, $x_2 := y_3$, so that $x_1 - x_2 = y_1$, $-x_1 + 2x_2 = y_2$ and $x_2 = y_3$, that is, $T_{\mathbf{A}}(\begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}}) = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathsf{T}}$.

Note: $\mathcal{I}(T_{\mathbf{A}})$ is a plane through the origin $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^{\mathsf{T}}$ in $\mathbb{R}^{3\times 1}$.

Complex Numbers

In our development of matrix theory, we have so far used real numbers as scalars, and we have considered matrices whose entries are real numbers. It turns out that all the concepts extend easily when the scalars allowed to be **complex numbers** instead of real numbers. We shall assume familiarity with the basics of complex numbers and just recall a few properties. The set of all complex numbers is denoted by $\mathbb{C}$. Let $z \in \mathbb{C}$. Then $z = x + iy$ for unique $x, y \in \mathbb{R}$. We call x the **real part** of z, and denote it by $\Re(z)$. Also, y is called the **imaginary part** of z, and it is denoted by $\Im(z)$. The complex number $x - iy$ is called the **conjugate** of $z = x + iy$, and it is denoted by $\bar{z}$. Let $a, b, c, d \in \mathbb{R}$. The addition and multiplication of complex numbers $a + ib$, $c + id$ is defined by

$$\begin{aligned} (a+ib)+(c+id) &= (a+c)+i(b+d) \\ (a+ib)(c+id) &= (ac-bd)+i(ad+bc). \end{aligned}$$

Recall also that the **absolute value** of a complex number $z = x + iy \in \mathbb{C}$ is defined by by

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2}.$$

Note that the **triangle inequality**

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

holds for all $z_1, z_2 \in \mathbb{C}$. In terms of the norm

$$\|(x, y)\| := \sqrt{x^2 + y^2} \quad \text{of a vector } (x, y) \in \mathbb{R}^2,$$

this corresponds to the inequality

$$\|(x_1, y_1) + (x_2, y_2)\| \leq \|(x_1, y_1)\| + \|(x_2, y_2)\|$$

for all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$. Also, note that

$$\max\{|\Re(z)|, |\Im(z)|\} \ \leq \ |z| \ \leq \ |\Re(z)|+|\Im(z)| \quad \text{for all } z \in \mathbb{C}.$$

We shall use complex numbers as scalars and consider matrices whose entries are complex numbers.

The set of all $m \times n$ matrices with entries in $\mathbb{C}$ is denoted by $\mathbb{C}^{m\times n}$. In particular, $\mathbb{C}^{1\times n}$ is the set of all row vectors of length n, while $\mathbb{C}^{m\times 1}$ is the set of all column vectors of length m.

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- A square matrix $\mathbf{A} = [a_{jk}]$ is called **skew-Hermitian** or **skew self-adjoint** if $a_{jk} = -\overline{a}_{kj}$ for all j, k.

Note: Every diagonal entry of a self-adjoint matrix is real since $a_{jj} = \overline{a_{jj}} \implies a_{jj} \in \mathbb{R}$ for $j = 1, \ldots, n$. On the other hand, the real part of every diagonal entry of a skew self-adjoint matrix

Note: If $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{C}^{n\times 1}$ is a column vector, then $\mathbf{x}^* = \begin{bmatrix} \overline{x_1} & \cdots & \overline{x_n} \end{bmatrix} \in \mathbb{C}^{1\times n}$ is a row vector, and $\mathbf{x}^*\mathbf{x} = |x_1|^2 + \cdots + |x_n|^2$. It follows that $\mathbf{x}^*\mathbf{x} = 0 \iff \mathbf{x} = \mathbf{0}$.

A matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ defines a linear transformation from $\mathbb{C}^{n\times 1}$ to $\mathbb{C}^{m\times 1}$, and every linear transformation from $\mathbb{C}^{n\times 1}$ to $\mathbb{C}^{m\times 1}$ can be represented by a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ (with respect to an ordered basis for $\mathbb{C}^{n\times 1}$ and an ordered basis for $\mathbb{C}^{m\times 1}$).

Similarly, we can consider vector subspaces of $\mathbb{C}^{n\times 1}$, and the concepts of linear dependence of vectors and of the span of a subset carry over to $\mathbb{C}^{n\times 1}$. The Fundamental Theorem for Linear Systems remains valid for matrices with complex entries.

Having thus completed our discussion of solution of a linear system, we shall turn to solution of an 'eigenvalue problem' associated with a matrix. In this development, the role of complex numbers will turn out to be important.

Matrix Eigenvalue Problem

The German word 'eigen' means 'belonging to itself'. The eigenvalue problem for a matrix consists of finding a nonzero vector which is sent to a scalar multiple of itself by the linear transformation defined by the matrix.

Eigenvalue problems come up frequently in many engineering branches, quantum mechanics, physical chemistry, biology, and even in economics and psychology.

Please refer to Section 8.2 of Kreyszig's book for applications of eigenvalue problems to stretching of elastic membranes, to vibrating mass-spring systems, to Markov processes and to population control models.

In the development that follows, we shall use either real numbers or complex numbers as scalars. To facilitate a general discussion which applies to both types of scalars, we shall write $\mathbb{K}$ to mean either $\mathbb{R}$ or $\mathbb{C}$. When we want to switch to a special treatment valid for only the real scalars, or only for the complex scalars, we shall specify $\mathbb{K} := \mathbb{R}$ or $\mathbb{K} := \mathbb{C}$.

Definition

Let $\mathbf{A}$ be an $n \times n$ matrix with entries in $\mathbb{K}$, that is, let $\mathbf{A} \in \mathbb{K}^{n\times n}$. A scalar $\lambda \in \mathbb{K}$ is called an **eigenvalue** *of $\mathbf{A}$ if*

$$\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x} \quad \textit{for some } \mathbf{x} \in \mathbb{K}^{n\times 1} \textit{ with } \mathbf{x} \neq \mathbf{0}.$$

Any nonzero vector $\mathbf{x} \in \mathbb{K}^{n\times 1}$ satisfying $\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}$ is called an **eigenvector** *of $\mathbf{A}$ corresponding to the eigenvalue λ. Further,*

$$\{\mathbf{x} \in \mathbb{K}^{n\times 1} : \mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}\} = \mathcal{N}(\mathbf{A} - \lambda\mathbf{I}).$$

is called the **eigenspace** *of $\mathbf{A}$ associated with λ.*

How to find eigenvalues

Let $\mathbf{A} = [a_{jk}] \in \mathbb{K}^{n \times n}$ and let $\lambda \in \mathbb{K}$. Clearly,

$$\begin{aligned} \lambda \text{ is an eigenvalue of } \mathbf{A} \quad &\iff \quad \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) \neq \{\mathbf{0}\} \\ &\iff \quad \operatorname{rank}(\mathbf{A} - \lambda \mathbf{I}) < n \\ &\iff \quad \det(\mathbf{A} - \lambda \mathbf{I}) = 0. \end{aligned}$$

The last condition suggests that we consider the polynomial

$$p_{\mathbf{A}}(t) := \det(\mathbf{A} - t\,\mathbf{I}) = \det \begin{bmatrix} a_{11} - t & \cdots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} - t \end{bmatrix}.$$

This is called the **characteristic polynomial** of $\mathbf{A}$. It is a polynomial of degree n with coefficients in $\mathbb{K}$ and for $\lambda \in \mathbb{K}$,

$$\lambda \text{ is an eigenvalue of } \mathbf{A} \iff \lambda \text{ is a root of } p_{\mathbf{A}}, \text{ i.e., } p_{\mathbf{A}}(\lambda) = 0.$$

In particular, an $n \times n$ matrix with entries in $\mathbb{K}$ has at most n eigenvalues in $\mathbb{K}$.

Algebraic and Geometric Multiplicities

Definition

Let $\mathbf{A} = [a_{jk}] \in \mathbb{K}^{n\times n}$ *and let* $\lambda \in \mathbb{K}$ *be an eigenvalue of* $\mathbf{A}$.

- *The* **algebraic multiplicity** *of* λ *(as an eigenvalue of* $\mathbf{A}$*) is the order* m *of the root* λ *of* $p_{\mathbf{A}}(t)$*, i.e.,* m *is the largest positive integer such that* $(t-\lambda)^m$ *divides* $p_{\mathbf{A}}(t)$.
- *The* **geometric multiplicity** *of* λ *(as an eigenvalue of* $\mathbf{A}$*) is the dimension of its eigenspace, i.e.,* $\dim \mathcal{N}(\mathbf{A}-\lambda\mathbf{I})$.

Observe that if $\lambda \in \mathbb{K}$ be an eigenvalue of $\mathbf{A}$, then

geometric multiplicity of $\lambda = \text{nullity}(\mathbf{A}-\lambda\mathbf{I}) = n-\text{rank}(\mathbf{A}-\lambda\mathbf{I})$.

This can be calculated by solving the homogeneous system $(\mathbf{A}-\lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ using, for instance, Gaussian elimination. In fact, GEM and back substitution will also give the *basic solutions*, i.e., a set of eigenvectors of $\mathbf{A}$ which forms a basis of the eigenspace of $\mathbf{A}$ associated to λ.

Examples

(i) Let $\mathbf{A} = \text{diag}(a_1, \ldots, a_n)$, i.e., let $\mathbf{A}$ be a diagonal matrix with diagonal entries $a_1, \ldots, a_n$ in that order. Clearly

$$p_{\mathbf{A}}(t) = (a_1 - t)(a_2 - t) \cdots (a_n - t).$$

Thus the eigenvalues of $\mathbf{A}$ are precisely $a_1, \ldots, a_n$. Note that this can also be seen directly since $\mathbf{A}\mathbf{e}_k = a_k \mathbf{e}_k$ for each $k = 1, \ldots, n$, where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the basic column vectors in $\mathbb{K}^{n \times 1}$. Observe that in this case the algebraic multiplicity of each eigenvalue is equal to its geometric multiplicity. Indeed, if $\lambda \in \{a_1, \ldots, a_n\}$, then the algebraic multiplicity of λ equals

$$m := \text{the number of } i \in \{1, \ldots, n\} \text{ such that } a_i = \lambda.$$

To find the geometric multiplicity of λ, consider $\mathbf{A} - \lambda \mathbf{I}$ and note that this is a diagonal matrix with exactly m rows of zeros and $n - m$ nonzero rows. So $\text{rank}(\mathbf{A} - \lambda \mathbf{I}) = n - m$ and hence

$$\text{geometric multiplicity of } \lambda = \text{nullity}(\mathbf{A} - \lambda \mathbf{I}) = m.$$

Examples Contd.

On the other hand, let $\lambda \in \mathbb{K}$ and $\lambda \notin \{a_1, \ldots, a_n\}$. Then $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0} \iff (a_1 - \lambda)x_1 = \cdots = (a_n - \lambda)x_n = 0$. Thus the linear system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ has only the zero solution, and so λ is not an eigenvalue of $\mathbf{A}$.

If $\lambda \in \mathbb{K}$ appears g times on the diagonal of the matrix $\mathbf{A}$, then the geometric multiplicity of the eigenvalue λ is equal to g since the number of zero rows in an REF of $\mathbf{A} - \lambda\mathbf{I}$ is g.

(ii) Let $\mathbf{A}$ be an upper triangular matrix with diagonal entries $a_{11}, \ldots, a_{nn}$. Again, the characteristic polynomials factors as

$$p_{\mathbf{A}}(t) = (a_{11} - t)(a_{22} - t)\cdots(a_{nn} - t).$$

So the eigenvalues of $\mathbf{A}$ are precisely $a_{11}, \ldots, a_{nn}$. The algebraic multiplicities can be found as in the previous example. However, they may not always coincide with the corresponding geometric multipliciities.

Examples Contd.

For example, consider the 2×2 matrix $\mathbf{A} := \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$. Clearly 3 is the only eigenvalue of $\mathbf{A}$ and its algebraic multiplicity is 2. On the other hand, the homogeneous linear system $(\mathbf{A} - 3\mathbf{I})\mathbf{x} = \mathbf{0}$ comprises of the equations $x_2 = 0$ and $0 = 0$. So the eigenspace of $\mathbf{A}$ associated with the eigenvalue 3 has $[1\ 0]^{\mathsf{T}}$ as its basis. Thus the geometric multiplicity of the eigenvalue 3 of $\mathbf{A}$ is 1.

Examples Contd.

Let $\mathbf{A}$ be an upper triangular matrix with diagonal entries $a_{11}, \ldots, a_{nn}$. Now a_{11} is an eigenvalue of $\mathbf{A}$ since $\mathbf{A}\mathbf{e}_1 = a_{11}\mathbf{e}_1$. In fact, each $a_{11}, \ldots, a_{nn}$ is an eigenvalue of $\mathbf{A}$. To see this, let $\lambda \in \{a_{11}, \ldots, a_{nn}\}$, and let k be the smallest positive integer such that $a_{kk} = \lambda$. Then $\mathbf{A} - \lambda\mathbf{I}$ is an upper triangular matrix with the kth diagonal entry zero and earlier diagonal entries nonzero, and so in an REF of $\mathbf{A} - \lambda\mathbf{I}$, the kth column cannot be pivotal. Hence $\text{rank}(\mathbf{A} - \lambda\mathbf{I}) < n$. This shows that λ is an eigenvalue of $\mathbf{A}$; eigenvectors of $\mathbf{A}$ corresponding to λ can be found by backward substitution.

On the other hand, if $\lambda \notin \{a_{11}, \ldots, a_{nn}\}$, then backward substitution shows that the linear system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ has only the zero solution, and so λ is not an eigenvalue of $\mathbf{A}$.

(iii) Let $\mathbf{A}$ be a lower triangular matrix.Then $\mathbf{A}^\mathsf{T}$ is an upper triangular matrix with the same diagonal elements as $\mathbf{A}$. Also, $\mathrm{rank}(\mathbf{A} - \lambda\mathbf{I}) = \mathrm{rank}(\mathbf{A}^\mathsf{T} - \lambda\mathbf{I})$ for any $\lambda \in \mathbb{K}$. It follows that each diagonal element of $\mathbf{A}$ is an eigenvalue of $\mathbf{A}$, and no other $\lambda \in \mathbb{K}$ is an eigenvalue of $\mathbf{A}$. Eigenvectors of $\mathbf{A}$ corresponding to λ can be found by forward substitution.

In general, solving an eigenvalue problem $\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}$ is much harder than finding solutions of a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$. In the latter case, the matrix $\mathbf{A}$ and the right side $\mathbf{b}$ are given. On the other hand, in the eigenvalue problem, the 'unknown' vector $\mathbf{x}$ appears on both sides of the equation, and additionally, an 'unknown' scalar λ appears on the right side.

We need to find an eigenvalue λ of $\mathbf{A}$ and a corresponding eigenvector $\mathbf{x}$ of $\mathbf{A}$ simultaneously. It is tough, but if one of them is known beforehand, then the other can be found easily. Suppose a scalar λ is known to be an eigenvalue of $\mathbf{A}$. Then all eigenvectors of $\mathbf{A}$ corresponding to λ can be obtained by finding the general solution of the homogeneous linear system $(\mathbf{A} - \lambda\mathbf{I})\,\mathbf{x} = \mathbf{0}$. Next, suppose a nonzero vector $\mathbf{x}$ is known to be an eigenvector of $\mathbf{A}$. Then one only needs to calculate $\mathbf{A}\mathbf{x}$ and observe that it is a scalar multiple of $\mathbf{x}$. This scalar is the corresponding eigenvalue of $\mathbf{A}$.

Examples

(i) Let $\mathbf{A} := \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$, and suppose somehow we know that $\lambda := -3$ is an eigenvalue of $\mathbf{A}$. Let

$$\mathbf{B} := \mathbf{A} - (-3)\mathbf{I} = \mathbf{A} + 3\mathbf{I} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix}.$$

By EROs, we can transform $\mathbf{B}$ to $\mathbf{B}' := \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Now the solution space of $\mathbf{B}'\mathbf{x} = \mathbf{0}$ is $\{\mathbf{x} \in \mathbb{R}^{3\times 1} : x_1 + 2x_2 - 3x_3 = 0\}$, which is also the solution space of $\mathbf{B}\mathbf{x} := (\mathbf{A} + 3\mathbf{I})\mathbf{x} = \mathbf{0}$, the basic solutions being $\mathbf{s}_2 := \begin{bmatrix} -2 & 1 & 0 \end{bmatrix}^\mathsf{T}$ and $\mathbf{s}_3 := \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}^\mathsf{T}$. Thus the eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue $\lambda = -3$ are the nonzero linear combinations of $\mathbf{s}_2$ and $\mathbf{s}_3$. The geometric multiplicity of the eigenvalue -3 is equal to 2.

(ii) Let $\mathbf{A} := \frac{1}{10}\begin{bmatrix} 29 & 6 & -1 \\ 29 & 16 & -11 \\ 25 & 10 & 15 \end{bmatrix}$, and suppose somehow we know that $\mathbf{x} := \begin{bmatrix} 1 & -1 & 3 \end{bmatrix}^{\mathsf{T}}$ is an eigenvector of $\mathbf{A}$. We easily find that $\mathbf{Ax} = 2\begin{bmatrix} 1 & -1 & 3 \end{bmatrix}^{\mathsf{T}}$. Hence 2 is the corresponding eigenvalue of $\mathbf{A}$.

We saw that eigenvalue problems for diagonal matrices are the easiest to solve. We wonder when a nondiagonal matrix would 'behave like a diagonal matrix'. To make this precise, we introduce the following notion.

Let $\mathbf{A}$, $\mathbf{B} \in \mathbb{K}^{n\times n}$. We say that $\mathbf{A}$ is **similar** to $\mathbf{B}$ (over $\mathbb{K}$) if there is an invertible $\mathbf{P} \in \mathbb{K}^{n\times n}$ such that $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$, that is, $\mathbf{AP} = \mathbf{PB}$. In this case, we write $\mathbf{A} \sim \mathbf{B}$.

Similarity of Square Matrices

Definition

Let $\mathbf{A}$, $\mathbf{B} \in \mathbb{K}^{n\times n}$. *We say that* $\mathbf{A}$ *is* **similar** *to* $\mathbf{B}$ *(over* $\mathbb{K}$*) if there is an invertible* $\mathbf{P} \in \mathbb{K}^{n\times n}$ *such that* $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, *that is,* $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{B}$. *In this case, we write* $\mathbf{A} \sim \mathbf{B}$.

One can easily check (i) $\mathbf{A} \sim \mathbf{A}$, (ii) if $\mathbf{A} \sim \mathbf{B}$ then $\mathbf{B} \sim \mathbf{A}$, and (iii) if $\mathbf{A} \sim \mathbf{B}$ and $\mathbf{B} \sim \mathbf{C}$, then $\mathbf{A} \sim \mathbf{C}$.

Examples:

(i) Let $\mathbf{A} := \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$. It is easily seen that $\mathbf{P} := \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ is invertible and . $\mathbf{P}^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. Hence $\mathbf{A}$ is similar to

$$\mathbf{B} := \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 6 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

which is a diagonal matrix.

(ii) Let $\mathbf{A} \in \mathbb{K}^{n\times n}$. Then $\mathbf{A} \sim \mathbf{I} \iff \mathbf{A} = \mathbf{I}$.

(iii) Let $\mathbf{A} \in \mathbb{K}^{n\times n}$, and let $\mathbf{E}$ be $n\times n$ an elementary matrix. Then $\mathbf{B} := \mathbf{E}\mathbf{A}\mathbf{E}^{-1}$ is similar to $\mathbf{A}$. Note: $\mathbf{EA}$ is obtained from $\mathbf{A}$ by an elementary row operation on $\mathbf{A}$, and $\mathbf{B}$ is obtained from $\mathbf{EA}$ by the 'reverse column operation' on $\mathbf{EA}$.

Similarity of matrices has the following characterization.

Proposition

Let $\mathbf{A}, \mathbf{B} \in \mathbb{K}^{n\times n}$. Then $\mathbf{A} \sim \mathbf{B}$ if and only if there is an ordered basis E for $\mathbb{K}^{n\times 1}$ such that $\mathbf{B}$ is the matrix of the linear transformation $T_{\mathbf{A}} : \mathbb{K}^{n\times 1} \to \mathbb{K}^{n\times 1}$ with respect to E.

In fact, $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ if and only if the columns of $\mathbf{P}$ form an ordered basis, say E, for $\mathbb{K}^{n\times 1}$ and $\mathbf{B} = \mathbf{M}_E^E(T_{\mathbf{A}})$.

A proof of this proposition will be outlined later.

Diagonalizability

Definition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ *is called* **diagonalizable** *(over* $\mathbb{K}$*) if* $\mathbf{A}$ *is similar to a diagonal matrix (over* $\mathbb{K}$*).*

Here is a useful characterization of diagonalizability.

Proposition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is diagonalizable if and only if there is a basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$. In fact,

$$\begin{aligned}
& \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}, \text{ where } \mathbf{P}, \mathbf{D} \in \mathbb{K}^{n\times n} \text{ are of the form} \\
& \mathbf{P} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix} \text{ and } \mathbf{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \\
\iff \quad & \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \text{ is a basis for } \mathbb{K}^{n\times 1} \text{ and} \\
& \mathbf{A}\mathbf{x}_k = \lambda_k \mathbf{x}_k \text{ for } k = 1, \ldots, n.
\end{aligned}$$

We shall see a proof of this result as well as many examples of diagonalizable and non-diagonalizable matrices later.

MA110: Lecture 11

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Spring 2025

Similarity of Square Matrices

Definition

Let $\mathbf{A}$, $\mathbf{B} \in \mathbb{K}^{n\times n}$. *We say that* $\mathbf{A}$ *is* **similar** *to* $\mathbf{B}$ *(over* $\mathbb{K}$*) if there is an invertible* $\mathbf{P} \in \mathbb{K}^{n\times n}$ *such that* $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$, *that is,* $\mathbf{AP} = \mathbf{PB}$. *In this case, we write* $\mathbf{A} \sim \mathbf{B}$.

One can easily check (i) $\mathbf{A} \sim \mathbf{A}$, (ii) if $\mathbf{A} \sim \mathbf{B}$ then $\mathbf{B} \sim \mathbf{A}$, and (iii) if $\mathbf{A} \sim \mathbf{B}$ and $\mathbf{B} \sim \mathbf{C}$, then $\mathbf{A} \sim \mathbf{C}$.

Examples:

(i) Let $\mathbf{A} := \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$. It is easily seen that $\mathbf{P} := \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ is invertible and . $\mathbf{P}^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. Hence $\mathbf{A}$ is similar to

$$\mathbf{B} := \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 6 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

which is a diagonal matrix.

More Examples and a Characterization of Similarity

(ii) Let $\mathbf{A} \in \mathbb{K}^{n\times n}$. Then $\mathbf{A} \sim \mathbf{I} \iff \mathbf{A} = \mathbf{I}$.

(iii) Let $\mathbf{A} \in \mathbb{K}^{n\times n}$, and let $\mathbf{E}$ be $n\times n$ an elementary matrix. Then $\mathbf{B} := \mathbf{EAE}^{-1}$ is similar to $\mathbf{A}$. Note: $\mathbf{EA}$ is obtained from $\mathbf{A}$ by an elementary row operation on $\mathbf{A}$, and $\mathbf{B}$ is obtained from $\mathbf{EA}$ by the 'reverse column operation' on $\mathbf{EA}$.

Diagonalizability

Definition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ *is called* **diagonalizable** *(over* $\mathbb{K}$*) if* $\mathbf{A}$ *is similar to a diagonal matrix (over* $\mathbb{K}$*).*

Examples

(i) Let $\mathbf{A} := \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$. Let $\mathbf{P} := \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, so that $\mathbf{P}^{-1} := \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. Then $\mathbf{A}$ is similar to the matrix

$$\mathbf{B} := \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 6 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

which is a diagonal matrix.

(ii) Let $\mathbf{A} \in \mathbb{K}^{n\times n}$. Then $\mathbf{A} \sim \mathbf{I} \iff \mathbf{A} = \mathbf{I}$.

(iii) Let $\mathbf{A} \in \mathbb{K}^{n\times n}$, and let $\mathbf{E}$ be $n\times n$ an elementary matrix. Then $\mathbf{B} := \mathbf{E}\mathbf{A}\mathbf{E}^{-1}$ is similar to $\mathbf{A}$. Note: $\mathbf{EA}$ is obtained from $\mathbf{A}$ by an elementary row operation on $\mathbf{A}$, and $\mathbf{B}$ is obtained from $\mathbf{EA}$ by the 'reverse column operation' on $\mathbf{EA}$.

Similarity of matrices has the following characterisation.

Proposition

Let $\mathbf{A}, \mathbf{B} \in \mathbb{K}^{n\times n}$. Then $\mathbf{A} \sim \mathbf{B}$ if and only if there is an ordered basis E for $\mathbb{K}^{n\times 1}$ such that $\mathbf{B}$ is the matrix of the linear transformation $T_{\mathbf{A}} : \mathbb{K}^{n\times 1} \to \mathbb{K}^{n\times 1}$ with respect to E.

In fact, $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ if and only if the columns of $\mathbf{P}$ form an ordered basis, say E, for $\mathbb{K}^{n\times 1}$ and $\mathbf{B} = \mathbf{M}_E^E(T_{\mathbf{A}})$.

Proof. Let $\mathbf{B} := [b_{jk}]$. Now $\mathbf{A} \sim \mathbf{B} \iff$ there is an invertible matrix $\mathbf{P}$ such that $\mathbf{AP} = \mathbf{PB}$. Let $\mathbf{x}_k$ be the k-th column of P. Then $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is an ordered basis for $\mathbb{K}^{n\times 1}$. Conversely, if $E = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is an ordered basis then the matrix $P = [\mathbf{x_1}, \ldots, \mathbf{x_n}]$ is invertible. Now $\mathbf{AP} = \mathbf{PB}$ implies:

$$\mathbf{A}\begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix}\begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \vdots & \vdots \\ b_{n1} & \cdots & b_{nn}\end{bmatrix}.$$

The kth column of LHS is $\mathbf{A}\mathbf{x}_k$ and the kth column of RHS is the linear combination of $\mathbf{x}_1, \ldots, \mathbf{x}_n$ with coefficients from the k column of $\mathbf{B}$. Thus $\mathbf{A}\mathbf{x}_k = b_{1k}\mathbf{x}_1 + \cdots + b_{nk}\mathbf{x}_n$ for $k = 1, \ldots, n$. This means the kth column of $\mathbf{M}_E^E(T_\mathbf{A})$ is the kth column $\begin{bmatrix} b_{1k} & \cdots & b_{nk} \end{bmatrix}^\mathsf{T}$ of $\mathbf{B}$, $k = 1, \ldots, n$, that is, $\mathbf{B} = \mathbf{M}_E^E(T_\mathbf{A})$.
Conversely, if $\mathbf{B} = \mathbf{M}_E^E(T_\mathbf{A})$ then again $\mathbf{A}\mathbf{x}_k = T_A(\mathbf{x_k}) = b_{1k}\mathbf{x}_1 + \cdots + b_{nk}\mathbf{x}_n$ for $k = 1, \ldots, n$, so $\mathbf{AP} = \mathbf{PB}$ where $P = [\mathbf{x_1}, \ldots, \mathbf{x_n}]$. $\square$

Proposition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is diagonalizable if and only if there is a basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$.

In fact, $\mathbf{X}^{-1}\mathbf{A}\mathbf{X} = \mathbf{D}$, where $\mathbf{X} := \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix}$ and $\mathbf{D} := \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \iff \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is a basis for $\mathbb{K}^{n\times 1}$ and $\mathbf{A}\mathbf{x}_k = \lambda_k \mathbf{x}_k$ for $k = 1, \ldots, n$.

Proof. $\mathbf{A}$ is diagonalizable $\iff$ there is an invertible matrix $\mathbf{X}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{AX} = \mathbf{XD}$. This is the case if and only if there is a basis $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ for $\mathbb{K}^{n\times 1}$ and there are $\lambda_1, \ldots, \lambda_n \in \mathbb{K}$ such that

$$\mathbf{A} \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix} \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

The LHS is just $[\mathbf{A}\mathbf{x}_\mathbf{1}, \cdots, \mathbf{A}\mathbf{x}_\mathbf{n}]$ and the RHS is just $[\lambda \mathbf{x}_1, \cdots, \lambda_n \mathbf{x}_n]$. Thus the proposition follows. $\square$

If $\mathbf{A} = \mathbf{X}\mathbf{D}\mathbf{X}^{-1}$, then for any $m \in \mathbb{N}$,

$$\mathbf{A}^m = (\mathbf{X}\mathbf{D}\mathbf{X}^{-1}) \cdots (\mathbf{X}\mathbf{D}\mathbf{X}^{-1}) = \mathbf{X}\mathbf{D}^m\mathbf{X}^{-1} = \mathbf{X}\,\mathrm{diag}(\lambda_1^m, \ldots, \lambda_n^m)\mathbf{X}^{-1}.$$

Example

We have seen that

$$\mathbf{A} := \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1}$$

and $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. Hence

$$\begin{aligned} \mathbf{A}^m &= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2^m & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 2^m 3 & 1 \\ 2^m 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2^m 3 - 2 & -2^m 3 + 3 \\ 2^m 2 - 2 & -2^m 2 + 3 \end{bmatrix} \quad \text{for } m \in \mathbb{N}. \end{aligned}$$

Similarity and Eigenvalues

Recall that $\lambda \in \mathbb{K}$ is an eigenvalue of $\mathbf{A} \in \mathbb{K}^{n \times n}$ if $\mathbf{A}\mathbf{x} = \lambda \mathbf{x}$ for some $\mathbf{x} \in \mathbb{K}^{n \times 1}$ with $\mathbf{x} \neq \mathbf{0}$. It turns out that similar matrices have the same eigenvalues. In fact, more is true.

Proposition

Let $\mathbf{A}$, $\mathbf{A}' \in \mathbb{K}^{n \times n}$ be similar. Then $p_{\mathbf{A}}(t) = p_{\mathbf{A}'}(t)$. In particular, $\lambda \in \mathbb{K}$ is an eigenvalue of $\mathbf{A}$ if and only if λ is an eigenvalue of $\mathbf{A}'$. Consequently, the algebraic multiplicity of λ as an eigenvalue of $\mathbf{A}$ is equal to the algebraic multiplicity of λ as an eigenvalue of $\mathbf{A}'$. Furthermore, the geometric multiplicity of λ as an eigenvalue of $\mathbf{A}$ is equal to the geometric multiplicity of λ as an eigenvalue of $\mathbf{A}'$.

Proof: SInce $\mathbf{A} \sim \mathbf{A}'$, there is an invertible $\mathbf{P} \in \mathbb{K}^{n \times n}$ such that $\mathbf{A}' = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. Writing $\mathbf{I} = \mathbf{P}^{-1}\mathbf{P}$, we see that

$$p_{\mathbf{A}'}(t) = \det(\mathbf{A}' - t\mathbf{I}) = \det(\mathbf{P}^{-1}\mathbf{A}\mathbf{P} - t\mathbf{I}) = \det(\mathbf{A} - t\mathbf{I}) = p_{\mathbf{A}}(t).$$

Proof Contd. It remains to prove the assertion about geometric multiplicity. Let λ be an eigenvalue of $\mathbf{A}$ and $\mathbf{x}$ be an eigenvector of $\mathbf{A}$ corresponding to λ. Then $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$. Since $\mathbf{A}' = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, we see that $\mathbf{x}' := \mathbf{P}^{-1}\mathbf{x}$ satisfies $\mathbf{A}'\mathbf{x}' = \lambda\mathbf{x}'$. Also $\mathbf{x}' \neq \mathbf{0}$ since $\mathbf{P}$ is invertible. Thus $\mathbf{x}'$ is an eigevector of $\mathbf{A}'$ corresponding to λ. Also, it is easy to check that $\{\mathbf{x}_1, \ldots, \mathbf{x}_g\}$ is a basis for $\mathcal{N}(\mathbf{A} - \lambda\mathbf{I})$ if and only if $\{\mathbf{P}^{-1}\mathbf{x}_1, \ldots, \mathbf{P}^{-1}\mathbf{x}_g\}$ is a basis for $\mathcal{N}(\mathbf{A}' - \lambda\mathbf{I})$. Hence the geometric multiplicity of λ as an eigenvalue of $\mathbf{A}$ is equal to the geometric multiplicity of λ as an eigenvalue of $\mathbf{A}'$. $\square$

Example

$$\mathbf{A} := \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \Rightarrow p_{\mathbf{A}}(t) = \det \begin{bmatrix} \lambda - t & 1 \\ 0 & \lambda - t \end{bmatrix} = (\lambda - t)^2.$$

Hence λ is the only eigenvalue of **A**, and its algebraic multiplicity is 2. But its geometric multiplicity is 1 since

$$\mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \implies \mathrm{rank}(\mathbf{A} - \lambda \mathbf{I}) = 1 \implies \mathrm{nullity}(\mathbf{A} - \lambda \mathbf{I}) = 1.$$

Note that in the above example, if **A** were similar to a diagonal matrix **D**, then we must have $\mathbf{D} = \mathrm{diag}(\lambda, \lambda)$, since eigenvalues and their algebraic multiplicities of **A** and **D** have to be the same. But the geometric multiplicity of λ as an eigenvalue of **D** is 2. This shows that **A** is not diagonalizable.

Example

$$\mathbf{A} := \begin{bmatrix} 3 & 0 & 0 \\ -2 & 4 & 2 \\ -2 & 1 & 5 \end{bmatrix} \Rightarrow p_{\mathbf{A}}(t) = \det \begin{bmatrix} 3-t & 0 & 0 \\ -2 & 4-t & 2 \\ -2 & 1 & 5-t \end{bmatrix}.$$

Computing the determinant, we find $p_{\mathbf{A}}(t) = (3-t)^2(6-t)$. Hence 3 is an eigenvalue of $\mathbf{A}$ of algebraic multiplicity 2, and 6 is an eigenvalue of $\mathbf{A}$ of algebraic multiplicity 1. Also,

$$\mathbf{A} - 3\mathbf{I} = \begin{bmatrix} 0 & 0 & 0 \\ -2 & 1 & 2 \\ -2 & 1 & 2 \end{bmatrix} \Rightarrow \operatorname{rank}(\mathbf{A} - 3\mathbf{I}) = 1.$$

So $\operatorname{nullity}(\mathbf{A} - 3\mathbf{I}) = 2$. In fact, $\left\{ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^{\mathsf{T}}, \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}^{\mathsf{T}} \right\}$ is a basis of the eigenspace of $\mathbf{A}$ corresponding to eigenvalue 3, and so its geometric multiplicity is equal to 2.

Relating geometric and algebraic multiplicities

Proposition

Let $\mathbf{A} \in \mathbb{K}^{n\times n}$, and let λ be an eigenvalue of $\mathbf{A}$. Then the geometric multiplicity of λ is less than or equal to its algebraic multiplicity.

Proof. Let g be the geometric multiplicity of λ. Let $(\mathbf{v}_1, \ldots, \mathbf{v}_g)$ be an ordered basis of the eigenspace of λ; extend it to an ordered basis $(\mathbf{v}_1, \ldots, \mathbf{v}_g, \mathbf{v}_{g+1}, \ldots, \mathbf{v}_n)$ of $\mathbb{K}^{n\times 1}$. Define $\mathbf{P} := \begin{bmatrix}\mathbf{v}_1 & \cdots & \mathbf{v}_n\end{bmatrix}$. Then $\mathbf{P}$ is invertible since its n columns $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent. Consider $\mathbf{A}' := \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. Since $\mathbf{A}\mathbf{v}_j = \lambda\mathbf{v}_j$ and $\mathbf{P}\mathbf{e}_j = \mathbf{v}_j$ for $j = 1, \ldots, g$, we see that the j th column of $\mathbf{A}'$ is given by

$$\mathbf{A}'\mathbf{e}_j = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{e}_j = \mathbf{P}^{-1}\mathbf{A}\mathbf{v}_j = \lambda\mathbf{P}^{-1}\mathbf{v}_j = \lambda\mathbf{e}_j.$$

Hence

$$\mathbf{A}' = \left[\begin{array}{ccc|c} \lambda & \cdots & 0 & \\ \vdots & \ddots & \vdots & \mathbf{C} \\ 0 & \cdots & \lambda & \\ \hline & \mathbf{O} & & \mathbf{D} \end{array}\right]$$

where $\mathbf{C} \in \mathbb{K}^{g\times(n-g)}$, $\mathbf{O} \in \mathbb{K}^{(n-g)\times g}$ and $\mathbf{D} \in \mathbb{K}^{(n-g)\times(n-g)}$. Expanding by the first column, we see that

$$\det(\mathbf{A}' - t\mathbf{I}) = (\lambda - t)^g q(t),$$

where $q(t)$ is a polynomial of degree $n - g$. Thus

$$p_{\mathbf{A}}(t) = p_{\mathbf{A}'}(t) = \det(\mathbf{A}' - t\mathbf{I}) = (\lambda - t)^g q(t).$$

Thus $(\lambda - t)^g$ divides the characteristic polynomial $p_{\mathbf{A}}(t)$ of $\mathbf{A}$. Since the algebraic multiplicity of λ is the largest natural number m such that $(\lambda - t)^m$ divides $p_{\mathbf{A}}(t)$, we obtain $g \leq m$. $\square$

Eigenvectors corresponding to distinct eigenvalues

Our next result is about the linear independence of eigenvectors corresponding to distinct eigenvalues of a matrix.

Lemma

Let $\mathbf{A} \in \mathbb{K}^{n\times n}$, and let $\lambda_1, \ldots, \lambda_k$ be distinct eigenvalues of $\mathbf{A}$. Let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ belong to the eigenspaces of $\mathbf{A}$ corresponding to $\lambda_1, \ldots, \lambda_k$ respectively. Then

$$\mathbf{x}_1 + \cdots + \mathbf{x}_k = \mathbf{0} \iff \mathbf{x}_1 = \cdots = \mathbf{x}_k = \mathbf{0}.$$

In particular, if $\mathbf{x}_1, \ldots, \mathbf{x}_k$ are eigenvectors of $\mathbf{A}$ corresponding to $\lambda_1, \ldots, \lambda_k$ respectively, then the set $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is linearly independent.

Proof. We use induction on the number k of distinct eigenvalues of $\mathbf{A}$. Clearly, the result holds for $k = 1$.

Let $k \geq 2$ and assume that the result holds for $k - 1$.

Suppose $\mathbf{x} := \mathbf{x}_1 + \cdots + \mathbf{x}_{k-1} + \mathbf{x}_k = \mathbf{0}$. Then $\mathbf{A}\mathbf{x} = \mathbf{0}$, that is, $\lambda_1 \mathbf{x}_1 + \cdots + \lambda_{k-1}\mathbf{x}_{k-1} + \lambda_k \mathbf{x}_k = \mathbf{0}$. Also, multiplying the first equation by λ_k, we obtain $\lambda_k \mathbf{x}_1 + \cdots + \lambda_k \mathbf{x}_{k-1} + \lambda_k \mathbf{x}_k = \mathbf{0}$. Subtraction gives $(\lambda_1 - \lambda_k)\mathbf{x}_1 + \cdots + (\lambda_{k-1} - \lambda_k)\mathbf{x}_{k-1} = \mathbf{0}$.

By the induction hypothesis, $(\lambda_1 - \lambda_k)\mathbf{x}_1 = \cdots = (\lambda_{k-1} - \lambda_k)\mathbf{x}_{k-1} = \mathbf{0}$. Since $\lambda_1 \neq \lambda_k, \ldots, \lambda_{k-1} \neq \lambda_k$, we obtain $\mathbf{x}_1 = \cdots = \mathbf{x}_{k-1} = \mathbf{0}$, and so $\mathbf{x}_k = \mathbf{0}$ as well.

Now let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ be eigenvectors. If $\alpha_1 \mathbf{x}_1 + \cdots + \alpha_k \mathbf{x}_k = \mathbf{0}$, then $\alpha_1 \mathbf{x}_1 = \cdots = \alpha_k \mathbf{x}_k = \mathbf{0}$. But $\mathbf{x}_1 \neq \mathbf{0}, \ldots, \mathbf{x}_k \neq \mathbf{0}$, so that $\alpha_1 = \cdots = \alpha_k = 0$. Thus $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is linearly independent.

Theorem

Let $\mathbf{A} \in \mathbb{K}^{n \times n}$, and let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of $\mathbf{A}$. Let g_j be the geometric multiplicity of λ_j for $j = 1, \ldots, k$. Then $g_1 + \cdots + g_k \leq n$. Further, $\mathbf{A}$ is diagonalizable if and only if $g_1 + \cdots + g_k = n$.

Proof. Let $p_A(t) = \prod_{j=1}^{k}(\lambda_j - t)^{n_j}$. Then n_j is the algebraic multiplicity of λ_j. We have already seen that $g_j \leq n_j$. Thus,

$$\sum_{j=1}^{k} g_j \leq \sum_{j=1}^{k} n_j = n.$$

If A is diagonalizable, that is, similar to a diagonal matrix D, then the distinct eigenvalues of D are exactly $\lambda_1, \ldots, \lambda_k$ and for each $j = 1, \ldots, k$, the geometric and algebraic multiplicities of λ_j are same for A and D. However as D is a diagonal matrix, the algebraic and geometric multiplicities for each eigenvalue are same, hence $g_j = n_j$ for each j, thus $\sum_{j=1}^{n} g_j = n$.

Conversely, suppose $\sum_{j=1}^{n} g_j = n$. Let V_j denote the eigenspace $\mathcal{N}(\mathbf{A} - \lambda_j \mathbf{I})$ of $\mathbb{K}^{n\times 1}$, and let E_j be a basis for V_j consisting of g_j eigenvectors of $\mathbf{A}$ corresponding to λ_j for $j = 1, \ldots, k$.

Proof contd.

We claim that the set $E := E_1 \cup \cdots \cup E_k$ is linearly independent. Let $\mathbf{x}$ be a linear combination of elements of E. Collate the elements of E_j for each $j = 1, \ldots, k$ and write $\mathbf{x} = \mathbf{x}_1 + \cdots + \mathbf{x}_k$, where $\mathbf{x}_j \in V_j$ for $j = 1, \ldots, k$. Suppose $\mathbf{x} = \mathbf{0}$. Then $\mathbf{x}_j = \mathbf{0}$ for $j = 1, \ldots, k$ by the previous lemma. For $j \in \{1, \ldots, k\}$, $\mathbf{x}_j$ is a linear combination of elements of the set E_j, and since the set E_j is linearly independent, every coefficient in this linear combination must be 0. Since this holds for each $j = 1, \ldots, k$, we see that every coefficient in the linear combination $\mathbf{x}$ of elements of E must be 0.

Now E_j has g_j many elements, so E has $\sum_j g_j = n$ many elements. As the dimension of $\mathbb{K}^n$ is n, E must be a basis. As every element of E is an eigenvector of A, by results already proved, A is diagonalizable. $\square$

MA 110:
Lecture 12

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Spring 2025

Diagonalizability Revisited

Recall the definition.

Definition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is called **diagonalizable** *(over $\mathbb{K}$) if $\mathbf{A}$ is similar to a diagonal matrix (over $\mathbb{K}$).*

We stated the following characterization of diagonalizability.

Proposition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is diagonalizable if and only if there is a basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$. In fact,

$$\begin{aligned} & \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}, \text{ where } \mathbf{P}, \mathbf{D} \in \mathbb{K}^{n\times n} \text{ are of the form} \\ & \mathbf{P} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix} \text{ and } \mathbf{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \\ \iff \quad & \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \text{ is a basis for } \mathbb{K}^{n\times 1} \text{ and} \\ & \mathbf{A}\mathbf{x}_k = \lambda_k \mathbf{x}_k \text{ for } k = 1, \ldots, n. \end{aligned}$$

Proof of a characterization of diagonalizability

Proof. The result is a consequence of the earlier characterization of similarity. It can also be seen as follows.

$$
\begin{aligned}
& \mathbf{A} \text{ is diagonalizable} \\
\iff \quad & \exists \text{ invertible matrix } \mathbf{P} \text{ and diagonal matrix } \mathbf{D} \text{ in } \mathbb{K}^{n\times n} \\
& \text{such that } \mathbf{AP} = \mathbf{PD} \\
\iff \quad & \exists \text{ a basis } \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \text{ for } \mathbb{K}^{n\times 1} \text{ and } \lambda_1, \ldots, \lambda_n \in \mathbb{K} \text{ such} \\
& \text{that } \mathbf{A}\begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix} \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \\
\iff \quad & \exists \text{ a basis } \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \text{ of } \mathbb{K}^{n\times 1} \text{ and } \lambda_1, \ldots, \lambda_n \in \mathbb{K} \text{ such} \\
& \text{that } \mathbf{A}\mathbf{x}_k = \lambda_k \mathbf{x}_k \text{ for } k = 1, \ldots, n. \qquad \square
\end{aligned}
$$

Application: If a matrix $\mathbf{A}$ is diagonalizable and we find invertible $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then any power of $\mathbf{A}$ can be found easily. This is seen as follows:

$$\mathbf{A}^m = (\mathbf{PDP}^{-1}) \cdots (\mathbf{PDP}^{-1}) = \mathbf{PD}^m\mathbf{P}^{-1} = \mathbf{P}\operatorname{diag}(\lambda_1^m, \ldots, \lambda_n^m)\mathbf{P}^{-1}.$$

Example: Consider $\mathbf{A} := \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$. Then

$$p_{\mathbf{A}}(t) = \det \begin{bmatrix} t-4 & 3 \\ -2 & t+1 \end{bmatrix} = (t-4)(t+1)+6 = (t-2)(t-1).$$

Thus 2 and 1 are the eigenvalues of $\mathbf{A}$ and it is easy to see that $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are corresponding eigenvectors. So $\mathbf{P} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ satisfies $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \operatorname{diag}(2, 1)$, which can be written as

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

Hence

$$\begin{aligned} \mathbf{A}^m &= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2^m & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 2^m 3 & 1 \\ 2^m 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2^m 3 - 2 & -2^m 3 + 3 \\ 2^m 2 - 2 & -2^m 2 + 3 \end{bmatrix} \quad \text{for } m \in \mathbb{N}. \end{aligned}$$

Eigenvectors corresponding to distinct eigenvalues

Our next result is about the linear independence of eigenvectors corresponding to distinct eigenvalues of a matrix.

Lemma

Let $\mathbf{A} \in \mathbb{K}^{n \times n}$, and let $\lambda_1, \ldots, \lambda_k$ be distinct eigenvalues of $\mathbf{A}$. Let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ belong to the eigenspaces of $\mathbf{A}$ corresponding to $\lambda_1, \ldots, \lambda_k$ respectively. Then

$$\mathbf{x}_1 + \cdots + \mathbf{x}_k = \mathbf{0} \iff \mathbf{x}_1 = \cdots = \mathbf{x}_k = \mathbf{0}.$$

In particular, if $\mathbf{x}_1, \ldots, \mathbf{x}_k$ are eigenvectors of $\mathbf{A}$ corresponding to $\lambda_1, \ldots, \lambda_k$ respectively, then the set $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is linearly independent.

Proof. We use induction on the number k of distinct eigenvalues of $\mathbf{A}$. Clearly, the result holds for $k = 1$.

Let $k \geq 2$ and assume that the result holds for $k - 1$.

Suppose $\mathbf{x} := \mathbf{x}_1 + \cdots + \mathbf{x}_{k-1} + \mathbf{x}_k = \mathbf{0}$. Then $\mathbf{A}\mathbf{x} = \mathbf{0}$, that is, $\lambda_1 \mathbf{x}_1 + \cdots + \lambda_{k-1}\mathbf{x}_{k-1} + \lambda_k \mathbf{x}_k = \mathbf{0}$. Also, multiplying the first equation by λ_k, we obtain $\lambda_k \mathbf{x}_1 + \cdots + \lambda_k \mathbf{x}_{k-1} + \lambda_k \mathbf{x}_k = \mathbf{0}$. Subtraction gives $(\lambda_1 - \lambda_k)\mathbf{x}_1 + \cdots + (\lambda_{k-1} - \lambda_k)\mathbf{x}_{k-1} = \mathbf{0}$.

By the induction hypothesis, $(\lambda_1 - \lambda_k)\mathbf{x}_1 = \cdots = (\lambda_{k-1} - \lambda_k)\mathbf{x}_{k-1} = \mathbf{0}$. Since $\lambda_1 \neq \lambda_k, \ldots, \lambda_{k-1} \neq \lambda_k$, we obtain $\mathbf{x}_1 = \cdots = \mathbf{x}_{k-1} = \mathbf{0}$, and so $\mathbf{x}_k = \mathbf{0}$ as well.

Now let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ be eigenvectors. If $\alpha_1 \mathbf{x}_1 + \cdots + \alpha_k \mathbf{x}_k = \mathbf{0}$, then $\alpha_1 \mathbf{x}_1 = \cdots = \alpha_k \mathbf{x}_k = \mathbf{0}$. But $\mathbf{x}_1 \neq \mathbf{0}, \ldots, \mathbf{x}_k \neq \mathbf{0}$, so that $\alpha_1 = \cdots = \alpha_k = 0$. Thus $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is linearly independent.

Theorem

Let $\mathbf{A} \in \mathbb{K}^{n\times n}$, and let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of $\mathbf{A}$. Let g_j be the geometric multiplicity of λ_j for $j = 1, \ldots, k$. Then $g_1 + \cdots + g_k \leq n$. Further, $\mathbf{A}$ is diagonalizable if and only if $g_1 + \cdots + g_k = n$.

Proof. Let V_j denote the eigenspace $\mathcal{N}(\mathbf{A} - \lambda_j \mathbf{I})$ of $\mathbb{K}^{n\times 1}$, and let E_j be a basis for V_j consisting of g_j eigenvectors of $\mathbf{A}$ corresponding to λ_j for $j = 1, \ldots, k$.

We claim that the set $E := E_1 \cup \cdots \cup E_k$ is linearly independent. Let $\mathbf{x}$ be a linear combination of elements of E. Collate the elements of E_j for each $j = 1, \ldots, k$ and write $\mathbf{x} = \mathbf{x}_1 + \cdots + \mathbf{x}_k$, where $\mathbf{x}_j \in V_j$ for $j = 1, \ldots, k$. Suppose $\mathbf{x} = \mathbf{0}$. Then $\mathbf{x}_j = \mathbf{0}$ for $j = 1, \ldots, k$ by the previous lemma. For $j \in \{1, \ldots, k\}$, $\mathbf{x}_j$ is a linear combination of elements of the set E_j, and since the set E_j is linearly independent, every coefficient in this linear combination must be 0. Since this holds for each $j = 1, \ldots, k$, we see that every coefficient in the linear combination $\mathbf{x}$ of elements of E must be 0. Hence O.K.

The number of elements in the linearly independent set E is $g_1 + \cdots + g_k$. Since E is a subset of the n dimensional vector space $\mathbb{K}^{n\times 1}$, it follows that $g_1 + \cdots + g_k \leq n$.

Now suppose $g_1 + \cdots + g_k = n$. Then E is a linearly independent subset of $\mathbb{K}^{n\times 1}$ having n elements. Thus E is a basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$. Hence $\mathbf{A}$ is diagonalizable.

Conversely, suppose $\mathbf{A}$ is diagonalizable. Then there is a basis for $\mathbb{K}^{n\times 1}$ consisting of n eigenvectors of $\mathbf{A}$. For $j = 1, \ldots, k$, let h_j elements of this basis belong to V_j. Since these elements form a linearly independent subset of V_j, we see that $h_j \leq g_j$ for $j = 1, \ldots, k$. Hence $n = h_1 + \cdots + h_k \leq g_1 + \cdots + g_k \leq n$. This shows that $g_1 + \cdots + g_k = n$. $\square$

Corollary

If $\mathbf{A} \in \mathbb{K}^{n\times n}$ has n distinct eigenvalues, then $\mathbf{A}$ is diagonalizable.

Proof. Clearly, $n = 1 + \cdots + 1 \leq g_1 + \cdots + g_n \leq n$, and so $g_1 + \cdots + g_n = n$. Hence the above theorem applies. $\square$

The case of $\mathbb{K} = \mathbb{C}$

In case $\mathbb{K} = \mathbb{C}$, then by the Fundamental Theorem of Algebra, every polynomial of degree n with coefficients in $\mathbb{C}$ has exactly n roots in $\mathbb{C}$, counting multiplicities. In particular, the characteristic polynomial $p_{\mathbf{A}}(t)$ of any $\mathbf{A} \in \mathbb{C}^{n\times n}$ has exactly n roots in $\mathbb{C}$, counting multiplicities. More precisely, we can factor

$$p_{\mathbf{A}}(t) = (\lambda_1 - t)^{m_1} \cdots (\lambda_k - t)^{m_k},$$

where $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$ are distinct and $m_1, \ldots, m_k \in \mathbb{N}$ satisfy $m_1 + \cdots + m_k = n$.

As an immediate consequence, we obtain the following result.

Theorem

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. Then there are distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ of $\mathbf{A}$ having algebraic multiplicities $m_1, \ldots, m_k$ such that $m_1 + \cdots + m_k = n$.

Proposition

Let $\mathbf{A} \in \mathbb{K}^{n \times n}$, and let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of $\mathbf{A}$. Let g_j and m_j be the geometric multiplicity and the algebraic multiplicity of λ_j respectively for $j = 1, \ldots, k$.

(i) If $\mathbf{A}$ be diagonalizable, then $g_j = m_j$ for $j = 1, \ldots, k$.

(ii) If $\mathbb{K} = \mathbb{C}$, and $g_j = m_j$ for $j = 1, \ldots, k$, then $\mathbf{A}$ is diagonalizable.

Proof. (i) Since $\mathbf{A}$ is diagonalizable, $g_1 + \cdots + g_k = n$. Hence

$$0 \leq (m_1 - g_1) + \cdots + (m_k - g_k) \leq n - n = 0.$$

But $m_j - g_j \geq 0$, and so $g_j = m_j$ for $j = 1, \ldots, k$.

(ii) Since $\mathbb{K} = \mathbb{C}$, $m_1 + \cdots + m_k = n$. Also, since $g_j = m_j$ for $j = 1, \ldots, k$, we see that $g_1 + \cdots + g_k = m_1 + \cdots + m_k = n$. Hence $\mathbf{A}$ is diagonalizable. $\square$

Remark

Part (ii) of the above proposition does not hold if $\mathbb{K} = \mathbb{R}$.

For example, let $\mathbf{A} := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$. Then $p_{\mathbf{A}}(t) = (1-t)(1+t^2)$, and the geometric multiplicity as well as the algebraic multiplicity of the only (real) eigenvalue 1 of $\mathbf{A}$ is equal to 1. Thus the 3×3 matrix $\mathbf{A}$ is not diagonalizable (over $\mathbb{R}$) since the sum of the geometric multiplicities of its eigenvalues is less than 3.

On the other hand, if $\mathbb{K} = \mathbb{C}$, then $p_{\mathbf{A}}(t) = (1-t)(t-i)(t+i)$, and for each of the eigenvalues $1, i, -i$ of $\mathbf{A}$, the geometric multiplicity as well as the algebraic multiplicity is equal to 1, and so $\mathbf{A}$ is diagonalizable (over $\mathbb{C}$).

Let $\mathbf{A}$ be a square matrix, and λ be an eigenvalue of $\mathbf{A}$. If the geometric multiplicity g of λ is less than the algebraic multiplicity m of λ, then the eigenvalue λ is called **defective**, and $m - g$ is called its **defect**. If a matrix does not have any defective eigenvalue, then the matrix is called **nondefective**.

The above proposition tells us that when $\mathbb{K} = \mathbb{C}$, a square matrix $\mathbf{A}$ is diagonalizable if and only if it is nondefective. We shall later show that if $\mathbb{K} = \mathbb{C}$, then every square matrix can be 'upper triangularized', that is, it is similar to an upper triangular matrix. In fact, we shall prove an even stronger result.

Existence and Location of Eigenvalues

Let $\mathbf{A} \in \mathbb{K}^{n \times n}$ and $\lambda \in \mathbb{K}$. Since λ is an eigenvalue of $\mathbf{A}$ if and only if λ is root of the characteristic polynomial of $\mathbf{A}$, and since this polynomial is of degree n, the matrix $\mathbf{A}$ can have at most n distinct eigenvalues. Let $k \in \mathbb{N}$ with $1 \leq k \leq n$. It is easy to see that the matrix $\mathbf{A} := \operatorname{diag}(1, 2, \ldots, k, k, \ldots, k)$ has exactly k distinct eigenvalues.

If $\mathbb{K} = \mathbb{R}$, then $\mathbf{A}$ may not have any eigenvalue if n is even, and $\mathbf{A}$ has at least one eigenvalue if n is odd. On the other hand, if $\mathbb{K} = \mathbb{C}$, then $\mathbf{A}$ has exactly n eigenvalues, if we count them according to their algebraic multiplicities.

Often, it is not enough to know that so many eigenvalues of $\mathbf{A}$ exist; one would like to know where they are located. In this connection, we give a 'localization' result.

Let $\mathbf{A} := [a_{jk}] \in \mathbb{K}^{n \times n}$. For $j \in \{1, \ldots, n\}$, define $r_j := \sum_{k \neq j} |a_{jk}|$, and let $D(a_{jj}, r_j) := \{a \in \mathbb{K} : |a - a_{jj}| \leq r_j\}$, which is a closed disk in $\mathbb{K}$ with centre at the jth diagonal entry of $\mathbf{A}$ and radius equal to the sum of the absolute values of the off-diagonal entries in the jth row of $\mathbf{A}$; it is called the jth **Gershgorin disk** of the matrix $\mathbf{A}$.

Proposition (Gershgorin circle theorem)

Let $\mathbf{A} \in \mathbb{K}^{n \times n}$. Every eigenvalue of $\mathbf{A}$ belongs in one of the Gerschgorin disks of $\mathbf{A}$.

Proof. Let $\mathbf{A} := [a_{jk}]$. Let $\lambda \in \mathbb{K}$ be an eigenvalue of $\mathbf{A}$, and let $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T}$ be a corresponding eigenvector.

Let $j \in \{1, \ldots, n\}$ be such that $|x_j| = \max\{|x_k| : k = 1, \ldots, n\}$. Then $x_j \neq 0$ since $\mathbf{x} \neq \mathbf{0}$. Multiplying $\mathbf{x}$ by $1/x_j$, we may assume that $x_j = 1$ and $|x_k| \leq 1$ for all $k \neq j$.

Comparing the jth components in the vector equation $\mathbf{Ax} = \lambda \mathbf{x}$, we obtain

$$\sum_{k \neq j} a_{jk} x_k + a_{jj} x_j = \lambda x_j, \text{ that is, } \sum_{k \neq j} a_{jk} x_k + a_{jj} = \lambda.$$

Now the triangle inequality for elements of $\mathbb{K}$ shows that

$$|a_{jj} - \lambda| = \left| \sum_{k \neq j} a_{jk} x_k \right| \leq \sum_{k \neq j} |a_{jk} x_k| \leq \sum_{k \neq j} |a_{jk}| = r_j.$$

Thus λ belongs to the jth Gershgorin disk of $\mathbf{A}$. $\square$

Example Let $\mathbf{A} := \begin{bmatrix} 10 & -1 & 0 & 1 \\ 0.2 & 8 & 0.3 & 0.1 \\ 1 & -1 & 2 & 1 \\ 1 & 0.5 & -1 & 11 \end{bmatrix}$. The centres of the Gerschgorin disks are 10, 8, 2, 11 with the respective radii 2, 0.6, 3, 2.5. Hence if λ is an eigenvalue of $\mathbf{A}$, then either $|\lambda - 10| \leq 2$ or $|\lambda - 8| \leq 0.6$ or $|\lambda - 2| \leq 3$ or $|\lambda - 11| \leq 2.5$.

Inner Product and Norm

Let $\mathbb{K} := \mathbb{R}$, the set of real numbers, or $\mathbb{K} := \mathbb{C}$, the set of complex numbers. For a scalar $\alpha \in \mathbb{K}$, we denote its conjugate by $\overline{\alpha}$. If $\alpha \in \mathbb{R}$, then of course, $\overline{\alpha} = \alpha$.

Consider column vectors $\mathbf{x} := \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{y} := \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ in $\mathbb{K}^{n \times 1}$.

The adjoint $\mathbf{x}^* := \begin{bmatrix} \overline{x}_1 & \cdots & \overline{x}_n \end{bmatrix}$ of $\mathbf{x}$ is a row vector in $\mathbb{K}^{1 \times n}$. The **inner product** of $\mathbf{x}$ with $\mathbf{y}$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^* \mathbf{y} = \overline{x}_1 y_1 + \cdots + \overline{x}_n y_n.$$

Note: If $\mathbb{K} = \mathbb{R}$, then $\langle \mathbf{x}, \mathbf{y} \rangle$ is just the scalar product of $\boldsymbol{x} := (x_1, \ldots, x_n)$ and $\boldsymbol{y} := (y_1, \ldots, y_n)$ in $\mathbb{R}^n$.

The inner product function $\langle \cdot\,,\,\cdot \rangle : \mathbb{K}^{n\times 1} \times \mathbb{K}^{n\times 1} \to \mathbb{K}$ has the following crucial properties. For $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{K}^{n\times 1}$ and $\alpha, \beta \in \mathbb{K}$,

1. $\langle \mathbf{x},\, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x},\, \mathbf{x} \rangle = 0 \iff \mathbf{x} = \mathbf{0}$ (positive definite),
2. $\langle \mathbf{x},\, \alpha\mathbf{y} + \beta\mathbf{z} \rangle = \alpha\langle \mathbf{x},\, \mathbf{y} \rangle + \beta\langle \mathbf{x},\, \mathbf{z} \rangle$ (linear in 2nd variable),
3. $\langle \mathbf{y},\, \mathbf{x} \rangle = \overline{\langle \mathbf{x},\, \mathbf{y} \rangle}$ (conjugate symmetric).

From the above three crucial properties, conjugate linearity in the 1st variable follows: $\langle \alpha\mathbf{x} + \beta\mathbf{y},\, \mathbf{z} \rangle = \overline{\alpha}\langle \mathbf{x},\, \mathbf{z} \rangle + \overline{\beta}\langle \mathbf{y},\, \mathbf{z} \rangle$.

Let $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{K}^{n\times 1}$. We define the **norm** of $\mathbf{x}$ by

$$\|\mathbf{x}\| := \langle \mathbf{x},\, \mathbf{x} \rangle^{1/2} = (|x_1|^2 + \cdots + |x_n|^2)^{1/2}.$$

For $n = 1$, the norm of $x \in \mathbb{K}$ is the absolute value $|x|$ of x.

Clearly, $\max\{|x_1|, \ldots, |x_m|\} \leq \|\mathbf{x}\| \leq |x_1| + \cdots + |x_m|$.

If $\mathbf{x} \in \mathbb{K}^{n\times 1}$ and $\|\mathbf{x}\| = 1$, then we say that $\mathbf{x}$ is a **unit vector**.

Theorem

Let $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. Then
(i) (Schwarz Inequality) $|\langle \mathbf{x}, \mathbf{y}\rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$.

(ii) (Triangle Inequality) $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$.

Proof. Suppose $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$ and $\mathbf{y} := \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}^{\mathsf{T}}$.

(i) If $\|\mathbf{x}\| = 0$ or $\|\mathbf{y}\| = 0$, then $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$. Hence we are done. Now let $\|\mathbf{x}\| \neq 0$ and $\|\mathbf{y}\| \neq 0$. Then

$$\frac{|\overline{x}_j|}{\|\mathbf{x}\|}\frac{|y_j|}{\|\mathbf{y}\|} \le \frac{1}{2}\Big(\frac{|x_j|^2}{\|\mathbf{x}\|^2} + \frac{|y_j|^2}{\|\mathbf{y}\|^2}\Big) \quad \text{for } j = 1, \ldots, n,$$

since $|\overline{\alpha}\beta| = |\alpha|\,|\beta| \le (|\alpha|^2 + |\beta|^2)/2$ for all $\alpha, \beta \in \mathbb{K}$. Hence

$$|\langle \mathbf{x}, \mathbf{y}\rangle| \le \sum_{j=1}^{n} |\overline{x}_j|\,|y_j| \le \frac{\|\mathbf{x}\|\|\mathbf{y}\|}{2}(1+1) = \|\mathbf{x}\|\|\mathbf{y}\|.$$

(ii) Since $\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle = 2\,\Re \langle \mathbf{x}, \mathbf{y} \rangle$, we see that

$$\begin{aligned} \|\mathbf{x}+\mathbf{y}\|^2 &= \langle \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y} \rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\,\Re \langle \mathbf{x}, \mathbf{y} \rangle \\ &\leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\,|\langle \mathbf{x}, \mathbf{y} \rangle| \\ &\leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\,\|\mathbf{x}\|\|\mathbf{y}\| \quad \text{(by the Schwarz inequality)} \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

Thus $\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$. $\square$

We observe that the norm function $\|\cdot\| : \mathbb{K}^{n\times 1} \to \mathbb{K}$ satisfies the following three crucial properties:

(i) $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{K}^{n\times 1}$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$,

(ii) $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$ for all $\alpha \in \mathbb{K}$ and $\mathbf{x} \in \mathbb{K}^{n\times 1}$,

(iii) $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$.

MA110:
Lecture 13

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Spring 2025

Inner Product and Norm

Let $\mathbb{K} := \mathbb{R}$, the set of real numbers, or $\mathbb{K} := \mathbb{C}$, the set of complex numbers. For a scalar $\alpha \in \mathbb{K}$, we denote its conjugate by $\overline{\alpha}$. If $\alpha \in \mathbb{R}$, then of course, $\overline{\alpha} = \alpha$.

Consider column vectors $\mathbf{x} := \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{y} := \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ in $\mathbb{K}^{n\times 1}$.

The conjugate transpose (or the adjoint) $\mathbf{x}^* := \begin{bmatrix} \overline{x}_1 & \cdots & \overline{x}_n \end{bmatrix}$ of $\mathbf{x}$ is a row vector in $\mathbb{K}^{1\times n}$. The **inner product** of $\mathbf{x}$ with $\mathbf{y}$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^*\mathbf{y} = \overline{x}_1 y_1 + \cdots + \overline{x}_n y_n.$$

Note: If $\mathbb{K} = \mathbb{R}$, then $\langle \mathbf{x}, \mathbf{y} \rangle$ is just the scalar product of $\boldsymbol{x} := (x_1, \ldots, x_n)$ and $\boldsymbol{y} := (y_1, \ldots, y_n)$ in $\mathbb{R}^n$.

The inner product function $\langle \cdot\,, \cdot \rangle : \mathbb{K}^{n\times 1} \times \mathbb{K}^{n\times 1} \to \mathbb{K}$ has the following crucial properties. For $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{K}^{n\times 1}$ and $\alpha, \beta \in \mathbb{K}$,

1. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0 \iff \mathbf{x} = \mathbf{0}$ (positive definite),
2. $\langle \mathbf{x}, \alpha\mathbf{y} + \beta\mathbf{z} \rangle = \alpha\langle \mathbf{x}, \mathbf{y} \rangle + \beta\langle \mathbf{x}, \mathbf{z} \rangle$ (linear in 2nd variable),
3. $\langle \mathbf{y}, \mathbf{x} \rangle = \overline{\langle \mathbf{x}, \mathbf{y} \rangle}$ (conjugate symmetric).

From the above three crucial properties, conjugate linearity in the 1st variable follows: $\langle \alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z} \rangle = \overline{\alpha}\langle \mathbf{x}, \mathbf{z} \rangle + \overline{\beta}\langle \mathbf{y}, \mathbf{z} \rangle$.

Let $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{K}^{n\times 1}$. We define the **norm** of $\mathbf{x}$ by

$$\|\mathbf{x}\| := \langle \mathbf{x}, \mathbf{x} \rangle^{1/2} = (|x_1|^2 + \cdots + |x_n|^2)^{1/2}.$$

For $n = 1$, the norm of $x \in \mathbb{K}$ is the absolute value $|x|$ of x.

Clearly, $\max\{|x_1|, \ldots, |x_m|\} \leq \|\mathbf{x}\| \leq |x_1| + \cdots + |x_m|$.

If $\mathbf{x} \in \mathbb{K}^{n\times 1}$ and $\|\mathbf{x}\| = 1$, then we say that $\mathbf{x}$ is a **unit vector**.

Theorem

Let $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. Then
(i) (Schwarz Inequality) $|\langle \mathbf{x}, \mathbf{y}\rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$.

(ii) (Triangle Inequality) $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$.

Proof. Suppose $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T}$ and $\mathbf{y} := \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}^\mathsf{T}$.

(i) If $\|\mathbf{x}\| = 0$ or $\|\mathbf{y}\| = 0$, then $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$. Hence we are done. Now let $\|\mathbf{x}\| \neq 0$ and $\|\mathbf{y}\| \neq 0$. Then

$$\frac{|\overline{x}_j|}{\|\mathbf{x}\|}\frac{|y_j|}{\|\mathbf{y}\|} \le \frac{1}{2}\Big(\frac{|x_j|^2}{\|\mathbf{x}\|^2} + \frac{|y_j|^2}{\|\mathbf{y}\|^2}\Big) \quad \text{for } j = 1, \ldots, n,$$

since $|\overline{\alpha}\beta| = |\alpha|\,|\beta| \le (|\alpha|^2 + |\beta|^2)/2$ for all $\alpha, \beta \in \mathbb{K}$. Hence

$$|\langle \mathbf{x}, \mathbf{y}\rangle| \le \sum_{j=1}^{n} |\overline{x}_j|\,|y_j| \le \frac{\|\mathbf{x}\|\|\mathbf{y}\|}{2}(1+1) = \|\mathbf{x}\|\|\mathbf{y}\|.$$

(ii) Since $\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle = 2\,\Re\,\langle \mathbf{x}, \mathbf{y} \rangle$, we see that

$$\begin{aligned} \|\mathbf{x}+\mathbf{y}\|^2 &= \langle \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y} \rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\,\Re\,\langle \mathbf{x}, \mathbf{y} \rangle \\ &\le \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\,|\langle \mathbf{x}, \mathbf{y} \rangle| \\ &\le \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\,\|\mathbf{x}\|\|\mathbf{y}\| \quad \text{(by the Schwarz inequality)} \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

Thus $\|\boldsymbol{x}+\boldsymbol{y}\| \le \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$. $\square$

We observe that the norm function $\|\cdot\| : \mathbb{K}^{n\times 1} \to \mathbb{K}$ satisfies the following three crucial properties:

(i) $\|\mathbf{x}\| \ge 0$ for all $\mathbf{x} \in \mathbb{K}^{n\times 1}$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$,

(ii) $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$ for all $\alpha \in \mathbb{K}$ and $\mathbf{x} \in \mathbb{K}^{n\times 1}$,

(iii) $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$.

The properties of the norm function allow us to define the distance between two vectors in $\mathbb{K}^{n\times 1}$. Let $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. Then the **distance** between $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|.$$

The distance function $d : \mathbb{K}^{n\times 1} \times \mathbb{K}^{n\times 1} \to \mathbb{K}$ has the following analogous properties.

(i) $d(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$, $d(\mathbf{x}, \mathbf{y}) = 0 \iff \mathbf{x} = \mathbf{y}$,

(ii) $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$,

(iii) $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{K}^{n\times 1}$.

The inner product defined earlier allows us to say when two column vectors are perpendicular to each other.

Orthogonality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. We say that $\mathbf{x}$ and $\mathbf{y}$ are **orthogonal** (to each other) if $\langle \mathbf{x}, \mathbf{y}\rangle = 0$, and then we write $\mathbf{x} \perp \mathbf{y}$.

Clearly, $\mathbf{x} \perp \mathbf{x} \iff \|\mathbf{x}\| = \mathbf{0} \iff \mathbf{x} = \mathbf{0}$.

Let E be a subset of $\mathbb{K}^{n\times 1}$, and define

$$E^{\perp} := \{\mathbf{y} \in \mathbb{K}^{n\times 1} : \mathbf{y} \perp \mathbf{x} \text{ for all } \mathbf{x} \in E\}.$$

It is easy to see that $E^{\perp}$ is a subspace of $\mathbb{K}^{n\times 1}$.

Proposition (Pythagoras Theorem)

Let $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. If $\mathbf{x} \perp \mathbf{y}$, then $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$.

Proof.

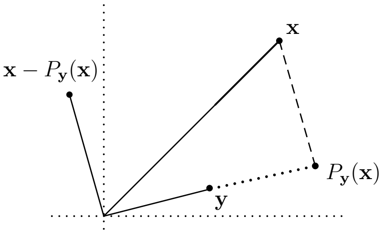
$$\begin{aligned}
\|\mathbf{x}+\mathbf{y}\|^2 &= \langle \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y}\rangle = \langle \mathbf{x}+\mathbf{y}, \mathbf{x}\rangle + \langle \mathbf{x}+\mathbf{y}, \mathbf{y}\rangle \\
&= \langle \mathbf{x}, \mathbf{x}\rangle + \langle \mathbf{y}, \mathbf{x}\rangle + \langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{y}, \mathbf{y}\rangle \\
&= \|\mathbf{x}\|^2 + 0 + 0 + \|\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2.
\end{aligned}$$

$\square$

We now introduce an important concept.
Let $\mathbf{y}$ be a nonzero vector in $\mathbb{K}^{n\times 1}$. For $\mathbf{x} \in \mathbb{K}^{n\times 1}$, define

$$P_{\mathbf{y}}(\mathbf{x}) := \frac{\langle \mathbf{y},\, \mathbf{x}\rangle}{\langle \mathbf{y},\, \mathbf{y}\rangle}\, \mathbf{y}.$$

It is called the (perpendicular) **projection** of the vector $\mathbf{x}$ in the direction of the vector $\mathbf{y}$. Note that $P_{\mathbf{y}} : \mathbb{K}^{n\times 1} \to \mathbb{K}^{n\times 1}$ is a linear map and its image space is one dimensional. Also, $P_{\mathbf{y}}(\mathbf{y}) = \mathbf{y}$, so that $(P_{\mathbf{y}})^2 := P_{\mathbf{y}} \circ P_{\mathbf{y}} = P_{\mathbf{y}}$.



Note that $P_{\mathbf{y}}(\mathbf{x})$ is a scalar multiple of $\mathbf{y}$ for every $\mathbf{x} \in \mathbb{K}^{n\times 1}$.

An important property of the projection of a vector in the direction of another (nonzero) vector is the following:

Proposition

Let $\mathbf{y} \in \mathbb{K}^{n\times 1}$ be nonzero. Then for every $\mathbf{x} \in \mathbb{K}^{n\times 1}$,

$$\big(\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\big) \perp \mathbf{y}.$$

Proof. Let $\mathbf{x} \in \mathbb{K}^{n\times 1}$. The result follows from

$$\langle \mathbf{y},\, \mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\rangle = \langle \mathbf{y},\, \mathbf{x}\rangle - \langle \mathbf{y},\, P_{\mathbf{y}}(\mathbf{x})\rangle = \langle \mathbf{y},\, \mathbf{x}\rangle - \frac{\langle \mathbf{y},\, \mathbf{x}\rangle}{\langle \mathbf{y},\, \mathbf{y}\rangle}\langle \mathbf{y},\, \mathbf{y}\rangle = 0. \qquad \square$$

Let E be a subset of $\mathbb{K}^{n\times 1}$. Then E is said to be **orthogonal** if any two (distinct) element of E are orthogonal (to each other), that is, $\mathbf{x} \perp \mathbf{y}$ for all $\mathbf{x}, \mathbf{y}$ in E with $\mathbf{x} \neq \mathbf{y}$.

For example, $E := \{\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 - \mathbf{e}_2\}$ is an orthogonal subset of $\mathbb{K}^{n\times 1}$.

Proposition

Let E be a subset of $\mathbb{K}^{n\times 1}$. If E is orthogonal and if $\mathbf{0} \notin E$, then E is linearly independent.

Proof. Let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ be distinct vectors in E, and let $\alpha_1, \ldots, \alpha_k$ be scalars such that $\alpha_1\mathbf{x}_1 + \cdots + \alpha_k\mathbf{x}_k = \mathbf{0}$. Fix $j \in \{1, \ldots, k\}$. Since $\langle \mathbf{x}_j, \mathbf{x}_i \rangle = 0$ for all $i \neq j$, we obtain

$$0 = \langle \mathbf{x}_j, \alpha_1\mathbf{x}_1 + \cdots + \alpha_k\mathbf{x}_k \rangle = \sum_{\ell=1}^{k} \alpha_\ell \langle \mathbf{x}_j, \mathbf{x}_\ell \rangle = \alpha_j \langle \mathbf{x}_j, \mathbf{x}_j \rangle.$$

But $\langle \mathbf{x}_j, \mathbf{x}_j \rangle \neq 0$ since $\mathbf{x}_j \neq \mathbf{0}$. Hence $\alpha_j = 0$. $\square$

The converse of the above proposition is not true, that is, a linear linearly independent subset of $\mathbb{K}^{n\times 1}$ need not be orthogonal. For example, the subset $\{\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2\}$ of $\mathbb{K}^{n\times 1}$ is linearly independent, but not orthogonal.

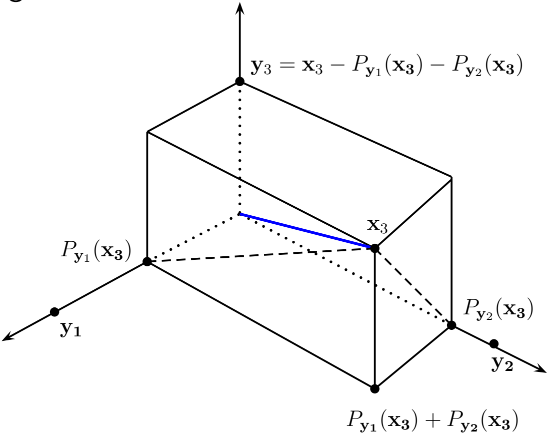
We now address the question: Can we modify a linearly independent set E to construct an orthogonal set, retaining the span of the elements in E at each step of the procedure?

Let E be an ordered linearly independent set of column vectors. Suppose $\mathbf{x}_1$ is the first vector in E. Then $\mathbf{x}_1 \neq 0$. Let $\mathbf{y}_1 := \mathbf{x}_1$. Let $\mathbf{x}_2$ be the second vector in E. If $\mathbf{x}_2$ is not orthogonal to $\mathbf{y}_1$, then it makes sense to subtract from $\mathbf{x}_2$, the projection of $\mathbf{x}_2$ in the direction of $\mathbf{y}_1$, so that $\mathbf{y}_2 := \mathbf{x}_2 - P_{\mathbf{y}_1}(\mathbf{x}_2)$ is orthogonal to $\mathbf{y}_1$. Also, in replacing $\mathbf{x}_2$ by $\mathbf{y}_2$, we do not alter the span of $\{\mathbf{x}_1, \mathbf{x}_2\}$ since $\mathbf{y}_2$ is a linear combination of $\mathbf{x}_2$ and $\mathbf{y}_1$, and $\mathbf{x}_2$ is a linear combination of $\mathbf{y}_2$ and $\mathbf{y}_1$, where $\mathbf{y}_1 = \mathbf{x}_1$.

Let $\mathbf{x}_3$ be the third vector in E. If $\mathbf{x}_3$ is not orthogonal to $\mathbf{y}_1$ and $\mathbf{y}_2$, then we may subtract from $\mathbf{x}_3$, the projections of $\mathbf{x}_3$ in the directions of $\mathbf{y}_1$ and $\mathbf{y}_2$. Then $\mathbf{y}_3 := \mathbf{x}_3 - P_{\mathbf{y}_1}(\mathbf{x}_3) - P_{\mathbf{y}_2}(\mathbf{x}_3)$ is orthogonal to $\mathbf{y}_1$ as well as to $\mathbf{y}_2$. We can see this as follows.

$$\langle \mathbf{y}_1,\, \mathbf{y}_3 \rangle = \langle \mathbf{y}_1,\, \mathbf{x}_3 - P_{\mathbf{y}_1}(\mathbf{x}_3) \rangle - \langle \mathbf{y}_1,\, P_{\mathbf{y}_2}(\mathbf{x}_3) \rangle = 0 - 0 = 0,$$

and similarly $\langle \mathbf{y}_3,\, \mathbf{y}_2 \rangle = 0$. We illustrate these vectors in the following figure.



This procedure can be continued to yield the famous

Gram-Schmidt Orthogonalization Process (G-S OP)

Let $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ be an ordered linearly independent set in $\mathbb{K}^{n\times 1}$. Define $\mathbf{y}_1 := \mathbf{x}_1$.
Let $1 \le j < k$. Suppose we have found $\mathbf{y}_1, \ldots, \mathbf{y}_j$ in $\mathbb{K}^{n\times 1}$ such that the set $\{\mathbf{y}_1, \ldots, \mathbf{y}_j\}$ is orthogonal, and also $\text{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j\} = \text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_j\}$. Define

$$\mathbf{y}_{j+1} := \mathbf{x}_{j+1} - P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) - \cdots - P_{\mathbf{y}_j}(\mathbf{x}_{j+1}).$$

Then $\text{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_{j+1}\} = \text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_{j+1}\}$ since $\mathbf{y}_{j+1} \in \text{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j, \mathbf{x}_{j+1}\} = \text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_j, \mathbf{x}_{j+1}\}$ and $\mathbf{x}_{j+1} \in \text{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j, \mathbf{y}_{j+1}\}$.

To show that the set $\{\mathbf{y}_1, \ldots, \mathbf{y}_{j+1}\}$ is orthogonal, it is enough to show that $\mathbf{y}_{j+1} \in \{\mathbf{y}_1, \ldots, \mathbf{y}_j\}^\perp$.

Let $i \in \{1, \ldots, j\}$. Then

$$\begin{aligned}
\langle \mathbf{y}_i,\, \mathbf{y}_{j+1} \rangle &= \langle \mathbf{y}_i,\, \mathbf{x}_{j+1} - P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) - \cdots - P_{\mathbf{y}_j}(\mathbf{x}_{j+1}) \rangle \\
&= \langle \mathbf{y}_i,\, \mathbf{x}_{j+1} \rangle - \langle \mathbf{y}_i,\, P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) \rangle - \cdots - \langle \mathbf{y}_i,\, P_{\mathbf{y}_j}(\mathbf{x}_{j+1}) \rangle \\
&= \langle \mathbf{y}_i,\, \mathbf{x}_{j+1} \rangle - \langle \mathbf{y}_i,\, P_{\mathbf{y}_i}(\mathbf{x}_{j+1}) \rangle \quad (\text{since } \mathbf{y}_i \perp \mathbf{y}_j,\ i \neq j) \\
&= \langle \mathbf{y}_i,\, \mathbf{x}_{j+1} - P_{\mathbf{y}_i}(\mathbf{x}_{j+1}) \rangle \\
&= 0 \quad (\text{by the important property of the projection}).
\end{aligned}$$

We remark that since the set $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is linearly independent, all vectors $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_k$ constructed in the G-S OP are nonzero: Clearly, $\mathbf{y}_1 = \mathbf{x}_1 \neq \mathbf{0}$. Also, if $\mathbf{y}_{j+1} = \mathbf{0}$ for some $j \in \{1, \ldots, k-1\}$, then $\mathbf{x}_{j+1}$ would belong to $\operatorname{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j\} = \operatorname{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_j\}$.

This completes the construction of the G-S OP.

An orthogonal set whose elements are unit vectors is called an **orthonormal set**.

Any orthogonal set whose elements are nonzero vectors can always be turned into an orthonormal set by dividing each element by its own norm.

Thus given an ordered linearly independent set $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$, we can construct an ordered orthogonal set $(\mathbf{y}_1, \ldots, \mathbf{y}_k)$ by the G-S OP, and if we let $\mathbf{u}_j := \mathbf{y}_j / \|\mathbf{y}_j\|$ for $j = 1, \ldots, k$, then $(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ is an ordered orthonormal set such that $\mathrm{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\} = \mathrm{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_k\} = \mathrm{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$.

Example

For $j = 1, \ldots, n$, let $\mathbf{x}_j := j(\mathbf{e}_1 + \cdots + \mathbf{e}_j)$. Then $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is an ordered linearly independent subset of $\mathbb{K}^{n \times 1}$. We claim that the G-S OP gives $\mathbf{y}_j := j\mathbf{e}_j$ for $j = 1, \ldots, n$. Indeed, $\mathbf{y}_1 := \mathbf{x}_1 = \mathbf{e}_1$. Also, assuming that $\mathbf{y}_j = j\mathbf{e}_j$, we see that

$$\begin{aligned}\mathbf{y}_{j+1} &= \mathbf{x}_{j+1} - P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) - \cdots - P_{\mathbf{y}_j}(\mathbf{x}_{j+1}) \\ &= (j+1)(\mathbf{e}_1 + \cdots + \mathbf{e}_{j+1}) - (j+1)\mathbf{e}_1 - \cdots - (j+1)\mathbf{e}_j \\ &= (j+1)\mathbf{e}_{j+1}.\end{aligned}$$

Hence our claim is justified. Since $\|\mathbf{y}_j\| = j$ for each j, we let $\mathbf{u}_j := \mathbf{y}_j/j$, so that $\mathbf{u}_j = \mathbf{e}_j$ for each $j = 1, \ldots, n$. Clearly, $(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ is an ordered orthonormal set in $\mathbb{K}^{n\times 1}$.

Let V be a subspace of $\mathbb{K}^{n\times 1}$. An **orthonormal basis** for V is a basis for V which is an orthonormal subset of V.

The G-S OP enables us to modify a given basis for a subspace of $\mathbb{K}^{n\times 1}$ to an orthonormal basis for that subspace.

Also, we can expand an orthonormal set in V to a possibly larger orthonormal set in V as follows.

Proposition

Let V be a subspace of $\mathbb{K}^{n\times 1}$, and let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be an orthonormal set in V. Then there is an orthonormal basis for V which contains $\mathbf{u}_1, \ldots, \mathbf{u}_k$.

Proof. If $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} = V$, then there is nothing to prove.

Now suppose $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \neq V$. Let $\dim V = r$. Since the set $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \neq V$ is linearly independent, there are $\mathbf{y}_{k+1}, \ldots, \mathbf{y}_r$ in V such that $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{y}_{k+1}, \ldots, \mathbf{y}_r\}$ is a basis for V. By the G-S OP, we can find $\mathbf{u}_{k+1}, \ldots, \mathbf{u}_r$ in V such that the set $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{u}_{k+1}, \ldots, \mathbf{u}_r\}$ is orthonormal and its span is equal to $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{y}_{k+1}, \ldots, \mathbf{y}_r\} = V$. $\square$

Corollary

Every nonzero vector subspace V has an orthonormal basis.

Proof. If $\mathbf{0} \neq \mathbf{x}_1 \in V$, then extend $\{\mathbf{x}_1/\|\mathbf{x}_1\|\}$ to an o. n. basis.

MA110:
Lecture 14

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Spring 2025

Review of the last lecture

- On $\mathbb{K}^{n\times 1}$, we have the inner product defined by
$$\langle \mathbf{x}, \mathbf{y}\rangle := \mathbf{x}^*\mathbf{y} = \overline{x}_1 y_1 + \cdots + \overline{x}_n y_n.$$
It is positive definite, linear in the 2nd variable, conjugate symmetric (hence conjugate linear in the 1st variable).
- The norm of $\mathbf{x} \in \mathbb{K}^{n\times 1}$ is defined by
$$\|\mathbf{x}\| := \langle \mathbf{x}, \mathbf{x}\rangle^{1/2} = (|x_1|^2 + \cdots + |x_n|^2)^{1/2}.$$
It satisfies the triangle inequality: $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$.
- The projection of a vector $\mathbf{x}$ in the direction of a nonzero vector $\mathbf{y}$ is a scalar multiple of $\mathbf{y}$ given by
$$P_{\mathbf{y}}(\mathbf{x}) := \frac{\langle \mathbf{y}, \mathbf{x}\rangle}{\langle \mathbf{y}, \mathbf{y}\rangle}\mathbf{y} \quad \text{and it satisfies} \quad \langle \mathbf{y}, \mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\rangle = 0.$$
- A subset E of $\mathbb{K}^{n\times 1}$ if orthogonal if $\langle \mathbf{x}, \mathbf{y}\rangle = 0$ for all $\mathbf{x}, \mathbf{y} \in E$ with $\mathbf{x} \neq \mathbf{y}$. If E is orthogonal and $\mathbf{0} \notin E$, then E is linearly independent.

Gram-Schmidt Orthogonalization Process

The Gram-Schmidt Orthogonalization Process (G-S OP) transforms a linearly independent set in $\mathbb{K}^{n\times 1}$ to an orthogonal set without altering the span. It is given by the following.

Let $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ be an ordered linearly independent set in $\mathbb{K}^{n\times 1}$. Define $\mathbf{y}_1 := \mathbf{x}_1$. Let $1 \le j < k$. Suppose we have found $\mathbf{y}_1, \ldots, \mathbf{y}_j$ in $\mathbb{K}^{n\times 1}$ such that the set $\{\mathbf{y}_1, \ldots, \mathbf{y}_j\}$ is orthogonal, and also $\operatorname{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j\} = \operatorname{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_j\}$. Define

$$\mathbf{y}_{j+1} := \mathbf{x}_{j+1} - P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) - \cdots - P_{\mathbf{y}_j}(\mathbf{x}_{j+1}).$$

Then $\operatorname{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_{j+1}\} = \operatorname{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_{j+1}\}$ since $\mathbf{y}_{j+1} \in \operatorname{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j, \mathbf{x}_{j+1}\} = \operatorname{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_j, \mathbf{x}_{j+1}\}$ and $\mathbf{x}_{j+1} \in \operatorname{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j, \mathbf{y}_{j+1}\}$.

To show that the set $\{\mathbf{y}_1, \ldots, \mathbf{y}_{j+1}\}$ is orthogonal, it is enough to show that $\mathbf{y}_{j+1} \in \{\mathbf{y}_1, \ldots, \mathbf{y}_j\}^{\perp}$.

Let $i \in \{1, \ldots, j\}$. Then

$$\begin{aligned}
\langle \mathbf{y}_i, \, \mathbf{y}_{j+1} \rangle &= \langle \mathbf{y}_i, \, \mathbf{x}_{j+1} - P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) - \cdots - P_{\mathbf{y}_j}(\mathbf{x}_{j+1}) \rangle \\
&= \langle \mathbf{y}_i, \, \mathbf{x}_{j+1} \rangle - \langle \mathbf{y}_i, \, P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) \rangle - \cdots - \langle \mathbf{y}_i, \, P_{\mathbf{y}_j}(\mathbf{x}_{j+1}) \rangle \\
&= \langle \mathbf{y}_i, \, \mathbf{x}_{j+1} \rangle - \langle \mathbf{y}_i, \, P_{\mathbf{y}_i}(\mathbf{x}_{j+1}) \rangle \quad (\text{since } \mathbf{y}_i \perp \mathbf{y}_j, \, i \neq j) \\
&= \langle \mathbf{y}_i, \, \mathbf{x}_{j+1} - P_{\mathbf{y}_i}(\mathbf{x}_{j+1}) \rangle \\
&= 0 \quad (\text{by the important property of the projection}).
\end{aligned}$$

We remark that since the set $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is linearly independent, all vectors $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_k$ constructed in the G-S OP are nonzero: Clearly, $\mathbf{y}_1 = \mathbf{x}_1 \neq \mathbf{0}$. Also, if $\mathbf{y}_{j+1} = \mathbf{0}$ for some $j \in \{1, \ldots, k-1\}$, then $\mathbf{x}_{j+1}$ would belong to $\text{span}\{\mathbf{y}_1, \ldots, \mathbf{y}_j\} = \text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_j\}$.

This completes the construction of the G-S OP.

Definition

An orthogonal set whose elements are unit vectors is called an **orthonormal set**.

Any orthogonal set whose elements are nonzero vectors can always be turned into an orthonormal set by dividing each element by its own norm.

Thus given an ordered linearly independent set $(\mathbf{x}_1, \dots, \mathbf{x}_k)$, we can construct an ordered orthogonal set $(\mathbf{y}_1, \dots, \mathbf{y}_k)$ by the G-S OP, and if we let $\mathbf{u}_j := \mathbf{y}_j / \|\mathbf{y}_j\|$ for $j = 1, \dots, k$, then $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ is an ordered orthonormal set such that $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \text{span}\{\mathbf{y}_1, \dots, \mathbf{y}_k\} = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$.

Example: For $j = 1, \dots, n$, let $\mathbf{x}_j := j(\mathbf{e}_1 + \cdots + \mathbf{e}_j)$. Then $E := (\mathbf{x}_1, \dots, \mathbf{x}_n)$ is an ordered linearly independent subset of $\mathbb{K}^{n\times 1}$. We claim that the G-S OP gives $\mathbf{y}_j := j\mathbf{e}_j$ for $j = 1, \dots, n$. Indeed, $\mathbf{y}_1 := \mathbf{x}_1 = \mathbf{e}_1$. Also, assuming that $\mathbf{y}_j = j\mathbf{e}_j$, we see that

$$\begin{aligned}
\mathbf{y}_{j+1} &= \mathbf{x}_{j+1} - P_{\mathbf{y}_1}(\mathbf{x}_{j+1}) - \cdots - P_{\mathbf{y}_j}(\mathbf{x}_{j+1}) \\
&= (j+1)(\mathbf{e}_1 + \cdots + \mathbf{e}_{j+1}) - (j+1)\mathbf{e}_1 - \cdots - (j+1)\mathbf{e}_j \\
&= (j+1)\mathbf{e}_{j+1}.
\end{aligned}$$

Hence our claim is justified. Since $\|\mathbf{y}_j\| = j$ for each j, we let $\mathbf{u}_j := \mathbf{y}_j/j$, so that $\mathbf{u}_j = \mathbf{e}_j$ for each $j = 1, \ldots, n$. Clearly, $(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ is an ordered orthonormal set in $\mathbb{K}^{n\times 1}$.

Definition

Let V be a subspace of $\mathbb{K}^{n\times 1}$. An **orthonormal basis** *for V is a basis for V which is an orthonormal subset of V.*

The G-S OP enables us to modify a given basis for a subspace of $\mathbb{K}^{n\times 1}$ to an orthonormal basis for that subspace.

Also, we can expand an orthonormal set in V to a possibly larger orthonormal set in V as follows.

Proposition

Let V be a subspace of $\mathbb{K}^{n\times 1}$, and let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be an orthonormal set in V. Then there is an orthonormal basis for V which contains $\mathbf{u}_1, \ldots, \mathbf{u}_k$.

Proof. If $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} = V$, then there is nothing to prove.

Now suppose $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \neq V$. Let $\dim V = r$. Since the set $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \neq V$ is linearly independent, there are $\mathbf{y}_{k+1}, \ldots, \mathbf{y}_r$ in V such that $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{y}_{k+1}, \ldots, \mathbf{y}_r\}$ is a basis for V. By the G-S OP, we can find $\mathbf{u}_{k+1}, \ldots, \mathbf{u}_r$ in V such that the set $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{u}_{k+1}, \ldots, \mathbf{u}_r\}$ is orthonormal and its span is equal to $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{y}_{k+1}, \ldots, \mathbf{y}_r\} = V$. $\square$

Corollary

Every nonzero vector subspace V has an orthonormal basis.

Proof. If $\mathbf{0} \neq \mathbf{x}_1 \in V$, then extend $\left\{\frac{\mathbf{x}_1}{\|\mathbf{x}_1\|}\right\}$ to an orthornormal basis, using the above proposition.

Example: Let W be the subspace of $\mathbb{K}^{4\times 1}$ spanned by the vectors $\mathbf{x}_1 := \begin{bmatrix}1 & 1 & 0 & 1\end{bmatrix}^\mathsf{T}$, $\mathbf{x}_2 := \begin{bmatrix}1 & -2 & 0 & 0\end{bmatrix}^\mathsf{T}$ and $\mathbf{x}_3 := \begin{bmatrix}1 & 0 & -1 & 2\end{bmatrix}^\mathsf{T}$.

Let us employ the G-S OP. Let $\mathbf{y}_1 := \mathbf{x}_1 = \begin{bmatrix}1 & 1 & 0 & 1\end{bmatrix}^\mathsf{T}$,

$$\begin{aligned}
\mathbf{y}_2 &:= \mathbf{x}_2 - P_{\mathbf{y}_1}(\mathbf{x}_2) = \begin{bmatrix}1 & -2 & 0 & 0\end{bmatrix}^\mathsf{T} + \frac{1}{3}\begin{bmatrix}1 & 1 & 0 & 1\end{bmatrix}^\mathsf{T} \\
&= \frac{1}{3}\begin{bmatrix}4 & -5 & 0 & 1\end{bmatrix}^\mathsf{T} \text{ and} \\
\mathbf{y}_3 &:= \mathbf{x}_3 - P_{\mathbf{y}_1}(\mathbf{x}_3) - P_{\mathbf{y}_2}(\mathbf{x}_3) \\
&= \begin{bmatrix}1 & 0 & -1 & 2\end{bmatrix}^\mathsf{T} - \frac{3}{3}\begin{bmatrix}1 & 1 & 0 & 1\end{bmatrix}^\mathsf{T} - \frac{1}{7}\begin{bmatrix}4 & -5 & 0 & 1\end{bmatrix}^\mathsf{T} \\
&= \frac{1}{7}\begin{bmatrix}-4 & -2 & -7 & 6\end{bmatrix}^\mathsf{T}.
\end{aligned}$$

Note that the subset $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ must be linearly independent since $\mathbf{y}_1$, $\mathbf{y}_2$, $\mathbf{y}_3$ are nonzero.

Further, let

$$\mathbf{u}_1 := \mathbf{y}_1/\sqrt{3},\ \mathbf{u}_2 := \sqrt{3}\,\mathbf{y}_2/\sqrt{14} \ \text{ and } \ \mathbf{u}_3 := \sqrt{7}\,\mathbf{y}_3/\sqrt{15}.$$

Then $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for the subspace W.

To extend $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ to an orthonormal basis for $V := \mathbb{K}^{4\times 1}$, we look for $\mathbf{y}_4 := \begin{bmatrix}\alpha_1 & \alpha_2 & \alpha_3 & \alpha_4\end{bmatrix}^{\mathsf{T}}$ which is orthogonal to the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$, that is,

$$\alpha_1 + \alpha_2 + \alpha_4 = 0,\ \alpha_1 - 2\alpha_2 = 0 \ \text{ and } \alpha_1 - \alpha_3 + 2\alpha_4 = 0.$$

Letting $\alpha_1 := 2$, we obtain $\mathbf{y}_4 := \begin{bmatrix}2 & 1 & -4 & -3\end{bmatrix}^{\mathsf{T}}$. Then $\mathbf{y}_4$ is orthogonal to $\text{span}\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ as well.

Now let $\mathbf{u}_4 := \mathbf{y}_4/\|\mathbf{y}_4\| = \mathbf{y}_4/\sqrt{30}$.

Then $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ is an orthonormal basis for $\mathbb{K}^{4\times 1}$ which extends the orthonormal subset $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ of $\mathbb{K}^{4\times 1}$.

We point out an advantage of working with an orthonormal basis. Suppose $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is a basis of $\mathbb{K}^{n\times 1}$. Then for $\mathbf{b} \in \mathbb{K}^{n\times 1}$, there are unique $\alpha_1, \ldots, \alpha_n$ such that $\mathbf{b} = \alpha_1\mathbf{x}_1 + \cdots + \alpha_n\mathbf{x}_n$. Finding these coefficients $\alpha_1, \ldots, \alpha_n$ is not always easy. In fact, $\begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^\mathsf{T}$ is the unique column vector satisfying the linear system

$$\mathbf{A}\begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^\mathsf{T} = \mathbf{b}, \text{ where } \mathbf{A} := \begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix},$$

and we would have to find this vector either by the GEM or by the Cramer Rule (involving $n+1$ determinants of size n).

On the other hand, suppose $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is an orthonormal basis of $\mathbb{K}^{n\times 1}$. If $\mathbf{b} \in \mathbb{K}^{n\times 1}$ and $\mathbf{b} = \alpha_1\mathbf{u}_1 + \cdots + \alpha_n\mathbf{u}_n$, then $\alpha_j = \langle \mathbf{u}_j, \mathbf{b}\rangle$ for $j = 1, \ldots, n$ by the orthonormality, so that

$$\mathbf{b} = \langle \mathbf{u}_1, \mathbf{b}\rangle\mathbf{u}_1 + \cdots + \langle \mathbf{u}_n, \mathbf{b}\rangle\mathbf{u}_n.$$

For instance, consider the ordered orthonormal basis $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4)$ of $\mathbb{K}^{4\times 1}$ which we have just constructed, where

$\mathbf{u}_1 := \begin{bmatrix} 1 & 1 & 0 & 1 \end{bmatrix}^{\mathsf{T}} / \sqrt{3}$,

$\mathbf{u}_2 := \begin{bmatrix} 4 & -5 & 0 & 1 \end{bmatrix}^{\mathsf{T}} / \sqrt{42}$,

$\mathbf{u}_3 := \begin{bmatrix} -4 & -2 & -7 & 6 \end{bmatrix}^{\mathsf{T}} / \sqrt{105}$,

$\mathbf{u}_4 := \begin{bmatrix} 2 & 1 & -4 & -3 \end{bmatrix}^{\mathsf{T}} / \sqrt{30}$.

Let $\mathbf{b} := \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}^{\mathsf{T}} \in \mathbb{K}^{4\times 1}$.

Then there are unique $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{K}$ such that $\mathbf{b} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \alpha_3\mathbf{u}_3 + \alpha_4\mathbf{u}_4$. In fact,

$\alpha_1 = \langle \mathbf{u}_1, \mathbf{b} \rangle = 3/\sqrt{3} = \sqrt{3}$,

$\alpha_2 = \langle \mathbf{u}_2, \mathbf{b} \rangle = 0$,

$\alpha_3 = \langle \mathbf{u}_3, \mathbf{b} \rangle = -7/\sqrt{105} = -\sqrt{7}/\sqrt{15}$,

$\alpha_4 = \langle \mathbf{u}_4, \mathbf{b} \rangle = -4/\sqrt{30}$.

Analogue for Row Vectors

We have defined an inner product as a function from $\mathbb{K}^{n\times 1} \times \mathbb{K}^{n\times 1}$ to $\mathbb{K}$. We can also define a similar function from $\mathbb{K}^{1\times n} \times \mathbb{K}^{1\times n}$ to $\mathbb{K}$ as follows.

Consider row vectors $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$, $\mathbf{y} := \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}$ in $\mathbb{K}^{1\times n}$. The **inner product** of $\mathbf{x}$ with $\mathbf{y}$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle := \overline{\mathbf{x}}\mathbf{y}^{\mathsf{T}} = \overline{x}_1 y_1 + \cdots + \overline{x}_n y_n.$$

Also, we can introduce the concepts of orthogonality and orthonormality of row vectors, and obtain a Gram-Schmidt Orthogonalization Process for row vectors.

After a detour of inner products and orthonormal sets, we come back to the matrix eigenvalue problem. We shall show that if the scalars are complex numbers, then every square matrix $\mathbf{A}$ can be 'upper triangularized', that is, it is similar to an upper triangular matrix $\mathbf{B} := \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. In fact, we shall show that the invertible matrix $\mathbf{P}$ can be chosen to be of a particularly nice type. We now describe what we mean by nice.

A matrix $\mathbf{U} \in \mathbb{K}^{n \times n}$ is called **unitary** if the columns of $\mathbf{U}$ form an orthonormal subset of $\mathbb{K}^{n \times 1}$. In that case, the columns are in fact an orthonormal basis for $\mathbb{K}^{n \times 1}$.

Proposition

A matrix is unitary if and only if it is invertible and its inverse is the same as its adjoint.

Proof. Let $\mathbf{U}$ be unitary. Then rank $\mathbf{U} = n$ since the n columns $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbf{U}$ form a basis for $\mathbb{K}^{n \times 1}$. Hence $\mathbf{U}$ is invertible. Further, because of the orthonormality,

$$\mathbf{U}^*\mathbf{U} = \begin{bmatrix} \mathbf{u}_1^* \\ \vdots \\ \mathbf{u}_n^* \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^*\mathbf{u}_1 & \cdots & \mathbf{u}_1^*\mathbf{u}_n \\ \vdots & \vdots & \vdots \\ \mathbf{u}_n^*\mathbf{u}_1 & \cdots & \mathbf{u}_n^*\mathbf{u}_n \end{bmatrix} = \mathbf{I}.$$

It follows that $\mathbf{U}\mathbf{U}^* = \mathbf{I}$ as well. Hence $\mathbf{U}^{-1} = \mathbf{U}^*$.

Conversely, the above calculation shows that if a square matrix $\mathbf{U}$ satisfies $\mathbf{U}^*\mathbf{U} = \mathbf{I}$, then its columns form an orthonormal set, that is, $\mathbf{U}$ is a unitary matrix. $\square$

We note that if $\mathbf{A}$ and $\mathbf{B}$ are unitary, then so is $\mathbf{AB}$ since $(\mathbf{AB})^*(\mathbf{AB}) = (\mathbf{B}^*\mathbf{A}^*)(\mathbf{AB}) = \mathbf{B}(\mathbf{A}^*\mathbf{A})\mathbf{B} = \mathbf{B}^*\mathbf{B} = \mathbf{I}$.

Examples

(i) The $n \times n$ identity matrix $\mathbf{I}_n := \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix}$ is unitary.

(ii) For $\theta \in \mathbb{R}$, the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$ is unitary.

It represents a rotation about the x_1-axis in $\mathbb{R}^{3\times 1}$.

(iii) The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is unitary. It represents reflection along the x_3-axis in $\mathbb{R}^{3\times 1}$.

(iv) If $\mathbb{K} = \mathbb{C}$, then the matrix $\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -i/\sqrt{2} & i/\sqrt{2} & 0 \\ 0 & 0 & i \end{bmatrix}$ is unitary.

(v) The matrix $\begin{bmatrix} 1/\sqrt{3} & 4/\sqrt{42} & -4/\sqrt{105} & 2/\sqrt{30} \\ 1/\sqrt{3} & -5/\sqrt{42} & -2/\sqrt{105} & 1/\sqrt{30} \\ 0 & 0 & -7/\sqrt{105} & -4/\sqrt{30} \\ 1/\sqrt{3} & 1/\sqrt{42} & 6/\sqrt{105} & -3/\sqrt{30} \end{bmatrix}$,

whose columns are obtained by orthonormalizing the column vectors $\begin{bmatrix} 1 & 1 & 0 & 1 \end{bmatrix}^{\mathsf{T}}$, $\begin{bmatrix} 1 & -2 & 0 & 0 \end{bmatrix}^{\mathsf{T}}$, $\begin{bmatrix} 1 & 0 & -1 & 2 \end{bmatrix}^{\mathsf{T}}$ and $\begin{bmatrix} 2 & 1 & -4 & -3 \end{bmatrix}^{\mathsf{T}}$ in $\mathbb{K}^{4\times 1}$ is unitary.

(vi) We shall consider an interesting unitary matrix later.

Let $\mathbf{A}, \mathbf{B} \in \mathbb{K}^{n \times n}$. We say that $\mathbf{A}$ is **unitarily similar** to $\mathbf{B}$ if there is a unitary matrix $\mathbf{U}$ such that $\mathbf{B} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$. Also, $\mathbf{A}$ is called **unitarily diagonalizable** if it is unitarily similar to a diagonal matrix.

We proved in Lecture 10 that a matrix $\mathbf{A}$ is similar to a matrix $\mathbf{B}$ if and only if there is an ordered basis E for $\mathbb{K}^{n \times 1}$ such that $\mathbf{B}$ is the matrix of the linear map $T_{\mathbf{A}} : \mathbb{K}^{n \times 1} \to \mathbb{K}^{n \times 1}$ with respect to E. We prove an analogous result below.

Proposition

Let $\mathbf{A}, \mathbf{B} \in \mathbb{K}^{n \times n}$. Then $\mathbf{A}$ is unitarily similar to $\mathbf{B}$ if and only if there is an ordered orthonormal basis E for $\mathbb{K}^{n \times 1}$ such that $\mathbf{B}$ is the matrix of the linear transformation $T_{\mathbf{A}} : \mathbb{K}^{n \times 1} \to \mathbb{K}^{n \times 1}$ with respect to E.
In fact, $\mathbf{B} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$ if and only if the columns of $\mathbf{U}$ form an ordered orthonormal basis, say E, for $\mathbb{K}^{n \times 1}$ and $\mathbf{B} = \mathbf{M}_E^E(T_{\mathbf{A}})$.

Proof. Let $\mathbf{B} := [b_{jk}]$. Now $\mathbf{A}$ is unitarily similar to $\mathbf{B}$ if and only if there is a unitary matrix $\mathbf{U}$ such that $\mathbf{AU} = \mathbf{UB}$. This is the case if and only if there is an ordered orthonormal basis $E := (\mathbf{u}_1, \ldots, \mathbf{u}_n)$ for $\mathbb{K}^{n\times 1}$ such that

$$\mathbf{A}\begin{bmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_n\end{bmatrix} = \begin{bmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_n\end{bmatrix}\begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \vdots & \vdots \\ b_{n1} & \cdots & b_{nn}\end{bmatrix}.$$

The kth column of LHS is $\mathbf{Au}_k$ and the kth column of RHS is the linear combination of $\mathbf{u}_1, \ldots, \mathbf{u}_n$ with coefficients from the k column of $\mathbf{B}$. Thus $\mathbf{Au}_k = b_{1k}\mathbf{u}_1 + \cdots + b_{nk}\mathbf{u}_n$ for $k = 1, \ldots, n$. This means the kth column of $\mathbf{M}_E^E(T_{\mathbf{A}})$ is the kth column $\begin{bmatrix} b_{1k} & \cdots & b_{nk}\end{bmatrix}^{\mathsf{T}}$ of $\mathbf{B}$, $k = 1, \ldots, n$, that is, $\mathbf{B} = \mathbf{M}_E^E(T_{\mathbf{A}})$. $\square$

The above result says that just as $\mathbf{A}$ is the matrix of the linear transformation $T_{\mathbf{A}}$ defined by $\mathbf{A}$ with respect to the standard ordered basis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ for $\mathbb{K}^{n\times 1}$, the matrix $\mathbf{B} := \mathbf{U}^{-1}\mathbf{AU}$ is the matrix of the same linear transformation $T_{\mathbf{A}}$ with respect to the ordered orthonormal basis for $\mathbb{K}^{n\times 1}$ consisting of the columns of $\mathbf{U}$.

In Lecture 10, we saw that a matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is diagonalizable (that is, $\mathbf{A}$ is similar to a diagonal matrix) if and only if there is a basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$. Similarly, we can prove the following result.

Proposition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is unitarily diagonalizable if and only if there is an orthonormal basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$.

In fact, $\mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \mathbf{D}$, where $\mathbf{U} := \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix}$ and $\mathbf{D} := \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \iff \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is an orthonormal basis for $\mathbb{K}^{n\times 1}$ and $\mathbf{A}\mathbf{u}_k = \lambda_k \mathbf{u}_k$ for $k = 1, \ldots, n$.

If $\mathbb{K} := \mathbb{C}$, then we can nicely characterise matrices that can be unitarily diagonalized. As a preparation, we prove a powerful result which says that every $\mathbf{A} \in \mathbb{C}^{n\times n}$ can be unitarily 'upper triangularized'.

Theorem (Schur)

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. Then $\mathbf{A}$ is unitarily similar to an upper triangular matrix.

Proof. We use induction on n. If $n = 1$, then there is nothing to prove. Let $n \geq 2$ and assume that the result holds for all $(n-1)\times(n-1)$ matrices with complex scalars.

We shall show that there is an $n\times n$ unitary matrix $\mathbf{U}$ and also an $n\times n$ upper triangular matrix $\mathbf{B}$ such that $\mathbf{A} = \mathbf{U}\mathbf{B}\mathbf{U}^*$.

By the Fundamental Theorem of Algebra, the characteristic polynomial of $\mathbf{A}$ has a root in $\mathbb{C}$. Hence there is $\lambda_1 \in \mathbb{C}$ and there is nonzero $\mathbf{x}_1 \in \mathbb{C}^{n\times 1}$ such that $\mathbf{A}\,\mathbf{x}_1 = \lambda_1\mathbf{x}_1$. We may assume WLOG that $\|\mathbf{x}_1\| = 1$.

We extend the orthonormal set $\{\mathbf{x}_1\}$ in $\mathbb{C}^{n\times 1}$ to an ordered orthonormal basis $E := (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)$ for $\mathbb{C}^{n\times 1}$.

Define $\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix}$. Then $\mathbf{X}$ is a unitary matrix, so that $\mathbf{X}^{-1} = \mathbf{X}^*$.

Consider the linear transformation $T_{\mathbf{A}} : \mathbb{C}^{n\times 1} \to \mathbb{C}^{n\times 1}$ defined by $T_{\mathbf{A}}(\mathbf{x}) := \mathbf{A}\,\mathbf{x}$. Let $\mathbf{C}$ denote the matrix $\mathbf{M}_E^E(T)$ of this linear map with respect to the basis E. Then $\mathbf{A}\,\mathbf{P} = \mathbf{P}\,\mathbf{C}$, that is, $\mathbf{A} = \mathbf{P}\,\mathbf{C}\,\mathbf{P}^*$. Also, since $\mathbf{A}\,\mathbf{x}_1 = \lambda_1 \mathbf{x}_1$, we obtain

$$\mathbf{C} = \left[\begin{array}{c|ccc} \lambda_1 & \alpha_2 & \cdots & \alpha_n \\ \hline 0 & & & \\ \vdots & & \mathbf{A}_1 & \\ 0 & & & \end{array}\right],$$

where $\alpha_2, \ldots, \alpha_n \in \mathbb{C}$ and $\mathbf{A}_1 \in \mathbb{C}^{(n-1)\times(n-1)}$. By the induction hypothesis, $\mathbf{A}_1 = \mathbf{P}_1 \mathbf{B}_1 \mathbf{P}_1^*$, where $\mathbf{P}_1$ is an $(n-1)\times(n-1)$ unitary matrix and $\mathbf{B}_1 = \mathbf{P}_1^* \mathbf{A}_1 \mathbf{P}_1$ is an $(n-1)\times(n-1)$ upper triangular matrix.
We now 'border' the unitary matrix $\mathbf{P}_1$ as follows.

Define
$$\mathbf{U}_1 := \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1 & \\ 0 & & & \end{array}\right].$$

Clearly, $\mathbf{U}_1$ is unitary. Now define $\mathbf{B} := \mathbf{U}_1^* \mathbf{C} \mathbf{U}_1$. Then

$$\mathbf{B} = \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1 & \\ 0 & & & \end{array}\right]^* \left[\begin{array}{c|ccc} \lambda_1 & \alpha_2 & \cdots & \alpha_n \\ \hline 0 & & & \\ \vdots & & \mathbf{A}_1 & \\ 0 & & & \end{array}\right] \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1 & \\ 0 & & & \end{array}\right]$$

that is,

$$\left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1^* & \\ 0 & & & \end{array}\right] \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{A}_1\mathbf{P}_1 & \\ 0 & & & \end{array}\right] = \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1^*\mathbf{A}_1\mathbf{P}_1 & \\ 0 & & & \end{array}\right],$$

where $\beta_2, \ldots, \beta_n \in \mathbb{C}$. Now the matrix

$$\mathbf{B} = \mathbf{U}_1^* \mathbf{C} \mathbf{U}_1 = \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1^* \mathbf{A}_1 \mathbf{P}_1 & \\ 0 & & & \end{array}\right] = \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{B}_1 & \\ 0 & & & \end{array}\right]$$

is upper triangular since $\mathbf{B}_1$ is upper triangular. Also, since $\mathbf{C} = \mathbf{U}_1 \mathbf{B}\, \mathbf{U}_1^*$, we see that

$$\mathbf{A} = \mathbf{P}\,\mathbf{C}\,\mathbf{P}^* = \mathbf{P}(\mathbf{U}_1 \mathbf{B}\, \mathbf{U}_1^*)\mathbf{P}^* = (\mathbf{P}\mathbf{U}_1)\mathbf{B}(\mathbf{P}\,\mathbf{U}_1)^*.$$

Because $\mathbf{P}$ and $\mathbf{U}_1$ are unitary, so is $\mathbf{U} := \mathbf{P}\mathbf{U}_1$, and we obtain $\mathbf{A} = \mathbf{U}\mathbf{B}\mathbf{U}^*$, as desired. □

Linear Algebra
Lecture 15

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Spring 2025

Theorem (Schur)

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. Then $\mathbf{A}$ is unitarily similar to an upper triangular matrix.

Proof. We use induction on n. If $n = 1$, then there is nothing to prove. Let $n \geq 2$ and assume that the result holds for all $(n-1)\times(n-1)$ matrices with complex scalars.

We shall show that there is an $n\times n$ unitary matrix $\mathbf{U}$ and also an $n\times n$ upper triangular matrix $\mathbf{B}$ such that $\mathbf{A} = \mathbf{UBU}^*$.

By the Fundamental Theorem of Algebra, the characteristic polynomial of $\mathbf{A}$ has a root in $\mathbb{C}$. Hence there is $\lambda_1 \in \mathbb{C}$ and there is nonzero $\mathbf{x}_1 \in \mathbb{C}^{n\times 1}$ such that $\mathbf{A}\,\mathbf{x}_1 = \lambda_1\mathbf{x}_1$. We may assume WLOG that $\|\mathbf{x}_1\| = 1$.

We extend the orthonormal set $\{\mathbf{x}_1\}$ in $\mathbb{C}^{n\times 1}$ to an ordered orthonormal basis $E := (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)$ for $\mathbb{C}^{n\times 1}$.

Define $\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix}$. Then $\mathbf{X}$ is a unitary matrix, so that $\mathbf{X}^{-1} = \mathbf{X}^*$.

Consider the linear transformation $T_{\mathbf{A}} : \mathbb{C}^{n\times 1} \to \mathbb{C}^{n\times 1}$ defined by $T_{\mathbf{A}}(\mathbf{x}) := \mathbf{A}\,\mathbf{x}$. Let $\mathbf{C}$ denote the matrix $\mathbf{M}_E^E(T)$ of this linear map with respect to the basis E. Then $\mathbf{A}\,\mathbf{P} = \mathbf{P}\,\mathbf{C}$, that is, $\mathbf{A} = \mathbf{P}\,\mathbf{C}\,\mathbf{P}^*$. Also, since $\mathbf{A}\,\mathbf{x}_1 = \lambda_1 \mathbf{x}_1$, we obtain

$$\mathbf{C} = \left[\begin{array}{c|ccc} \lambda_1 & \alpha_2 & \cdots & \alpha_n \\ \hline 0 & & & \\ \vdots & & \mathbf{A}_1 & \\ 0 & & & \end{array}\right],$$

where $\alpha_2, \ldots, \alpha_n \in \mathbb{C}$ and $\mathbf{A}_1 \in \mathbb{C}^{(n-1)\times(n-1)}$. By the induction hypothesis, $\mathbf{A}_1 = \mathbf{P}_1 \mathbf{B}_1 \mathbf{P}_1^*$, where $\mathbf{P}_1$ is an $(n-1)\times(n-1)$ unitary matrix and $\mathbf{B}_1 = \mathbf{P}_1^* \mathbf{A}_1 \mathbf{P}_1$ is an $(n-1)\times(n-1)$ upper triangular matrix.
We now 'border' the unitary matrix $\mathbf{P}_1$ as follows.

Define
$$\mathbf{U}_1 := \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1 & \\ 0 & & & \end{array}\right].$$

Clearly, $\mathbf{U}_1$ is unitary. Now define $\mathbf{B} := \mathbf{U}_1^*\mathbf{C}\mathbf{U}_1$. Then

$$\mathbf{B} = \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1 & \\ 0 & & & \end{array}\right]^* \left[\begin{array}{c|ccc} \lambda_1 & \alpha_2 & \cdots & \alpha_n \\ \hline 0 & & & \\ \vdots & & \mathbf{A}_1 & \\ 0 & & & \end{array}\right] \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1 & \\ 0 & & & \end{array}\right]$$

that is,

$$\left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1^* & \\ 0 & & & \end{array}\right] \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{A}_1\mathbf{P}_1 & \\ 0 & & & \end{array}\right] = \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1^*\mathbf{A}_1\mathbf{P}_1 & \\ 0 & & & \end{array}\right],$$

where $\beta_2, \ldots, \beta_n \in \mathbb{C}$. Now the matrix

$$\mathbf{B} = \mathbf{U}_1^* \mathbf{C} \mathbf{U}_1 = \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{P}_1^* \mathbf{A}_1 \mathbf{P}_1 & \\ 0 & & & \end{array}\right] = \left[\begin{array}{c|ccc} \lambda_1 & \beta_2 & \cdots & \beta_n \\ \hline 0 & & & \\ \vdots & & \mathbf{B}_1 & \\ 0 & & & \end{array}\right]$$

is upper triangular since $\mathbf{B}_1$ is upper triangular. Also, since $\mathbf{C} = \mathbf{U}_1 \mathbf{B}\, \mathbf{U}_1^*$, we see that

$$\mathbf{A} = \mathbf{P}\,\mathbf{C}\,\mathbf{P}^* = \mathbf{P}(\mathbf{U}_1 \mathbf{B}\, \mathbf{U}_1^*)\mathbf{P}^* = (\mathbf{P}\mathbf{U}_1)\mathbf{B}(\mathbf{P}\,\mathbf{U}_1)^*.$$

Because $\mathbf{P}$ and $\mathbf{U}_1$ are unitary, so is $\mathbf{U} := \mathbf{P}\mathbf{U}_1$, and we obtain $\mathbf{A} = \mathbf{U}\mathbf{B}\mathbf{U}^*$, as desired. $\square$

It may seem that we have cracked the eigenvalue problem for $n \times n$ matrices if we admit complex scalars: Find an upper triangular matrix $\mathbf{B} := \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$ which is (unitarily) similar to $\mathbf{A}$. Then the diagonal entries of $\mathbf{B}$ are the eigenvalues of $\mathbf{A}$. Also, we can find all eigenvectors of $\mathbf{B}$ corresponding to an eigenvalue λ by back substitution, and observe that $\mathbf{y}$ is an eigenvector of $\mathbf{B}$ if and only if $\mathbf{U}\mathbf{y}$ is an eigenvector of $\mathbf{A}$.

The above argument sounds convincing, but it is of no practical use since no algorithm is known for finding an upper triangular matrix which is similar to a given matrix. Note that our proof of the Schur theorem is based on induction.

However, there are well-known constructive methods (like the QR-algorithm) which 'upper triangularize' a given matrix approximately, that is, makes the subdiagonal entries arbitrarily small. These can be studied in a course on Numerical Analysis.

The Schur Theorem does not hold for real scalars. For example, the matrix $\mathbf{A} := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is not similar to any upper triangular matrix $\mathbf{B}$; otherwise the diagonal entries of $\mathbf{B}$ would be the eigenvalues of $\mathbf{A}$, but we have seen that $\mathbf{A}$ does not have any real eigenvalue.

The Schur theorem is of great theoretical importance.

Corollary

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$, and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $\mathbf{A}$ counted according to their algebraic multiplicities. Then (i) $\operatorname{trace} \mathbf{A} = \sum_{j=1}^{n} \lambda_j$, (ii) $\det \mathbf{A} = \prod_{j=1}^{n} \lambda_j$, and (iii) $p(\lambda_j) = 0$ for $j = 1, \ldots, n$, whenever $p(t)$ is a polynomial satisfying $p(\mathbf{A}) = \mathbf{O}$.

Proof. By Schur's theorem, there is an upper triangular matrix $\mathbf{B}$, and also a unitary matrix $\mathbf{U}$ such that $\mathbf{A} = \mathbf{U}\mathbf{B}\mathbf{U}^{-1}$.

Since the characteristic polynomial of $\mathbf{B}$ is the same as the characteristic polynomial of $\mathbf{A}$, the eigenvalues of $\mathbf{B}$ are the same as those of $\mathbf{A}$, counted according to their algebraic multiplicities. Hence

(i)
$\operatorname{trace} \mathbf{A} = \operatorname{trace} \mathbf{U}\mathbf{B}\mathbf{U}^{-1} = \operatorname{trace} \mathbf{U}^{-1}\mathbf{U}\mathbf{B} = \operatorname{trace} \mathbf{B} = \sum_{j=1}^{n} \lambda_j$.

(ii)
$\det \mathbf{A} = \det \mathbf{U}\mathbf{B}\mathbf{U}^{-1} = \det \mathbf{U} \det \mathbf{B} \det \mathbf{U}^{-1} = \det \mathbf{B} = \prod_{j=1}^{n} \lambda_j$.

(iii) Suppose $p(t)$ is a polynomial such that $p(\mathbf{A}) = \mathbf{O}$. Since $\mathbf{B}^k = (\mathbf{U}^{-1}\mathbf{A}\mathbf{U})^k = \mathbf{U}^{-1}\mathbf{A}^k\mathbf{U}$ for all $k \in \mathbb{N}$, we see that $p(\mathbf{B}) = \mathbf{U}^{-1}p(\mathbf{A})\mathbf{U} = \mathbf{O}$.

Let $j \in \{1, \ldots, n\}$. Then λ_j is the (j, j)th entry of $\mathbf{B}$, and since $\mathbf{B}$ is upper triangular, λ_j^k is the (j, j)th entry of $\mathbf{B}^k$ for all $k \in \mathbb{N}$. It follows that $p(\lambda_j)$ is the (j, j)th entry of $p(\mathbf{B})$. It must be equal to 0 because $p(\mathbf{B}) = \mathbf{O}$. $\square$

The 3 results given in the above corollary are useful in determining eigenvalues of $\mathbf{A}$ in some cases.

After a detour of inner products and orthonormal sets, we come back to the matrix eigenvalue problem. We shall show that if the scalars are complex numbers, then every square matrix be 'upper triangularized'. For this purpose, it is convenient to talk about eigenvalues and eigenvectors of a linear transformation.

Let V be a vector subspace of dimension n (possibly contained in a higher dimensional space of column vectors). Let $T : V \to V$ be a linear transformation. We say that $\lambda \in \mathbb{K}$ is an **eigenvalue** of T if there is nonzero $\mathbf{x} \in V$ such that $T(\mathbf{x}) = \lambda\,\mathbf{x}$, and then such a nonzero vector is called an **eigenvector** of T corresponding to λ.

Recall that if $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is an ordered basis for V, then the linear transformation T can be represented by the $n \times n$ matrix $\mathbf{A} := \mathbf{M}_E^E(T)$ whose kth column is $\begin{bmatrix} a_{1k} & \cdots & a_{nk} \end{bmatrix}^\mathsf{T}$, where $T(\mathbf{x}_k) = a_{1k}\mathbf{x}_1 + \cdots + a_{nk}\mathbf{x}_n$.

Lemma

Let V be an n dimensional vector subspace, and let E be an ordered basis for V. Suppose $T : V \to V$ is a linear map. Then $\lambda \in \mathbb{K}$ is an eigenvalue of T if and only if λ is an eigenvalue of $M_E^E(T)$.
In particular, if $\mathbb{K} := \mathbb{C}$, then T has an eigenvalue.

Proof. Let $E := (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ and $\mathbf{A} := M_E^E(T) = [a_{jk}]$.
Consider $\lambda \in \mathbb{K}$. Let $\mathbf{x} := \alpha_1\mathbf{x}_1 + \cdots + \alpha_n\mathbf{x}_n \in V$. Then

$$T(\mathbf{x}) = \lambda\,\mathbf{x} \iff \mathbf{A}\begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^{\mathsf{T}} = \lambda\begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^{\mathsf{T}}.$$

Also, $\mathbf{x} \neq \mathbf{0} \iff \begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^{\mathsf{T}} \neq \mathbf{0}$. Hence $\mathbf{x}$ is an eigenvector of $T \iff \begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^{\mathsf{T}}$ is an eigenvector of $\mathbf{A}$.
Let $\mathbb{K} = \mathbb{C}$. Since $\mathbf{A}$ has an eigenvalue, so does T $\square$

Proposition

Let $\mathbb{K} := \mathbb{C}$, and let V be a vector subspace of dimension n. Let $T : V \to V$ be a linear map. Then there is an orthonormal basis E for V such that $\mathbf{M}_E^E(T)$ is upper triangular.

Proof. Let $F = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ be any othonormal basis of V, and let $\mathbf{B} = \mathbf{M}_F^F(T)$. This means,
$T[\mathbf{x}_1, \ldots, \mathbf{x}_n] = [\mathbf{x}_1, \ldots, \mathbf{x}_n]\mathbf{B}$. By Schur's theorem, there is a unitary matrix $\mathbf{U}$ such that $\mathbf{A} = \mathbf{U}^{-1}\mathbf{B}\mathbf{U}$ is upper triangular.
Now $T[\mathbf{x}_1, \ldots, \mathbf{x}_1]\mathbf{U} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]\mathbf{B}\mathbf{U} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]\mathbf{U}\mathbf{A}$.
Write $[\mathbf{x}_1, \ldots, \mathbf{x}_n]\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_n]$. As $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is an orthonormal basis, $[\mathbf{x}_1, \ldots, \mathbf{x}_n]$ is unitary. As $\mathbf{U}$ is also unitary, so is $[\mathbf{u}_1, \ldots, \mathbf{u}_n]$, which means $E := (\mathbf{u}_1, \ldots, \mathbf{u}_n)$ is an orthonormal basis. Again, the above implies,
$T[\mathbf{u}_1, \ldots, \mathbf{u}_n] = [\mathbf{u}_1, \ldots, \mathbf{u}_n]\mathbf{A}$, which means $\mathbf{M}_E^E(T) = \mathbf{A}$, which is upper triangular. $\square$

Examples

(1) Let $\mathbf{A} \in \mathbb{C}^{3\times 3}$ be such that $\mathbf{A}^2 = 6\mathbf{A}$ and $\operatorname{trace} \mathbf{A} = 12$. Let us determine the eigenvalues of $\mathbf{A}$.

Consider the polynomial $p(t) = t^2 - 6t$. Since $p(\mathbf{A}) = \mathbf{O}$, we see that $p(\lambda) = \lambda^2 - 6\lambda = \lambda(\lambda - 6) = 0$ for every eigenvalue λ of $\mathbf{A}$. Since $\operatorname{trace} \mathbf{A} = 12$, the sum of the eigenvalues of $\mathbf{A}$ (counting algebraic multiplicities) is equal to 12. Hence the eigenvalues of $\mathbf{A}$ are $6, 6, 0$, that is, 6 is an eigenvalue of $\mathbf{A}$ of algebraic multiplicity 2, and 0 is an eigenvalue of $\mathbf{A}$ of algebraic multiplicity 1.

(ii) Let $\mathbf{A} := \begin{bmatrix} \mathbf{e}_2 & \mathbf{e}_3 & \cdots & \mathbf{e}_n & \mathbf{e}_1 \end{bmatrix}$, where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the basic column vectors in $\mathbb{C}^{n\times 1}$.
Then $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_2, \ldots, \mathbf{A}\mathbf{e}_{n-1} = \mathbf{e}_n$ and $\mathbf{A}\mathbf{e}_n = \mathbf{e}_1$. Hence
$\mathbf{A} \begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^{\mathsf{T}} = \begin{bmatrix} x_n & x_1 & x_2 & \cdots & x_{n-1} \end{bmatrix}^{\mathsf{T}}$
for $\begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^{\mathsf{T}} \in \mathbb{C}^{n\times 1}$.

The matrix $\mathbf{A}$ represents a **cyclic shift to the right**. Observe that $\mathbf{A}^n = \mathbf{I}$ and also that $\mathbf{A}$ is a unitary matrix given by

$$\mathbf{A} := \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & \cdots & 0 \\ \vdots & \vdots & \cdots & \cdots & \cdots & \cdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Let λ be an eigenvalue of $\mathbf{A}$. Since $\mathbf{A}^n = \mathbf{I}$, we see that $\lambda^n = 1$. Let $\omega := e^{2\pi i/n}$. Then the nth roots of 1 are $1, \omega, \omega^2, \ldots, \omega^{n-1}$. Thus $\lambda \in \{1, \omega, \omega^2, \ldots, \omega^{n-1}\}$. Conversely, we show that each $1, \omega, \omega^2, \ldots, \omega^{n-1}$ is an eigenvalue of $\mathbf{A}$ by finding a corresponding eigenvector.

Let $\mathbf{x} := \begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^\mathsf{T} \in \mathbb{C}^{n\times 1}$. Then $\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}$ means $x_1 = \lambda x_2, x_2 = \lambda x_3, \ldots, x_{n-1} = \lambda x_n$ and $x_n = \lambda x_1$.

For $j = 0, \ldots, n-1$, define

$$\mathbf{x}_j := \begin{bmatrix} 1 & 1/\omega^j & 1/\omega^{2j} & \cdots & 1/\omega^{(n-1)j} \end{bmatrix}^\mathsf{T}.$$

Now $1/\omega^{(n-1)j} = \omega^j$ since $\omega^n = 1$, and so

$$\mathbf{x}_j := \begin{bmatrix} 1 & 1/\omega^j & 1/\omega^{2j} & \cdots & 1/\omega^{(n-2)j} & \omega^j \end{bmatrix}^\mathsf{T}.$$

Hence $\mathbf{A}\mathbf{x}_j = \begin{bmatrix} \omega^j & 1 & 1/\omega^j & 1/\omega^{2j} & \cdots & 1/\omega^{(n-2)j} \end{bmatrix}^\mathsf{T} = \omega^j \mathbf{x}_j$. Thus $\mathbf{x}_j$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue ω^j for $j = 0, 1, \ldots, n-1$.

Remark: $\mathbf{A}$ is an example of a circulant matrix. In general a circulant matrix is a square matrix each of whose rows are obtained by a cyclic shift to the right of the preceding rows.

Linear Algebra
Lecture 16

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Spring 2025

Examples

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Consider the polynomial $p(t) = t^2 - 6t$. Since $p(\mathbf{A}) = \mathbf{O}$, we see that $p(\lambda) = \lambda^2 - 6\lambda = \lambda(\lambda - 6) = 0$ for every eigenvalue λ of $\mathbf{A}$. Since $\operatorname{trace} \mathbf{A} = 12$, the sum of the eigenvalues of $\mathbf{A}$ (counting algebraic multiplicities) is equal to 12. Hence the eigenvalues of $\mathbf{A}$ are $6, 6, 0$, that is, 6 is an eigenvalue of $\mathbf{A}$ of algebraic multiplicity 2, and 0 is an eigenvalue of $\mathbf{A}$ of algebraic multiplicity 1.

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Then $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_2, \ldots, \mathbf{A}\mathbf{e}_{n-1} = \mathbf{e}_n$ and $\mathbf{A}\mathbf{e}_n = \mathbf{e}_1$. Hence
$\mathbf{A} \begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^\mathsf{T} = \begin{bmatrix} x_n & x_1 & x_2 & \cdots & x_{n-1} \end{bmatrix}^\mathsf{T}$
for $\begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^\mathsf{T} \in \mathbb{C}^{n\times 1}$.

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Let λ be an eigenvalue of $\mathbf{A}$. Since $\mathbf{A}^n = \mathbf{I}$, we see that $\lambda^n = 1$. Let $\omega := e^{2\pi i/n}$. Then the nth roots of 1 are $1, \omega, \omega^2, \ldots, \omega^{n-1}$. Thus $\lambda \in \{1, \omega, \omega^2, \ldots, \omega^{n-1}\}$.

Conversely, we show that each $1, \omega, \omega^2, \ldots, \omega^{n-1}$ is an eigenvalue of $\mathbf{A}$ by finding a corresponding eigenvector.

Let $\mathbf{x} := \begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^\mathsf{T} \in \mathbb{C}^{n\times 1}$. Then $\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}$ means $x_1 = \lambda x_2, x_2 = \lambda x_3, \ldots, x_{n-1} = \lambda x_n$ and $x_n = \lambda x_1$.

For $j = 0, \ldots, n-1$, define

$$\mathbf{x}_j := \begin{bmatrix} 1 & 1/\omega^j & 1/\omega^{2j} & \cdots & 1/\omega^{(n-1)j} \end{bmatrix}^\mathsf{T}.$$

Now $1/\omega^{(n-1)j} = \omega^j$ since $\omega^n = 1$, and so

$$\mathbf{x}_j := \begin{bmatrix} 1 & 1/\omega^j & 1/\omega^{2j} & \cdots & 1/\omega^{(n-2)j} & \omega^j \end{bmatrix}^\mathsf{T}.$$

Hence $\mathbf{A}\mathbf{x}_j = \begin{bmatrix} \omega^j & 1 & 1/\omega^j & 1/\omega^{2j} & \cdots & 1/\omega^{(n-2)j} \end{bmatrix}^\mathsf{T} = \omega^j \mathbf{x}_j$. Thus $\mathbf{x}_j$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue ω^j for $j = 0, 1, \ldots, n-1$.

Remark: $\mathbf{A}$ is an example of a circulant matrix. In general a circulant matrix is a square matrix each of whose rows are obtained by a cyclic shift to the right of the preceding rows.

Unitarily Diagonalizable Matrix

In the last lecture we saw that if the scalars are complex numbers, then every matrix can be unitarily 'upper triangularized'. Now we take up the question: 'Which matrices can be unitarily diagonalized?'. We saw that a matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is unitarily diagonalizable if and only if there is an orthonormal basis for $\mathbb{K}^{n\times 1}$ consisting of eigenvectors of $\mathbf{A}$. Let us investigate this condition further.

Suppose $\mathbf{A} \in \mathbb{K}^{n\times n}$ is unitarily diagonalizable, and let eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbf{A}$ form an orthonormal basis for $\mathbb{K}^{n\times 1}$. Let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues of $\mathbf{A}$, so that $\mathbf{A}\mathbf{u}_j = \lambda_j \mathbf{u}_j$ for $j = 1, \ldots, n$. Let $\mathbf{x} \in \mathbb{K}^{n\times 1}$. Then

$$\mathbf{x} = \sum_{j=1}^{n} \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{u}_j \quad \text{and } \mathbf{A}\mathbf{x} = \sum_{j=1}^{n} \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{A}\mathbf{u}_j = \sum_{j=1}^{n} \lambda_j \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{u}_j .$$

Thus for any $\mathbf{x} \in \mathbb{K}^{n\times 1}$, the column vector $\mathbf{A}\,\mathbf{x}$ is completely determined by the inner products $\langle \mathbf{u}_1, \mathbf{x}\rangle, \ldots, \langle \mathbf{u}_n, \mathbf{x}\rangle$, and by the eigenvalues and eigenvectors of $\mathbf{A}$. The above representation of $\mathbf{A}\,\mathbf{x}$ can be used for various purposes. For this reason, we would like to find necessary and/or sufficient conditions under which a square matrix can be unitarily diagonalized. We introduce a new class of matrices.

Definition

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is called **normal** *if it commutes with its adjoint, that is, $\mathbf{A}^*\mathbf{A} = \mathbf{A}\,\mathbf{A}^*$.*

Examples (i) If $\mathbf{A}$ is self-adjoint (i.e., $\mathbf{A}^* = \mathbf{A}$), or skew self-adjoint (i.e., $\mathbf{A}^* = -\mathbf{A}$), or unitary, then $\mathbf{A}$ is normal.

(ii) The matrix $\mathbf{A} := \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \in \mathbb{K}^{2\times 2}$ is normal, but it is neither self-adjoint, nor skew self-adjoint, nor unitary. But not every matrix is normal, as the example $\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ shows.

Proposition

Let $\mathbf{A} \in \mathbb{K}^{n\times n}$. Then

$\mathbf{A}$ *is normal* $\iff \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y}\rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y}\rangle$ *for all* $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$.

Proof. For $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$, $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y}\rangle = (\mathbf{A}\mathbf{x})^*\mathbf{A}\mathbf{y} = \mathbf{x}^*\mathbf{A}^*\mathbf{A}\mathbf{y}$ and $\langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y}\rangle = (\mathbf{A}^*\mathbf{x})^*\mathbf{A}^*\mathbf{y} = \mathbf{x}^*\mathbf{A}\,\mathbf{A}^*\mathbf{y}$. This shows that if $\mathbf{A}$ is normal, then $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y}\rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. Conversely, suppose $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y}\rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. Then letting $\mathbf{x} := \mathbf{e}_j$ and $\mathbf{y} := \mathbf{e}_k$, we see that $\mathbf{e}_j^*\mathbf{A}^*\mathbf{A}\,\mathbf{e}_k = \mathbf{e}_j^*\mathbf{A}\,\mathbf{A}^*\mathbf{e}_k$, that is, the (j, k)th entries of $\mathbf{A}^*\mathbf{A}$ and $\mathbf{A}\,\mathbf{A}^*$ are the same for all $j, k = 1, \ldots, n$. Hence $\mathbf{A}^*\mathbf{A} = \mathbf{A}\,\mathbf{A}^*$, that is, $\mathbf{A}$ is normal. $\square$

The above condition for normality of a matrix $\mathbf{A}$ says that the lengths of the vectors $\mathbf{A}\mathbf{x}$ and $\mathbf{A}^*\mathbf{x}$ should be the same, and the angle between $\mathbf{A}\mathbf{x}$ and $\mathbf{A}\mathbf{y}$ should be the same as the angle between $\mathbf{A}^*\mathbf{x}$ and $\mathbf{A}^*\mathbf{y}$ for all $\mathbf{x}$, $\mathbf{y} \in \mathbb{K}^{n\times 1}$.

Corollary

Let $\mathbf{A} \in \mathbb{K}^{n \times n}$ be normal. Then $\|\mathbf{A}\mathbf{x}\| = \|\mathbf{A}^*\mathbf{x}\|$ for all $\mathbf{x}$ in $\mathbb{K}^{n \times 1}$. Let $\mathbf{x}$ be an eigenvector of $\mathbf{A}$ corresponding to an eigenvalue λ. Then $\mathbf{x}$ itself is an eigenvector of $\mathbf{A}^*$ corresponding to the eigenvalue $\overline{\lambda}$ of $\mathbf{A}^*$. Further, if $\mathbf{y}$ is an eigenvector of $\mathbf{A}$ corresponding to an eigenvalue $\mu \neq \lambda$, then $\mathbf{y}$ is orthogonal to $\mathbf{x}$.

Proof. Let $\mathbf{x} \in \mathbb{K}^{n \times 1}$. Since $\mathbf{A}$ is normal,

$$\|\mathbf{A}\mathbf{x}\|^2 = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x} \rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{x} \rangle = \|\mathbf{A}^*\mathbf{x}\|^2 \quad \text{for all } \mathbf{x} \in \mathbb{K}^{n \times 1}.$$

Next, let $\mathbf{x}$ be an eigenvector of $\mathbf{A}$ corresponding to an eigenvalue λ. Then $\|\mathbf{A}^*\mathbf{x} - \overline{\lambda}\mathbf{x}\| = \|\mathbf{A}\mathbf{x} - \lambda\mathbf{x}\| = 0$. Hence $\mathbf{x}$ itself is an eigenvector $\mathbf{A}^*$ corresponding to the eigenvalue $\overline{\lambda}$. Finally, let $\mathbf{y}$ be an eigenvector of $\mathbf{A}$ corresponding to an eigenvalue $\mu \neq \lambda$. Then

$$\mu \langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mu\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{y} \rangle = \langle \overline{\lambda}\mathbf{x}, \mathbf{y} \rangle = \lambda \langle \mathbf{x}, \mathbf{y} \rangle.$$

Since $\mu \neq \lambda$, we see that $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, that is, $\mathbf{x} \perp \mathbf{y}$. □

We now characterize a diagonal matrix in terms of its upper triangularity and normality.

Lemma

A matrix $\mathbf{A} \in \mathbb{K}^{n \times n}$ is diagonal if and only if it is upper triangular and normal.

Proof. Let $\mathbf{A} := \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then $\mathbf{A}$ is upper triangular. Also, it is normal since $\mathbf{A}^*\mathbf{A} = \mathrm{diag}(|\lambda_1|^2, \ldots, |\lambda_n|^2) = \mathbf{A}\,\mathbf{A}^*$.

Conversely, suppose $\mathbf{A} := [a_{jk}]$ is upper triangular and normal. Let $\mathbf{B} := \mathbf{A}^*\mathbf{A} = [b_{jk}]$ and $\mathbf{C} := \mathbf{A}\mathbf{A}^* = [c_{jk}]$. Since $\mathbf{A}$ is upper triangular, $a_{jk} = 0$ if $j > k$, and so for $k = 1, \ldots, n$,

$$b_{kk} = \sum_{\ell=1}^{n} \overline{a}_{\ell k} a_{\ell k} = \sum_{\ell=1}^{k} |a_{\ell k}|^2 \quad \text{and} \quad c_{kk} = \sum_{\ell=1}^{n} a_{k\ell} \overline{a}_{k\ell} = \sum_{\ell=1}^{n} |a_{k\ell}|^2.$$

Since $\mathbf{A}$ is normal, we see that $b_{kk} = c_{kk}$ for $k = 1, \ldots, n$.

Let $k = 1$. Then
$|a_{11}|^2 = b_{11} = c_{11} = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2$.
Hence $\boxed{a_{12} = \cdots = a_{1n} = 0}$.

Next, let $k = 2$. Then
$|a_{22}|^2 = |a_{12}|^2 + |a_{22}|^2 = b_{22} = c_{22} = |a_{22}|^2 + |a_{23}|^2 + \cdots + |a_{2n}|^2$.
Hence $\boxed{a_{23} = \cdots = a_{2n} = 0}$.

Proceeding in this manner, for $k = n - 1$, we obtain
$|a_{(n-1)(n-1)}|^2 = |a_{1(n-1)}|^2 + \cdots + |a_{(n-1)(n-1)}|^2 = b_{(n-1)(n-1)} = c_{(n-1)(n-1)} = |a_{(n-1)(n-1)}|^2 + |a_{(n-1)n}|^2$.
Hence $\boxed{a_{(n-1)n} = 0}$.

Thus $a_{jk} = 0$ if $j < k$, that is, $\mathbf{A}$ is lower triangular. Since $\mathbf{A}$ is given to be upper triangular, it is is in fact diagonal. $\square$

We are now ready to state and prove necessary conditions as well as sufficient conditions for diagonalizing a matrix unitarily.

Proposition (Spectral Theorem for Normal Matrices)

(i) If $\mathbf{A} \in \mathbb{K}^{n\times n}$ is unitarily diagonalizable, then $\mathbf{A}$ is normal.

(ii) If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is normal, then $\mathbf{A}$ is unitarily diagonalizable.

Proof. (i) Suppose $\mathbf{A} \in \mathbb{K}^{n\times n}$ is unitarily diagonalizable. Then $\mathbf{D} = \mathbf{U}^*\mathbf{A}\mathbf{U}$ for some unitary matrix $\mathbf{U}$ and diagonal matrix $\mathbf{D}$. Hence $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$, and so $\mathbf{A}^* = \mathbf{U}\mathbf{D}^*\mathbf{U}^*$. Now

$$\mathbf{A}^*\mathbf{A} = (\mathbf{U}\mathbf{D}^*\mathbf{U}^*)(\mathbf{U}\mathbf{D}\mathbf{U}^*) = \mathbf{U}(\mathbf{D}^*\mathbf{D})\mathbf{U}^*$$

whereas

$$\mathbf{A}\,\mathbf{A}^* = (\mathbf{U}\mathbf{D}\mathbf{U}^*)(\mathbf{U}\mathbf{D}^*\mathbf{U}^*) = \mathbf{U}(\mathbf{D}\mathbf{D}^*)\mathbf{U}^*.$$

Since $\mathbf{D}$ is diagonal, $\mathbf{D}^*\mathbf{D} = \mathbf{D}\mathbf{D}^*$. Consequently, $\mathbf{A}$ is normal.

(ii) Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. By Schur's theorem, there is a unitary matrix $\mathbf{U}$ and an upper triangular matrix $\mathbf{B}$ with $\mathbf{B} = \mathbf{U}^*\mathbf{A}\mathbf{U}$.

Suppose $\mathbf{A}$ is normal, that is, $\mathbf{A}^*\mathbf{A} = \mathbf{A}\,\mathbf{A}^*$. Then using $\mathbf{B} = \mathbf{U}^*\mathbf{A}\mathbf{U}$, we easily see that $\mathbf{B}^*\mathbf{B} = \mathbf{B}\,\mathbf{B}^*$, that is, $\mathbf{B}$ is normal. Now, since $\mathbf{B}$ is upper triangular and normal, $\mathbf{B}$ is diagonal by the previous lemma. This proves that $\mathbf{A}$ is unitarily diagonalizable. □

Remarks

(i) The unitary matrix $\mathbf{U}$ and the diagonal matrix $\mathbf{B}$ such that $\mathbf{A} = \mathbf{U}\mathbf{B}\mathbf{U}^*$ are not unique. We shall give some examples later.

(ii) Part (ii) of the proposition is not true for real scalars, that is, even if $\mathbf{A} \in \mathbb{R}^{n\times n}$ is normal, there may be no unitary $\mathbf{U} \in \mathbb{R}^{n\times n}$ and diagonal $\mathbf{D} \in \mathbb{R}^{n\times n}$ such that $\mathbf{D} = \mathbf{U}^*\mathbf{A}\mathbf{U}$. For example, $\mathbf{A} := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 2}$ is normal, but it is not diagonalizable using real scalars since it has no real eigenvalue.

Self-adjoint matrices

Recall that $\mathbf{A} \in \mathbb{K}^{n\times n}$ is called **self-adjoint** (or **Hermitian**) if $\mathbf{A}^* = \mathbf{A}$. We shall now prove a "spectral theorem" for self-adjoint matrices. For this purpose, the following lemma is useful.

Lemma

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is diagonal with all diagonal entries real if and only if it is upper triangular and self-adjoint.

Proof. Let $\mathbf{A} := \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$. Then clearly $\mathbf{A}$ is upper triangular. Also, it is self-adjoint since $\mathbf{A}^* = \operatorname{diag}(\overline{\lambda}_1, \ldots, \overline{\lambda}_n) = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \mathbf{A}$.

Conversely, suppose $\mathbf{A}$ is upper triangular and self-adjoint. Then $\mathbf{A}^*$ is lower triangular. Since $\mathbf{A}^* = \mathbf{A}$, we see that $\mathbf{A}$ is in fact diagonal with all diagonal entries real. $\square$

Just as we have proved that a matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$ is normal if and only if it is unitarily diagonalizable, we prove a similar result for self-adjoint matrices.

Proposition (Spectral Theorem for Self-Adjoint Matrices)

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$. Then $\mathbf{A}$ is self-adjoint if and only if $\mathbf{A}$ is unitarily diagonalizable and all eigenvalues of $\mathbf{A}$ are real.

Proof. Suppose $\mathbf{A}$ is unitarily diagonalizable and all eigenvalues of $\mathbf{A}$ are real. Let $\mathbf{U}$ be a unitary matrix and let $\mathbf{D}$ be a diagonal matrix such that $\mathbf{D} = \mathbf{U}^*\mathbf{A}\mathbf{U}$. Then the diagonal entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$, and so they are real. Hence $\mathbf{D}^* = \mathbf{D}$. Consequently,

$$\mathbf{A}^* = (\mathbf{U}\mathbf{D}\mathbf{U}^*)^* = \mathbf{U}\mathbf{D}^*\mathbf{U}^* = \mathbf{U}\mathbf{D}\mathbf{U}^* = \mathbf{A}.$$

Thus $\mathbf{A}$ is self-adjoint.

Conversely, suppose $\mathbf{A}$ is self-adjoint. By Schur's theorem, there is a unitary matrix $\mathbf{U}$, and also an upper triangular matrix $\mathbf{B}$ such that $\mathbf{B} = \mathbf{U}^*\mathbf{A}\mathbf{U}$. Then

$$\mathbf{B}^* = (\mathbf{U}^*\mathbf{A}\mathbf{U})^* = \mathbf{U}^*\mathbf{A}^*\mathbf{U} = \mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{B},$$

so that $\mathbf{B}$ is self-adjoint. Since $\mathbf{B}$ is upper triangular and self-adjoint, the previous lemma shows that $\mathbf{B}$ is in fact diagonal with all diagonal entries real. Thus $\mathbf{A}$ is unitarily diagonalizable. Also, all eigenvalues of $\mathbf{A}$ are real, since they are the diagonal entries of the matrix $\mathbf{B}$. $\square$

Our short proof of the spectral theorem for self-adjoint matrices is based on Schur's theorem. This result can also be deduced from part (ii) of the spectral theorem for normal matrices since every self-adjoint matrix in normal, provided we independently show that every eigenvalue of a self-adjoint matrix is real. The latter statement can be easily proved.

Proposition

If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is self-adjoint, then every eigenvalue of $\mathbf{A}$ is real.

Proof.
Let λ be an eigenvalue of a self-adjoint matrix $\mathbf{A}$, and let $\mathbf{x}$ be a corresponding unit eigenvector. Then

$$\lambda = \lambda\,\mathbf{x}^*\mathbf{x} = \mathbf{x}^*\mathbf{A}\,\mathbf{x} = \mathbf{x}^*\mathbf{A}^*\mathbf{x} = (\mathbf{A}\,\mathbf{x})^*\mathbf{x} = (\lambda\,\mathbf{x})^*\mathbf{x} = \overline{\lambda}\,\mathbf{x}^*\mathbf{x} = \overline{\lambda}.$$

Hence λ is real. $\square$

Finally, let us consider a real symmetric matrix $\mathbf{A}$, that is, $\mathbf{A} \in \mathbb{R}^{n\times n}$ such that $\mathbf{A}^\mathsf{T} = \mathbf{A}$. We shall prove a spectral theorem for $\mathbf{A}$ which involves only real scalars.

A unitary matrix with real entries is also called an **orthogonal matrix**. Thus $\mathbf{C} \in \mathbb{R}^{n\times n}$ is orthogonal if its columns form an orthonormal subset of $\mathbb{R}^{n\times 1}$. Clearly, an orthogonal matrix is invertible and its inverse is the same as its transpose.

A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called **orthogonally diagonalizable** if there is an orthogonal matrix $\mathbf{C} \in \mathbb{R}^{n \times n}$ and a diagonal matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ such that $\mathbf{D} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$. We prove Jacobi's theorem.

Proposition (Spectral Theorem for Real Symmetric Matrices)

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then $\mathbf{A}$ is symmetric if and only if $\mathbf{A}$ is orthogonally diagonalizable. In this case, $\mathbf{A}$ has n real eigenvalues counted according to their algebraic multiplicities.

Proof.

Suppose $\mathbf{A}$ is orthogonally diagonalizable. Let $\mathbf{C} \in \mathbb{R}^{n \times n}$ be an orthogonal matrix, and let $\mathbf{D} \in \mathbb{R}^{n \times n}$ be a diagonal matrix such that $\mathbf{D} = \mathbf{C}^{\mathsf{T}}\mathbf{A}\mathbf{C}$. Since $\mathbf{D}^{\mathsf{T}} = \mathbf{D}$,

$$\mathbf{A}^{\mathsf{T}} = (\mathbf{C}\mathbf{D}\mathbf{C}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{C}\mathbf{D}^{\mathsf{T}}\mathbf{C}^{\mathsf{T}} = \mathbf{C}\mathbf{D}\mathbf{C}^{\mathsf{T}} = \mathbf{A}.$$

Thus $\mathbf{A}$ is symmetric.

Conversely, suppose $\mathbf{A}$ is symmetric. Since $\mathbb{R}$ can be considered as a subset of $\mathbb{C}$, we treat $\mathbf{A}$ as a matrix with complex entries. Then $\mathbf{A}^* = \mathbf{A}$, that is, $\mathbf{A}$ is self-adjoint, and so all eigenvalues of $\mathbf{A}$ are real.

By the spectral theorem for self-adjoint matrices, there is a unitary matrix $\mathbf{U} \in \mathbb{C}^{n \times n}$ and a diagonal matrix $\mathbf{D} \in \mathbb{C}^{n \times n}$ such that $\mathbf{D} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$, that is, $\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{D}$. Since the diagonal entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$, we see that $\mathbf{D} \in \mathbb{R}^{n \times n}$.

The n columns of the unitary matrix $\mathbf{U}$ form an orthonormal set in $\mathbb{C}^{n \times n}$, and each column is an eigenvector of $\mathbf{A}$ corresponding to an eigenvalue of $\mathbf{A}$.

Let λ be an eigenvalue of $\mathbf{A}$. Then $\lambda \in \mathbb{R}$. Using the Gauss Elimination Method, we may find the basic solutions of the homogeneous linear system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$. Their entries are real since all entries of $\mathbf{A}$ are real and $\lambda \in \mathbb{R}$. These solutions form a basis for the eigenspace of $\mathbf{A}$ corresponding to λ.

Further, we can use the Gram-Schmidt Orthogonalization Process for these basic solutions to obtain an orthonormal basis for the eigenspace of $\mathbf{A}$ corresonding to λ. In this process, the entries of the basis vectors remain real.

We replace the n columns of the unitary matrix $\mathbf{U}$ by n eigenvectors of $\mathbf{A}$ which form an orthonormal set in $\mathbb{R}^{n\times n}$. Let $\mathbf{C} \in \mathbb{R}^{n\times n}$ denote the matrix with these columns arranged in the same order as the columns of $\mathbf{U}$. Then $\mathbf{D} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$. □

The spectral theorem says that given a normal matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ or a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, there is a unitary matrix $\mathbf{U}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$, that is, $\mathbf{AU} = \mathbf{UD}$. Let us write the matrix $\mathbf{U}$ in terms of its n columns $\mathbf{U} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix}$ and let $\mathbf{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

Then the equation

$$\mathbf{A} \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

means $\mathbf{A}\mathbf{u}_1 = \lambda_1 \mathbf{u}_1, \ldots, \mathbf{A}\mathbf{u}_n = \lambda_n \mathbf{u}_n$, so that $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are eigenvectors of $\mathbf{A}$ corresponding to the eigenvalues $\lambda_1, \ldots, \lambda_n$ respectively. We may list the eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in some other order to form another unitary matrix and correspondingly change the ordering of the eigenvalues $\lambda_1, \ldots, \lambda_n$ to form another diagonal matrix which will serve the same purpose.

Since the eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathbf{A}$ may not be distinct, we may pool together all eigenvectors among $\mathbf{u}_1, \ldots, \mathbf{u}_n$ corresponding to the same eigenvalue.

Also, since $\mathbf{A}$ is diagonalizable, the geometric multiplicity of any eigenvalue of $\mathbf{A}$ is equal to its algebraic multiplicity.

Thus if we know the eigenvalues of $\mathbf{A}$, then we may use the following procedure to form a unitary matrix $\mathbf{U}$ whose n columns are eigenvectors of $\mathbf{A}$.

1. Let $\mu_1, \ldots, \mu_k$ be the distinct eigenvalues of $\mathbf{A}$. These are the distinct roots of the characteristic polynomial of $\mathbf{A}$. Let the algebraic multiplicity of μ_j be m_j, so that $m_1 + \cdots + m_k = n$. Also,the geometric multiplicity g_j of μ_j is equal to m_j, and so $g_1 + \cdots + g_k = n$.

If in fact, $\mathbf{A}$ is self-adjoint, then each μ_j is real.

2. For each $j = 1, \ldots, k$, find a basis for the null space $\mathcal{N}(\mathbf{A} - \mu_j \mathbf{I})$ consisting of g_j elements by solving the homogeneous linear system $(\mathbf{A} - \mu_j \mathbf{I})\mathbf{x} = \mathbf{0}$ using the Gauss Elimination Method.

3. For each $j = 1, \ldots, k$, obtain an ordered orthonormal basis $(\mathbf{u}_{j,1}, \ldots, \mathbf{u}_{j,g_j})$ for $\mathcal{N}(\mathbf{A} - \mu_j \mathbf{I})$ using the Gram-Schmidt Orthonormalization Process.

$\boxed{4.}$ Form an $n \times n$ matrix $\mathbf{U}$ as follows:

$$\mathbf{U} := \begin{bmatrix} \mathbf{u}_{11} & \ldots & \mathbf{u}_{1g_1} & \mathbf{u}_{21} & \ldots & \mathbf{u}_{2g_2} & \ldots & \ldots & \mathbf{u}_{k1} & \ldots \mathbf{u}_{kg_k} \end{bmatrix}$$

Now since $\mathbf{A}$ is a normal matrix, $\mathbf{u}_{ik} \perp \mathbf{u}_{j\ell}$ if $i \neq j$. Thus the n columns of $\mathbf{U}$ form an orthonormal set. Hence $\mathbf{U}$ is unitary.

Form an $n \times n$ diagonal matrix $\mathbf{D}$ as follows:

$$\mathbf{D} := \operatorname{diag}(\lambda_{11}, \ldots, \lambda_{1g_1}, \lambda_{21}, \ldots, \lambda_{2g_2}, \ldots, \ldots, \lambda_{k1}, \ldots, \lambda_{kg_k}),$$

where $\lambda_{11} = \cdots = \lambda_{1g_1} = \mu_1, \ldots, \lambda_{k1} = \cdots = \lambda_{kg_k} = \mu_k$.

Then $\boxed{\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{D}}$.

A similar procedure works for a real symmetric matrix $\mathbf{A}$, and so we can find an orthogonal matrix $\mathbf{C} \in \mathbb{R}^{n \times n}$ and a diagonal matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ such that $\boxed{\mathbf{C}^\mathsf{T}\mathbf{A}\mathbf{C} = \mathbf{D}}$.

Linear Algebra
Lecture 17

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Spring 2025

Recall: In the last lecture, we proved

Proposition (Spectral Theorem for Normal Matrices)

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. Then $\mathbf{A}$ is normal if and only if $\mathbf{A}$ is unitarily diagonalizable.

We then turned to self-adjoint matrices and proved:

Lemma

A matrix $\mathbf{A} \in \mathbb{K}^{n\times n}$ is diagonal with all diagonal entries real if and only if it is upper triangular and self-adjoint.

Proof. Let $\mathbf{A} := \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$. Then clearly $\mathbf{A}$ is upper triangular. Also, it is self-adjoint since $\mathbf{A}^* = \operatorname{diag}(\overline{\lambda}_1, \ldots, \overline{\lambda}_n) = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \mathbf{A}$.

Conversely, suppose $\mathbf{A}$ is upper triangular and self-adjoint. Then $\mathbf{A}^*$ is lower triangular. Since $\mathbf{A}^* = \mathbf{A}$, we see that $\mathbf{A}$ is in fact diagonal with all diagonal entries real. $\square$

Proposition (Spectral Theorem for Self-Adjoint Matrices)

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$. Then $\mathbf{A}$ is self-adjoint if and only if $\mathbf{A}$ is unitarily diagonalizable and all eigenvalues of $\mathbf{A}$ are real.

Proof. Suppose $\mathbf{A}$ is unitarily diagonalizable and all eigenvalues of $\mathbf{A}$ are real. Then $\mathbf{D} = \mathbf{U}^*\mathbf{A}\mathbf{U}$ for some unitary matrix $\mathbf{U}$ and diagonal matrix $\mathbf{D}$. The diagonal entries entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$, and so they are real. Hence $\mathbf{D}^* = \mathbf{D}$. Consequently, $\mathbf{A}^* = (\mathbf{U}\mathbf{D}\mathbf{U}^*)^* = \mathbf{U}\mathbf{D}^*\mathbf{U}^* = \mathbf{U}\mathbf{D}\mathbf{U}^* = \mathbf{A}$. Thus $\mathbf{A}$ is self-adjoint.

Conversely, suppose $\mathbf{A}$ is self-adjoint. By Schur's theorem, $\mathbf{B} = \mathbf{U}^*\mathbf{A}\mathbf{U}$ for some unitary matrix $\mathbf{U}$, and upper triangular matrix $\mathbf{B}$. Now $\mathbf{B}^* = (\mathbf{U}^*\mathbf{A}\mathbf{U})^* = \mathbf{U}^*\mathbf{A}^*\mathbf{U} = \mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{B}$. So $\mathbf{B}$ is self-adjoint. Hence by the Lemma above, $\mathbf{B}$ is diagonal with all diagonal entries real. This proves that $\mathbf{A}$ is unitarily diagonalizable and all eigenvalues of $\mathbf{A}$ are real.

Our short proof of the spectral theorem for self-adjoint matrices is based on Schur's theorem. This result can also be deduced from part (ii) of the spectral theorem for normal matrices since every self-adjoint matrix in normal, provided we independently show that every eigenvalue of a self-adjoint matrix is real. The latter statement can be easily proved.

Proposition

If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is self-adjoint, then every eigenvalue of $\mathbf{A}$ is real.

Proof. Let λ be an eigenvalue of a self-adjoint matrix $\mathbf{A}$, and let $\mathbf{x}$ be a corresponding unit eigenvector. Then

$$\lambda = \lambda\,\mathbf{x}^*\mathbf{x} = \mathbf{x}^*\mathbf{A}\,\mathbf{x} = \mathbf{x}^*\mathbf{A}^*\mathbf{x} = (\mathbf{A}\,\mathbf{x})^*\mathbf{x} = (\lambda\,\mathbf{x})^*\mathbf{x} = \overline{\lambda}\,\mathbf{x}^*\mathbf{x} = \overline{\lambda}.$$

Hence λ is real. $\square$

Finally, let us consider a real symmetric matrix $\mathbf{A}$, that is, $\mathbf{A} \in \mathbb{R}^{n \times n}$ such that $\mathbf{A}^\mathsf{T} = \mathbf{A}$. We shall prove a spectral theorem for $\mathbf{A}$ which involves only real scalars.

A unitary matrix with real entries is also called an **orthogonal matrix**. Thus $\mathbf{C} \in \mathbb{R}^{n \times n}$ is orthogonal if its columns form an orthonormal subset of $\mathbb{R}^{n \times 1}$. Clearly, an orthogonal matrix is invertible and its inverse is the same as its transpose.

Definition

A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ *is called* **orthogonally diagonalizable** *if there is an orthogonal matrix* $\mathbf{C} \in \mathbb{R}^{n \times n}$ *and a diagonal matrix* $\mathbf{D} \in \mathbb{R}^{n \times n}$ *such that* $\mathbf{D} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$.

Proposition (Spectral Theorem for Real Symmetric Matrices)

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then $\mathbf{A}$ is symmetric if and only if $\mathbf{A}$ is orthogonally diagonalizable. In this case, $\mathbf{A}$ has n real eigenvalues counted according to their algebraic multiplicities.

Proof of the Spectral Theorem for Real Symmetric Matrices:
Suppose $\mathbf{A}$ is orthogonally diagonalizable. Then $\mathbf{D} = \mathbf{C}^\mathsf{T}\mathbf{A}\mathbf{C}$ for some orthogonal matrix $\mathbf{C}$ and diagonal matrix $\mathbf{D}$. in $\mathbb{R}^{n\times n}$. Now $\mathbf{D}^\mathsf{T} = \mathbf{D} \implies \mathbf{A}^\mathsf{T} = (\mathbf{C}\mathbf{D}\mathbf{C}^\mathsf{T})^\mathsf{T} = \mathbf{C}\mathbf{D}^\mathsf{T}\mathbf{C}^\mathsf{T} = \mathbf{C}\mathbf{D}\mathbf{C}^\mathsf{T} = \mathbf{A}$. Thus $\mathbf{A}$ is symmetric.

Conversely, suppose $\mathbf{A}$ is symmetric. By the spectral theorem for self-adjoint matrices, there is a unitary matrix $\mathbf{U} \in \mathbb{C}^{n\times n}$ and a diagonal matrix $\mathbf{D} \in \mathbb{C}^{n\times n}$ such that $\mathbf{D} = \mathbf{U}^{-\mathbf{1}}\mathbf{A}\mathbf{U}$, that is, $\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{D}$. Since the diagonal entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$, we see that $\mathbf{D} \in \mathbb{R}^{n\times n}$.

Let λ be an eigenvalue of $\mathbf{A}$. Then $\lambda \in \mathbb{R}$. Using the Gauss Elimination Method, we may find the basic solutions of the homogeneous linear system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$. Their entries are real since all entries of $\mathbf{A}$ are real and $\lambda \in \mathbb{R}$. These solutions form a basis for the eigenspace of $\mathbf{A}$ corresponding to λ.

Further, we can use the Gram-Schmidt Orthogonalization Process for these basic solutions to obtain an orthonormal basis for the eigenspace of $\mathbf{A}$ corresonding to λ. In this process, the entries of the basis vectors remain real. Putting together the orthonormal bases for the eigenspaces of $\mathbf{A}$ corresponding to distinct eigenvalues, we get an orthonormal set (since eigenvectors corresponding to distinct eigenvalues of a normal matrix are orthogonal); moreover, this set contains exactly n vectors (since the sum of geometric multiplicities of eigenvalues of $\mathbf{A}$ is n because $\mathbf{A}$ is diagonalisable). So the matrix $\mathbf{C} \in \mathbb{R}^{n \times n}$ formed by these n vectors as its columns is orthogonal and moreover $\mathbf{C}^{-1}\mathbf{A}\mathbf{C}$ is a diagonal matrix. This proves that $\mathbf{A}$ is orthogonally diagonalizable. $\square$

Remark: The above proof suggests a constructive method to orthogonally diagonalize a real symmetric matrix (and similarly, to unitary diagonalize a complex self-adjoint matrix), provided we know its eigenvalues. We illustrate this with an example.

Example: Let $\mathbf{A} := \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$. Clearly $\mathbf{A}$ is real symmetric.

1. $p_{\mathbf{A}}(t) = \det(\mathbf{A} - t\mathbf{I}) = (3-t)^2(3+t)$. So the eigenvalues of $\mathbf{A}$ are $\mu_1 = 3$ with $m_1 = g_1 = 2$, and $\mu_2 = -3$ with $m_2 = g_2 = 1$.

2. (i) $(\mathbf{A} - 3\mathbf{I})\mathbf{x} = \mathbf{0}$, that is,

$$\begin{bmatrix} -2 & 2 & -2 \\ 2 & -2 & 2 \\ -2 & 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \iff \begin{bmatrix} -2 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\iff x_1 - x_2 + x_3 = 0$ by the Gauss Elimination Method.

Hence $\boxed{\mathbf{x}_{11} := \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T}$ and $\boxed{\mathbf{x}_{12} := \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}^T}$ form a basis for the null space $\mathcal{N}(\mathbf{A} - 3\mathbf{I})$.

(ii) Similarly, $\boxed{\mathbf{x}_{21} := \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}^T}$ forms a basis for the null space $\mathcal{N}(\mathbf{A} + 3\mathbf{I})$.

3. Gram-Schmidt Orthogonalization Process gives $\mathbf{u}_{11} := \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}^{\mathsf{T}}$ and $\mathbf{u}_{12} := \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix}^{\mathsf{T}}$, which form an orthonormal basis for $\mathcal{N}(\mathbf{A} - 3\mathbf{I})$.

Also, $\mathbf{u}_{21} := \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}^{\mathsf{T}}$ forms an orthonormal basis for $\mathcal{N}(\mathbf{A} + 3\mathbf{I})$.

4. Let $\mathbf{U} := \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$

and $\mathbf{D} := \operatorname{diag}(3, 3, -3)$. Then $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{D}$. Note that since $\mathbf{U}$ is real (and unitary), we can also say that $\mathbf{U}$ is orthogonal and $\mathbf{U}^{\mathsf{T}}\mathbf{A}\mathbf{U} = \mathbf{D}$.

We now show that the unitary matrix $\mathbf{U}$ and the diagonal matrix $\mathbf{D}$ satisfying $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{D}$ we have found are not unique.

For example, let us interchange the order of the columns of $\mathbf{U}$ and make a corresponding interchange in the diagonal entries of $\mathbf{D}$.

Thus $\mathbf{U} := \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & -1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & 2/\sqrt{6} \end{bmatrix}$

and $\mathbf{D} := \mathrm{diag}(3, -3, 3)$ would also satisfy $\mathbf{U}^*\mathbf{AU} = \mathbf{D}$.

Further, we could have chosen $\mathbf{x}_{11} := \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^\mathsf{T}$ and $\mathbf{x}_{12} := \begin{bmatrix} -1 & 1 & 2 \end{bmatrix}^\mathsf{T}$ as basis vectors for the null space $\mathcal{N}(\mathbf{A} - 3\mathbf{I})$, and orthonormalized them to obtain $\mathbf{u}_{11} := \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}^\mathsf{T}$ & $\mathbf{u}_{12} := \begin{bmatrix} -2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \end{bmatrix}^\mathsf{T}$.

Then $\mathbf{U} := \begin{bmatrix} 0 & -2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$

and $\mathbf{D} := \mathrm{diag}(3, 3, -3)$ would also satisfy $\mathbf{U}^*\mathbf{AU} = \mathbf{D}$.

Final Comments

The spectral theorems proved in this lecture say the following.

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be normal. Then there is an orthonormal set of eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbf{A}$ corresponding to the eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$.

If $\mathbf{x} \in \mathbb{C}^{n\times 1}$, then $\mathbf{x} = \langle \mathbf{u}_1, \mathbf{x}\rangle \mathbf{u}_1 + \cdots + \langle \mathbf{u}_n, \mathbf{x}\rangle \mathbf{u}_n$ since $\mathbf{u}_1, \ldots, \mathbf{u}_n$ is an orthonormal basis for $\mathbb{C}^{n\times 1}$. Hence

$$\mathbf{A}\,\mathbf{x} = \lambda_1 \langle \mathbf{u}_1, \mathbf{x}\rangle \mathbf{u}_1 + \cdots + \lambda_n \langle \mathbf{u}_n, \mathbf{x}\rangle \mathbf{u}_n \quad \text{for all } \mathbf{x} \in \mathbb{C}^{n\times 1}.$$

In particular, if $\mathbf{A}$ is self-adjoint, then $\lambda_1, \ldots, \lambda_n$ are real. This also holds if $\mathbf{A} \in \mathbb{R}^{n\times n}$ is symmetric (with $\mathbf{u}_1, \ldots, \mathbf{u}_n$ having all real entries).

The above formula gives a spectral representation of $\mathbf{A}$. This is useful in evaluating the action of powers of $\mathbf{A}$ or more generally, polynomials in $\mathbf{A}$, on column vectors.

Let $k \in \mathbb{N}$. Then $\mathbf{A}^k(\mathbf{u}_j) = \lambda_j^k \mathbf{u}_j$ for each $j = 1, \ldots, n$, and so

$$\mathbf{A}^k \mathbf{x} = \sum_{j=1}^{n} \lambda_j^k \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{u}_j \quad \text{for all } \mathbf{x} \in \mathbb{C}^{n \times 1}.$$

More generally, if $p(t)$ is any polynomial, then

$$p(\mathbf{A})\, \mathbf{x} = \sum_{j=1}^{n} p(\lambda_j) \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{u}_j \quad \text{for all } \mathbf{x} \in \mathbb{C}^{n \times 1}.$$

In the example which we have just worked out, we obtain

$$\mathbf{A}^k \mathbf{x} = 3^k \langle \mathbf{u}_{11}, \mathbf{x} \rangle \mathbf{u}_{11} + 3^k \langle \mathbf{u}_{12}, \mathbf{x} \rangle \mathbf{u}_{12} + (-3)^k \langle \mathbf{u}_{21}, \mathbf{x} \rangle \mathbf{u}_{21}$$

for all $\mathbf{x} \in \mathbb{K}^{3 \times 1}$ and all $k \in \mathbb{N}$, where

$$\begin{aligned}
\langle \mathbf{u}_{11}, \mathbf{x}\rangle \mathbf{u}_{11} &= \frac{1}{2}(x_1 + x_2)\begin{bmatrix}1 & 1 & 0\end{bmatrix}^\mathsf{T} \\
\langle \mathbf{u}_{12}, \mathbf{x}\rangle \mathbf{u}_{12} &= \frac{1}{6}(-x_1 + x_2 + 2x_3)\begin{bmatrix}-1 & 1 & 2\end{bmatrix}^\mathsf{T} \\
\langle \mathbf{u}_{21}, \mathbf{x}\rangle \mathbf{u}_{21} &= \frac{1}{3}(x_1 - x_2 + x_3)\begin{bmatrix}1 & -1 & 1\end{bmatrix}^\mathsf{T}.
\end{aligned}$$

For instance, if $\mathbf{x} := \begin{bmatrix}1 & 2 & 3\end{bmatrix}^\mathsf{T}$, then

$$\mathbf{A}^k\mathbf{x} = 3^k\frac{3}{2}\begin{bmatrix}1 & 1 & 0\end{bmatrix}^\mathsf{T} + 3^k\frac{7}{6}\begin{bmatrix}-1 & 1 & 2\end{bmatrix}^\mathsf{T} + (-3)^k\frac{2}{3}\begin{bmatrix}1 & -1 & 1\end{bmatrix}^\mathsf{T}$$

for each $k \in \mathbb{N}$.

Real Quadratic Forms

Let $n \in \mathbb{N}$. A **real n-ary quadratic form** Q is a homogeneous polynomial of degree 2 in n variables with coefficients in $\mathbb{R}$. Thus

$$\begin{aligned} Q(x_1, \ldots, x_n) &:= \sum_{j=1}^{n} \sum_{k=1}^{n} \alpha_{jk} x_j x_k \\ &= \sum_{j=1}^{n} \alpha_{jj} x_j^2 + \sum_{1 \le j < k \le n} (\alpha_{jk} + \alpha_{kj}) x_j x_k, \end{aligned}$$

where $\alpha_{jk} \in \mathbb{R}$ for $j, k = 1, \ldots, n$.

Examples Let $a, b, c, a', b', c' \in \mathbb{R}$.

$n = 1 : Q(x) := a\,x^2$ (unary quadratic form)

$n = 2 : Q(x, y) := a\,x^2 + b\,y^2 + a'xy$ (binary quadratic form)

$n = 3 : Q(x, y, z) := a\,x^2 + b\,y^2 + c\,z^2 + a'xy + b'yz + c'zx$ (ternary quadratic form)

For $n \in \mathbb{N}$, consider an $n \times n$ real matrix $\mathbf{A} := [a_{jk}]$.
Then for $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$,

$$\mathbf{x}^{\mathsf{T}} \mathbf{A}\, \mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} \sum_{k=1}^{n} a_{1k} x_k \\ \vdots \\ \sum_{k=1}^{n} a_{nk} x_k \end{bmatrix} = \sum_{j=1}^{n} \Big(\sum_{k=1}^{n} a_{jk} x_k \Big) x_j$$

$$= \sum_{j=1}^{n} a_{jj} x_j^2 + \sum_{1 \le j < k \le n} (a_{jk} + a_{kj}) x_j x_k,$$

which is an n-ary quadratic form.
In fact, $Q(x_1, \ldots, x_n) = \mathbf{x}^{\mathsf{T}} \mathbf{A}\, \mathbf{x}$ for all $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$ in $\mathbb{R}^{n \times 1}$ if and only if

$$\alpha_{jk} + \alpha_{kj} = a_{jk} + a_{kj} \quad \text{for all } j, k = 1, \ldots, n.$$

In general, many $n \times n$ matrices induce the same quadratic form. For example, the matrices $\begin{bmatrix} 3 & 4 \\ 6 & 2 \end{bmatrix}, \begin{bmatrix} 3 & 5 \\ 5 & 2 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ 11 & 2 \end{bmatrix}$ induce the same binary quadratic form.

But if we require the matrix $\mathbf{A} := [a_{jk}]$ inducing the quadratic form Q to be symmetric, that is, $a_{jk} = a_{kj}$ for all j, k, then

$$a_{jk} = \frac{1}{2}(\alpha_{jk} + \alpha_{kj}) \quad \text{for all } j, k = 1, \ldots, n.$$

Thus given an n-ary quadratic form Q, there is a unique $n \times n$ real symmetric matrix $\mathbf{A}$ such that $Q(x_1 \ldots, x_n) = \mathbf{x}^\mathsf{T} \mathbf{A}\, \mathbf{x}$ for all $\mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{n \times 1}$; in fact

$$\mathbf{A} := [a_{jk}], \quad \text{where } a_{jk} := \frac{1}{2}(\alpha_{jk} + \alpha_{kj}),\ j, k = 1, \ldots, n.$$

This real symmetric matrix $\mathbf{A}$ is called the **matrix associated with** the quadratic form Q, and we write $Q = Q_{\mathbf{A}}$.

A real n-ary quadratic form Q is said to be a **diagonal quadratic form** if there are $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ such that

$$Q(x_1, \ldots, x_n) = \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2.$$

It is clear that a quadratic form Q is diagonal if and only if Q is associated with a diagonal matrix $\mathbf{D}$, that is, $Q = Q_{\mathbf{D}}$. Using the spectral theorem for real symmetric matrices, we show that every quadratic form can be orthogonally transformed to a diagonal quadratic form.

Theorem (Principle Axis Theorem)

Let Q be a real quadratic form and $\mathbf{A} \in \mathbb{R}^{n\times n}$ be the symmetric matrix associated with Q. If $\mathbf{C}$ is an orthogonal matrix such that the matrix $\mathbf{D} := \mathbf{C}^{\mathsf{T}}\mathbf{A}\,\mathbf{C}$ is diagonal, then $Q(\mathbf{x}) = Q_{\mathbf{D}}(\mathbf{y})$, where $\mathbf{y} := \mathbf{C}^{\mathsf{T}}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^{n\times 1}$.

Proof. Let $\mathbf{x} \in \mathbb{R}^{n\times 1}$ and $\mathbf{y} := \mathbf{C}^{\mathsf{T}}\mathbf{x} = \mathbf{C}^{-1}\mathbf{x}$. Then $\mathbf{x} = \mathbf{C}\mathbf{y}$ and

$$Q_{\mathbf{A}}(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\,\mathbf{x} = (\mathbf{C}\mathbf{y})^{\mathsf{T}}\mathbf{A}(\mathbf{C}\mathbf{y}) = \mathbf{y}^{\mathsf{T}}(\mathbf{C}^{\mathsf{T}}\mathbf{A}\mathbf{C})\mathbf{y} = \mathbf{y}^{\mathsf{T}}\mathbf{D}\mathbf{y} = Q_{\mathbf{D}}(\mathbf{y}).$$

To diagonalise a real n-ary quadratic form Q, we first write down the (real symmetric) matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ associated with Q. We then find an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ consisting of eigenvectors of $\mathbf{A}$ corresponding to its eigenvalues $\lambda_1, \ldots, \lambda_n$ counted according to their algebraic multiplicities. If we let

$$\mathbf{C} := \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

Then $Q(\mathbf{x}) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$, where $\mathbf{y} := \mathbf{C}^\mathsf{T}\mathbf{x} = \begin{bmatrix} \mathbf{u}_1^\mathsf{T} \\ \vdots \\ \mathbf{u}_n^\mathsf{T} \end{bmatrix} \mathbf{x}$.

Example

Let us transform the quadratic form $Q(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3 - 4x_3x_1$ to a diagonal form. Here $\mathbf{A} := \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$ is the associated matrix.

We have seen before that
$\mathbf{u}_1 := \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}^{\mathsf{T}}$, $\mathbf{u}_2 := \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix}^{\mathsf{T}}$ and $\mathbf{u}_3 := \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}^{\mathsf{T}}$ are eigenvectors of $\mathbf{A}$ corresponding to the eigenvalues 3, 3 and -3 respectively, and they form an orthonormal basis for $\mathbb{R}^{3\times 1}$. Hence let

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Then $\mathbf{C}^{\mathsf{T}} \mathbf{A}\, \mathbf{C} = \mathbf{D}$, and so $Q(\mathbf{x}) = 3(y_1^2 + y_2^2 - y_3^2)$, where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} := \mathbf{C}^{\mathsf{T}}\mathbf{x} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

that is, $y_1 = (x_1 + x_2)/\sqrt{2}$, $y_2 = (-x_1 + x_2 + 2x_3)/\sqrt{6}$ and $y_3 = (x_1 - x_2 + x_3)/\sqrt{3}$.

MA110: Lecture 18

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Spring 2025

Real Quadratic Forms

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$$\begin{aligned} Q(x_1, \ldots, x_n) &:= \sum_{j=1}^{n} \sum_{k=1}^{n} \alpha_{jk} x_j x_k \\ &= \sum_{j=1}^{n} \alpha_{jj} x_j^2 + \sum_{1 \le j < k \le n} (\alpha_{jk} + \alpha_{kj}) x_j x_k, \end{aligned}$$

where $\alpha_{jk} \in \mathbb{R}$ for $j, k = 1, \ldots, n$.

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For $n \in \mathbb{N}$, consider an $n \times n$ real matrix $\mathbf{A} := [a_{jk}]$.
Then for $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T}$,

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$$= \sum_{j=1}^n a_{jj} x_j^2 + \sum_{1 \le j < k \le n} (a_{jk} + a_{kj}) x_j x_k,$$

which is an n-ary quadratic form.
In fact, $Q(x_1, \ldots, x_n) = \mathbf{x}^\mathsf{T} \mathbf{A}\, \mathbf{x}$ for all $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$ in $\mathbb{R}^{n \times 1}$ if and only if

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Thus given an n-ary quadratic form Q, there is a unique $n \times n$ real symmetric matrix $\mathbf{A}$ such that $Q(x_1 \ldots, x_n) = \mathbf{x}^{\mathsf{T}} \mathbf{A}\, \mathbf{x}$ for all $\mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{n \times 1}$; in fact

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It is clear that a quadratic form Q is diagonal if and only if Q is associated with a diagonal matrix $\mathbf{D}$, that is, $Q = Q_{\mathbf{D}}$. Using the spectral theorem for real symmetric matrices, we show that every quadratic form can be orthogonally transformed to a diagonal quadratic form.

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Let Q be a real quadratic form and $\mathbf{A} \in \mathbb{R}^{n\times n}$ be the symmetric matrix associated with Q. If $\mathbf{C}$ is an orthogonal matrix such that the matrix $\mathbf{D} := \mathbf{C}^{\mathsf{T}}\mathbf{A}\,\mathbf{C}$ is diagonal, then $Q(\mathbf{x}) = Q_{\mathbf{D}}(\mathbf{y})$, where $\mathbf{y} := \mathbf{C}^{\mathsf{T}}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^{n\times 1}$.

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To diagonalise a real n-ary quadratic form Q, we first write down the (real symmetric) matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ associated with Q. We then find an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ consisting of eigenvectors of $\mathbf{A}$ corresponding to its eigenvalues $\lambda_1, \dots, \lambda_n$ counted according to their algebraic multiplicities. If we let

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Then $Q(\mathbf{x}) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$, where $\mathbf{y} := \mathbf{C}^\mathsf{T}\mathbf{x} = \begin{bmatrix} \mathbf{u}_1^\mathsf{T} \\ \vdots \\ \mathbf{u}_n^\mathsf{T} \end{bmatrix} \mathbf{x}$.

Example

Let us transform the quadratic form $Q(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3 - 4x_3x_1$ to a diagonal form. Here $\mathbf{A} := \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$ is the associated matrix.

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$$\mathbf{C} := \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \operatorname{diag}(3, 3, -3).$$

Then $\mathbf{C}^{\mathsf{T}}\mathbf{A}\,\mathbf{C} = \mathbf{D}$, and so $Q(\mathbf{x}) = 3(y_1^2 + y_2^2 - y_3^2)$, where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} := \mathbf{C}^{\mathsf{T}}\mathbf{x} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

that is, $y_1 = (x_1 + x_2)/\sqrt{2}$, $y_2 = (-x_1 + x_2 + 2x_3)/\sqrt{6}$ and $y_3 = (x_1 - x_2 + x_3)/\sqrt{3}$.

Conic Sections

A **conic section** is the locus in $\mathbb{R}^2$ of an equation

$$a\,x^2 + b\,y^2 + c\,xy + a'x + b'y + c' = 0,$$

where $a, b, c, a', b', c' \in \mathbb{R}$ and at least one among a, b, c is nonzero. We assume WLOG that not both a and b are negative. It can be proved that the conic is one of these:
(i) the empty set
(ii) a single point
(iii) one or two straight lines
(iv) an ellipse
(v) a hyperbola
(vi) a parabola.

Terms of the second degree on the LHS of the equation give

$$Q(x, y) := a\,x^2 + b\,y^2 + c\,xy.$$

It is a binary quadratic form. It determines the type of the conic.

The (real symmetric) matrix associated with Q is

$$\mathbf{A} := \begin{bmatrix} a & c/2 \\ c/2 & b \end{bmatrix}.$$

Hence the equation of the given conic section becomes

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & c/2 \\ c/2 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a' & b' \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + c' = 0,$$

that is,

$\begin{bmatrix} x & y \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y \end{bmatrix}^{\mathsf{T}} + \begin{bmatrix} a' & b' \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix}^{\mathsf{T}} + c' = 0.$

Let $\mathbf{C} := [\mathbf{u}_1, \mathbf{u}_2]$ be an orthogonal matrix whose columns $\mathbf{u}_1$ and $\mathbf{u}_2$ are eigenvectors of $\mathbf{A}$ with corresponding eigenvalues λ_1 and λ_2, and let $\mathbf{D} := \mathrm{diag}(\lambda_1, \lambda_2)$ so that $\mathbf{C}^{\mathsf{T}} \mathbf{A}\, \mathbf{C} = \mathbf{D}$.

We use the change of variables $\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{C} \begin{bmatrix} u \\ v \end{bmatrix}$ to transform the quadratic form $Q(x, y)$ to a diagonal form as follows.

$$\begin{aligned} Q(x,y) &= \begin{bmatrix} x & y \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y \end{bmatrix}^{\mathsf{T}} \\ &= \begin{bmatrix} u & v \end{bmatrix} \mathbf{C}^{\mathsf{T}} \mathbf{A}\, \mathbf{C} \begin{bmatrix} u & v \end{bmatrix}^{\mathsf{T}} \\ &= \begin{bmatrix} u & v \end{bmatrix} \mathbf{D} \begin{bmatrix} u & v \end{bmatrix}^{\mathsf{T}} \\ &= \lambda_1 u^2 + \lambda_2 v^2 = Q_{\mathbf{D}}(u,v). \end{aligned}$$

The ordered orthonormal basis $(\mathbf{u}_1, \mathbf{u}_2)$ determines a new set of coordinate axes, so that the locus of the original equation is given by

$$\begin{aligned} &\begin{bmatrix} u & v \end{bmatrix} \operatorname{diag}(\lambda_1, \lambda_2) \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} a' & b' \end{bmatrix} \mathbf{C} \begin{bmatrix} u \\ v \end{bmatrix} + c' \\ = \;& \lambda_1 u^2 + \lambda_2 v^2 + \begin{bmatrix} a' & b' \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + c' = 0. \end{aligned}$$

If the conic so determined is not degenerate, that is, if it does not reduce to an empty set, a point, or line(s), then the signs of λ_1 and λ_2 determine the type of the conic section as follows.

The equation represents

1. ellipse if $\lambda_1\lambda_2 > 0$, that is, both λ_1 and λ_2 are positive,

2. hyperbola if $\lambda_1\lambda_2 < 0$, that is, one of λ_1, λ_2 is positive and the other is negative,

3. parabola if $\lambda_1\lambda_2 = 0$, that is, one of λ_1, λ_2 is zero.

Note: Since $Q(x, y) := a\,x^2 + b\,y^2 + c\,xy = \lambda_1 u^2 + \lambda_2 v^2$, where not both a and b are negative, it follows that not both λ_1 and λ_2 can be negative, and since the conic is assumed to be nondegenerate, not both λ_1 and λ_2 can be equal to zero.

Examples

$\boxed{1.}$ Consider the conic section given by $2x^2 + 4xy + 5y^2 + 4x + 13y - 1/4 = 0$, and the binary quadratic form $Q(x, y) := 2x^2 + 4xy + 5y^2$.

Then the associated symmetric matrix $\mathbf{A} := \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$ has eigenvalues $\lambda_1 := 1$ and $\lambda_2 := 6$, and the corresponding eigenvectors $\mathbf{u}_1 := \begin{bmatrix} 2 & -1 \end{bmatrix}^{\mathsf{T}} / \sqrt{5}$ and $\mathbf{u}_2 := \begin{bmatrix} 1 & 2 \end{bmatrix}^{\mathsf{T}} / \sqrt{5}$ form an orthonormal basis for $\mathbb{R}^{2\times 1}$. Hence let

$$\mathbf{C} := \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \mathrm{diag}(1, 6).$$

Then $\mathbf{C}^{\mathsf{T}}\mathbf{A}\mathbf{C} = \mathbf{D}$. So $Q(x, y) = Q_{\mathbf{D}}(u, v) = u^2 + 6v^2$, where $\begin{bmatrix} u \\ v \end{bmatrix} := \mathbf{C}^{\mathsf{T}} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$, that is, $u = (2x - y)/\sqrt{5}$ and $v = (x + 2y)/\sqrt{5}$.

Since $\begin{bmatrix} x \\ y \end{bmatrix} := \mathbf{C} \begin{bmatrix} u \\ v \end{bmatrix}$, substituting $x = (2u + v)/\sqrt{5}$ and $y = (-u + 2v)/\sqrt{5}$ in the given equation of the conic section, we obtain

$$u^2 + 6v^2 - \sqrt{5}u + 6\sqrt{5}v - \frac{1}{4} = 0.$$

Completing the squares, we see that

$$\left(u - \frac{\sqrt{5}}{2}\right)^2 + 6\left(v + \frac{\sqrt{5}}{2}\right)^2 = 9.$$

This is an equation of an ellipse with its centre at $(\sqrt{5}/2, -\sqrt{5}/2)$ in the uv-plane, where the u-axis and the v-axis are determined by the eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$.

2. Consider the conic section given by $2x^2 - 4xy - y^2 - 4x + 10y - 13 = 0$, and the binary quadratic form $Q(x, y) := 2x^2 - 4xy - y^2$.

The associated symmetric matrix $\mathbf{A} := \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix}$ has eigenvalues $\lambda_1 = 3, \lambda_2 = -2$, and the corresponding eigenvectors $\mathbf{u}_1 := \begin{bmatrix} 2 & -1 \end{bmatrix}^{\mathsf{T}} / \sqrt{5}$ and $\mathbf{u}_2 := \begin{bmatrix} 1 & 2 \end{bmatrix}^{\mathsf{T}} / \sqrt{5}$ form an orthonormal basis for $\mathbb{R}^{2 \times 1}$. Hence let

$$\mathbf{C} := \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \operatorname{diag}(3, -2).$$

Then $\mathbf{C}^{\mathsf{T}} \mathbf{A}\, \mathbf{C} = \mathbf{D}$. Hence $Q(x, y) = Q_{\mathbf{D}}(u, v) = 3u^2 - 2v^2$,

$$\text{where } \begin{bmatrix} u \\ v \end{bmatrix} := \mathbf{C}^{\mathsf{T}} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix},$$

that is, $u = (2x - y)/\sqrt{5}$ and $v = (x + 2y)/\sqrt{5}$.

Thus substituting $x = (2u + v)/\sqrt{5}$ and $y = (-u + 2v)/\sqrt{5}$ in the given equation of the conic section, we obtain

$$3u^2 - 2v^2 - \frac{4}{\sqrt{5}}(2u + v) + \frac{10}{\sqrt{5}}(-u + 2v) - 13 = 0,$$

that is,

$$3u^2 - 2v^2 - \frac{1}{\sqrt{5}}(18u - 16v) - 13 = 0.$$

Completing the squares, we see that

$$\frac{(u - 3/\sqrt{5})^2}{4} - \frac{(v - 4/\sqrt{5})^2}{6} = 1.$$

This is an equation of a hyperbola with its centre $(3/\sqrt{5}, 4/\sqrt{5})$ in the uv-plane, where the u-axis and the v-axis are determined by the eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$.

3. Consider the conic section given by $9x^2 + 24xy + 16y^2 - 20x + 15y = 0$, and the binary quadratic form $Q(x, y) := 9x^2 + 24xy + 16y^2$. Then the associated symmetric matrix $\mathbf{A} := \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$ has eigenvalues $\lambda_1 := 25$ and $\lambda_2 := 0$, and the corresponding eigenvectors $\mathbf{u}_1 := \begin{bmatrix} 3 & 4 \end{bmatrix}^\mathsf{T}/5$ and $\mathbf{u}_2 := \begin{bmatrix} -4 & 3 \end{bmatrix}^\mathsf{T}/5$ form an orthonormal basis for $\mathbb{R}^{2\times 1}$. Hence let $\mathbf{C} := \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$ and $\mathbf{D} := \text{diag}(25, 0)$. Then $\mathbf{C}^\mathsf{T}\mathbf{A}\,\mathbf{C} = \mathbf{D}$. Thus $Q(x, y) = Q_\mathbf{D}(u, v) = 25u^2$, where

$$\begin{bmatrix} u \\ v \end{bmatrix} := \mathbf{C}^\mathsf{T} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix},$$

that is, $u = (3x + 4y)/5$ and $v = (-4x + 3y)/5$. Substituting $x = (3u - 4v)/5$ and $y = (4u + 3v)/5$ in the equation of the conic, we obtain $25u^2 + 25v = 0$, i.e., $u^2 = -v$, which is an equation of a parabola with its vertex at $(0, 0)$ in the uv-plane.

Quadric Surfaces

A **quadric surface** is the locus in $\mathbb{R}^3$ of an equation

$$a\,x^2 + b\,y^2 + c\,z^2 + a'xy + b'yz + c'zx + a''x + b''y + c''z + d = 0,$$

where $a, b, c, a', b', c', a'', b'', c'', d \in \mathbb{R}$. We assume WLOG that not all three a, b and c are negative.

Terms of the second degree on the LHS of the equation give

$$Q(x, y, z) := a\,x^2 + b\,y^2 + c\,z^2 + a'xy + b'yz + c'zx.$$

It is a ternary quadratic form. It determines the type of the quadric surface. The real symmetric matrix associated with the quadratic form Q is

$$\mathbf{A} := \begin{bmatrix} a & a'/2 & c'/2 \\ a'/2 & b & b'/2 \\ c'/2 & b'/2 & c \end{bmatrix}.$$

Hence the equation of the given quadric surface becomes

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & a'/2 & c'/2 \\ a'/2 & b & b'/2 \\ c'/2 & b'/2 & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} a'' & b'' & c'' \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + d = 0,$$

that is,

$$\begin{bmatrix} x & y & z \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y & z \end{bmatrix}^{\mathsf{T}} + \begin{bmatrix} a'' & b'' & c'' \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}^{\mathsf{T}} + d = 0.$$

Let $\mathbf{C} := \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix}$ be an orthogonal matrix whose columns $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are eigenvectors of $\mathbf{A}$ with corresponding eigenvalues $\lambda_1, \lambda_2, \lambda_3$, and let $\mathbf{D} := \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$.

We use the change of variables $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{C} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$ to transform the quadratic form $Q(x, y, z)$ to a diagonal form as follows.

$$\begin{aligned} Q(x,y,z) &= \begin{bmatrix} x & y & z \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y & z \end{bmatrix}^{\mathsf{T}} \\ &= \begin{bmatrix} u & v & w \end{bmatrix} \mathbf{C}^{\mathsf{T}} \mathbf{A}\, \mathbf{C} \begin{bmatrix} u & v & w \end{bmatrix}^{\mathsf{T}} \\ &= \begin{bmatrix} u & v & w \end{bmatrix} \mathbf{D} \begin{bmatrix} u & v & w \end{bmatrix}^{\mathsf{T}} \\ &= \lambda_1 u^2 + \lambda_2 v^2 + \lambda_3^2 w^2 = Q_{\mathbf{D}}(u,v,w). \end{aligned}$$

The ordered orthonormal basis $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ determines a new set of coordinate axes, so that the locus of the original equation is given by

$$\begin{bmatrix} u & v & w \end{bmatrix} \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3) \begin{bmatrix} u \\ v \\ w \end{bmatrix} + \begin{bmatrix} a'' & b'' & c'' \end{bmatrix} \mathbf{C} \begin{bmatrix} u \\ v \\ w \end{bmatrix} + d = 0.$$

Leaving aside the degenerate cases, the primary cases are:

Equation	Surface	Eigenvalues of A
$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$	ellipsoid	all three positive
$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z}{c}=0$	elliptic paraboloid	two positive, one zero
$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$	elliptic cone	two positive, one negative
$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$	1-sheeted hyperboloid	two positive, one negative
$\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$	2-sheeted hyperboloid	one positive, two negative
$\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z}{c}=0$	hyperbolic paraboloid	one positive, one negative, one zero.

Pictures of these surfaces can be found on the Internet by searching with their names.

Example Consider the quadric surface given by

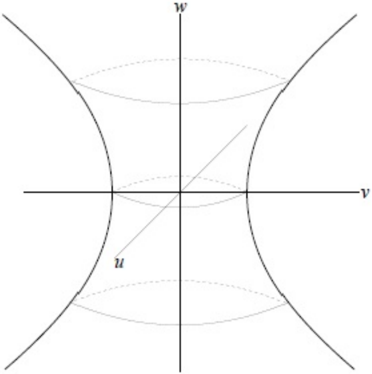
$$x^2 + y^2 + z^2 + 4xy + 4yz - 4zx - 27 = 0,$$

and the associated ternary quadratic form
$Q(x, y, z) := x^2 + y^2 + z^2 + 4xy + 4yz - 4zx$.

We have already transformed the associated symmetric matrix
$\mathbf{A} := \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$ to a diagonal form, and have obtained
$Q(x, y, z) = Q_{\mathbf{D}}(u, v, w) = 3(u^2 + v^2 - w^2)$ (with x_1, x_2, x_3 and y_1, y_2, y_3 in place of x, y, z and u, v, w),
where $\mathbf{D} := \mathrm{diag}(3, 3, -3)$ and
$u = (x+y)/\sqrt{2}$, $v = (-x+y+2z)/\sqrt{6}$, $w = (x-y+z)/\sqrt{3}$.

Under this change of coordinates, the quadric surface reduces to $u^2 + v^2 - w^2 = 9$.

This is an equation of a one-sheeted hyperboloid in the *uvw*-space, as shown in the following figure, where the *u*-axis, the *v*-axis and the *w*-axis are determined by the eigenvectors $\mathbf{u}_1 := \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}^\mathsf{T}$, $\mathbf{u}_2 := \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix}^\mathsf{T}$ and $\mathbf{u}_3 := \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}^\mathsf{T}$. (See Lecture 16.)



Orthogonal Projection

Let $\mathbb{K}$ denote either $\mathbb{R}$ or $\mathbb{C}$. Recall that in Lecture 13, we have defined the (perpendicular) projection of $\mathbf{x} \in \mathbb{K}^{n\times 1}$ in the direction of nonzero $\mathbf{y} \in \mathbb{K}^{n\times 1}$ as follows:

$$P_{\mathbf{y}}(\mathbf{x}) := \frac{\langle \mathbf{y}, \mathbf{x}\rangle}{\langle \mathbf{y}, \mathbf{y}\rangle}\,\mathbf{y}.$$

In particular, if $\mathbf{y}$ is a unit vector, then $P_{\mathbf{y}}(\mathbf{x}) = \langle \mathbf{y}, \mathbf{x}\rangle \mathbf{y}$.

We noted that the vector $P_{\mathbf{y}}(\mathbf{x})$ is a scalar multiple of the vector $\mathbf{y}$, and proved the important relation

$$\big(\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\big) \perp \mathbf{y}.$$

As a consequence,

$$\begin{aligned}
\|\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\|^2 &= \langle \mathbf{x} - P_{\mathbf{y}}(\mathbf{x}), \mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\rangle \\
&= \langle \mathbf{x}, \mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\rangle \\
&= \|\mathbf{x}\|^2 - \langle \mathbf{x}, P_{\mathbf{y}}(\mathbf{x})\rangle.
\end{aligned}$$

More generally, let Y be a nonzero subspace of $\mathbb{K}^{n\times 1}$. We would like to find a (perpendicular) projection of $\mathbf{x} \in \mathbb{K}^{n\times 1}$ into Y, that is, we want to find $\mathbf{y} \in Y$ such that $(\mathbf{x} - \mathbf{y}) \in Y^{\perp}$. (This $\mathbf{y}$ is 'the foot of the perpendicular' from $\mathbf{x}$ into Y.)

If $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is an orthonormal basis for the subspace Y, then a vector belongs to $Y^{\perp}$ if and only if it is orthogonal to each $\mathbf{u}_j$ for $j = 1, \ldots, k$. As we saw while studying G-S OP, the vector

$$\tilde{\mathbf{y}} := \mathbf{x} - P_{\mathbf{u}_1}(\mathbf{x}) - \cdots - P_{\mathbf{u}_k}(\mathbf{x}) = \mathbf{x} - \langle \mathbf{u}_1, \mathbf{x}\rangle \mathbf{u}_1 - \cdots - \langle \mathbf{u}_k, \mathbf{x}\rangle \mathbf{u}_k$$

is orthogonal to each $\mathbf{u}_j$ for $j = 1, \ldots, k$, and so $\tilde{\mathbf{y}} \in Y^{\perp}$.

Since the vector $\mathbf{y} := \langle \mathbf{u}_1, \mathbf{x}\rangle \mathbf{u}_1 + \cdots + \langle \mathbf{u}_k, \mathbf{x}\rangle \mathbf{u}_k$ belongs to Y, it is a (perpendicular) projection of $\mathbf{x}$ in Y.

The following result shows that this is the only vector in Y that works!

MA110
Lecture 19

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Spring 2025

Orthogonal Projection

Let $\mathbb{K}$ denote either $\mathbb{R}$ or $\mathbb{C}$. Recall that in Lecture 13, we have defined the (perpendicular) projection of $\mathbf{x} \in \mathbb{K}^{n\times 1}$ in the direction of nonzero $\mathbf{y} \in \mathbb{K}^{n\times 1}$ as follows:

$$P_{\mathbf{y}}(\mathbf{x}) := \frac{\langle \mathbf{y},\, \mathbf{x}\rangle}{\langle \mathbf{y},\, \mathbf{y}\rangle}\,\mathbf{y}.$$

In particular, if $\mathbf{y}$ is a unit vector, then $P_{\mathbf{y}}(\mathbf{x}) = \langle \mathbf{y},\, \mathbf{x}\rangle \mathbf{y}$.

We noted that the vector $P_{\mathbf{y}}(\mathbf{x})$ is a scalar multiple of the vector $\mathbf{y}$, and proved the important relation

$$\big(\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\big) \perp \mathbf{y}.$$

As a consequence,

$$\begin{aligned} \|\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\|^2 &= \langle \mathbf{x} - P_{\mathbf{y}}(\mathbf{x}),\, \mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\rangle \\ &= \langle \mathbf{x},\, \mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\rangle \\ &= \|\mathbf{x}\|^2 - \langle \mathbf{x},\, P_{\mathbf{y}}(\mathbf{x})\rangle. \end{aligned}$$

More generally, let Y be a nonzero subspace of $\mathbb{K}^{n\times 1}$. We would like to find a (perpendicular) projection of $\mathbf{x} \in \mathbb{K}^{n\times 1}$ into Y, that is, we want to find $\mathbf{y} \in Y$ such that $(\mathbf{x} - \mathbf{y}) \in Y^{\perp}$. (This $\mathbf{y}$ is 'the foot of the perpendicular' from $\mathbf{x}$ into Y.)

If $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is an orthonormal basis for the subspace Y, then a vector belongs to $Y^{\perp}$ if and only if it is orthogonal to each $\mathbf{u}_j$ for $j = 1, \ldots, k$. As we saw while studying G-S OP, the vector

$$\tilde{\mathbf{y}} := \mathbf{x} - P_{\mathbf{u}_1}(\mathbf{x}) - \cdots - P_{\mathbf{u}_k}(\mathbf{x}) = \mathbf{x} - \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 - \cdots - \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k$$

is orthogonal to each $\mathbf{u}_j$ for $j = 1, \ldots, k$, and so $\tilde{\mathbf{y}} \in Y^{\perp}$.

Since the vector $\mathbf{y} := \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \cdots + \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k$ belongs to Y, it is a (perpendicular) projection of $\mathbf{x}$ in Y.

The following result shows that this is the only vector in Y that works!

Proposition (Projection Theorem)

Let Y be a subspace of $\mathbb{K}^{n\times 1}$. Then for every $\mathbf{x} \in \mathbb{K}^{n\times 1}$, there are unique $\mathbf{y} \in Y$ and $\tilde{\mathbf{y}} \in Y^{\perp}$ such that $\mathbf{x} = \mathbf{y} + \tilde{\mathbf{y}}$, that is, $\mathbb{K}^{n\times 1} = Y \oplus Y^{\perp}$. The map $P_Y : \mathbb{K}^{n\times 1} \to \mathbb{K}^{n\times 1}$ given by $P_Y(\mathbf{x}) = \mathbf{y}$ is linear and satisfies $(P_Y)^2 = P_Y$.

In fact, if $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is an orthonormal basis for Y, then

$$P_Y(\mathbf{x}) = \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \cdots + \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k.$$

Proof. If $Y = \{\mathbf{0}\}$, then every $\mathbf{x} \in Y^{\perp}$, and so $\mathbf{x} = \mathbf{0} + \mathbf{x}$.

Suppose $Y \neq \{\mathbf{0}\}$, and let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be an orthonormal basis for Y. For $\mathbf{x} \in \mathbb{K}^{n\times 1}$, define

$$P_Y(\mathbf{x}) := \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \cdots + \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k.$$

Clearly, $\mathbf{y} := P_Y(\mathbf{x}) \in Y$. Define $\tilde{\mathbf{y}} := \mathbf{x} - P_Y(\mathbf{x})$. Then

$$\begin{aligned}\langle \mathbf{u}_j, \tilde{\mathbf{y}}\rangle &= \langle \mathbf{u}_j, \mathbf{x} - P_Y(\mathbf{x})\rangle \\ &= \langle \mathbf{u}_j, \mathbf{x}\rangle - \sum_{\ell=1}^{k}\langle \mathbf{u}_\ell, \mathbf{x}\rangle\langle \mathbf{u}_j, \mathbf{u}_\ell\rangle \\ &= \langle \mathbf{u}_j, \mathbf{x}\rangle - \langle \mathbf{u}_j, \mathbf{x}\rangle = 0\end{aligned}$$

for $j = 1, \ldots, k$ by orthonormality. Hence $\tilde{\mathbf{y}} \in Y^\perp$. Thus $\mathbf{x} = \mathbf{y} + \tilde{\mathbf{y}}$ with $\mathbf{y} \in Y$ and $\tilde{\mathbf{y}} \in Y^\perp$. This proves existence.

To prove uniqueness, let $\mathbf{z} \in Y$ and $\tilde{\mathbf{z}} \in Y^\perp$ be such that $\mathbf{x} = \mathbf{z} + \tilde{\mathbf{z}}$. Then $(\mathbf{y} - \mathbf{z}) \in Y$ and also $\mathbf{y} - \mathbf{z} = (\tilde{\mathbf{z}} - \tilde{\mathbf{y}}) \in Y^\perp$, so that $(\mathbf{y} - \mathbf{z}) \in Y \cap Y^\perp$. Hence $(\mathbf{y} - \mathbf{z}) \perp (\mathbf{y} - \mathbf{z})$, and so $\mathbf{y} - \mathbf{z} = \mathbf{0}$. Thus $\mathbf{z} = \mathbf{y}$, and in turn, $\tilde{\mathbf{z}} = \tilde{\mathbf{y}}$.

The map P_Y is linear since the inner product is linear in the second variable. Also, $P_Y(\mathbf{u}_j) = \mathbf{u}_j$ for all $j = 1, \ldots, k$. Hence $P_Y(\mathbf{x}) = \mathbf{x}$ if and only if $\mathbf{x} \in Y$. As a consequence, $P_Y^2(\mathbf{x}) = P_Y(P_Y(\mathbf{x})) = P_Y(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{K}^{n\times 1}$. □

Let Y be a subspace of $\mathbb{K}^{n\times 1}$. Then the linear map $P_Y : \mathbb{K}^{n\times 1} \to \mathbb{K}^{n\times 1}$ whose existence and uniqueness is proved in the above result is called the **orthogonal projection map** of $\mathbb{K}^{n\times 1}$ onto the subspace Y.

Given $\mathbf{x} \in \mathbb{K}^{n\times 1}$, we shall show that its orthogonal projection $P_Y(\mathbf{x})$ is the unique vector in Y which is closest to $\mathbf{x}$.

Definition

Let E be a nonempty subset of $\mathbb{K}^{n\times 1}$ and let $\mathbf{x} \in \mathbb{K}^{n\times 1}$. A **best approximation** *to $\mathbf{x}$ from E is an element $\mathbf{y}_0 \in E$ such that $\|\mathbf{x} - \mathbf{y}_0\| \leq \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{y} \in E$.*

Simple examples show that a best approximation to a vector $\mathbf{x}$ from a nonempty subset E of $\mathbb{K}^{n\times 1}$ may not exist, and if it exists, it may not be unique. However, if E is in fact a subspace of $\mathbb{K}^{n\times 1}$, then the following noteworthy result holds.

Proposition

Let Y be a subspace of $\mathbb{K}^{n\times 1}$ and let $\mathbf{x} \in \mathbb{K}^{n\times 1}$. Then there is a unique best approximation to $\mathbf{x}$ from Y, namely, $P_Y(\mathbf{x})$.

Further, $P_Y(\mathbf{x})$ is the unique element of Y such that $\mathbf{x} - P_Y(\mathbf{x})$ is orthogonal to Y. Also, the square of the distance from $\mathbf{x}$ to its best approximation from Y is

$$\|\mathbf{x} - P_Y(\mathbf{x})\|^2 = \|\mathbf{x}\|^2 - \langle \mathbf{x},\, P_Y(\mathbf{x})\rangle.$$

Proof. We note that $P_Y(\mathbf{x}) \in Y$ and $\big(\mathbf{x} - P_Y(\mathbf{x})\big) \in Y^\perp$.

Let $\mathbf{y} \in Y$. Then $P_Y(\mathbf{x}) - \mathbf{y}$ also belongs to Y. Hence by the Pythagorus theorem,

$$\begin{aligned}
\|\mathbf{x} - \mathbf{y}\|^2 &= \big\|\big(\mathbf{x} - P_Y(\mathbf{x})\big) + \big(P_Y(\mathbf{x}) - \mathbf{y}\big)\big\|^2 \\
&= \|\mathbf{x} - P_Y(\mathbf{x})\|^2 + \|P_Y(\mathbf{x}) - \mathbf{y}\|^2 \\
&\geq \|\mathbf{x} - P_Y(\mathbf{x})\|^2,
\end{aligned}$$

where equality holds if and only if $\mathbf{y} = P_Y(\mathbf{x})$. This shows that

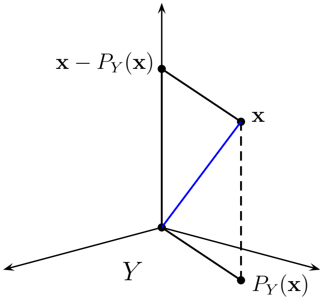
$P_Y(\mathbf{x})$ is the unique best approximation to $\mathbf{x}$ from Y.

Further, let $\mathbf{y} \in Y$ be such that $(\mathbf{x} - \mathbf{y}) \in Y^\perp$. Then $\mathbf{x} = \mathbf{y} + (\mathbf{x} - \mathbf{y})$, and since $\mathbb{K}^{n\times 1} = Y \oplus Y^\perp$, it follows that $\mathbf{y} = P_Y(\mathbf{x})$.

Finally, since $P_Y(\mathbf{x}) \in Y$ and $\big(\mathbf{x} - P_Y(\mathbf{x})\big) \in Y^\perp$,

$$\begin{aligned} \|\mathbf{x} - P_Y(\mathbf{x})\|^2 &= \langle \mathbf{x} - P_Y(\mathbf{x}),\, \mathbf{x} - P_Y(\mathbf{x})\rangle \\ &= \langle \mathbf{x},\, \mathbf{x} - P_Y(\mathbf{x})\rangle \\ &= \|\mathbf{x}\|^2 - \langle \mathbf{x},\, P_Y(\mathbf{x})\rangle. \end{aligned}$$

□



Remark

As we have seen before, if $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is an orthonormal basis for Y, then $P_Y(\mathbf{x}) = \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \cdots + \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k$, and so

$$\left\| \mathbf{x} - P_Y(\mathbf{x}) \right\|^2 = \|\mathbf{x}\|^2 - \langle \mathbf{x}, P_Y(\mathbf{x}) \rangle = \|\mathbf{x}\|^2 - \sum_{j=1}^{k} |\langle \mathbf{x}, \mathbf{u}_j \rangle|^2.$$

We now give an application of the above considerations to a linear system $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A} \in \mathbb{K}^{m \times n}$ and $\mathbf{b} \in \mathbb{K}^{m \times 1}$.

Let $\mathbf{A} = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_n \end{bmatrix}$ in terms of its n columns. Now the above system has a solution $\mathbf{x} := \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^{\mathsf{T}}$ if and only if $x_1 \mathbf{c}_1 + \cdots + x_n \mathbf{c}_n = \mathbf{b}$, that is, $\mathbf{b}$ belongs to the column space $\mathcal{C}(\mathbf{A})$ of $\mathbf{A}$.

Suppose $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$. We would like to find the best approximation to $\mathbf{b}$ from the subspace $\mathcal{C}(\mathbf{A})$ of $\mathbb{K}^{m \times 1}$.

Starting with the first column $\mathbf{c}_1$ of $\mathbf{A}$, and using the G-S OP, we find an ordered orthonormal basis $(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ of $\mathcal{C}(\mathbf{A})$, where $k \leq n$. The best approximation $\mathbf{a}$ to $\mathbf{b}$ from $\mathcal{C}(\mathbf{A})$ is

$$\mathbf{a} := P_{\mathcal{C}(\mathbf{A})}(\mathbf{b}) = \sum_{j=1}^{k} \langle \mathbf{u}_j, \mathbf{b} \rangle \mathbf{u}_j = \sum_{j=1}^{k} (\mathbf{u}_j^* \mathbf{b}) \mathbf{u}_j.$$

Example

Let $\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{b} := \begin{bmatrix} 1 \\ 0 \\ -5 \end{bmatrix}$. Then $\mathcal{C}(\mathbf{A})$ is the span of the 2 column vectors $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^\mathsf{T}$ and $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^\mathsf{T}$ of $\mathbf{A}$. Then $\mathbf{u}_1 := \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}^\mathsf{T}$ and $\mathbf{u}_2 := \begin{bmatrix} 1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix}^\mathsf{T}$ form an orthonormal basis for $\mathcal{C}(\mathbf{A})$. Hence the best approximation to $\mathbf{b}$ from $\mathcal{C}(\mathbf{A})$ is

$$(\mathbf{u}_1^* \mathbf{b}) \mathbf{u}_1 + (\mathbf{u}_2^* \mathbf{b}) \mathbf{u}_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^\mathsf{T} - \frac{3}{2} \begin{bmatrix} 1 & -1 & 2 \end{bmatrix}^\mathsf{T} = \begin{bmatrix} -1 & 2 & -3 \end{bmatrix}^\mathsf{T}.$$

Let us calculate the distance from $\mathbf{b}$ to its best approximation $\mathbf{a}$ from $\mathcal{C}(\mathbf{A})$. Clearly, it is equal to

$$\begin{aligned}\|\mathbf{b}-\mathbf{a}\| &= \| \begin{bmatrix}1 & 0 & -5\end{bmatrix}^{\mathsf{T}} - \begin{bmatrix}-1 & 2 & -3\end{bmatrix}^{\mathsf{T}} \| \\ &= \| \begin{bmatrix}2 & -2 & -2\end{bmatrix}^{\mathsf{T}} \| = 2\sqrt{3}.\end{aligned}$$

Also, according to an earlier formula, the square of this distance is equal to

$$\|\mathbf{b}\|^2 - |\langle \mathbf{b},\, \mathbf{u}_1\rangle|^2 - |\langle \mathbf{b},\, \mathbf{u}_2\rangle|^2 = 26 - \frac{1}{2} - \frac{27}{2} = 12,$$

and so the desired distance is equal to $\sqrt{12} = 2\sqrt{3}$, as obtained above.

We now explain an alternative approach for finding the best approximation $\mathbf{y}_0$ to $\mathbf{b}$ from $\mathcal{C}(\mathbf{A})$. It avoids the G-S OP.

By the previous proposition, $\mathbf{y}_0$ is the unique element of $\mathcal{C}(\mathbf{A})$ such that $(\mathbf{b} - \mathbf{y}_0) \in \mathcal{C}(\mathbf{A})^\perp$, that is, $\mathbf{c}_j^*(\mathbf{b} - \mathbf{y}_0) = 0$, where $\mathbf{c}_j$ is the jth column of $\mathbf{A}$, for $j = 1, \ldots, n$, which means $\mathbf{A}^*(\mathbf{b} - \mathbf{y}_0) = \mathbf{0}$.

Thus we need to find $\mathbf{x}_0 \in \mathbb{K}^{n\times 1}$ such that $\mathbf{A}^*(\mathbf{b} - \mathbf{A}\mathbf{x}_0) = \mathbf{0}$. Hence $\mathbf{y}_0$ can be obtained by finding a solution $\mathbf{x}_0$ of the **normal equations** $\boxed{\mathbf{A}^*\mathbf{A}\,\mathbf{x} = \mathbf{A}^*\mathbf{b}}$, and then letting $\mathbf{y}_0 := \mathbf{A}\mathbf{x}_0$. Since $\mathbf{y}_0 \in \mathcal{C}(\mathbf{A})$, this system of n equations in n unknowns always has a solution.

Suppose $\operatorname{rank} \mathbf{A} = n$, that is, the columns of the matrix $\mathbf{A}$ are linearly independent. Then $\operatorname{rank} \mathbf{A}^*\mathbf{A} = n$, and the unique solution of the normal equations is given by $\mathbf{x}_0 := (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}$. In this case, the best approximation to $\mathbf{b}$ from $\mathcal{C}(\mathbf{A})$ is $\mathbf{y}_0 := \mathbf{A}\mathbf{x}_0 = \mathbf{A}(\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}$. In particular, if $m = n$, then $\mathbf{A}$ is invertible, and $\mathbf{y}_0 = \mathbf{A}\mathbf{A}^{-1}(\mathbf{A}^*)^{-1}\mathbf{A}^*\mathbf{b} = \mathbf{b}$.

Next, suppose rank $\mathbf{A} < n$, that is, the columns of $\mathbf{A}$ are linearly dependent. Then rank $\mathbf{A}^*\mathbf{A} < n$, and the normal equations have infinitely many solutions, but if $\mathbf{x}_0$ is any one of them, then $\mathbf{y}_0 = \mathbf{A}\mathbf{x}_0$.

Example

Let $\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{K}^{3\times 2}$ and $\mathbf{b} := \begin{bmatrix} 1 \\ 0 \\ -5 \end{bmatrix} \in \mathbb{K}^{3\times 1}$. We find the best approximation to $\mathbf{b}$ from the column space of $\mathbf{A}$ using normal equations. Here $\mathbf{A}^*\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{A}^*\mathbf{b} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$.

Now the normal equations $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$, that is, $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$ have a unique solution, namely $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$. Hence $\mathbf{A}\mathbf{x} = \begin{bmatrix} -1 & 2 & -3 \end{bmatrix}^{\mathsf{T}}$ as before.

Next, let $\mathbf{B} := \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \end{bmatrix}$ and $\mathbf{b} := \begin{bmatrix} 1 \\ 0 \\ -5 \end{bmatrix}$. Then $\mathcal{C}(\mathbf{B}) = \mathcal{C}(\mathbf{A})$ since the third column of $\mathbf{B}$ is the average of the first two columns of $\mathbf{A}$, and so the best approximation to $\mathbf{b}$ from $\mathcal{C}(\mathbf{B})$ is the same column vector $\begin{bmatrix} -1 & 2 & -3 \end{bmatrix}^{\mathsf{T}}$ as before. But here

$$\mathbf{B}^*\mathbf{B} = \begin{bmatrix} 2 & 1 & 3/2 \\ 1 & 2 & 3/2 \\ 3/2 & 3/2 & 3/2 \end{bmatrix} \quad \text{and} \quad \mathbf{B}^*\mathbf{b} = \begin{bmatrix} 1 \\ -4 \\ -3/2 \end{bmatrix}.$$

Now the normal equations $\mathbf{B}^*\mathbf{B}\mathbf{x} = \mathbf{B}^*\mathbf{b}$ have many solutions such as $\begin{bmatrix} 2 & -3 & 0 \end{bmatrix}^{\mathsf{T}}$ or $\begin{bmatrix} 3 & -2 & -2 \end{bmatrix}^{\mathsf{T}}$, since the columns of $\mathbf{B}$ are linearly dependent. However,

$$\mathbf{B} \begin{bmatrix} 2 & -3 & 0 \end{bmatrix}^{\mathsf{T}} = \mathbf{B} \begin{bmatrix} 3 & -2 & -2 \end{bmatrix}^{\mathsf{T}} = \begin{bmatrix} -1 & 2 & -3 \end{bmatrix}^{\mathsf{T}} \text{ as before.}$$

Least Squares Approximation

Suppose a large number of data points $(a_1, b_1), \ldots, (a_n, b_n)$ in $\mathbb{R}^2$ are given, and we want to find a straight line passing through these points. If all these points are collinear, then we may join any two of them by a straight line, which will work for us. If, however, these points are not collinear (which is most often the case), then we want to find a straight line $t = x_1 + s\,x_2$ in the st-plane which is most suitable in the following sense.

Consider the 'error' $|x_1 + a_j x_2 - b_j|$ caused because of the straight line $t = x_1 + s\,x_2$ not passing through the data point (a_j, b_j) for $j = 1, \ldots, n$. We collect these errors and attempt to find $x_1, x_2 \in \mathbb{R}$ such that the **least squares error**

$$\left(\sum_{j=1}^{n} |x_1 + a_j x_2 - b_j|^2\right)^{1/2}.$$

is minimized. This is known as the **least squares problem**.

To solve this problem, let

$$\mathbf{A} := \begin{bmatrix} 1 & a_1 \\ 1 & a_2 \\ \vdots & \vdots \\ 1 & a_n \end{bmatrix}, \quad \mathbf{x} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \mathbf{b} := \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Then $\mathbf{Ax} = \begin{bmatrix} x_1 + a_1x_2 & x_1 + a_2x_2 & \cdots & x_1 + a_nx_2 \end{bmatrix}^{\mathsf{T}}$.

The least squares problem is to find $\mathbf{x} \in \mathbb{R}^{2\times 1}$ such that

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \sum_{j=1}^{n} |x_1 + a_jx_2 - b_j|^2$$

is minimised, that is, to find the best approximation to the vector $\mathbf{b}$ from the column space $\mathcal{C}(\mathbf{A}) = \{\mathbf{Ax} : \mathbf{x} \in \mathbb{R}^{2\times 1}\}$ of $\mathbf{A}$.

As we have seen, this is done either by orthonormalizing the columns of $\mathbf{A}$, or by finding a vector $\mathbf{x}_0 := \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathsf{T}}$ such that $\mathbf{b} - \mathbf{Ax}_0$ is orthogonal to $\mathcal{C}(\mathbf{A})$ using the normal equations.

Instead of fitting a straight line $t = x_1 + s\,x_2$ to the data points, we may attempt to fit a curve given by a polynomial $t = x_1 + s\,x_2 + s^2 x_3 + \cdots + s^k x_{k+1}$ of degree $k \in \mathbb{N}$, to the data points $(a_1, b_1), \ldots, (a_n, b_n)$. To solve this problem, let

$$\mathbf{A} := \begin{bmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^k \\ 1 & a_2 & a_2^2 & \cdots & a_2^k \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^k \end{bmatrix}, \quad \mathbf{x} := \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k+1} \end{bmatrix} \quad \text{and} \quad \mathbf{b} := \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Then $\mathbf{Ax}$ is equal to
$\begin{bmatrix} x_1 + a_1 x_2 + \cdots + a_1^k x_{k+1} & \cdots & x_1 + a_n x_2 + \cdots + a_n^k x_{k+1} \end{bmatrix}^{\mathsf{T}}$.

As before, the least squares problem is to minimize

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \sum_{j=1}^{n} |x_1 + a_j x_2 + \cdots + a_j^k x_{k+1} - b_j|^2$$

by finding the best approximation to the vector $\mathbf{b}$ from the column space $\mathcal{C}(\mathbf{A}) = \{\mathbf{Ax} : \mathbf{x} \in \mathbb{R}^{(k+1)\times 1}\}$ of $\mathbf{A}$.

Example

Let us find a straight line $t = x_1 + s\,x_2$ that best fits the data points $(-1, 1), (1, 1), (2, 3)$ in the least squares sense. For this purpose, we find the orthogonal projection of the vector $\mathbf{b} := \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ into the column space of the matrix $\mathbf{A} := \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$.

Then $\mathbf{A}^*\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$ and $\mathbf{A}^*\mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$.

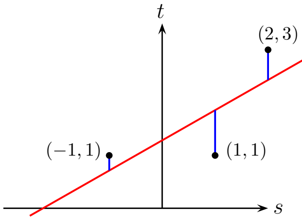
Hence the unique solution of the normal equations $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$ is $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^\mathsf{T} := \begin{bmatrix} 9/7 & 4/7 \end{bmatrix}^\mathsf{T}$. Thus the straight line which best fits the data is $7t = 9 + 4s$.

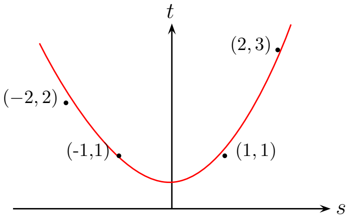
Next, we find a parabola $t = x_1 + s\,x_2 + s^2 x_3$ that best fits the data points $(-1, 1), (1, 1), (2, 3), (-2, 2)$ in the least squares sense.

Let $\mathbf{A} := \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & -2 & 4 \end{bmatrix}$ and $\mathbf{b} := \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix}$.

Then $\mathbf{A}^*\mathbf{A} = \begin{bmatrix} 4 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix}$ and $\mathbf{A}^*\mathbf{b} = \begin{bmatrix} 7 & 2 & 22 \end{bmatrix}$.

The unique solution of the normal equations $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$ is $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^\mathsf{T} := \begin{bmatrix} 1/2 & 1/5 & 1/2 \end{bmatrix}^\mathsf{T}$. Thus the parabola which best fits the data is $10t = 5 + 2s + 5s^2$.





Abstract Vector Spaces

Throughout the course so far, we have dealt with row vectors, column vectors and matrices. We have introduced many interesting concepts such as linear independence of vectors, span of a set of vectors, a subspace of vectors, a basis for a subspace, the dimension of a subspace, the nullity and the rank of a matrix, linear transformations induced by matrices, inner product of two vectors, orthogonality of vectors, orthonormal basis for a subspace, orthogonal projection onto a subspace, and so on.

Based on these concepts, we have proved some important theorems like the Rank-Nullity theorem, the Fundamental Theorem for Linear Systems, the Spectral Theorem, the Projection Theorem.

Now we discuss these concepts in a more abstract setting.

Abstract Vector Space

Definition Let $\mathbb{K}$ denote $\mathbb{R}$ or $\mathbb{C}$ as usual. A **vector space** over $\mathbb{K}$ is a nonempty set V together with the operation of **addition** (i.e., a map $V \times V \to V$ given by $(u, v) \mapsto u + v$) and of **scalar multiplication** (i.e., a map $\mathbb{K} \times V \to V$ given by $(\alpha, v) \mapsto \alpha v$) satisfying the following properties.

I Closure axioms

1. $u + v \in V$ for all $u, v \in V$.
2. $\alpha \cdot v \in V$ for all $\alpha \in \mathbb{K}$ and $v \in V$.

(We shall write αv instead of $\alpha \cdot v \in V$ hence onward.)

II Axioms for addition

1. $u + v = v + u$ for all $u, v \in V$. (commutativity)
2. $u+(v+w)=(u+v)+w$ for all $u, v, w \in V$. (associativity)
3. There is (unique) $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
4. For $v \in V$, there is (unique) $u \in V$ such that $v + u = 0$.

(We shall write this element u as $-v$.)

III Axioms for scalar multiplication

1. $\alpha(\beta v) = (\alpha\beta)v$ for all $\alpha, \beta \in \mathbb{K}$ and $v \in V$.
2. $\alpha(u + v) = \alpha u + \alpha v$ for all $\alpha \in \mathbb{K}$ and $u, v \in V$.
3. $(\alpha + \beta)v = \alpha v + \beta v$ for all $\alpha, \beta \in \mathbb{K}$ and $v \in V$.
4. $1v = v$ for all $v \in V$.

An element of a vector space is called a **vector**.

Examples: $\boxed{1}$ $\mathbb{K}^n = \mathbb{K}^{n\times 1}$ is a vector space over $\mathbb{K}$. More generally, every vector subspace of $\mathbb{K}^n$ is a vector space over $\mathbb{K}$. Likewise for $\mathbb{K}^{1\times n}$.

$\boxed{2}$ $\mathbb{K}^{m\times n}$, the space of all $m \times n$ matrices with entries in $\mathbb{K}$, is a vector space over $\mathbb{K}$ with respect to the addition and scalar multiplication of matrices.

$\boxed{3}$ The set $\mathbb{K}[x]$ of all polynomials in the indeterminate x with coefficients in $\mathbb{K}$ is a vector space over $\mathbb{K}$ with respect to the usual addition and scalar multiplication of polynomials.

$\boxed{4}$ The set $\mathcal{P}_n$ of polynomials in $\mathbb{K}[x]$ of degree $\leq n$ is a vector space with addition and scalar multiplication as in $\boxed{3}$.

5 Let $a, b \in \mathbb{R}$ with $a < b$. Then the space $C[a, b]$ of all continuous functions from $[a, b]$ to $\mathbb{R}$ a vector space over $\mathbb{R}$ with respect to pointwise addition and pointwise scalar multiplication of functions.

6 Let $a, b \in \mathbb{R}$ with $a < b$. Then the space $C^1[a, b]$ of all continuously differentiable functions from $[a, b]$ to $\mathbb{R}$ a vector space over $\mathbb{R}$ with addition and scalar multiplication as in 5.

7 The space c of all convergent sequences of real numbers is a vector space over $\mathbb{R}$ with respect to pointwise addition and pointwise scalar multiplication of sequences.

Exercise: Verify that the above spaces are indeed vector spaces, i.e., all the axioms in the definition are satisfied.

Definition

Let V be a vector space (over $\mathbb{K}$). A nonempty subset W of V is called a **subspace** *of V if $v + w \in W$ for all $v, w \in W$ and $\alpha w \in W$ for all $\alpha \in \mathbb{K}$ and $w \in W$.*

Definition

Let V be a vector space (over $\mathbb{K}$), and let $n \in \mathbb{N}$. Given $v_1, \ldots, v_n \in V$ and $\alpha_1, \ldots, \alpha_n \in \mathbb{K}$, the element

$$\alpha_1 v_1 + \cdots + \alpha_n v_n$$

of V is called the **linear combination** *of the vectors $v_1, \ldots, v_n$ with* **coefficients** *$\alpha_1, \ldots, \alpha_n$.*

Let V be a vector space. If W is a subspace of V, then clearly every linear combination of the elements of W belongs W.

Let $W_1. W_2$ be subspaces of V. Then it is easy to see that:

- $W_1 \cap W_2$ is a subspace of V. In fact it is the largest subspace of V which is contained in both W_1 and W_2.
- $W_1 \cup W_2$ need not be a subspace of V.
- $W_1 + W_2 := \{w_1 + w_2 : w_1 \in W_1 \text{ and } w_2 \in W_2\}$ is a subspace of V. In fact it is the smallest subspace of V containing both W_1 and W_2.

The notions in $\mathbb{R}^n$ of span of a set of vectors, linear dependence and independence of vectors, the dimension of a subspace of vectors carry over to an abstract vector space without any difficulty. Let V be a vector space (over $\mathbb{K}$).

Definition

Let $S \subset V$. The set of all linear combinations of elements of S is called the **span** *of S, and we denote it by* $\boxed{\text{span } S}$.

Definition

A subset S of V is called **linearly dependent** *if there are $v_1, \ldots, v_m$ in S and there are $\alpha_1, \ldots, \alpha_m \in \mathbb{K}$, not all zero, satisfying*

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0.$$

It can be seen that S is linearly dependent $\iff$ either $\mathbf{0} \in S$ or a vector in S is a linear combination of other vectors in S.

Definition

A subset S of V is called **linearly independent** *if it is not linearly dependent, that is,*

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0 \implies \alpha_1 = \cdots = \alpha_m = 0,$$

whenever $v_1, \ldots, v_m \in S$ and $\alpha_1, \ldots, \alpha_m \in \mathbb{K}$.

Clearly, S is linearly independent if and only if $\mathbf{0} \notin S$ and no elements of S is a linear combination of other elements of S.

The following crucial result was proved for the case $V := \mathbb{R}^{n\times 1}$. Exactly the same proof works in the general case.

Proposition

Let S be a subset of s elements and R be a set of r elements of V. If $S \subset \operatorname{span} R$ and $s > r$, then S is linearly dependent.

Examples

$\boxed{1}$. Let $m, n \in \mathbb{N}$, and let $V := \mathbb{K}^{m\times n}$ be set of all $m\times n$ matrices with entries in $\mathbb{K}$ with entry-wise addition and scalar multiplication. For $j = 1, \ldots, m$ and $k = 1, \ldots, n$, let $\mathbf{E}_{jk}$ denote the $m\times n$ matrix whose (j, k)th entry is equal to 1 and all other entries are equal to zero. Then the set $S := \{\mathbf{E}_{jk} : 1 \le j \le m,\ 1 \le k \le, n\}$ is linearly independent.

$\boxed{2}$. Let $V := c_0$ denote the set of all sequences in $\mathbb{K}$ which converge to 0. For $j \in \mathbb{N}$, let e_j denote the element of S whose j-th term is equal to 1 and all other terms are equal to 0. Then the set $S := \{e_j : j \in \mathbb{N}\}$ is linearly independent. Next, let $S_1 := S \cup \{e\}$, where the nth entry of the sequence e is equal to $1/n$ for $n \in \mathbb{N}$. Then the set S_1 is also linearly independent since e is not a (finite) linear combination of elements of S.

3. Let $V := \mathbb{K}[x]$ denote the set of all polynomials in the indeterminate x with coefficients in $\mathbb{K}$. Then the set $S := \{x^j : j = 0, 1, 2, \ldots\}$ is linearly independent.
Next, let $S_1 := S \cup \{p\}$, where $p \in \mathbb{K}[x]$. Then the set S_1 is linearly dependent since $p \in \operatorname{span} S$.

4. Let $V := C[-\pi, \pi]$. For $n \in \mathbb{N}$, let

$$u_n(t) := \cos nt \quad \text{and} \quad v_n(t) := \sin nt \qquad \text{for } t \in [-\pi, \pi].$$

Then the set $S := \{u_1, u_2, \ldots\} \cup \{v_1, v_2, \ldots\}$ is linearly independent. (Note that the zero element of this vector space is the function having all its values on $[-\pi, \pi]$ equal to 0.) To prove this, use the idea that if $\alpha \cos t + \beta \sin t = 0$, then by differentiating, we also have $-\alpha \sin t + \beta \cos t = 0$.

Next, let $S_1 := S \cup \{w\}$, where $w(t) := t$ for $t \in [-\pi, \pi]$. Then the set S_1 is also linearly independent, since $w(\pi) \neq w(-\pi)$, and so $w \notin \operatorname{span} S$.

Definition

A vector space V is said to be **finite dimensional** *if there is a finite subset S of V such that $V = \operatorname{span} S$; otherwise the vector space V is said to be* **infinite dimensional**.

If a vector space V is infinite dimensional, then V is larger than the span of any finite subset of V, and so V must contain an infinite linearly independent subset. Conversely, if V contains an infinite linearly independent subset, then V must be infinite dimensional.

Examples: Let $n, m \in \mathbb{N}$. The vector spaces $\mathbb{K}^{n\times 1}$, $\mathbb{K}^{1\times n}$ and $\mathbb{K}^{m\times n}$ are finite dimensional, and so is the vector space $\mathcal{P}_n$ of all polynomials in the indeterminate x having degree less than or equal to n. But the vector spaces $\mathbb{K}[x]$, $C[-\pi, \pi]$, c, and c_0 are infinite dimensional.

MA110
Lecture 20

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Spring 2025

Cayley-Hamilton theorem

Theorem. Let $\mathbf{A} \in \mathbb{K}^{n\times n}$ and let $p(t)$ be its characteristic polynomial. Then $p(\mathbf{A}) = \mathbf{0}$.

Proof. Let $\mathbf{B}$ be the adjugate of the matrix $(t\mathbf{I}_n - \mathbf{A})$. Then we have the following formula for determinant, where $p(t)$ is the characteristic polynomial:

$$(t\mathbf{I}_n - \mathbf{A})\mathbf{B} = \det(t\mathbf{I}_n - \mathbf{A})\mathbf{I}_n = p(t)\mathbf{I}_n.$$

Now the (j,k)-th entry of the adjugate of $(t\mathbf{I}_n - \mathbf{A})$ is obtained by taking determinant of an $(n-1)\times(n-1)$-submatrix. Thus, the (j,k)-th entry of $\mathbf{B}$ is a polynomial in t of degree $\leq n-1$. Thus we can say, $\mathbf{B} = \mathbf{B}_0 + t\mathbf{B}_1 + \dots t^{n-1}\mathbf{B}_{n-1}$ for matrices $\mathbf{B}_j \in \mathbb{K}^{n\times n}$.

$$
\begin{aligned}
p(t)\mathbf{I}_n &= (t\mathbf{I}_n - \mathbf{A})\mathbf{B} && (1)\\
&= (tI_n - \mathbf{A})\sum_{i=0}^{n-1} t^i \mathbf{B}_i && (2)\\
&= \sum_{i=0}^{n-1} t\mathbf{I}_n \cdot t^i \mathbf{B}_i - \sum_{i=0}^{n-1} \mathbf{A} \cdot t^i \mathbf{B}_i && (3)\\
&= \sum_{i=0}^{n-1} t^{i+1}\mathbf{B}_i - \sum_{i=0}^{n-1} t^i \mathbf{A}\mathbf{B}_i && (4)\\
&= t^n \mathbf{B}_{n-1} + \sum_{i=1}^{n-1} t^i (\mathbf{B}_{i-1} - \mathbf{A}\mathbf{B}_i) - \mathbf{A}\mathbf{B}_0. && (5)
\end{aligned}
$$

It is clear that $p(A) = 0$. $\square$

Recall: We have defined the notion of (abstract) vector space over $\mathbb{K}$ and the notion of subspace of a vector space. We looked at some of the examples of vector spaces such as

- Subspaces of $\mathbb{K}^{n\times 1}$
- The space $\mathbb{K}^{m\times n}$ of all $m \times n$ matrices with entries in $\mathbb{K}$
- The spaces $\mathbb{K}[X]$ and $\mathcal{P}_n$ of polynomials in one variable
- The spaces $C[a, b]$ and $C^1[a, b]$ of functions $[a, b] \to \mathbb{K}$
- The space c of convergent sequences of real numbers

We have also seen that several notions and results discussed over $\mathbb{R}^n$ have a straightforward analogue in the context of a general vector space V over $\mathbb{K}$. These are as follows.

- Linear combination
- span
- Linear dependence
- Linear independence

Proposition (Crucial Result)

Let S be a subset of s elements and R be a set of r elements of V. If $S \subset \operatorname{span} R$ and $s > r$, then S is linearly dependent.

Examples

$\boxed{1}$. Let $m, n \in \mathbb{N}$, and let V be the vector space $\mathbb{K}^{m\times n}$. of all $m\times n$ matrices with entries in $\mathbb{K}$. For $j = 1, \ldots, m$ and $k = 1, \ldots, n$, let $\mathbf{E}_{jk}$ denote the $m\times n$ matrix whose (j,k)th entry is equal to 1 and all other entries are equal to zero. Then the set $S := \left\{\mathbf{E}_{jk} : 1 \le j \le m,\, 1 \le k \le, n\right\}$ is linearly independent. Moreover S spans V.

$\boxed{2}$. Let $V := c_0$ be the subpace of c consisting of all sequences in $\mathbb{R}$ which converge to 0. For $j \in \mathbb{N}$, let e_j denote the element of S whose jth term is equal to 1 and all other terms are equal to 0. Then the set $S := \left\{e_j : j \in \mathbb{N}\right\}$ is linearly independent. However, S does not span V. To see this, let $e = (1/n)$ be the sequence whose nth term is $1/n$ for $n \in \mathbb{N}$. Then $e \in c_0$, but e is not a (finite) linear combination of elements of S. Thus $S_1 := S \cup \{e\}$ is also linearly independent

.

$\boxed{3}$. Let $V := \mathbb{K}[x]$ be the vector space of all polynomials in the indeterminate x with coefficients in $\mathbb{K}$. Then the set $S := \{x^j : j = 0, 1, 2, \ldots\}$ is linearly independent. Moreover S spans V. For a fixed $n \in \mathbb{N}$, the set $S_n := \{x^j : 0 \leq j \leq n\}$ is linearly independent and it spans the subspace $\mathcal{P}_n$ of $\mathbb{K}[X]$.

$\boxed{4}$. Let $V := C[-\pi, \pi]$. For $n \in \mathbb{N}$, define $u_n, v_n \in V$ by

$$u_n(t) := \cos nt \quad \text{and} \quad v_n(t) := \sin nt \qquad \text{for } t \in [-\pi, \pi].$$

Then the set $S := \{u_1, u_2, \ldots\} \cup \{v_1, v_2, \ldots\}$ is linearly independent. (Note that the zero element of this vector space is the function having all its values on $[-\pi, \pi]$ equal to 0.)

But S doesn't span V. To see this, consider $w(t) := t$ for $t \in [-\pi, \pi]$. Then the set $S_1 := S \cup \{w\}$ is also linearly independent, since $w(\pi) \neq w(-\pi)$, and so $w \notin \operatorname{span} S$.

Definition

A vector space V is said to be **finite dimensional** *if there is a finite subset S of V such that $V = \operatorname{span} S$; otherwise the vector space V is said to be* **infinite dimensional**.

If a vector space V is infinite dimensional, then V is larger than the span of any finite subset of V, and so V must contain an infinite linearly independent subset. Conversely, if V contains an infinite linearly independent subset, then V must be infinite dimensional.

Examples: Let $n, m \in \mathbb{N}$. The vector spaces $\mathbb{K}^{n\times 1}$, $\mathbb{K}^{1\times n}$ and $\mathbb{K}^{m\times n}$ are finite dimensional, and so is the vector space $\mathcal{P}_n$ of all polynomials in the indeterminate x having degree less than or equal to n. But the vector spaces $\mathbb{K}[x]$, $C[-\pi, \pi]$, c, and c_0 are infinite dimensional.

Definition

Any linearly independent subset of a finite dimensional vector space V which spans V is called a **basis** *for V.*

Here is the most important result about finite dimensional vector spaces. The proof is similar to that in the case of subspaces of $\mathbb{K}^n$.

Proposition

Let V be a finite dimensional vector space over $\mathbb{K}$. Then:

- *V has a basis.*
- *Every set that spans V has a subset which is a basis of V.*
- *Every linearly independent subset of V can be extended to a basis of V.*
- *Any two bases of V have the same cardinality, called the* **dimension** *of V and denoted by* $\dim V$.

Linear Transformations

Definition

Let V and W be vector spaces over $\mathbb{K}$. A **linear transformation** *or a* **linear map** *from V to W is a function $T : V \to W$ which 'preserves' the operations of addition and scalar multiplication, that is, for all $u, v \in V$ and $\alpha \in \mathbb{K}$,*

$$T(u+v) = T(u) + T(v) \quad \textit{and} \quad T(\alpha\, v) = \alpha\, T(v).$$

It is clear that if $T : V \to W$ is linear, then $T(0) = 0$. Also, T 'preserves' linear combinations of elements of V:

$$T(\alpha_1 v_1 + \cdots + \alpha_k v_k) = \alpha_1 T(v_1) + \cdots + \alpha_k T(v_k)$$

for all $k \in \mathbb{N}$, $v_1, \ldots, v_k \in V$ and $\alpha_1, \ldots, \alpha_k \in \mathbb{K}$.

Remark: A linear transformation from a vector space V to itself is often called a linear operator on V.

Examples

$\boxed{1}$. Let $\mathbf{A}$ be an $m \times n$ matrix with entries in $\mathbb{K}$. Then the map $T : \mathbb{K}^{n\times 1} \to \mathbb{K}^{m\times 1}$ defined by $T(\mathbf{x}) := \mathbf{A}\,\mathbf{x}$ is linear. Similarly, the map $T' : \mathbb{K}^{1\times m} \to \mathbb{K}^{1\times n}$ defined by $T'(\mathbf{y}) := \mathbf{yA}$ is linear. More generally, the map

$$T : \mathbb{K}^{n\times p} \to \mathbb{K}^{m\times p} \quad \text{defined by} \quad T(\mathbf{X}) := \mathbf{A}\,\mathbf{X}$$

is linear, and the map

$$T' : \mathbb{K}^{p\times m} \to \mathbb{K}^{p\times n} \quad \text{defined by} \quad T'(\mathbf{Y}) := \mathbf{YA}$$

is linear.

$\boxed{2}$. $T : \mathbb{K}^{m\times n} \to \mathbb{K}^{n\times m}$ defined by $T(\mathbf{A}) := \mathbf{A}^{\mathsf{T}}$ is linear.

$\boxed{3}$. The map $T : \mathbb{K}^{n\times n} \to \mathbb{K}$ defined by $T(\mathbf{A}) := \operatorname{trace}\mathbf{A}$ is linear. But $\mathbf{A} \longmapsto \det\mathbf{A}$ does not define a linear map.

$\boxed{4}$. The map $T : \mathbb{K}[X] \to \mathbb{K}$ defined by $T(p(X)) = p(0)$ is linear.

5. Let $V := c_0$, the set of all sequences in $\mathbb{K}$ which converge to 0. Then the map $T : V \to V$ defined by

$$T(x_1, x_2, \dots) := (0, x_1, x_2, \dots)$$

is linear, and so is the map $T' : V \to V$ defined by

$$T'(x_1, x_2, \dots) := (x_2, x_3, \dots).$$

Note that $T' \circ T$ is the identity map on V, but $T \circ T'$ is not the identity map on V. The map T is called the **right shift operator** and T' is called the **left shift operator** on V.

6. Let $V := C^1([a, b])$, the set of all real-valued continuously differentiable functions, and let $W := C([a, b])$, the set of all real-valued continuous functions on $[a, b]$. Then the map $T' : V \to W$ defined by $T'(f) = f'$ is linear. Also, the map

$$T : W \to V \text{ defined by } \ T(f)(x) := \int_a^x f(t)dt \ \text{ for } x \in [a, b],$$

is linear. [Question. What are $T' \circ T$ and $T' \circ T$?]

Let V and W be vector spaces over $\mathbb{K}$, and let $T : V \to W$ be a linear map. Two important subspaces associated with T are

(i) $\mathcal{N}(T) := \{v \in V : T(v) = 0\}$, the **null space** of T, which is a subspace of V,

(ii) $\mathcal{I}(T) := \{T(v) : v \in V\}$, the **image space** of T, which is a subspace of W.

Suppose V is finite dimensional, and let $\dim V = n$. Since $\mathcal{N}(T)$ is a subspace of V, it is finite dimensional and $\dim \mathcal{N}(T) \leq n$

Let $v_1, \ldots, v_n$ be a basis for V. If $v \in V$, then there are $\alpha_1, \ldots, \alpha_n \in \mathbb{K}$ such that $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$, so that $T(v) = \alpha_1 T(v_1) + \cdots + \alpha_n T(v_n)$. This shows that $\mathcal{I}(T) = \text{span}\{T(v_1), \ldots, T(v_n)\}$. Hence $\mathcal{I}(T)$ is also finite dimensional and $\dim \mathcal{I}(T) \leq n$.

Definition

The dimension of $\mathcal{N}(T)$ is called the **nullity** *of the linear map T, and the dimension of $\mathcal{I}(T)$ is called the* **rank** *of T.*

The Rank-Nullity Theorem for a matrix $\mathbf{A}$ that we proved earlier is a special case of the following result.

Proposition (Rank-Nullity Theorem for Linear Maps)

Let V and W be vector spaces over $\mathbb{K}$, and let $T : V \to W$ be a linear map. Suppose $\dim V = n \in \mathbb{N}$. Then

$$\mathrm{rank}(T) + \mathrm{nullity}(T) = n.$$

Proof (Sketch): Let $s := \mathrm{nullity}(T)$ and let $\{u_1, \ldots, u_s\}$ be a basis of $\mathcal{N}(T)$. Extend the linearly independent set $\{u_1, \ldots, u_s\}$ to a basis $\{u_1, \ldots, u_s, u_{s+1}, \ldots, u_n\}$ of V. Check that the set $\{T(u_{s+1}), \ldots, T(u_n)\}$ is a basis of $\mathcal{I}(T)$. $\square$

Corollary

Let V, W be finite dimensional vector spaces with $\dim V = n$ and $\dim W = m$. Also, let $T : V \to W$ be a linear map. Then

$$T \textit{ is one-one} \iff \operatorname{rank}(T) = n.$$

In particular, if T is one-one, then $n \leq m$. Further,

$$\textit{if } m = n, \textit{ then} \quad T \textit{ is one-one} \iff T \textit{ is onto.}$$

Proof. The first assertion follows from the Rank-Nullity Theorem since

$$T \text{ is one-one} \iff \mathcal{N}(T) = \{0\} \iff \operatorname{nullity}(T) = 0.$$

If T is one-one, then $n = \operatorname{rank}(T) = \dim \mathcal{I}(T) \leq \dim W = m$. Further, if $m = n$, then $\operatorname{rank}(T) = n \iff T$ is onto. $\square$

MA110
Lecture 21

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Spring 2025

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Corollary

Let V, W be finite dimensional vector spaces with $\dim V = n$ and $\dim W = m$. Also, let $T : V \to W$ be a linear map. Then

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Proof. The first assertion follows from the Rank-Nullity Theorem since

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If T is one-one, then $n = \operatorname{rank}(T) = \dim \mathcal{I}(T) \leq \dim W = m$. Further, if $m = n$, then $\operatorname{rank}(T) = n \iff T$ is onto. $\square$

As another application of the Rank-Nullity Theorem, we find an interesting relation between dimensions of finite dimensional subspaces of a vector space.

Proposition

Let W_1 and W_2 be finite dimensional subspaces of a vector space V. Then

$$\dim W_1 + \dim W_2 = \dim(W_1 \cap W_2) + \dim(W_1 + W_2).$$

Proof. The dimension of the vector space $W_1 \times W_2$ is equal to $\dim W_1 + \dim W_2$. Define $T : W_1 \times W_2 \to W_1 + W_2$ by $T(w_1, w_2) := w_1 - w_2$. Then T is linear, and

$$\mathcal{N}(T) = \{(w, w) : w \in W_1 \cap W_2\} \text{ and } \mathcal{I}(T) = W_1 + W_2.$$

Hence by the Rank-Nullity Theorem,

$$\dim(W_1 + W_2) + \dim(W_1 \cap W_2) = \dim(W_1 \times W_2). \quad \square$$

To a linear map from a finite dimensional vector space to another finite dimensional space, one can associate a matrix exactly as before.

Let V be a vector space of dimension n, and let $E := (v_1, \ldots, v_n)$ be an ordered basis for V. Also, let W be a vector space of dimension m, and let $F := (w_1, \ldots, w_m)$ be an ordered basis for W. Let $T : V \to W$ be a linear map. Then for each $k = 1, \ldots, n$, we can uniquely write

$$T(v_k) = a_{1k} w_1 + \cdots + a_{mk} w_m = \sum_{j=1}^{m} a_{jk} w_j \quad \text{for some } a_{jk} \in \mathbb{K}.$$

The $m \times n$ matrix $\mathbf{A} := [a_{jk}]$ is called the **matrix of the linear transformation** $T : V \to W$ with respect to the ordered basis $E := (v_1, \ldots, v_n)$ of V and the ordered basis $F := (w_1, \ldots, w_m)$ of W. It is denoted by $\boxed{\mathbf{M}_F^E(T)}$.

Examples 1. Define $T : \mathcal{P}_n \to \mathcal{P}_{n-1}$ by $T(p) = p'$, the derivative of p. Consider the ordered bases $E := (1, t, \ldots, t^n)$ and $F := (1, t, \ldots, t^{n-1})$ of $\mathcal{P}_n$ and $\mathcal{P}_{n-1}$ respectively. Then the $n \times (n+1)$ matrix of the linear map T with respect to these bases is

$$\mathbf{M}_F^E(T) := \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & n \end{bmatrix}.$$

2. Let $T : \mathbb{K}^{2\times 2} \to \mathbb{K}^{2\times 2}$ be the linear transformation defined by $T(A) = A^{\mathsf{T}}$. Then the matrix of T with respect to the basis $E := \{\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22}\}$ is $\mathbf{M}_E^E(T) := \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Eigenvalues and Eigenvectors of Linear Operators

Definition

Let V be a vector space over $\mathbb{K}$, and let $T : V \to V$ be a **linear operator**. *A scalar $\lambda \in \mathbb{K}$ is called an* **eigenvalue** *of T if there is a nonzero $v \in V$ such that $T(v) = \lambda v$, and then v is called an* **eigenvector** *or an* **eigenfunction** *of T corresponding to λ, and the subspace $\mathcal{N}(T - \lambda I)$ is called the* **eigenspace** *of T.*

Example: Let V denote the vector space $C^\infty(\mathbb{R})$ of all real-valued infinitely differentiable functions on $\mathbb{R}$. Define $T(f) = f'$ for $f \in V$. Then T is a linear operator on V.

Given $\lambda \in \mathbb{R}$, consider $f_\lambda(t) := e^{\lambda t}$ for $t \in \mathbb{R}$. Then $f_\lambda \in V$, $f_\lambda \neq 0$ and $T(f_\lambda) = \lambda\, f_\lambda$. Thus every $\lambda \in \mathbb{R}$ is an eigenvalue of T with f_λ as a corresponding eigenfunction. In fact, any eigenfunction of T corresponding to λ is a scalar multiple of f_λ.

We now consider a vector space V of finite dimension. Let E be an ordered basis for V, and let $\mathbf{A} := \mathbf{M}_E^E(T)$, the matrix of the linear operator T with respect to E. We remark that if F is another ordered basis for V, and $\mathbf{B} := \mathbf{M}_F^F(T)$, the matrix of the linear operator T with respect to F, then $\mathbf{B}$ is similar to $\mathbf{A}$; in fact we have seen that $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, where $\mathbf{P} = \mathbf{M}_F^E(I)$, and where $I : V \to V$ is the identity map.

Definition

The **geometric multiplicity** *of an eigenvalue of T is the dimension of the corresponding eigenspace. It equals the geometric multiplicity of λ as an eigenvalue of the associated matrix* $\mathbf{A}$*. The* **characteristic polynomial** *of T is defined to be the characteristic polynomials of* $\mathbf{A}$*. Further, T is called* **diagonalizable** *if the matrix* $\mathbf{A}$ *is diagonalizable.*

Definition

The **algebraic multiplicity** *of an eigenvalue of the linear operator T is defined to be the algebraic multiplicity of the associated matrix* $\mathbf{A}$.

The relationships between the geometric multiplicity and the algebraic multiplicity of an eigenvalue of a square matrix hold for a linear operator as well.

The above definitions do not depend on the choice of the ordered basis E for V because if F is any other ordered basis of V, then the matrix $\mathbf{B} := \mathbf{M}_F^F(T)$ is similar to the matrix $\mathbf{A} := \mathbf{M}_E^E(T)$ as we have seen earlier.

Results about the linear independence of eigenvectors corresponding to distinct eigenvalues hold in the general case.

Inner Product Spaces

Definition

Let V be a vector space over $\mathbb{K}$. An **inner product** *on V is a function $\langle \cdot\,, \cdot\rangle : V \times V \to \mathbb{K}$ satisfying the following properties. For $u, v, w \in V$ and $\alpha, \beta \in \mathbb{K}$,*

1. $\langle v,\ v\rangle \geq 0$ *and* $\langle v,\ v\rangle = 0 \iff v = 0$, *(positive definite)*
2. $\langle u,\ \alpha v + \beta w\rangle = \alpha\langle u,\ v\rangle + \beta\langle u,\ w\rangle$, *(linear in 2nd variable)*
3. $\langle v,\ u\rangle = \overline{\langle u,\ v\rangle}$. *(conjugate symmetric)*

From the above properties, conjugate linearity in the 1st variable follows: $\langle \alpha u + \beta v,\ w\rangle = \overline{\alpha}\langle u,\ w\rangle + \overline{\beta}\langle v,\ w\rangle$.

A vector space V over $\mathbb{K}$ with a prescribed inner product on it is called an **inner product space**.

If $u, v \in V$ and $\langle u,\ v\rangle = 0$, then we say that u and v are **orthogonal**, and we write $u \perp v$.

For $v \in V$, we define the **norm** of v by $\|v\| := \langle v,\ v\rangle^{1/2}$.

If $v \in V$ and $\|v\| = 1$, then we say that v is a **unit vector** or a **unit function**. The set $\{v \in V : \|v\| \le 1\}$ is called the **unit ball** of V.

Examples

$\boxed{1}$. We have already studied the primary example, namely $V := \mathbb{K}^{n\times 1}$ with the **usual inner product** $\langle \mathbf{x},\ \mathbf{y}\rangle := \mathbf{x}^*\mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$. There are other inner products on $\mathbb{K}^{n\times 1}$. For example, let $w_1, \dots, w_n$ be positive real numbers, and define

$$\langle \mathbf{x},\ \mathbf{y}\rangle := w_1\overline{x}_1 y_1 + \cdots + w_n \overline{x}_n y_n \quad \text{for } \mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}.$$

On the other hand, the function on $\mathbb{R}^{4\times 1} \times \mathbb{R}^{4\times 1}$ defined by

$$\langle \mathbf{x},\ \mathbf{y}\rangle_M := x_1y_1 + x_2y_2 + x_3y_3 - x_4y_4 \quad \text{for } \mathbf{x}, \mathbf{y} \in \mathbb{R}^{4\times 1}$$

is not an inner product on $\mathbb{R}^{4\times 1}$. Note that for $\mathbf{x} \in \mathbb{R}^{4\times 1}$, $\langle \mathbf{x},\ \mathbf{x}\rangle_M = x_1^2 + x_2^2 + x_3^2 - x_4^2$. (This is used in defining the **Minkowski space**)

2. Let $V := C([a, b])$, the vector space of all continuous $\mathbb{K}$-valued functions on $[a, b]$. Define

$$\langle f, g \rangle := \int_a^b \overline{f(t)} g(t) dt \quad \text{for } f, g \in V.$$

It is easy to check that this is an inner product on V. We shall call this inner product the **usual inner product** on $C([a, b])$.

In this case, the norm of $f \in V$ is $\|f\| := \left(\int_a^b |f(t)|^2 dt \right)^{1/2}$.

This example gives a continuous analogue of the usual inner product on $\mathbb{K}^{n \times 1}$.

There are other inner products on V. For example, let $w : [a, b] \to \mathbb{R}$ be positive function, and define

$$\langle f, g \rangle := \int_a^b w(t) \overline{f(t)} g(t) dt \quad \text{for } f, g \in V.$$

Let w be a nonzero element of V. As earlier, define

$$P_w(v) := \frac{\langle w,\, v\rangle}{\langle w,\, w\rangle}\, w \quad \text{for } v \in V.$$

It is called the (orthogonal) **projection** of v in the direction of w. It is easy to see that $P_w : V \to V$ is a linear map and its image space is one dimensional. Also, $P_w(w) = w$, so that $(P_w)^2 := P_w \circ P_w = P_w$.

Two important properties of the projection of a vector in the direction of another (nonzero) vector are as follows.

Proposition

Let $w \in V$ be nonzero. Then for every $v \in V$,

(i) $\big(v - P_w(v)\big) \perp w$ and (ii) $\|P_w(v)\| \le \|v\|$.

The proof of (i) is an easy verification, and (ii) follows from the formula $\|v\|^2 = \|P_w(v)\|^2 + \|v - P_w(v)\|^2$, which is a consequence of (i).

Theorem

Let $\langle \cdot\,, \cdot \rangle$ be an inner product on a vector space V, and let $v, w \in V$. Then
(i) (Schwarz Inequality) $|\langle v,\, w\rangle| \leq \|v\|\|w\|$.

(ii) (Triangle Inequality) $\|v + w\| \leq \|v\| + \|w\|$.

Proof. (i) First, suppose $w = 0$. Then for any $v \in V$, $\langle v,\, w\rangle = \langle v,\, 0\rangle = \langle v,\, 0 + 0\rangle = 2\langle v,\, 0\rangle$, and so $\langle v,\, w\rangle = 0$. Also, $\|w\| = 0$. Hence we are done.
Now suppose $w \neq 0$. Then by (ii) of the previous proposition,

$$\left\| \frac{\langle w,\, v\rangle}{\langle w,\, w\rangle} w \right\| = \|P_w(v)\| \leq \|v\|,$$

that is,

$$|\langle w,\, v\rangle| \|w\| \leq \|v\| \langle w,\, w\rangle = \|v\|\|w\|^2.$$

Hence $|\langle v,\, w\rangle| \leq \|v\|\|w\|$.

(ii) Since $\langle v,\ w\rangle + \langle w,\ v\rangle = 2\,\Re\,\langle v,\ w\rangle$, we see that

$$\begin{aligned}
\|v+w\|^2 &= \langle v+w,\ v+w\rangle = \|v\|^2 + \|w\|^2 + 2\,\Re\,\langle v,\ w\rangle \\
&\le \|v\|^2 + \|w\|^2 + 2\,|\langle v,\ w\rangle| \\
&\le \|v\|^2 + \|w\|^2 + 2\,\|v\|\|w\| \quad \text{(by (i) above)} \\
&= (\|v\| + \|w\|)^2.
\end{aligned}$$

Thus $\|v+w\| \le \|v\| + \|w\|$. $\square$

We observe that the norm function $\|\cdot\| : V \to \mathbb{K}$ satisfies the following three crucial properties:

(i) $\|v\| \ge 0$ for all $v \in V$ and $\|v\| = 0 \iff v = 0$,

(ii) $\|\alpha v\| = |\alpha|\|v\|$ for all $\alpha \in \mathbb{K}$ and $v \in V$,

(iii) $\|v+w\| \le \|v\| + \|w\|$ for all $v, w \in V$.

Let V be an inner product space. Let E be a subset of V. Define

$$E^{\perp} := \{w \in V : w \perp v \text{ for all } v \in E\}.$$

It is easy to see that $E^{\perp}$ is a subspace of V.

The set E is said to be **orthogonal** if any two (distinct) elements of E are orthogonal (to each other), that is, $v \perp w$ for all v, w in E with $v \neq w$. An orthogonal set whose elements are unit vectors is called an **orthonormal set**.

If E is orthogonal and does not contain 0, then E is linearly independent. For example, let $V := C[-\pi, \pi]$ and $E := \{\cos nt : n \in \mathbb{N}\} \cup \{\sin nt : n \in \mathbb{N}\}$. Since E is orthogonal and $0 \notin E$, the set E is linearly independent.

If we are given a sequence of linearly independent elements of V, then we can construct an orthogonal subset of V not containing 0, retaining the span of the elements so constructed at every stepby the Gram-Schmidt Orthogonalization Process (G-S OP), just as discussed earlier.

Let (v_n) be a sequence of linearly independent elements in V. Define $w_1 := v_1$, and for $j \in \mathbb{N}$, define

$$\begin{aligned} w_{j+1} &:= v_{j+1} - P_{w_1}(v_{j+1}) - \cdots - P_{w_j}(v_{j+1}) \\ &= v_{j+1} - \frac{\langle w_1,\ v_{j+1}\rangle}{\langle w_1,\ w_1\rangle} w_1 - \cdots - \frac{\langle w_j,\ v_{j+1}\rangle}{\langle w_j,\ w_j\rangle} w_j. \end{aligned}$$

Then $\text{span}\{w_1, \ldots, w_{j+1}\} = \text{span}\{v_1, \ldots, v_{j+1}\}$, and the set $\{w_1, \ldots, w_{j+1}\}$ is orthogonal.

Now let $u_j := w_j/\|w_j\|$ for $j \in \mathbb{N}$, then $(u_1, u_2, \ldots)$ is an ordered orthonormal set such that for each $j \in \mathbb{N}$,

$$\text{span}\{v_1, \ldots, v_j\} = \text{span}\{w_1, \ldots, w_j\} = \text{span}\{u_1, \ldots, u_j\}.$$

Example

Let V be the set of all real-valued polynomial functions on $[-1, 1]$ along with the inner product defined by

$$\langle p,\ q\rangle := \int_{-1}^{1} p(t)q(t)dt \quad \text{for } p, q \in V.$$

For $j = 0, 1, 2, \ldots$, let $p_j(t) := t^j$, $t \in [-1, 1]$. Let us orthogonalize the set $\{p_0, p_1, p_2, p_3\}$. Define $q_0 := p_0$, and

$$q_1 := p_1 - \frac{\langle q_0,\ p_1\rangle}{\langle q_0,\ q_0\rangle} q_0 = p_1 - \left(\frac{1}{2}\int_{-1}^{1} t\,dt\right) p_0 = p_1.$$

Next, define

$$
\begin{aligned}
q_2 &:= p_2 - \frac{\langle q_0,\ p_2\rangle}{\langle q_0,\ q_0\rangle} q_0 - \frac{\langle q_1,\ p_2\rangle}{\langle q_1,\ q_1\rangle} q_1 \\
&= p_2 - \left(\frac{1}{2}\int_{-1}^{1} t^2 dt\right) q_0 - \left(\frac{3}{2}\int_{-1}^{1} t^3 dt\right) q_1 \\
&= p_2 - \frac{1}{3} p_0,
\end{aligned}
$$

and similarly,

$$
\begin{aligned}
q_3 &:= p_3 - \frac{\langle q_0,\ p_3\rangle}{\langle q_0,\ q_0\rangle} q_0 - \frac{\langle q_1,\ p_3\rangle}{\langle q_1,\ q_1\rangle} q_1 - \frac{\langle q_2,\ p_3\rangle}{\langle q_2,\ q_2\rangle} q_2 \\
&= p_3 - \frac{3}{5} p_1.
\end{aligned}
$$

.

Further, $\|q_0\| = \sqrt{2}$, $\|q_1\| = \sqrt{2}/\sqrt{3}$, $\|q_2\| = 2\sqrt{2}/3\sqrt{5}$ and $\|q_3\| = 2\sqrt{2}/5\sqrt{7}$.

Hence we obtain the following orthonormal subset of V having the same span as $\text{span}\{p_0, p_1, p_2, p_3\}$, namely all real-valued polynomial functions of degree at most 3:

$$u_0(t) := \frac{\sqrt{2}}{2}, \quad u_1(t) := \frac{\sqrt{6}}{2}\, t,$$
$$u_2(t) := \frac{\sqrt{10}}{4}\,(3t^2 - 1), \quad u_3(t) := \frac{\sqrt{14}}{4}\,(5t^3 - 3t).$$

The sequence of orthonormal polynomials thus obtained by orthonormalizing the monomials by the G-S OP is known as the sequence of **Legendre polynomials**. These are of much use in many contexts.

Let V be a finite dimensional inner product space. An **orthonormal basis** for V is a basis for V which is an orthonormal subset of V.

We have proved the following results for subspaces of $\mathbb{K}^{n\times 1}$. Their proofs remain valid for any inner product space.

If $u_1, \ldots, u_k$ is an orthonormal set in V, then we can extend it to an orthonormal basis. As a consequence, every nonzero vector subspace V has an orthonormal basis.

The G-S OP enables us to improve the quality of a given basis for V by orthonormalizing it. For instance, if $\{u_1, \ldots, u_n\}$ is an orthonormal basis for V, and $v \in V$, then it is extremely easy to write v as a linear combination of $u_1, \ldots, u_n$; in fact

$$v = \langle u_1,\ v\rangle u_1 + \cdots + \langle u_n,\ v\rangle u_n.$$

Orthogonal Projections

Let W be a subspace of a finite dimensional inner product space V. The **Orthogonal Projection Theorem** says that for every $v \in V$, there are unique $w \in W$ and $\tilde{w} \in W^{\perp}$ such that $v = w + \tilde{w}$, that is, $V = W \oplus W^{\perp}$. The map $P_W : V \to V$ given by $P_W(v) = w$ is linear and satisfies $(P_W)^2 = P_W$. It is called the **orthogonal projection map** of V onto the subspace W.

In fact, if $u_1, \ldots, u_k$ is an orthonormal basis for W, then

$$P_W(v) = \langle u_1,\, v\rangle u_1 + \cdots + \langle u_k,\, v\rangle u_k \quad \text{for} \quad v \in V.$$

Given $v \in V$, its orthogonal projection $P_W(v)$ is the **unique best approximation to v from W**.

Further, $P_W(v)$ is the unique element of W such that $v - P_W(v)$ is orthogonal to W.

Definition

Suppose V is an inner product space of dimension n. For a linear operator $T : V \to V$, define its **adjoint** $T^* : V \to V$ *by*

$$\langle T(u), v\rangle = \langle u, T^*(v)\rangle \quad \textit{for all } u, v \in V.$$

Define T to be **Hermitian** *or* **self-adjoint** *if $T = T^*$, and* **skew-Hermitian** *or* **skew self-adjoint** *if $T = -T^*$.*

Thus, T is **Hermitian** if

$$\langle T(u), v\rangle = \langle u, T(v)\rangle \quad \text{for all } u, v \in V,$$

and T is **skew-Hermitian** if

$$\langle T(u), v\rangle = -\langle u, T(v)\rangle \quad \text{for all } u, v \in V,$$

Note that for $\mathbf{A} \in \mathbb{K}^{n\times n}$ and $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n\times 1}$,

$$\langle \mathbf{A}\,\mathbf{x},\ \mathbf{y}\rangle = (\mathbf{A}\,\mathbf{x})^*\mathbf{y} = \mathbf{x}^*(\mathbf{A}^*\mathbf{y}) = \langle \mathbf{x},\ \mathbf{A}^*\mathbf{y}\rangle.$$

Hence a matrix $\mathbf{A}$ is self-adjoint, that is, $\mathbf{A}^* = \mathbf{A}$ if and only if $\langle \mathbf{A}\,\mathbf{x},\ \mathbf{y}\rangle = \langle \mathbf{x},\ \mathbf{A}\mathbf{y}\rangle$. The following result, therefore, is natural.

Proposition

Let V be a finite dimensional inner product space, and let $T : V \to V$ be a linear operator. Then T is Hermitian if and only if the matrix of T with respect to any ordered orthonormal basis of V is self-adjoint.

An operator T which commutes with its adjoint T^* will be called **normal operator** on V. One can prove the spectral theorem for a normal operator on a finite dimensional inner product space V just as before.

THE END